DATA SHEET

(DOC No. HX8399-A-DS)

HX8399-A

1200RGB x 1920 dot, 16.7M color, LTPS Mobile Single Chip Driver

Temporary Version 01 June, 2013

Himax Technologies, Inc.
http://www.himax.com.tw

Revision History

June, 2013

Version	Date	Description of Changes
01	2013/06/13	New setup
01.01	2013/06/13	Modify OTP flow in P.128~P.132

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1. General Description

This document describes Himax's HX8399-A supports Full HD resolution driving controller. The HX8399-A is designed to provide a single-chip solution that combines a source driver, gate driver control, power supply circuit to drive a LTPS dot matrix LCD with 1200RGBx1920 dots at maximum.

The HX8399-A can be operated in low-voltage condition for the interface and integrated internal boosters that produce the liquid crystal voltage, breeder resistance and the voltage follower circuit for liquid crystal driver. In addition, The HX8399-A also supports various functions to reduce the power consumption of a LCD system via software control.

The HX8399-A supports MIPI DSI (Display Serial Interface) interface mode, I2C and SPI. The interface mode is selected by the external hardware pins IM2~0.

The HX8399-A is suitable for any small portable battery-driven and long-term driving products, such as small PDAs, digital cellular phones and bi-directional pagers.

2. Features

2.1 Display

- Single chip solution for a Full HD LTPS type LCD display
- Resolution:
  - 1200RGB x 1920
  - 1080RGB x 1920
  - 1080RGB x 1440
  - 1024RGB x 1600
  - 1024RGB x 1280
  - 960RGB x 1600
  - 960RGB x 1440
  - 960RGB x 1280
  - 900RGB x 1600
  - 900RGB x 1440
  - 800RGB x 1280
  - 720RGB x 1280
- Display color modes
  - Full color mode:
    - 16.7M colours (24-bit 8(R):8(G):8(B))
  - Reduce color mode:
    - 262k colours (18-bit 6(R):6(G):6(B))
    - 65k colours (16-bit 5(R):6(G):5(B))
    - 8 colors (Idle mode on): 8 colors (3-bit binary mode)

2.2 Display module

- Support 1202 source channel outputs
- Internal level shifter for Gate Driver control
- Supports 1-dot / 2-dot / 4-dot / 8-dot / column / pixel column inversion and Zig-Zag inversion
- Gamma correction
- On module VCOM control (-4 to -0.3V common electrode output voltage range)
- On module DC/DC converter
  - VSP_REG=4.6 to 6.1V
  - VSN_REG=-4.6 to -6.1V
  - Positive source output voltage level: VSPR=3.1V to 6.1V
  - Negative source output voltage level: VSNR=-3.1V to -6.1V
  - VCL=-2.5~-3.1V or –VDD3
  - VGH=7.5V to 15V
  - VGL=-7.5V to -15V
  - Positive gate driver output voltage level: VGHO=6V to 12.3V
  - Negative gate driver output voltage level: VGLO=-5V to -11.3V
  - VCOM=-4V to -0.3V, a step=10mV

2.3 Display / Control interface

- Display interface types supported
  - MIPI-DSI (Display Serial Interface) interface
    - Support DSI Version 1.1
    - Support D-PHY Version 1.1
  - MIPI-DSI + I2C interface
  - MIPI-DSI + DBI TypeC(Option1/Option3) interface

2.4 Input power

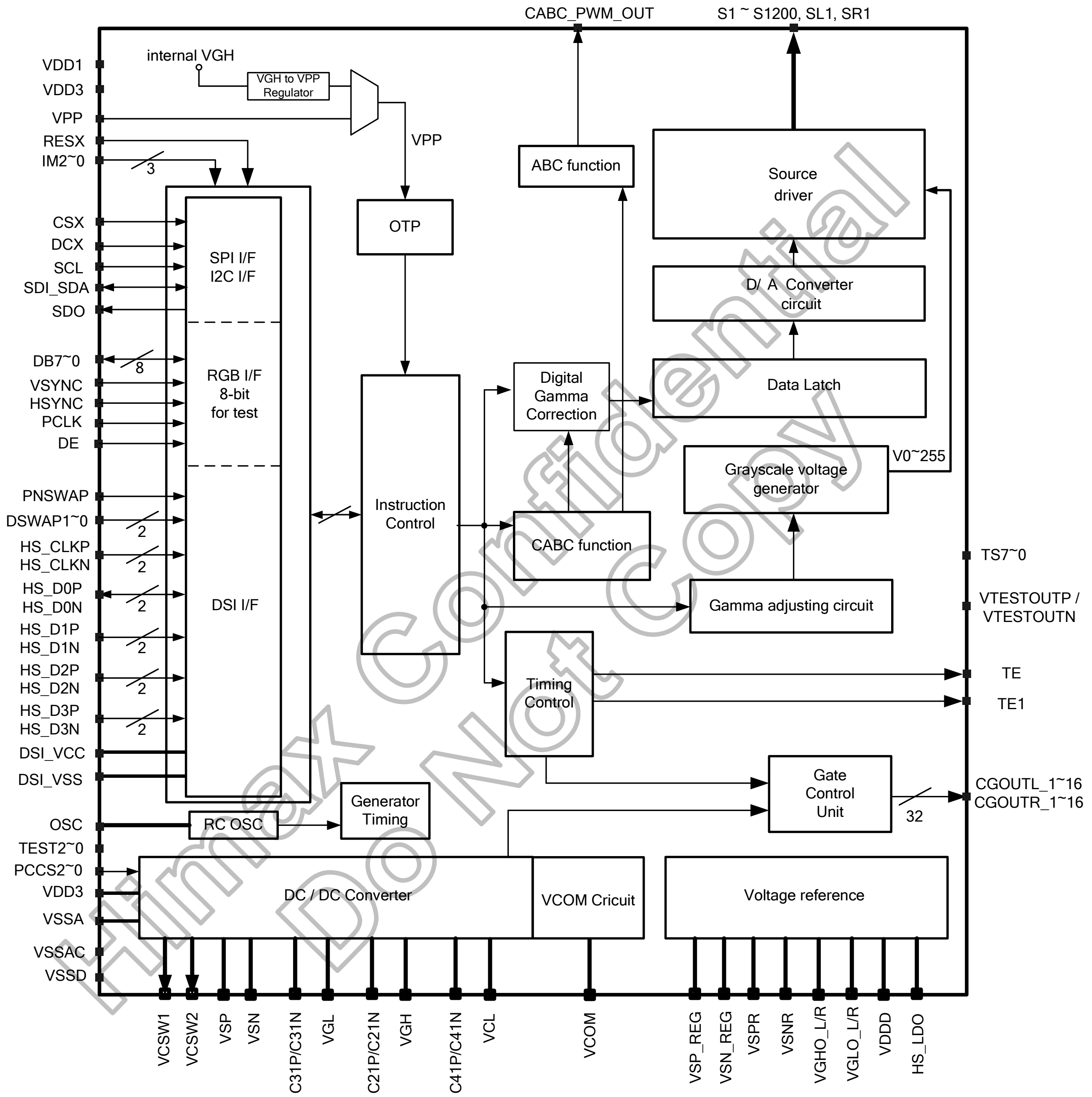
- I/O and Logic power supply (VDD1): 1.65V to 3.3V
- Analog power supply (VDD3): 2.5V to 3.6V
- High speed interface power supply (HS_VCC): 1.65V to 3.3V
- OTP programming voltage (VPP): 7.5V ± 0.2V
- Positive source driver power (External input mode VSP): 4.8 to 6.75V
- Negative source driver power (External input mode VSN): -4.8 to -6.75V

2.5 Miscellaneous

- Software programmable color depth mode
- Oscillator for display clock generation
- Low power consumption, suitable for battery operated systems
- CMOS compatible inputs
- Proprietary multi phase driving for lower power consumption
- GAS function for preventing image sticking when abnormal power off
- Optimized layout for COG assembly
- DC/DC converter for source
- Support DC COM driving
- VCOM voltage generator
- On-chip OTP program voltage generator
- OTP memory to store initialization register settings
- 3 times MTP for VCOM and ID setting
- Support CABC (Content Adaptive Brightness Control) function
- Support Color Enhancement function
- Support DGC (Digital Gamma Correction) function
- Support SLR (Sun Light Readability)
- Operation temperature range: -40 to +85 ℃

3. Device Overview

3.1 Block diagram



3.2 Pin description

<table>
<tr><th colspan="5">Host interface pins</th></tr>
<tr><th>Signals</th><th>I/O</th><th>Pin no.</th><th>Connected with</th><th>Description</th></tr>
<tr><td>IM2 ~ IM0</td><td>I</td><td>3</td><td>VDD1 / VSSD</td><td>These pins must be connected to VDD1 or VSSD.

Interface mode is selected as listed below:
<table>
<tr><th>IM2</th><th>IM1</th><th>IM0</th><th>interface mode</th><th>DB pins</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>DSI Video mode + I2C</td><td>HS_D0P, HS_D0N, HS_D1P, HS_D1N, HS_D2P, HS_D2N, HS_D3P, HS_D3N, SDA</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>Reserved</td><td>Reserved</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>DSI Video mode + DBI TypeC Option1</td><td>HS_D0P, HS_D0N, HS_D1P, HS_D1N, HS_D2P, HS_D2N, HS_D3P, HS_D3N, SDI, SDO</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>DSI Video mode + DBI TypeC Option3</td><td>HS_D0P, HS_D0N, HS_D1P, HS_D1N, HS_D2P, HS_D2N, HS_D3P, HS_D3N, SDI, SDO</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>Reserved</td><td>Reserved</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>Reserved</td><td>Reserved</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>DSI Video mode</td><td>HS_D0P, HS_D0N, HS_D1P, HS_D1N, HS_D2P, HS_D2N, HS_D3P, HS_D3N</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>Reserved</td><td>Reserved</td></tr>
</table>
Pixel format (RGB565 / RGB666 / RGB888) is selected by DCS command (0x3Ah).</td></tr>
<tr><td>PNSWAP, DSWAP1~0</td><td>I</td><td>3</td><td>VDD1 / VSSD</td><td>These pins must be connected to VDD1 or VSSD to set 1 or 0.
PNSWAP and DSWAP1~0 are used for the combination of polarity swap and data lane swap of DSI.
<table>
<tr><th>PNSWAP</th><th>DSWAP [1:0]</th><th>HS_D2P</th><th>HS_D2N</th><th>HS_D1P</th><th>HS_D1N</th><th>HS_CKP</th><th>HS_CKN</th><th>HS_D0P</th><th>HS_D0N</th><th>HS_D3P</th><th>HS_D3N</th></tr>
<tr><td rowspan="4">0</td><td>00</td><td>D3-</td><td>D3+</td><td>D2-</td><td>D2+</td><td>CLK-</td><td>CLK+</td><td>D1-</td><td>D1+</td><td>D0-</td><td>D0+</td></tr>
<tr><td>01</td><td>D3-</td><td>D3+</td><td>D0-</td><td>D0+</td><td>CLK-</td><td>CLK+</td><td>D1-</td><td>D1+</td><td>D2-</td><td>D2+</td></tr>
<tr><td>10</td><td>D0-</td><td>D0+</td><td>D1-</td><td>D1+</td><td>CLK-</td><td>CLK+</td><td>D2-</td><td>D2+</td><td>D3-</td><td>D3+</td></tr>
<tr><td>11</td><td>D2-</td><td>D2+</td><td>D1-</td><td>D1+</td><td>CLK-</td><td>CLK+</td><td>D0-</td><td>D0+</td><td>D3-</td><td>D3+</td></tr>
<tr><td rowspan="4">1</td><td>00</td><td>D3+</td><td>D3-</td><td>D2+</td><td>D2-</td><td>CLK+</td><td>CLK-</td><td>D1+</td><td>D1-</td><td>D0+</td><td>D0-</td></tr>
<tr><td>01</td><td>D3+</td><td>D3-</td><td>D0+</td><td>D0-</td><td>CLK+</td><td>CLK-</td><td>D1+</td><td>D1-</td><td>D2+</td><td>D2-</td></tr>
<tr><td>10</td><td>D0+</td><td>D0-</td><td>D1+</td><td>D1-</td><td>CLK+</td><td>CLK-</td><td>D2+</td><td>D2-</td><td>D3+</td><td>D3-</td></tr>
<tr><td>11</td><td>D2+</td><td>D2-</td><td>D1+</td><td>D1-</td><td>CLK+</td><td>CLK-</td><td>D0+</td><td>D0-</td><td>D3+</td><td>D3-</td></tr>
</table></td></tr>
<tr><td>CSX</td><td>I</td><td>1</td><td>MPU</td><td>Chip select pin.
Low: Chip can be accessed.
High: Chip can not be accessed.
If not use, please connect it to VSSD or VDD1.</td></tr>
<tr><td>RESX</td><td>I</td><td>1</td><td>MPU or reset circuit</td><td>Reset pin. Setting either pin low initializes the LSI. Must be reset after power is supplied (Must be connected to VSSD or VDD1).</td></tr>
<tr><td>DCX</td><td>I</td><td>1</td><td>MPU</td><td>Command/parameter selection for DBI TypeC Option3.
If not use, please connect it to VSSD or VDD1.</td></tr>
<tr><td>SCL</td><td>I</td><td>1</td><td>MPU</td><td>Serves as serial clock in serial bus interface.
Serves as serial iput/output clock in I2C interface.
If not use, please connect it to VSSD or VDD1.</td></tr>
<tr><td>DB7~0</td><td>I/O</td><td>8</td><td>MPU</td><td>Only for internal test.
Let the unused pins open or connect it to VSSD or VDD1.</td></tr>
<tr><td>SDO</td><td>O</td><td>1</td><td>MPU</td><td>Serial data output pin.
If not used, let it open .</td></tr>
<tr><td>SDI_SDA</td><td>I</td><td>3</td><td>MPU</td><td>SDI: Serial data input pin in serial bus interface.
SDA: Serial input/output data in I2C bus interface.
If not use, please connect it to VSSD or VDD1.</td></tr>
<tr><td>HSYNC</td><td>I</td><td>1</td><td>MPU</td><td>Only for internal test.</td></tr>
</table>

				Please connect it to VSSD or VDD1.
DE	**I**	**1**	**MPU**	Only for internal test. Please connect it to VSSD or VDD1.
VSYNC	**I**	**1**	**MPU**	Only for internal test. Please connect it to VSSD or VDD1.
PCLK	**I**	**1**	**MPU**	Only for internal test. Please connect it to VSSD or VDD1.
Source driver output pins				
S1 to S1200 SL1, SR1	**O**	**1202**	**LCD**	Output voltages applied to the liquid crystal. If source output less than 1200, please let unused source pins open. SL1 and SR1 are for Zig-Zag inversion used.

H_RES[2:0]	V_RES[1:0]	Resolution	Source channels
000	00	1080RGBx1920	S1 ~ S540 , S661 ~S1200
000	10	1080RGBx1440	S1 ~ S540 , S661 ~S1200
001	11	1024RGBx1280	S1 ~ S512 , S689 ~ S1200
001	01	1024RGBx1600	S1 ~ S512 , S689 ~ S1200
010	01	960RGBx1600	S1 ~ S480 , S721 ~ S1200
010	10	960RGBx1440	S1 ~ S480 , S721 ~ S1200
010	11	960RGBx1280	S1 ~ S480 , S721 ~ S1200
011	01	900RGBx1600	S1 ~ S450 , S751 ~ S1200
011	10	900RGBx1440	S1 ~ S450 , S751 ~ S1200
100	00	1200RGBx1920	S1 ~ S1200
101	11	720RGBx1280	S1 ~ S360 , S841 ~ S1200
110	11	800RGBx1280	S1 ~ S400 , S801 ~ S1200
111	-	Inhibited	-

Gate Driver control singal				
CGOUTL_1~16, CGOUTR_1~16	**O**	**64**	**LCD**	These are pins are LTPS control signal, the function can be selected by register setting. These pins output high/low level VGHO/VGLO.
Power supply pins				
VDD1	**I**	**21**	**Power supply**	A power supply for the I/O circuit and logic power. VDD1=1.65 to 3.3V
VDD3	**I**	**30**	**Power supply**	A power supply for the analog power. VDD3=2.5V to 3.6V
VPP	**I**	**4**	**Power supply**	External high voltage pin used in OTP mode and operates at 7.5V. If not used, let it open.
VSSA	**P**	**48**	**Power supply**	Analoge ground. VSSA=0V. When using the COG method, connect to VSSD on the FPC to prevent noise.
VSSAC	**P**	**6**	**Power supply**	Analoge ground. Must connect to VSSA on the FPC.
VSSD	**P**	**63**	**Power supply**	Ground for the internal logic. VSSD=0V. When using the COG method, connect to VSSA on the FPC to prevent noise.
Power supply pins				
VSP	**I**	**15**	**Stabilizing capacitor**	Input voltage generated from HX5186, PFM or external (4.8V to 6.75V).
VSN	**I**	**15**	**Stabilizing capacitor**	Input voltage generated from HX5186, PFM or external (-4.8V to -6.75V).
VSP_REG	**O**	**30**	**Stabilizing capacitor**	Output regulated voltage for panel voltage. It is generated from VSP. Connect to a stabilizing capacitor between VSSA and VSP_REG.
VSN_REG	**O**	**30**	**Stabilizing capacitor**	Output regulated voltage for panel voltage. It is generated from VSN. Connect to a stabilizing capacitor between VSSA and VSN_REG.
VCL	**O**	**24**	**Stabilizing capacitor**	Output voltage from the set-up circuit (-VDD3 or -2.5~-3.1V). it is generated from VDD3.
VSPR	**O**	**1**	**-**	Positive regulated voltage output (3.1V to 6.1V) for Gamma.
VSNR	**O**	**1**	**-**	Negative regulated voltage output (-3.1V to -6.1V) for Gamma.
VDDD	**O**	**12**	**Stabilizing capacitor**	Internal logic voltage output(1.35V)
VGH	**O**	**6**	**Stabilizing capacitor**	Output voltage from the step-up circuit. Connect to a stabilizing capacitor between VSSA and VGH.
VGL	**O**	**6**	**Stabilizing capacitor**	Output voltage from the step-up circuit. Connect to a stabilizing capacitor between VSSA and VGL.
VGHO_L, VGHO_R	**O**	**6, 3**	**Stabilizing capacitor**	Output regulated voltage for panel voltage. It is generated from VGH. Connect to VGHO_L and VGHO_R on FPC and with a stabilizing capacitor between VSSA and VGHO.

VGLO_L, VGLO_R	**O**	**3, 6**	**Stabilizing capacitor**	Output regulated voltage for panel voltage. It is generated from VGL. Connect VGLO_L and VGLO_R on FPC and with a stabilizing capacitor between VSSA and VGLO
VCOM	**O**	**45**	**Stabilizing capacitor**	The power supply of common voltage in DC com driving. The voltage range is set between -4V to -0.3V. It must be connected a stabilizing capacitor 2.2uF to VSSD.
DC/DC pumping				
PCCS2 PCCS1 PCCS0	**I**	**1 2 2**	**VDD1 / VSP/ VSSD**	These pins are for analog power source mode selection. Note: PCCS2 high level is VDD1; PCCS1 & PCCS0 high level is VSP.

PCCS[2:0]	Power Source	Driving Mode
000	VDD3 / VDD1	PFM Type C
001	VDD3 / VDD1	HX5186-A/B/C
010	VSP / VDD1	PFM Type B(for VSN)
011	VSP / VSN / VDD1	External-1
101	VDD3 /VDD1	PFM Type A
111	VDD3 / VSP / VSN / VDD1	External-2
Others	Reserved	Reserved

C31P, C31N	**I/O**	**9,9**	**Step-up Capacitor**	Connect to the step-up capacitor according to the DC/DC pumping factor by pumping the VGL voltage.
C21P, C21N	**I/O**	**9,8**	**Step-up Capacitor**	Connect to the step-up capacitor according to the DC/DC pumping factor by pumping the VGH voltage.
C41P, C41N	**I/O**	**6,6**	**Step-up Capacitor**	Connect to the step-up capacitor according to the DC/DC pumping factor by pumping the VCL voltage.
VCSW1, VCSW2	**O**	**2,3**	**-**	**In external input power mode:** Not used, Please open these pin. **In HX5186 mode:** VCSW1 and VCSW2 connect to HX5186-A/B/C. **In PFM mode:** VCSW1 and VCSW2 connect to external MOS. Detail connection, please refer to DC/DC conveter circuit
CABC & ABC & Ambient light sensor(general purpose input and output)				
CABC_PWM_OUT	**O**	**1**	**LED driver**	Backlight on/fff control pin. If use CABC function, the pin can connect to external LED driver IC. The output voltage range=0 to VDD1.
TE	**O**	**1**	**-**	Serves TE (Tearing Effect) output pin. Ouput signal can be selected by register setting, please refere to SETGPO.
TE1	**O**	**1**	**-**	Serves TE (Tearing Effect) pin of each scan line. Ouput signal can be selected by register setting, please refere to SETGPO.
High speed interface parts				
HS_D0P, HS_D0N	**I/O**	**6,6**	**DSI Host**	MIPI-DSI Data differential signal input pins. (Data lane 0) if not used , Please connected to VSSD or open..
HS_CLKP, HS_CLKN	**I**	**6,6**	**DSI Host**	MIPI-DSI CLOCK differential signal input pins. if not used , Please connected to VSSD or open..
HS_D1P, HS_D1N	**I**	**6,6**	**DSI Host**	MIPI-DSI Data differential signal input pins. (Data lane 1) if not used , Please connected to VSSD or open.
HS_D2P, HS_D2N	**I**	**6,6**	**DSI Host**	MIPI-DSI Data differential signal input pins. (Data lane 2) if not used , Please connected to VSSD or open..
HS_D3P, HS_D3N	**I**	**6,6**	**DSI Host**	MIPI-DSI Data differential signal input pins. (Data lane 3) if not used , Please connected to VSSD or open..
HS_VCC	**P**	**9**	**Power Supply**	Power supply for the MIPI DSI analog power. HS_VCC=1.65V to 3.3V
HS_VSS	**P**	**21**	**Ground**	MIPI DSI analog ground. HS_VSS=0V. When using the COG method, connect to VSSA on the FPC to prevent noise.
HS_LDO	**O**	**6**	**Capacitor**	DSI I/F: DSI regulator output pin. (1.5V) Connect to a stabilizing capacitor between HS_LDO and HS_VSS If not used, please open these pins.
Test Pins				
OSC	**I**	**1**	**Open**	Oscillator input for test purpose. If not used, please let it open or connected to VSSD.(weak pull low)
TEST2~0	**I**	**3**	**Open**	Test pins. These pins are for internal logic function test.These pina can

				output on FPC. If not used, let it open or connected to VSSD.(weak pull low)
TS7~0	**O**	**8**	**Open**	Test pins. Disconnect these pins.
VTESTOUT_P	**O**	**1**	**Open**	A test pin. Disconnect it. This pin can output on FPC.
VTESTOUT_N	**O**	**1**	**Open**	A test pin. Disconnect it. This pin can output on FPC.
DUMMYR1 **DUMMYR2**	**-**	**2**	**Open**	Dummy pads. Available for measuring the COG contact resistance. They are short-circuited within the chip.
DUMMY	**-**	**219**	**Open**	Not used. Let it open.

4. Interface

4.1 System interface

The HX8399-A supports I2C interface and MIPI interfaces: DBI (Display Bus Interface), DSI (Display Serial Interface). Where DBI supports Serial interface (Type C Option1 and Option3). The interface mode can be selected by IM2-0 pins setting as show in Table 4.1.

IM2	IM1	IM0	interface mode	DB pins	Note
0	0	0	DSI Video mode + I2C	SDA, HS_D0P/N~HS_D3P/N	-
0	1	0	DSI Video mode + DBI TypeC Option1	SDI, SDO, HS_D0P/N~HS_D3P/N	-
0	1	1	DSI Video mode + DBI TypeC Option3	SDI, SDO, HS_D0P/N~HS_D3P/N	-
1	1	0	DSI Video mode	HS_D0P/N~HS_D3P/N	-
Other setting			Reserved	Reserved	-

Table 4.1: Interface selection

Interface	CSX	SCL	DCX	Input/Output pin
DSI Video mode + I2C	CSX	SCL	Unused	SDA, HS_CLKP, HS_CLKN, HS_D0P, HS_D0N, HS_D1P, HS_D1N, HS_D2P, HS_D2N, HS_D3P, HS_D3N
DSI Video mode + DBI TypeC Option1	CSX	SCL	Unused	SDI, SDO, HS_CLKP, HS_CLKN, HS_D0P, HS_D0N, HS_D1P, HS_D1N, HS_D2P, HS_D2N, HS_D3P, HS_D3N
DSI Video mode + DBI TypeC Option3	CSX	SCL	DCX	SDI, SDO, HS_CLKP, HS_CLKN, HS_D0P, HS_D0N, HS_D1P, HS_D1N, HS_D2P, HS_D2N, HS_D3P, HS_D3N
DSI Video mode	Unused	Unused	Unused	HS_CLKP, HS_CLKN, HS_D0P, HS_D0N, HS_D1P, HS_D1N, HS_D2P, HS_D2N, HS_D3P, HS_D3N

Table 4.2: Pin connection based on different interface

4.2 Serial data transfer interface (MIPI DBI-TypeC)

The HX8399-A supports two type serial data transfer interface, the interface selection by setting IM2-0 pins. The IM2-0 set “010” is select 3-wire Option1 serial bus. The IM2-0 is set “011” when select 4-wire Option3 serial bus.

The 3-wire serial bus is use: chip select line (CSX), serial input/output data (SDI and SDO) and the serial transfer clock line (SCL).The 4-wire serial bus is use: chip select line (CSX), data/command select (DCX), serial input/output data (SDI and SDO) and the serial transfer clock line (SCL).

4.2.1 Serial data write mode

The 3-pin serial data packet contains a control bit D/CX and a transmission byte and in 4-pin serial case, data packet contains just transmission byte and control signal D/CX is transferred by DCX pin. If DCX is low, the transmission byte is command byte. If D/CX is high, the transmission byte is stored in to command register. The MSB is transmitted first. The serial interface is initialized when CSX is high. In this state, SCL clock pulse or serial input/output data (SDI and SDO) have no effect. A falling edge on CSX enables the serial interface and indicates the start of data transmission.

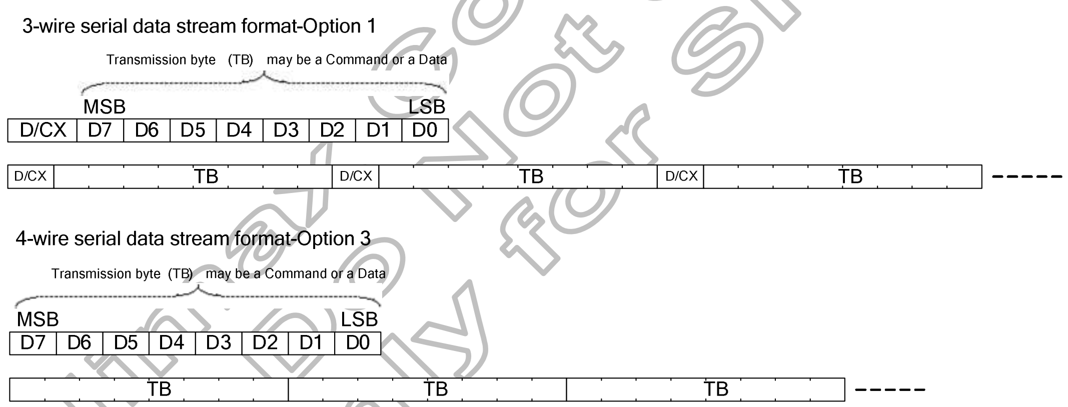
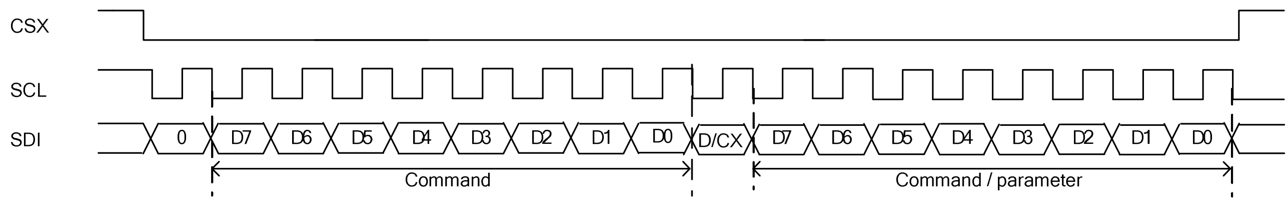


Figure 4.1: Serial data stream, write mode

DBI Type C: Interface protocol-Option 1 (3-wire)



DBI Type C: Interface protocol Option 3 (4-wire)

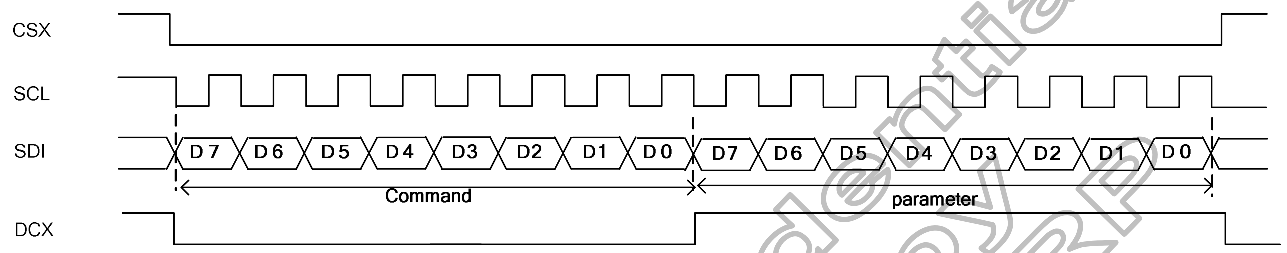
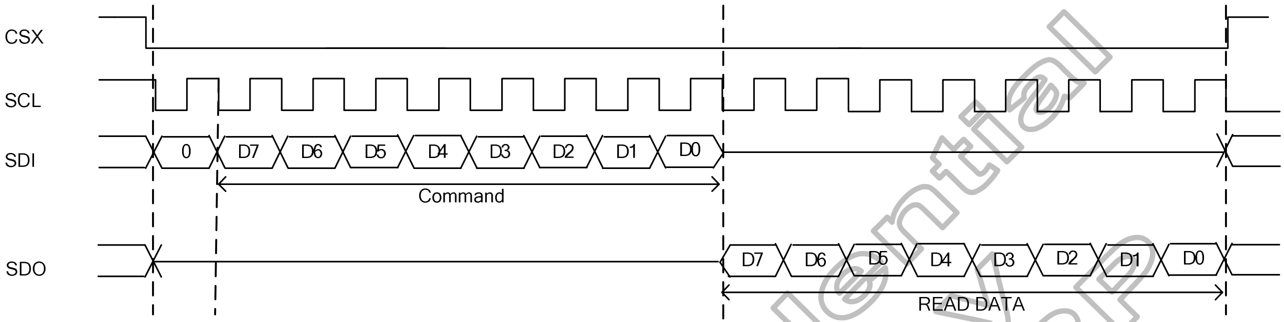


Figure 4.2: DBI Type C: Serial interface protocol 3-wire/4-wire, write mode

4.2.2 Serial data read mode

The micro-controller first has to send a command and then the following byte is transmitted in the opposite direction. The 3-wire serial read data format which just needs 8-bit.

DBI Type C: Interface protocol-Option 1 (3-wire)



DBI Type-C Interface Protocol - Option 3 (4 wire)

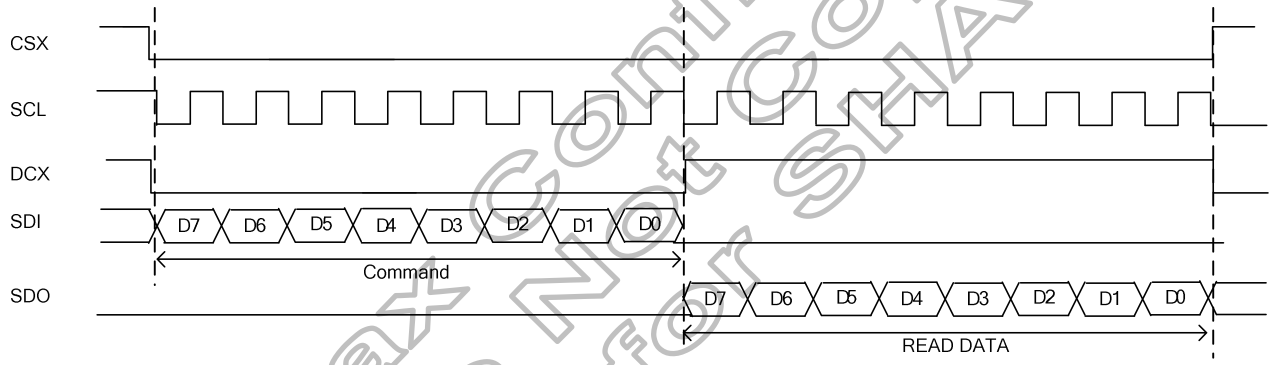


Figure 4.3: Type C:Serial interface protocol 3-wire/4-wire read mode

If there is a break on data transmission when transmit a command before a whole byte has been completed, then the display module will have reset the interface such that it will be ready to receive the same byte re-transmitted when the chip select line (CSX) is next activated. See the following figure.

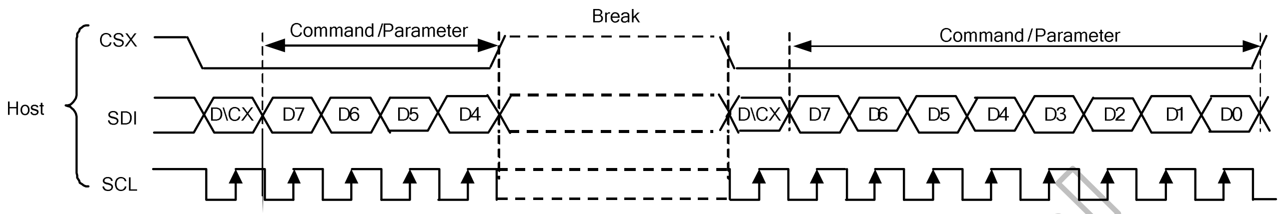


Figure 4.4: Display module data transfer recovery

4.3 I2C interface

The HX8399-A supports I2C interface, the interface selection by setting IM[2:0] pins. The IM[2:0] set “000” is select I2C interface.

I2C interface 2 hardware pin – serial data (SDA) and serial clock (SCL), carry information between the devices connected to the bus. Each device is recognized by a unique address — whether it’s a microcontroller, LCD driver, memory or keyboard interface — and can operate as either a transmitter or receiver, depending on the function of the device. Both SDA and SCL are needed connected to a positive supply voltage via a pull-up resistor. The pull-up resistor should connect to VDD1. When the bus is free, both lines are HIGH.

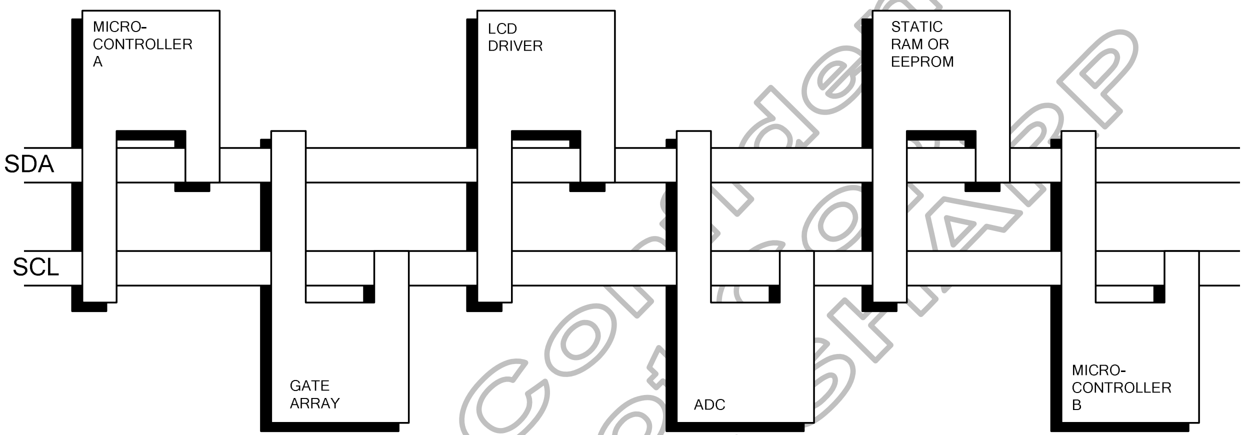


Figure 4.5: I2C connection diagram

4.3.1 I2C protocol

The data on the SDA line must be stable during the HIGH period of the clock. The HIGH or LOW state of the data line can only change when the clock signal on the SCL line is LOW.

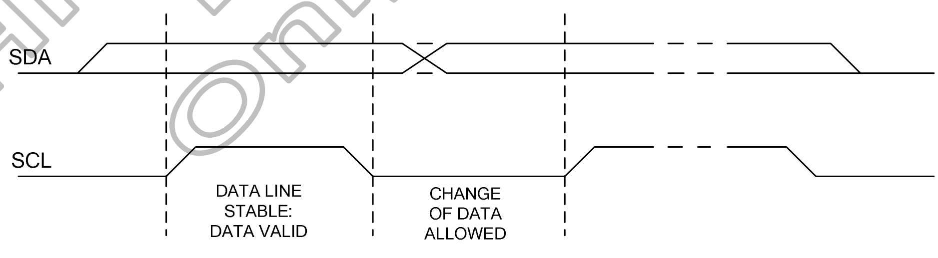


Figure 4.6: I^2C Signal timing

Within the procedure of the I^2C-bus, unique situations arise which are defined as START and STOP conditions. A HIGH to LOW transition on the SDA line while SCL is HIGH is one such unique case. This situation indicates a START condition. A LOW to HIGH transition on the SDA line while SCL is HIGH defines a STOP condition. START and STOP conditions are always generated by the master. The I^2C bus is considered to be busy after the START condition. The I^2C bus is considered to be free again a certain time after the STOP condition.

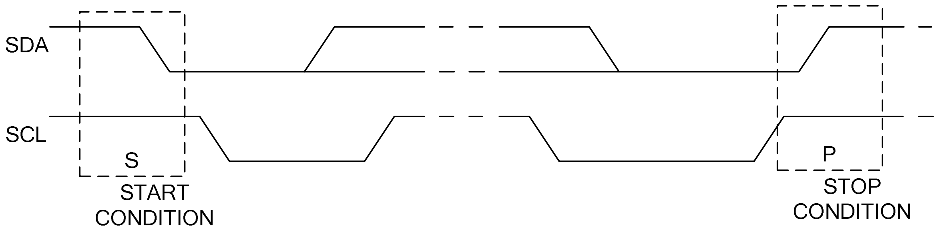


Figure 4.7: I^{2}C START/STOP

Every byte put on the SDA line must be 8-bits long. The number of bytes that can be transmitted per transfer is unrestricted. Each byte has to be followed by an acknowledge bit. Data is transferred with the most significant bit (MSB) first.

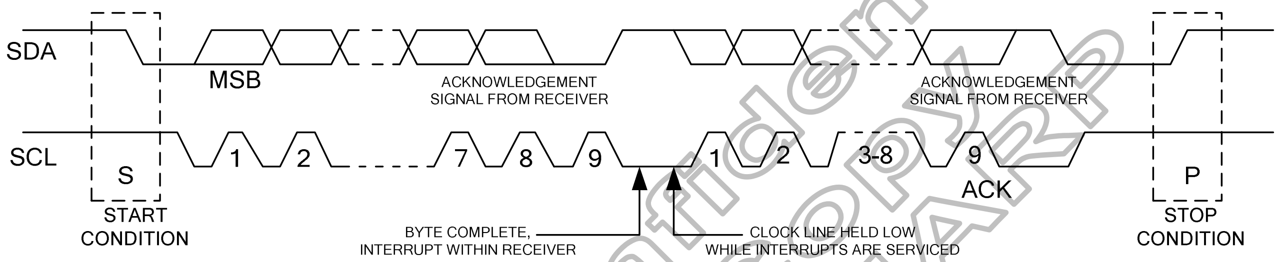


Figure 4.8: I^{2}C data transfer

4.3.2 I2C slave address

HX8399-A support many slave address could be select by register setting in RE8h. The slave address is defined a follow table.

I2C_SA[6 :0]	Slave address (A6-A0)
000_0000	000_0000
000_0001~000_0111	Reserved
000_1000	000_1000
:	:
111_0110	111_0110
111_0111	111_0111
111_1xxx	Reserved

Table 4.3: I^{2}C Slave Address table

4.3.3 I2C interface write mode

HX8399-A support I2C to write data to register. The write flow is described as below

1. Send Start condition followed by I2C 7 bits slave address and 1 bit '0'(write flag)
2. Sned High byte of 16bits address, then IC feedback Ack.
3. Sned Low byte of 16bits address, then IC feedback Ack.
4. Sned 8bits register data, MSB first , ADD[7] send first, the IC feedback Ack
5. Send Stop condition

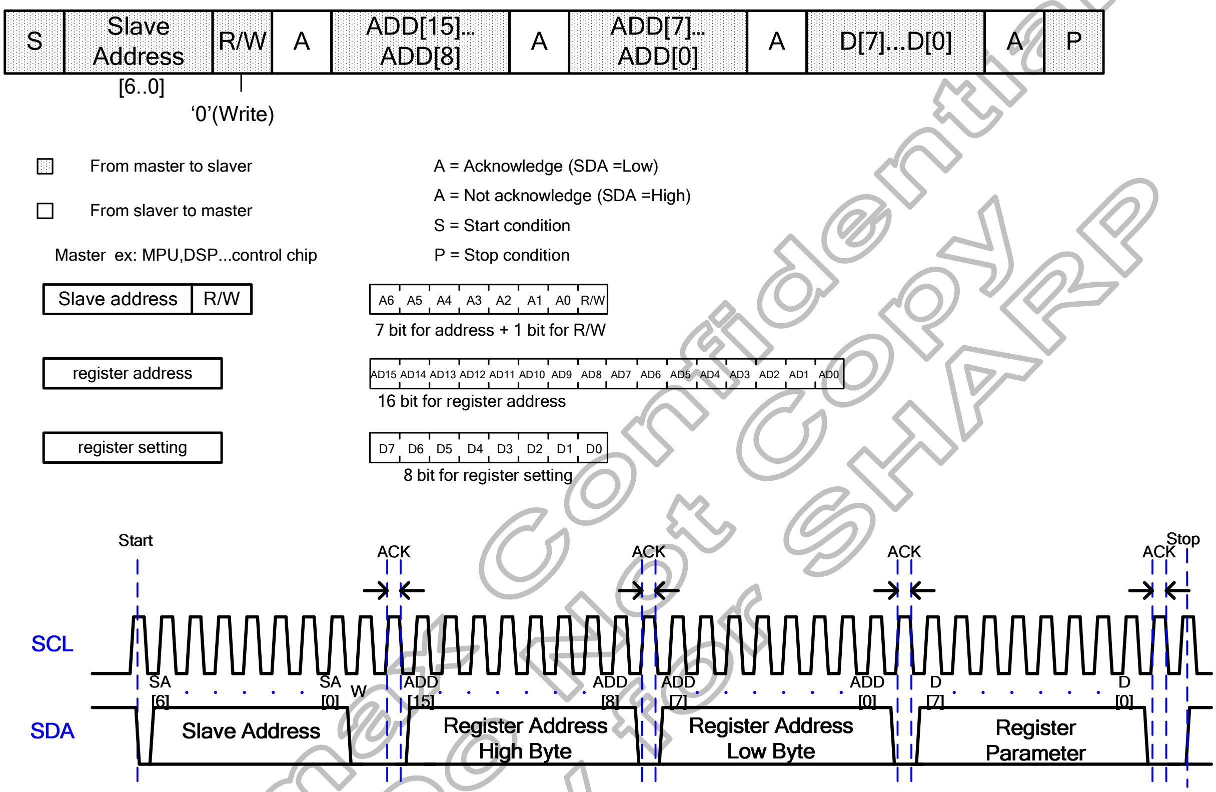


Figure 4.9: I²C interface register write flow

4.3.4 I2C interface read mode

HX8399-A also support I2C to read data from register. The write flow is described as below

1. Send Start condition followed by I2C 7 bits slave address and 1 bit '0'(write flag)
2. Sned High byte of 16bits address, then IC feedback Ack.
3. Sned Low byte of 16bits address, then IC feedback Ack.
4. Send restart condition followed by I2C 7 bits slave address and 1 bit '1'(read flag)
5. IC send register data to baseband, and followed by an non-ack.
6. Send Stop condition

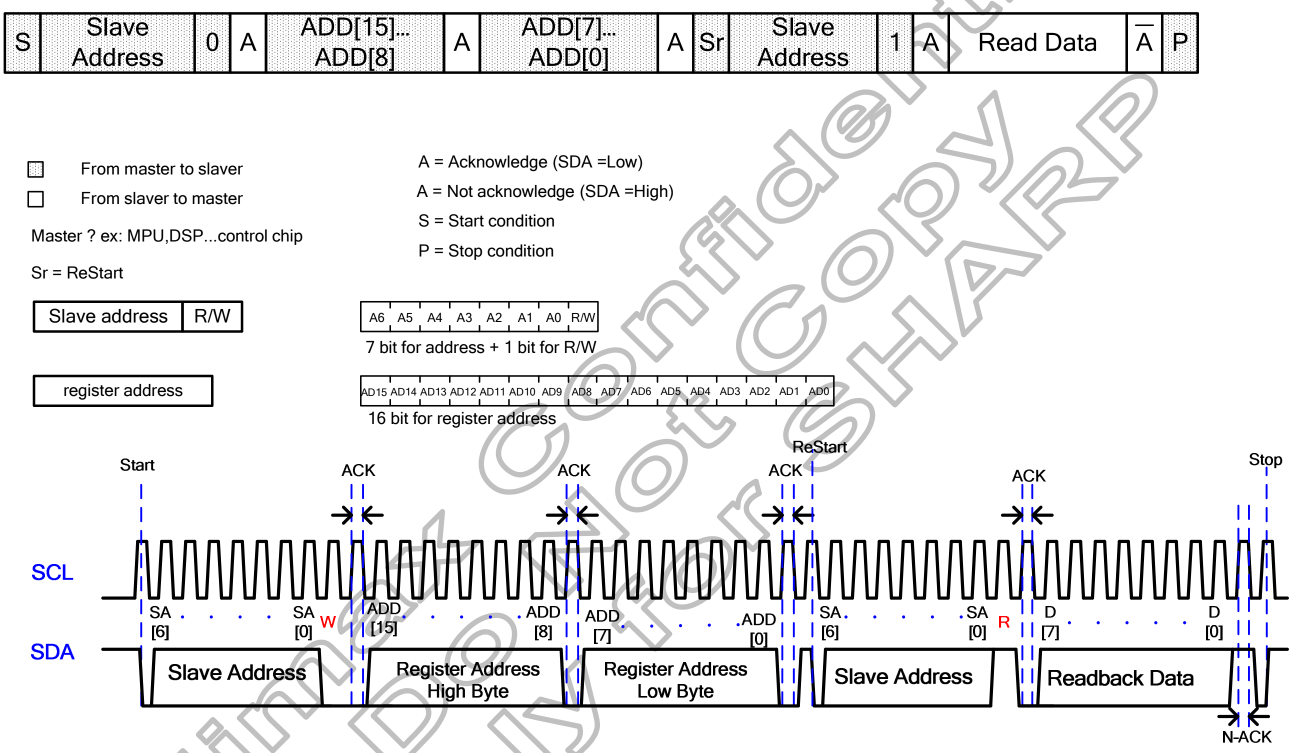


Figure 4.10: I^{2}C interface register read flow

4.4 DSI system interface

The DSI specifies the interface between a host processor and a peripheral such as a display module. Figure 4.11 shows a simplified DSI interface. From a conceptual viewpoint, a DSI-compliant interface also sends pixels or commands to the peripheral, and can read back status or pixel information from the peripheral. The main difference is that DSI serializes all pixel data, commands, and events that. DSI-compliant peripherals support Command Mode. Which mode is used depends on the architecture and capabilities of the peripheral. The mode definitions reflect the primary intended use of DSI for display.

Command Mode refers to operation in which transactions primarily take the form of sending Commands and data to a peripheral, such as a display module, that incorporates a display controller. Systems using Command Mode write to, and read from, the registers. The host processor indirectly controls activity at the peripheral by sending commands, parameters and data to the display controller. The host processor can also read display module status information. Command Mode operation requires a bidirectional interface.

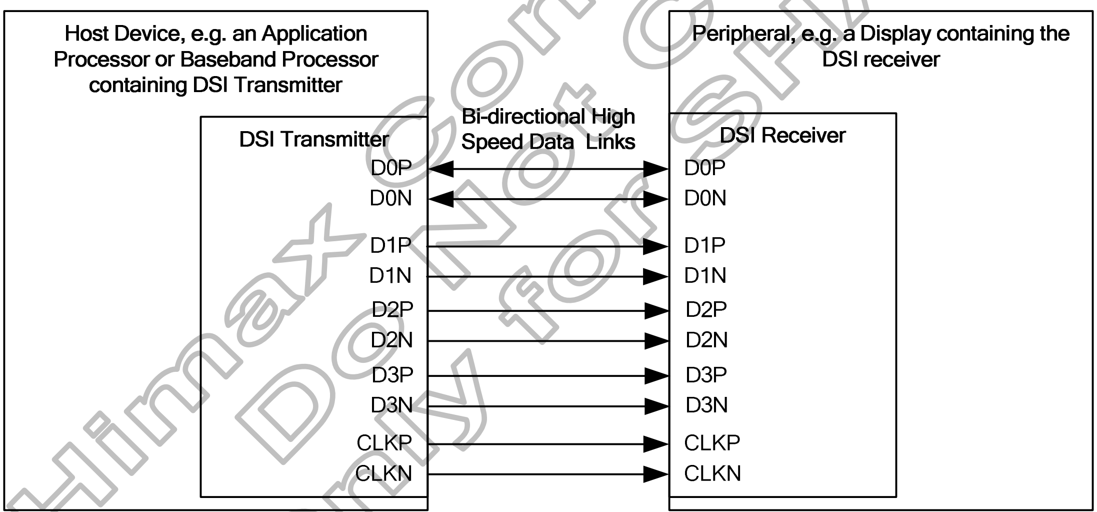


Figure 4.11: DSI transmitter and receiver interface

Please refer to "**DRAFT MIPI Alliance Standard for DSI**" for DSI detailed specifications. The data lane number select by internal register(RBAh).

4.4.1 DSI layer definitions

According Figure 4.12 DSI transmitter and Receiver interface to understand simple interface block diagram. Then under diagram is internal block for DSI which include four types: PHY Layer, Lane Management Layer, Low level protocol and Application Layer.

The PHY Layer specifies the characteristics of transmission medium and electrical parameters for signaling the timing relationship between clock and Data Lanes.

The Lane Management Layer specifies DSI is Lane-scalable for increased performance. The data signals maybe transmission through one or more channel depending on the bandwidth requirements of the application.

The Protocol Layer specifies at the lowest level, DSI protocol specifies the sequence and value of bits and bytes traversing the interface. It specifies how bytes are organized into defined groups called packets.

The Application Layer describes higher-level encoding and interpretation of data contained in the data stream. The DSI specification describes the mapping of pixel values, commands and command's parameters to bytes in the packet assembly.

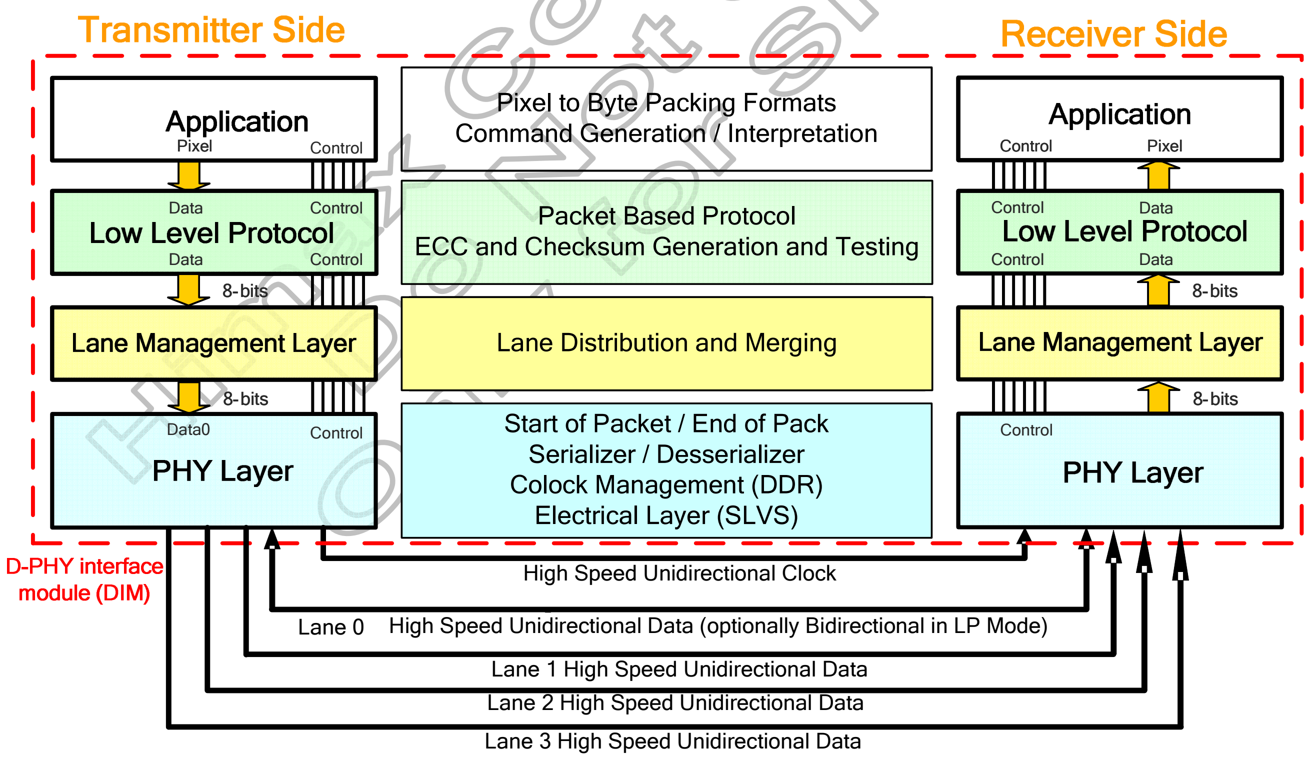


Figure 4.12: DSI transmitter and receiver interface

4.4.1.1 Lane States

The HX8399-A uses Data Lane and Clock Lane differential pairs for DSI. Both differential lane pairs can be driven LP (Low Power) or HS (High Speed) mode.

LP mode means each line of the differential pairs are used in independently and single-ended. In LP mode differential receiver is disable(termination resistor of the receiver is disable). In LP mode there are four possible Low-Power Lane states (LP-00, LP-01, LP-10, LP-11).

HS mode means the differential pairs are not used in single-end and termination resistor of the receiver is enable. There are different modes and protocol in each mode when transfer display data frim MCU to the display module.

The state code of HS and LP Lane pair are defined as below:

State Code	Line Voltage Levels		High-Speed	Low-Power	
	Dp-Line	Dn-Line	Burst Mode	Control Mode	Escape Mode
HS-0	HS Low	HS High	Differential-0	Note1	Note1
HS-1	HS High	HS Low	Differential-1	Note1	Note1
LP-00	LP Low	LP Low	N/A	Bridge	Space
LP-01	LP Low	LP High	N/A	HS-Rqst	Mark-0
LP-10	LP High	LP Low	N/A	LP-Rqst	Mark-1
LP-11	LP High	LP High	N/A	Stop	Note2

Note1: During High-Speed transmission the Low-Power Receivers observe LP-00 on the Lines.
Note2: If LP-11 occurs during Escape mode the Lane returns to Stop state (Control Mode LP-11)

4.4.1.2 Clock Lane Mode

Figure 4.13 shows the state diagram for Clock Lane Mode. The Clock Lane has three different power modes: Low Power Stop State, Ultra Low Power State(ULPS) and High Speed clock transmission.

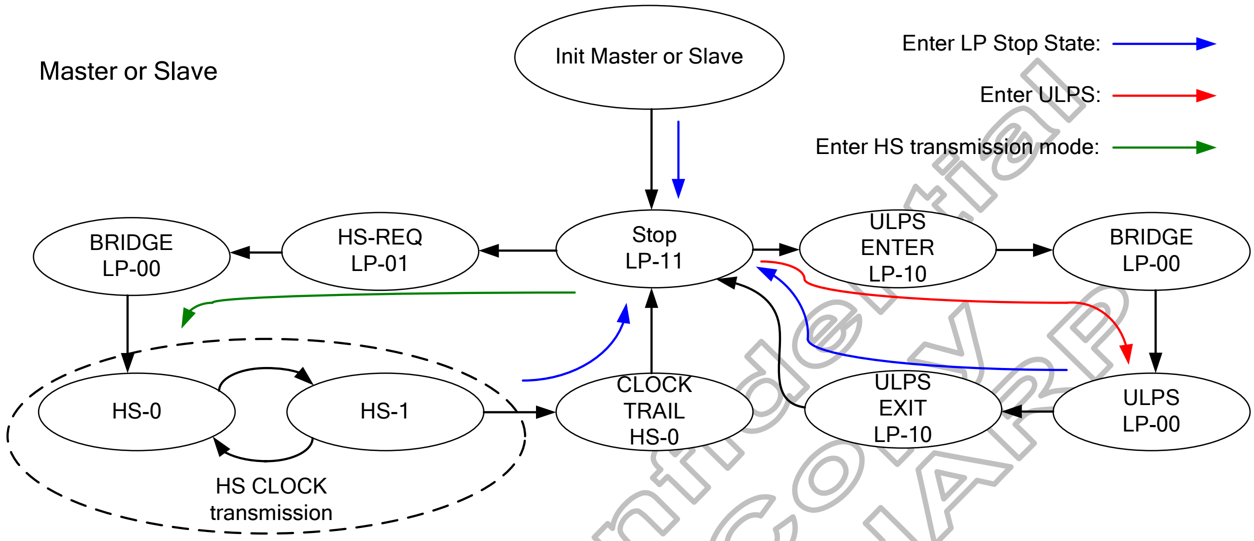
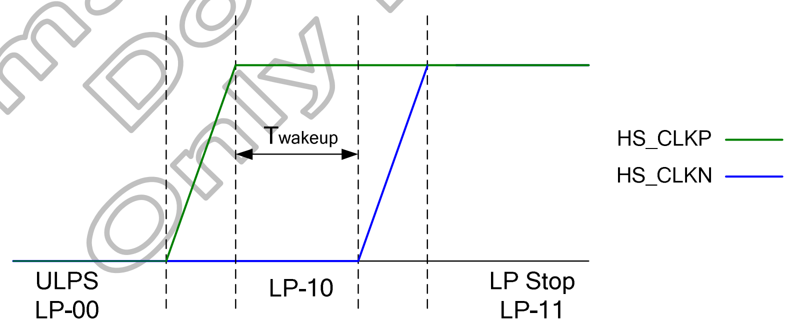


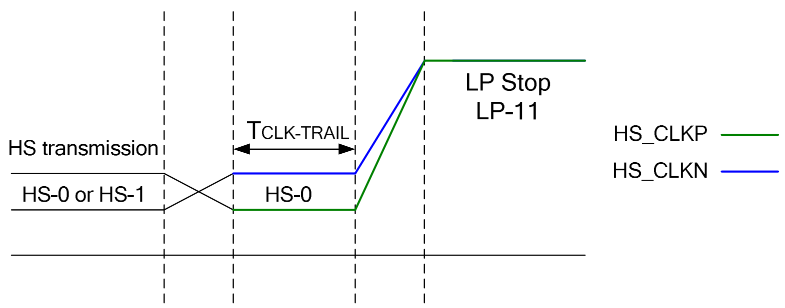
Figure 4.13: Clock Lane Mode State diagram

Clock Lane can be driven LP-11 to enter Low Power Stop State. There are three ways to enter Lower Power Stop State:

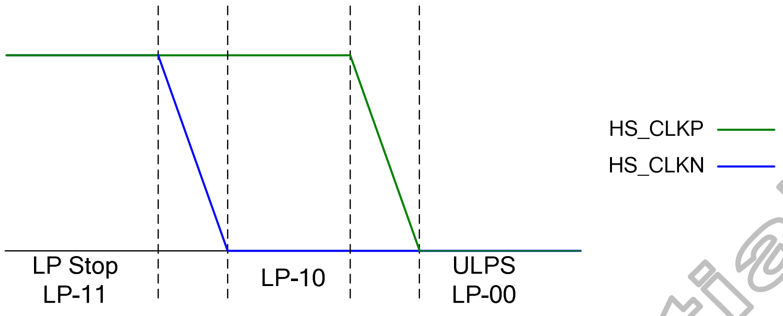
(1) After Initial state(HW reset, SW reset, Power on sequence).
(2) Leaving ULPS: ULPS LP-00 -> LP-10 -> Low Power Stop State LP-11.



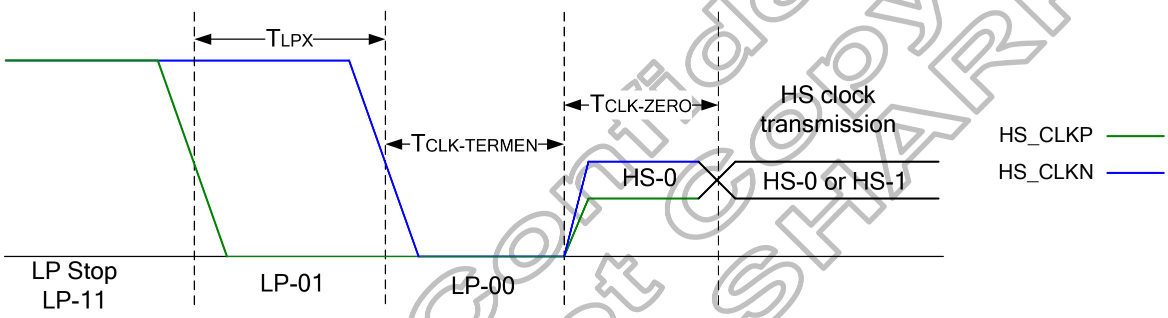
(3) Leaving HS clock transmission mode: HS mode (HS-0 or HS-1) -> HS-0 -> Low Power Stop State LP-11.



Clock Lane can be driven LP-00 to enter Ultra Low Power State from Low Power Stop State. The flow is Low Power Stop State LP-11 -> LP-10 -> ULPS LP-00.



Clock Lane can be High Speed Clock transmission State from Low Power Stop State. The flow is Low Power Stop State LP-11 -> LP-01 -> LP-00 -> HS-0/1.



4.4.1.3 Data Lane Mode

Figure 4.14 shows the operational flow diagram for Data Lane Mode. There are three operating modes in Data Lane: Escape mode, High-Speed transmission mode and Turnaround.

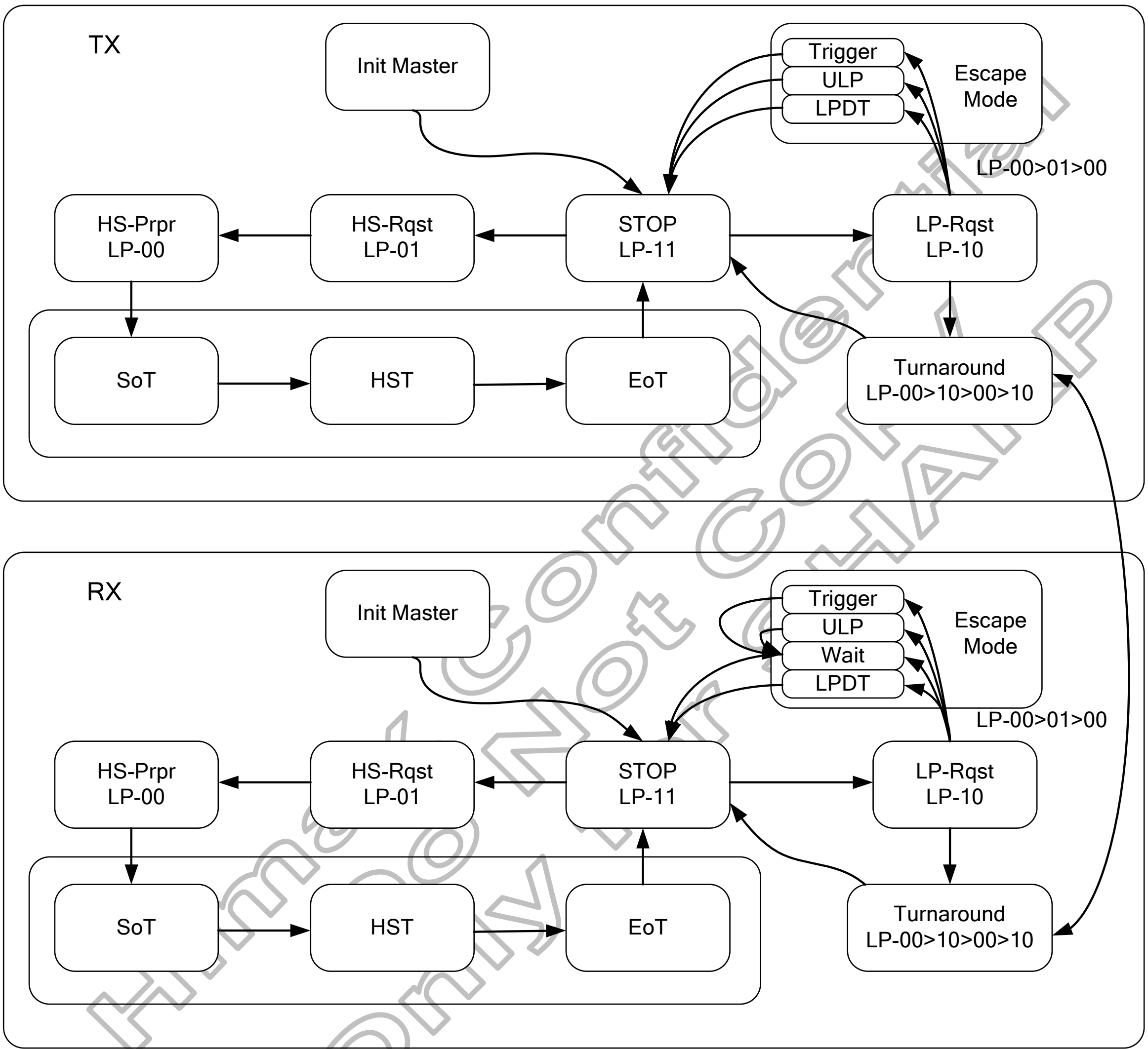


Figure 4.14: Data Lane Mode State diagram

4.4.1.3.1 Escape Mode

Data Lane0 is used in Escape Mode when data lane in LP mode. Data Lane shall enter Escape mode via LP-11 -> LP-10 -> LP-00 -> LP-01 -> LP-00 and exit Escape mode via LP-10 -> LP-11.

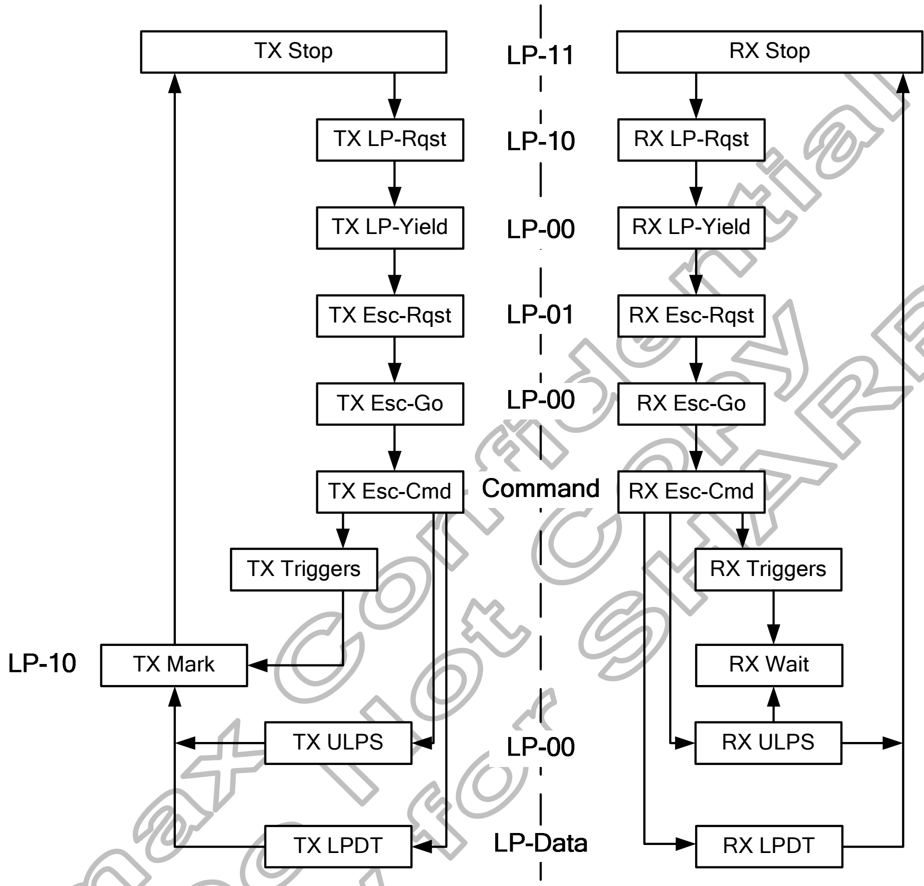


Figure 4.15: Escape Mode State Machine

Once Escape mode is entered, the transmitter shall send an 8-bit entry code to indicate the requested action. The Entry Code as follows:

(1) Trigger (Reset-Trigger(46h), Tearing effect(BAh), Acknowledge(84h))
(2) Drive Data Lane to Ultra Low Power State(78h)
(3) Send Low Power Data Transmission(87h)

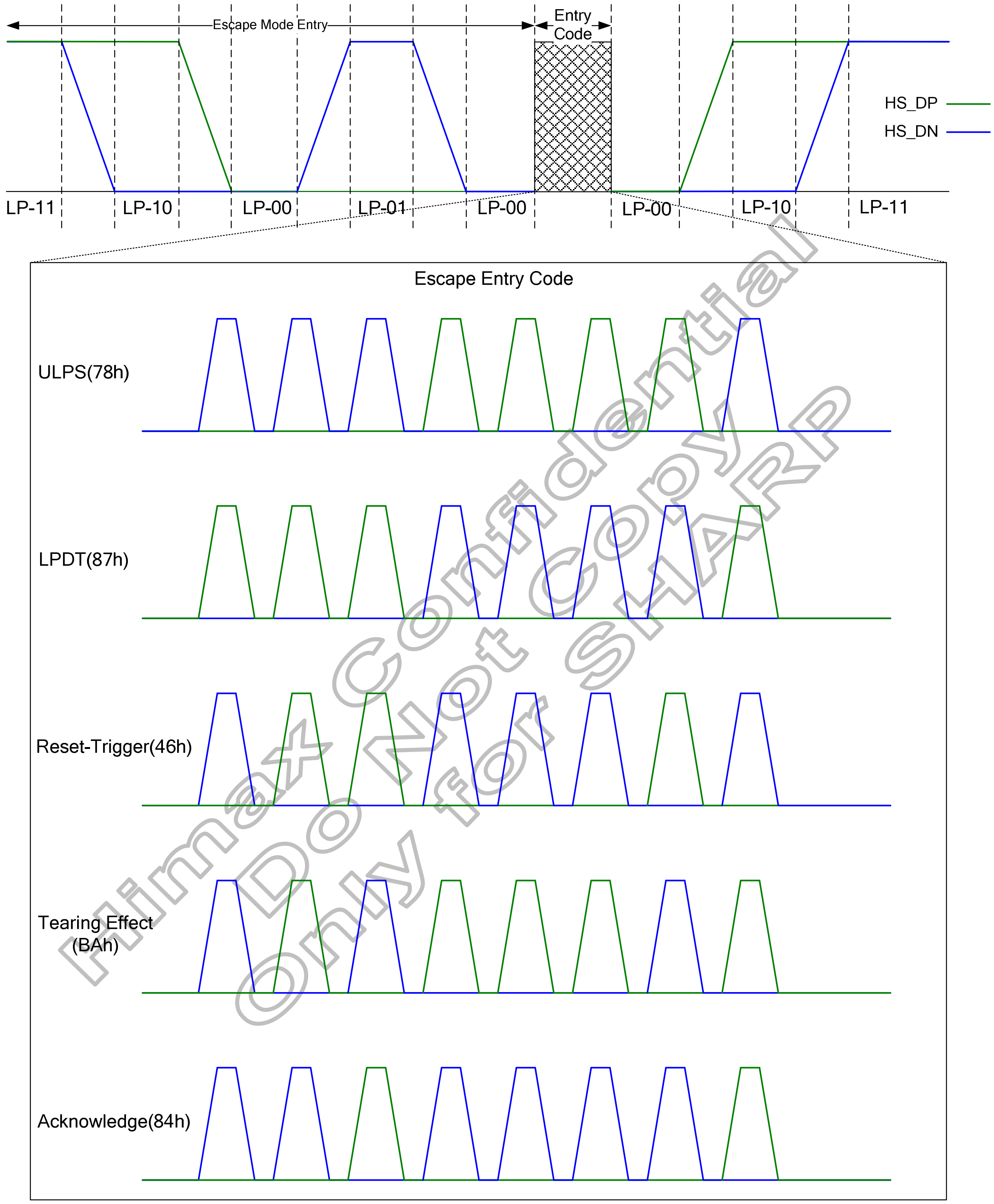


Figure 4.16: Escape Mode timing sequence

4.4.1.3.2 High Speed Data Trnsmission

The display module can enter High Speed Data Transimission when Clock Lane in the High Speed Clock Mode. All Data Lane enter High Speed Data Tranmission synchronously but may end at different time. Data Lane enter High Speed Data Transmission flow: LP-11 -> LP-01 -> LP-00 -> SoT(0001_1101). And exit High Speed Data Transmission flow: Toggles differential state immediately after last payload data bit and keeps that state for a time $T_{HS-TRAIL}$.

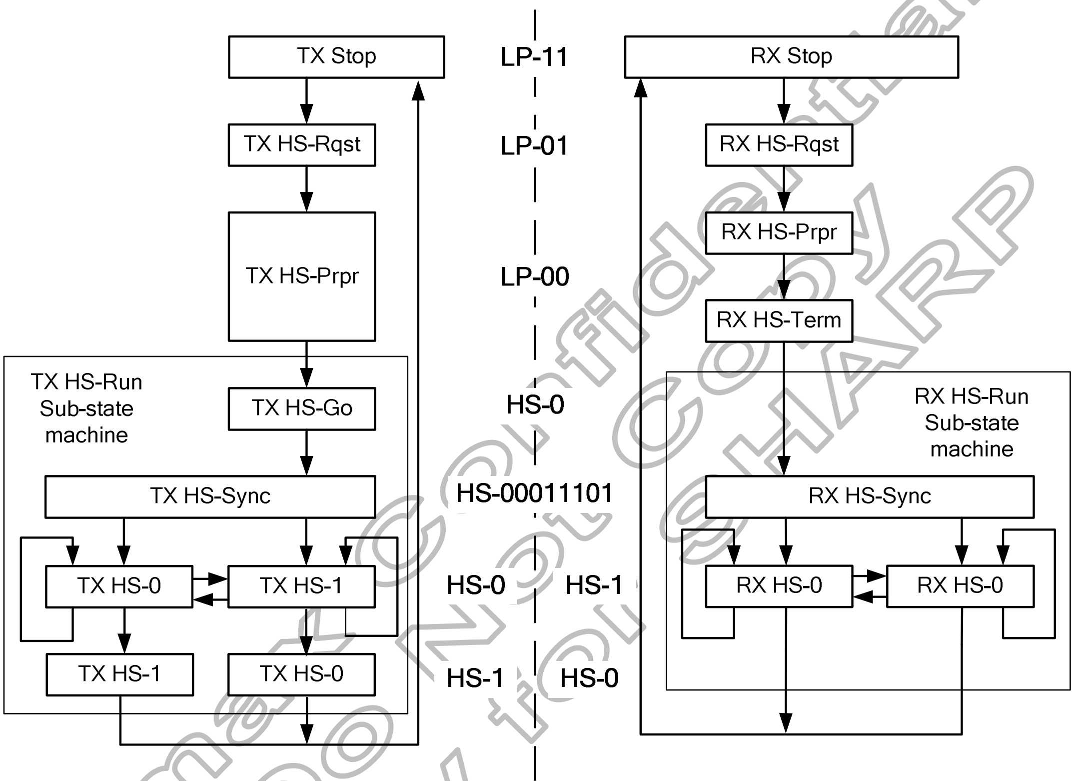


Figure 4.17: High Speed Data Transmission State Machine

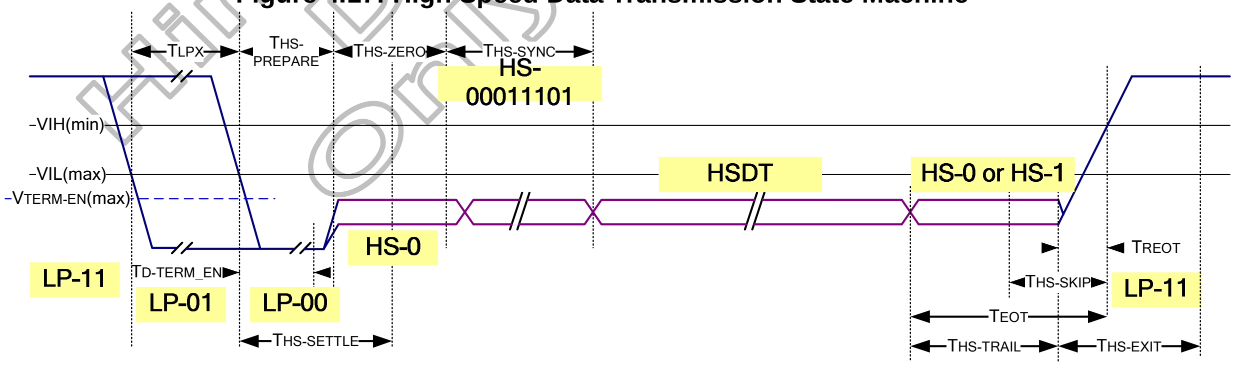


Figure 4.18: High Speed Data Transmission timing sequence

4.4.1.3.3 Bi-directional Data Lane Turnaround

The transmission direction od a bi-directional Data Lane can be swapped by means of a Link Turnaround procedure. This procedure enables information transfer in the opposite direction of the current direction. The procedure is the same for either a change from Forward-to-Reverse direction or Reverse-to-Forward direction.

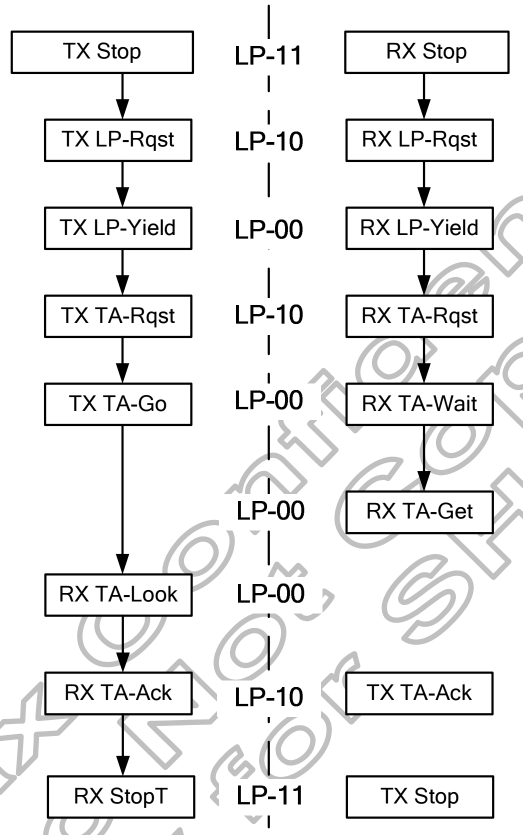


Figure 4.19: Turnaround State Machine

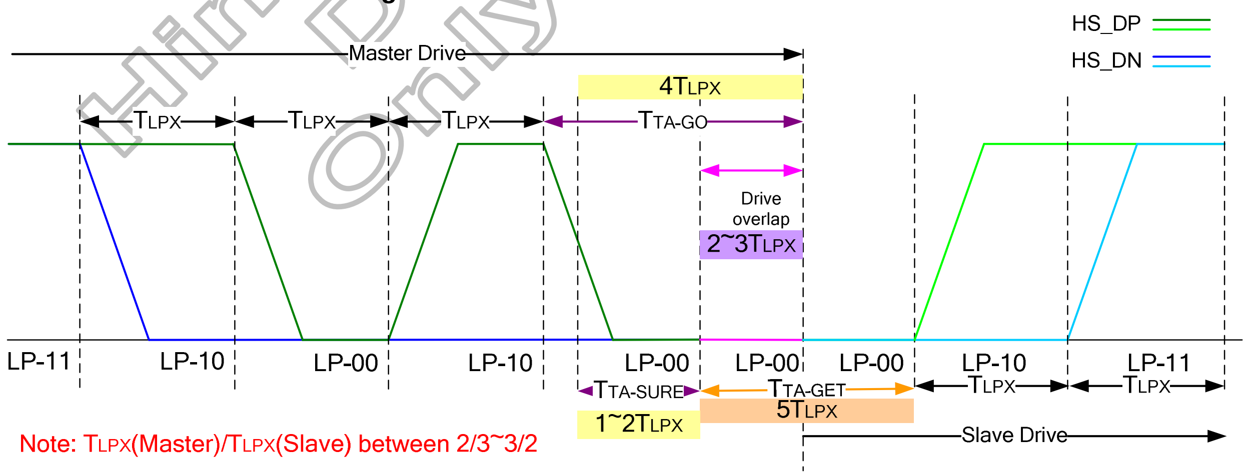
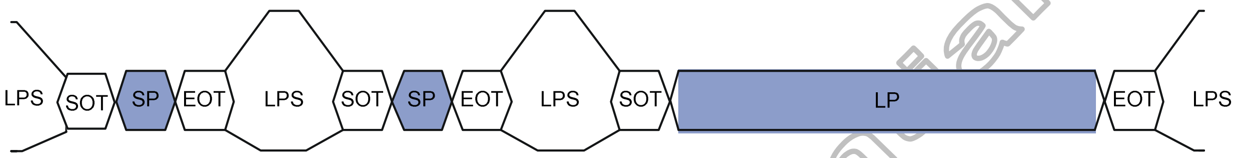


Figure 4.20: Turnaround timing sequence

4.4.2 DSI protocol

The protocol layer appends packet-protocol information and headers. The receiver side of a DSI Link performs the converse of the transmitter side, decomposing the packet into parallel data, signal events and commands. The DSI protocol permits multiple packets which is useful for events such as peripheral initialization, where many registers may be loaded separate write commands at system startup. Figure 4.21 illustrates multiple HS Transmission packets.

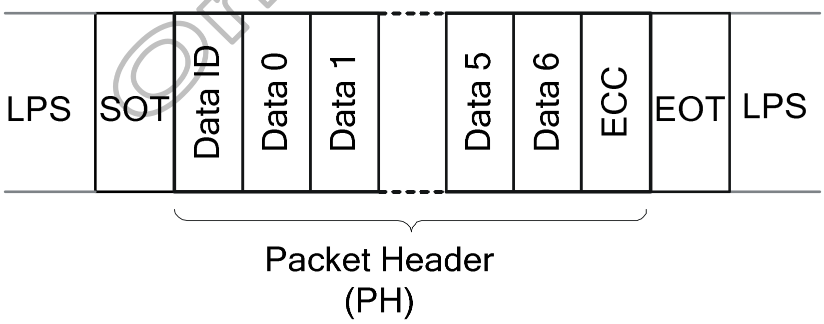


LPS：Low power state
SOT：Start of Transmission
SP：Short Packet
LP：Long Packet
EOT：End of Transmission

Figure 4.21: Multiple HS transmission packets

The packet includes two types which are Long packet and short packet. The first byte of the packet, the Data Identifier (DI), includes information specifying the length of the packet. Command Mode systems send commands and an associated set of parameters, with the number of parameters depending on the command type.

Short packets specify the payload length using the Data Type field and are from two to nine bytes in length. Short packet is used for most Command Mode commands and associated parameters. Where short packets format include an 8-bit Data ID followed by zero to seven bytes and an 8-bit ECC. Figure 4.22 shows the structure of the Short packet.

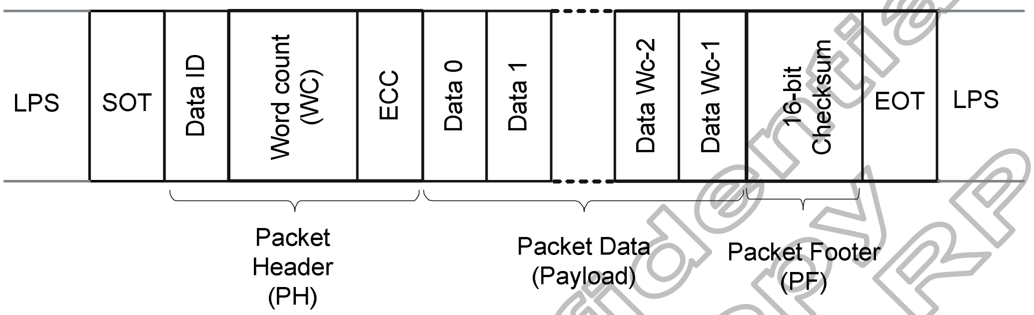


DI(Data ID)：Contain Virtual Channel Identifier and Data Type.
ECC(Error Correction Code)：The Error Correction Code allows single-bit errors to be corrected and 2-bit errors to be detected in the Packet Header.

Figure 4.22: Structure of the short packet

Long packets specify the payload length using a two-byte Word Count field and then the payload maybe from 0 to 65,541 bytes in length. Long packets permit transmission of large blocks of pixel or other data. Figure 4.23 shows the structure of the Long packet. Long Packet Header composed of three elements: an 8-bit Data Identifier, a 16-bit Word Count, and 8-bit ECC. The Packet Footer has one element, a 16-bit checksum. Long packets can be from 6 to 65,541 bytes in length.

Where 65,541 bytes = (2^{16}-1) + 4 bytes PH + 2 bytes PF



DI (Data ID)：Contain Virtual Channel Identifier and Data Type.
WC (Word Count)：The receiver use WC to define packet end.
ECC (Error Correction Code)：The Error Correction Code allows single-bit errors to be corrected and 2-bit errors to be detected in the Packet Header.
PF(Packet Footer)：Mean 16-bit Checksum.

Figure 4.23: Structure of the long packet

According to packet form, basic elements include DI and ECC. Figure 4.24 the shows format of Data ID.

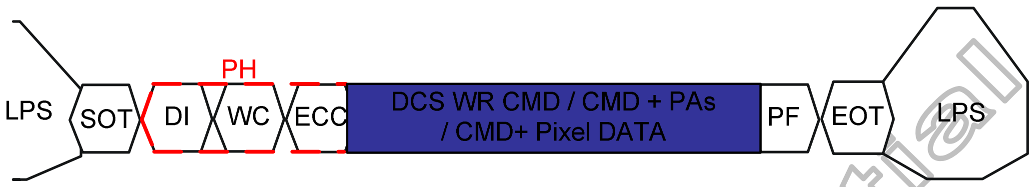
DI7	DI6	DI5	DI4	DI3	DI2	DI1	DI0
VC (Virtual Channel)		DT (Data Type)					

DI[7:6] ➔ These two bits identify the data as directed to one of four virtual channels.
DI[5:0]: These six bits specify the Data Type, which specifies the size, format and, in some cases, the interpretation of the packet contents.

Figure 4.24: The format of data ID.

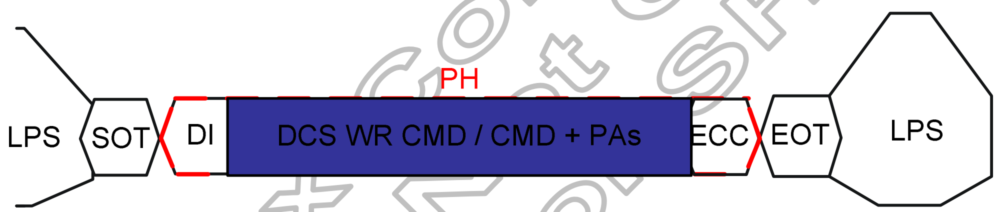
Due to Data Type (DT) mean format of transmission type, Figure 4.25 show Short- / Long-packet transmission command sequence.

Long packet write Command / Parameters / Pixel Datas



DI ➔ Write suitable Data type.
WC ➔ Write number of Payload Data.
Ex: One CMD write, WC setting as 1.
CMD + PAs write, WC setting as number of (CMD+PAs).
CMD + DATA write, WC setting as number of (CMD + Pixel DATA).

Short packet write Command / Parameters



DI ➔ Write suitable Data type.
Ex: One CMD write, DI + DCS WR CMD
CMD + PAs write, DI + DCS WR CMD + PAs

Figure 4.25: show Short- / Long-packet transmission command sequence

4.4.3 Processor to peripheral direction packets data types

The set of transaction types sent from the host processor to a peripheral, such as a display module, are shown in Table 4.4 Data Types for Processor-sourced Packets.

Data type, hex	Data type, binary	Description packet	Size
01h	00 0001	Sync Event, V Sync Start	Short
11h	01 0001	Sync Event, V Sync End	Short
21h	10 0001	Sync Event, H Sync Start	Short
31h	11 0001	Sync Event, H Sync End	Short
08h	00 1000	End of Transmission packet(EoTp)	Short
22h	10 0010	Shut Down Peripheral Command	Short
32h	11 0010	Turn On Peripheral Command	Short
05h	000101	DCS WRITE, no parameter	Short
15h	010101	DCS WRITE, 1 parameter	Short
06h	00 0110	DCS READ, no parameters	Short
37h	11 0111	Set Maximum Return Packet Size	Short
09h	00 1001	Null Packet, no data	Long
19h	01 1001	Blanking Packet, no data	Long
39h	11 1001	DCS Long Write/write_LUT Command Packet	Long
0Eh	00 1110	Packed Pixel Stream, 16-bit RGB, 5-6-5 Format	Long
1Eh	01 1110	Packed Pixel Stream, 18-bit RGB, 6-6-6 Format	Long
2Eh	10 1110	Loosely Packed Pixel Stream, 18-bit RGB, 6-6-6 Format	Long
3Eh	11 1110	Packed Pixel Stream, 24-bit RGB, 8-8-8 Format	Long
X0h and XFh, unspecified	xx 0000 xx 1111	DO NOT USE All unspecified codes are reserved	-

Table 4.4: Data types for processor-sourced packets

Under tables list all detail function of all data types

Sync event (H start, H end, V start, V end), data type=xx 0001 (x1h)		
Data type, hex	**Function description**	**Number of bytes**
01h	V Sync start, Start of VSA pulse.	4 bytes (DI+Data0+Data1+ECC)
11h	V Sync End, End of VSA pulse.	
21h	H Sync Start, Start of HSA pulse.	
31h	H Sync End, End of HSA pulse.	
Note: V Sync Start and V Sync End event represents the start and end of the VSA, respectively. Similarly H Sync Start and H Sync End event represents the start and end of the HSA, respectively.		

Display status (shutdown command, turn-on command)		
Data type, hex	**Function description**	**Number of bytes**
22h	Shutdown Peripheral command that turns off the display in a Video Mode display for power saving.	4 bytes (DI+Data0+Data1+ECC)
32h	Turn On Peripheral command that turns on the display in Video Mode display for normal display.	
Note: When use shutdown command, interface shall remain powered in order to receive the turn-on, or wake-up, command.		

Color mode status (Color Mode On, Color Mode Off)		
Data type, hex	**Function description**	**Number of bytes**
05h and 15h	DCS Short Write command, 0 or 1 parameter, Data Types = 00 0101(05h), 01 0101 (15h), Respectively.	4 bytes (DI+Data0+Data1+ECC)
NOTE: (1) For write part, If DCS Short Write command, followed by BTA, the peripheral shall respond with **ACK** when without error was detected in the transmission (Host → Slave). Unless an error was detected, the peripheral shall respond with **Acknowledge with Error Report**.		

For example: 05h DCS WRITE for no parameter command set.

05h	CMD	0	ECC

Ex. 05h, 29h, 00, 1Ch — Display On(29h)

For example: 15h DCS WRITE for only one parameter command set.

15h	CMD	Par	ECC

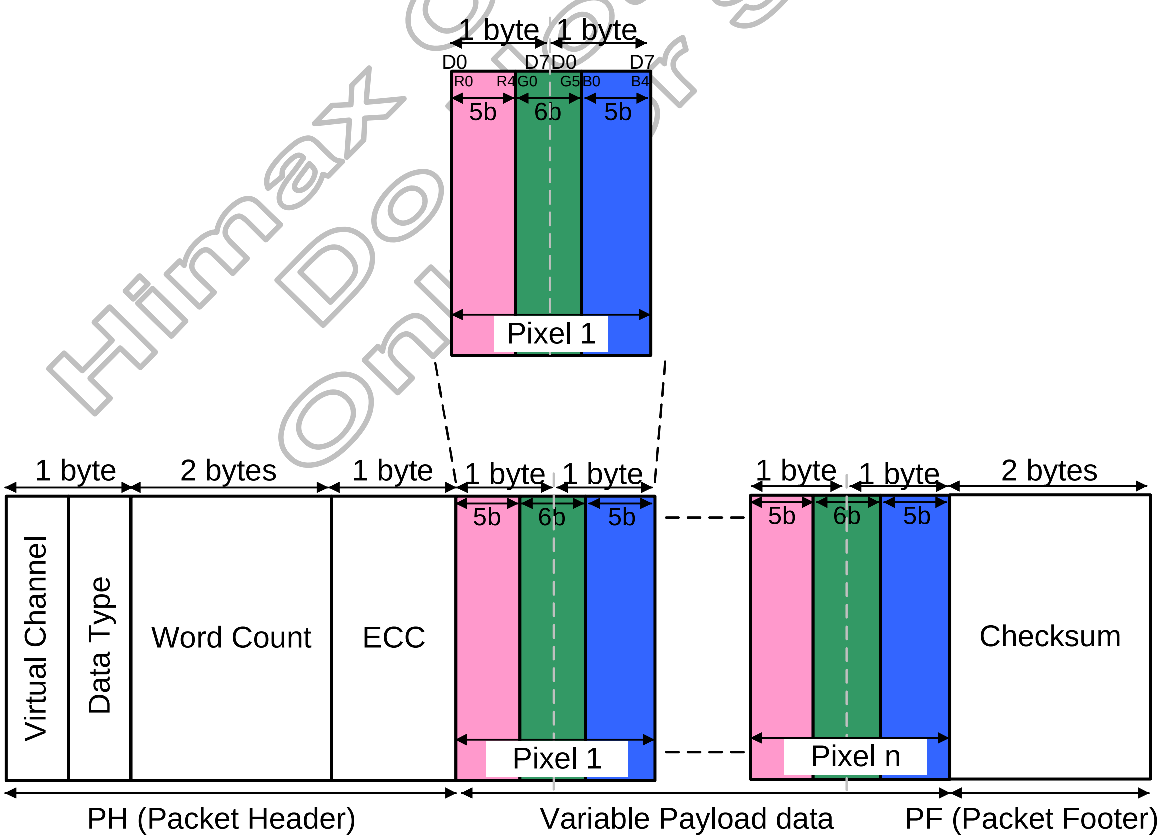
Ex. 15h, 36h, 08h, 11h — MADCTL(36h)-BGR bit=1

DCS command setting		
Data type, hex	**Function description**	**Number of bytes**
06h	DCS Read command, the returned data may be of Short or Long packet format.	4 bytes (DI+Data0+Data1+ECC)
39h	DCS Long Write/ Write _ LUT Command is used to send larger blocks of data to a display module that implements the Display Command Set.	Up to 65541 bytes (DI + WC + ECC + DCS CMD. + Payload DATA + PF)
NOTE: (1) When use DCS Read Command, the **Set Max Return Packet Size** command will limit the size of returning packets. (2) The peripheral shall respond to DCS Read Command Request in one of the following ways: ◆ If an error was detected by the peripheral, it shall send *Acknowledge with Error Report*. So the peripheral shall transmit the requested READ data packet with suitable ECC in the same transmission. ◆ If no error was detected by the peripheral, it shall send the requested READ packet (Short or Long) with appropriate ECC and Checksum, if either or both features are enabled. (3) **One byte <= Length of payload DATA <= 2^{WC}-1**		

Return packet size setting		
Data type, hex	**Function description**	**Number of bytes**
37h	Set Maximum Return Packet Size that specifies the maximum size of the payload in a Long packet transmitted from peripheral back to the host processor.	4 bytes (DI + WC + ECC)
Note: The two-byte value is transmitted with LS byte first. And during a power-on or Reset sequence, the Maximum Return Packet Size shall be set by the peripheral to a default value of one.		

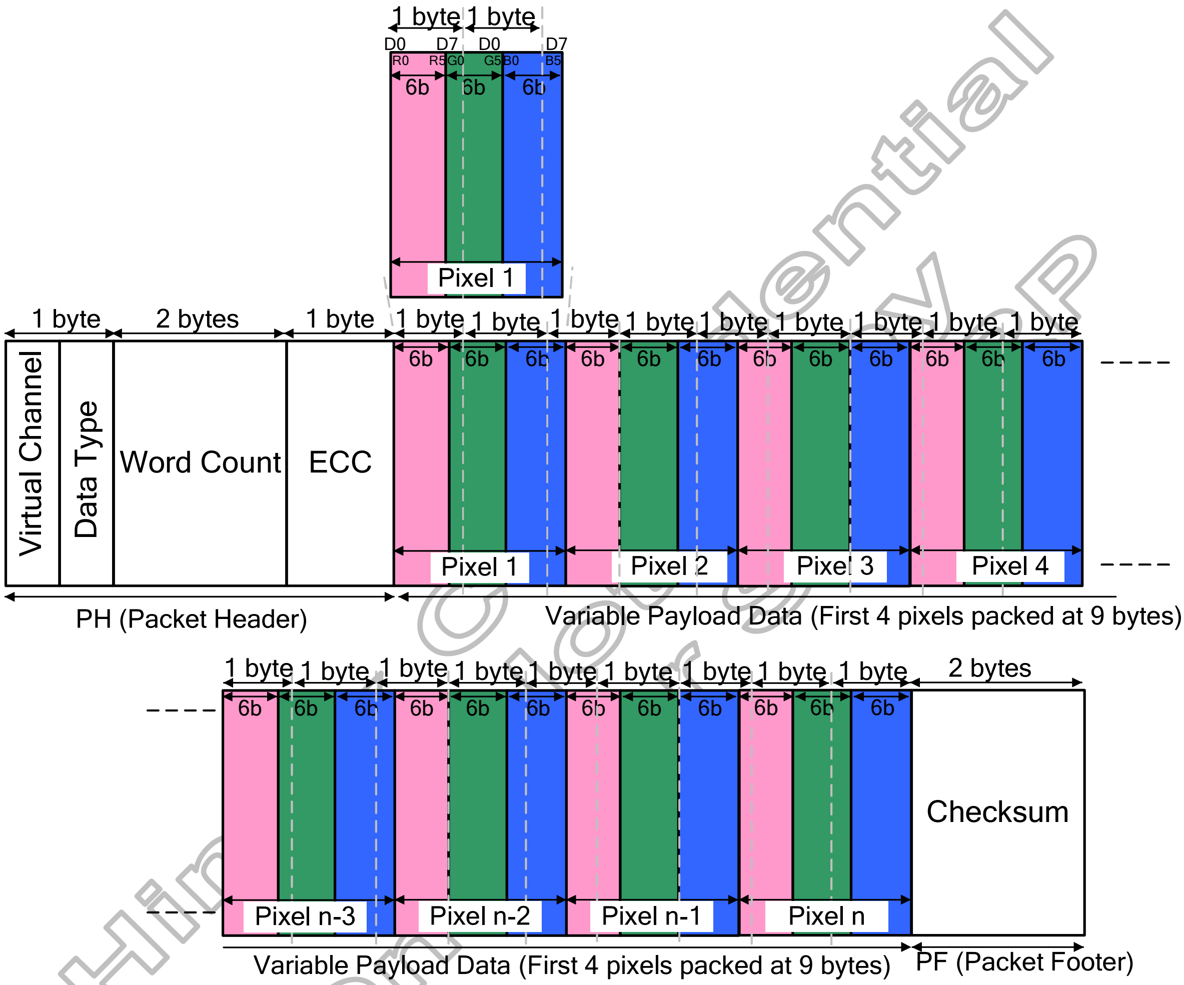
Variable data packet		
Data type, hex	**Function description**	**Number of bytes**
09h	Null Packet is a mechanism for keeping the serial Data Lane(s) in High-Speed mode while sending dummy data.	Up to 65541 bytes (DI + WC + ECC + DCS CMD. + Payload DATA + PF)
19h	Blanking packet is used to convey blanking timing information in a Long packet.	
Note: (1) When **Null Packet**, the Payload Data belong “null” Data, actual data values sent are irrelevant because the peripheral does not capture or store the data. (2) When **Blanking packet**, the packet represents a period between active scan lines of a Video Mode display,		

Data stream format		
Data type, hex	**Function description**	**Number of bytes**
0Eh	Packed Pixel Stream 16-Bit Format is used to transmit image data formatted as 16-bit pixels to a Video Mode display module. Pixel format is “(5 bits) red, (6 bits) green and (5 bits) blue”.	Up to 65541 bytes (DI + WC + ECC + DCS CMD. + Payload DATA + PF)



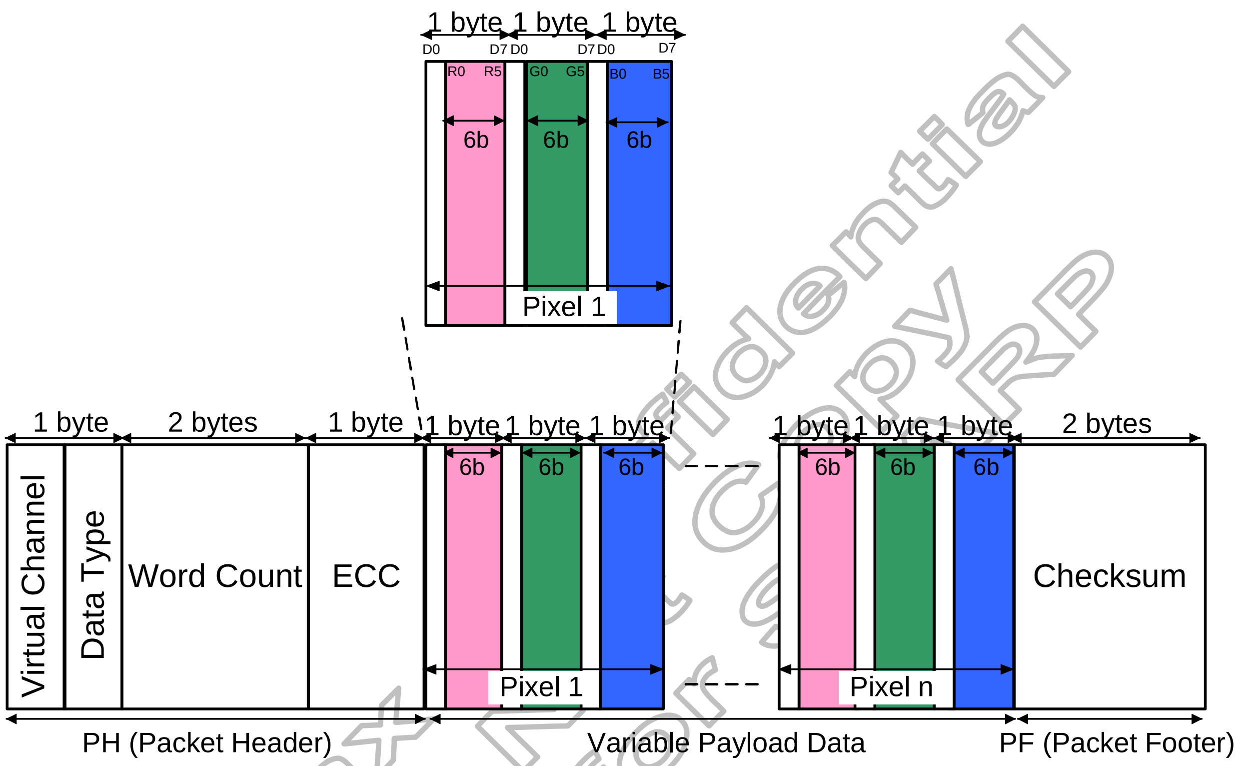
Note: Within a color component, the “LSB is sent first, the MSB last “.

Data stream format		
Data type, hex	**Function description**	**Number of bytes**
1Eh	Packed Pixel Stream 18-Bit Format is used to transmit image data formatted as 18-bit pixels to a Video Mode display module. Pixel format is "(6 bits) red, (6 bits) green and (6 bits) blue".	Up to 65541 bytes (DI + WC + ECC + DCS CMD. + Payload DATA + PF)



Note: Within a color component, the LSB is sent first and the MSB last and pixel boundaries only line up with byte boundaries every four pixels (nine bytes). Preferably, display modules employing this format have a horizontal extent (width in pixels) evenly divisible by four, so no partial bytes remain at the end of the display line data. It is possible to send pixel data that represent a line width that is not a multiple of four pixels, but display logic on the receiver end shall dispose of the extra bits of the partial byte at the end of active display and ensure a "clean start" for the next line.

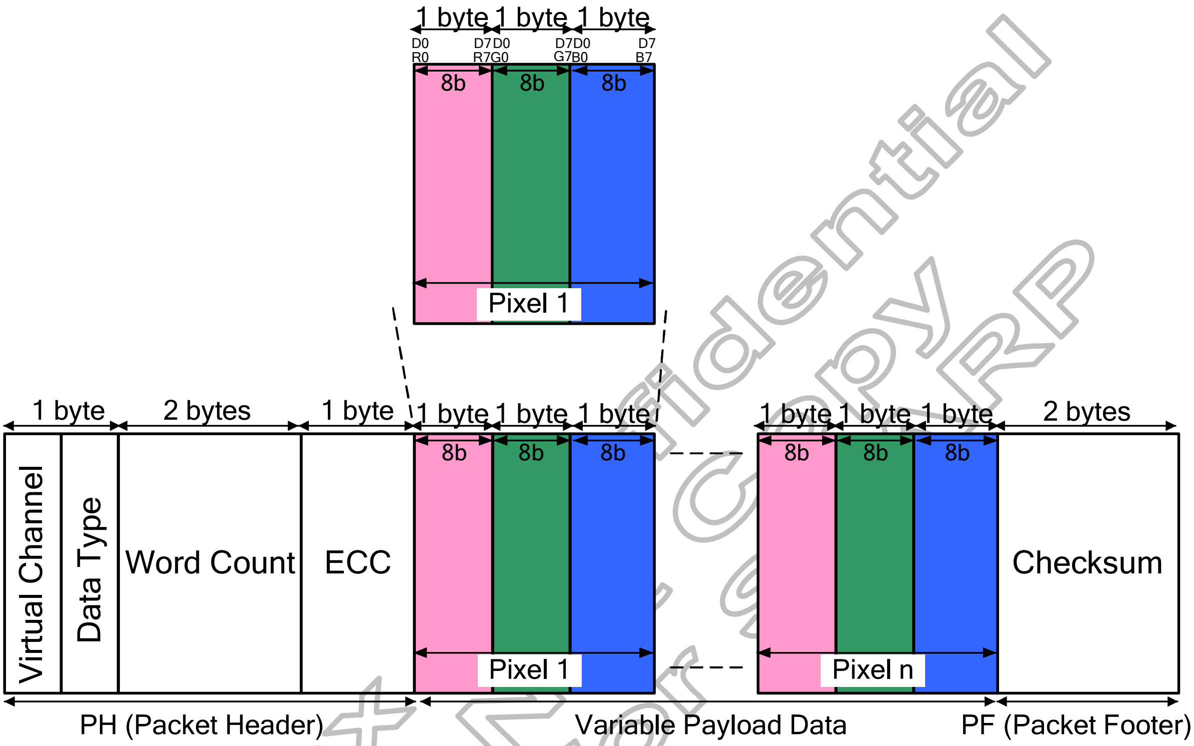
Data stream format		
Data type, hex	**Function description**	**Number of bytes**
2Eh	Packed Pixel Stream 18-Bit Format, each R, G, or B color component is one byte form, but the valid pixel bits occupy bits [7:2] and bits [1:0] of are ignored. Pixel format is “(6 bits) red, (6 bits) green and (6 bits) blue”.	Up to 65541 bytes (DI + WC + ECC + DCS CMD. + Payload DATA + PF)



Note: Within a color component, the LSB is sent first, the MSB last and With this format, pixel boundaries line up with byte boundaries every three bytes.

Packed pixel stream, 24-bit format

Data type, hex	Function description	Number of bytes
3Eh	Packed Pixel Stream 24-Bit Format is used to transmit image data formatted as 24-bit pixels to a Video Mode display module. Pixel format is (8 bits) red, (8 bits) green and (8 bits) blue.	Up to 65541 bytes (DI + WC + ECC + DCS CMD. + Payload DATA + PF)



Note: Within a color component, the LSB is sent first, the MSB last and With this format, pixel boundaries line up with byte boundaries every three bytes.

4.4.4 Peripheral to processor (reverse direction)

All Command Mode systems require bidirectional capability for returning READ data, ACK or error information to the host processor. Command Mode that use DCS shall have a bidirectional data path. Short packets and the header of Long packets may use ECC and Checksum to provide a higher level of data integrity. The Checksum feature enables detection of errors in the payload of Long packets. The packet structure for peripheral-to-processor transactions is the same as for the processor-to-peripheral direction.

Peripheral-to-processor transactions are of four basic types:

A. *Tearing Effect* is a Trigger message sent to convey display timing information to the host processor. Trigger messages ate signal byte packets sent by a peripheral's PHY layer in response to a signal form the DSI protocol layer.
B. *Acknowledge* is a Trigger Message sent when the current transmission, as well as all preceding transmissions since the last peripheral to host communication.
C. *Acknowledge and Error Report* is a Short packet sent if any errors were detected in preceding transmission from the host processor. Once reported, accumulated errors in the error register are cleared.
D. *Response to Read Request* may be Short or Long packet that returns data requested by the preceding READ command from the processor.

In general, if the host processor completes a transmission to the peripheral with BTA asserted, the peripheral shall respond with one or more appropriate packet(s), and then return bus ownership to the host processor. If BTA is not asserted following a transmission from the host processor, the peripheral shall not communicate an Acknowledge or other error information back to the host processor.

The processor-to-peripheral transactions with BTA asserted, can contain under form.

A. Following a **non-Read command** in which no error was detected, the peripheral shall respond with Acknowledge.
B. Following a **Read request** in which no error was detected, the peripheral shall send the requested READ data.
C. Following a **Read request in which the ECC error** was detected and corrected, the Peripheral shall send the requested READ data in a Long or Short packet, followed by a 4-byte (Acknowledge with Error Report*)* packet in the same LP transmission. The Error Report shall have the ECC Error flag set.
D. Following a **non-Read command in which the ECC error** was detected and corrected, the peripheral shall proceed to execute the command, and shall respond to BTA by sending a 4-byte (Acknowledge with Error Report*)* packet, the Error Report shall have the ECC Error flag set.
E. Following any command in which **SoT Error, SoT Sync Error, EoT Sync Error, LP Transmit Sync Error, checksum error or DSI VC ID Invalid** was detected, or the DSI command was not recognized, the peripheral shall send a 4-byte Acknowledge with Error Report response, with the appropriate error flags set in the two-byte error field. Only the ACK/Error Report packet shall be transmitted; no read or write accesses shall take place on the peripheral in response.

Which,

A. “Acknowledge” includes 2 bytes which are DI (VC + Acknowledge Data Type) and ECC.
B. “Acknowledge with Error Report” include 4 bytes which are DI, 2 bytes Error report and ECC.
C. “Response to Read Request” contains 2 types which are Short packet and long packet.

An error report is comprised of two bytes following the DI byte, with an ECC byte following the error report bytes. Table 4.5 shows the Error Report Bit Definitions. And Table 4.6 list complete set of peripheral-to-processor Data Types.

Bit	Description
0	SoT Error
1	SoT Sync Error
2	reserved
3	Escape Mode Entry Command Error
4	Low-Power Transmit Sync Error
5	LP-TX Timeout Error
6	reserved
7	reserved
8	ECC Error, single-bit (detected and corrected)
9	ECC Error, multi-bit (detected, not corrected)
10	Checksum Error (long packet only)
11	DSI Data Type Not Recognized
12	DSI VC ID Invalid
13	reserved
14	reserved
15	reserved

Table 4.5: Shows the error report bit definitions.

Data type, hex	Data type, binary	Description packet	Size
02h	00 0010	Acknowledge with Error Report	Short
1Ch	01 1100	DCS Long READ Response	Long
Others (00h→3Fh)		Reserved	-

Table 4.6: The complete set of peripheral-to-processor data types.

Acknowledge types		
Data type, hex	**Function description**	**Number of bytes**
02	Get Acknowledge with Error report when Error occurs from processor transmission.	4 bytes
Note: When processor transmits complete Payload, following signal by BTA, peripheral must respond to processor. With error→Acknowledge with error report, Without error→ Acknowledge.		

DCS Read types		
Data type, hex	**Function description**	**Number of bytes**
1Ch	This is the long-packet response to DCS Long Read Request.	Up to 65541 bytes (DI + WC + ECC + DCS CMD. + Payload DATA + PF)
Note: If the peripheral is Checksum capable, is shall return a calculated two-byte Checksum appended to the N-byte payload data. If the peripheral does not support Checksum, it shall return 0000h. If the DCS command itself is possibly corrupt, due to an uncorrectable ECC error, SoT or SoT Sync error, the requested READ data packet shall not be sent after the Acknowledge with Error Report packet be sent.		

5. Function Description

5.1 Tearing effect output line

The Tearing Effect output line supplies to the MPU a Panel synchronization signal. This signal can be enabled or disabled by the Tearing Effect Line Off & On commands. The mode of the Tearing Effect signal is defined by the parameter of the Tearing Effect Line On command.

Tearing Effect Line Modes

Mode 1, the Tearing Effect Output signal consists of V-Blanking Information only:

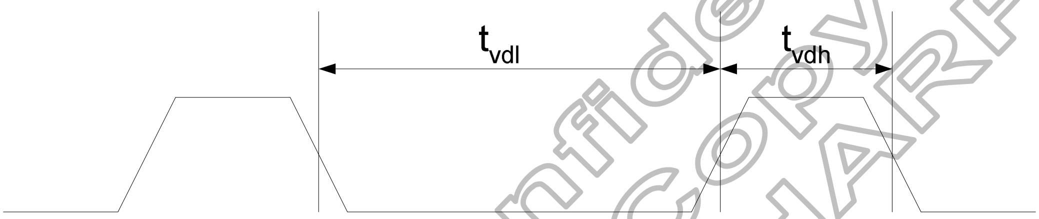


Figure 5.1: Tearing Effect Output signal mode 1

Under Mode1, the TE output timing will be defined by TEP[10:0] setting.

Ex: 1. VFB + VS + VBP= 6 line .

TEP[10:0]=0, then TE signal will output after last line finished.

TEP[10:0]=7, then TE signal will output at second line start.

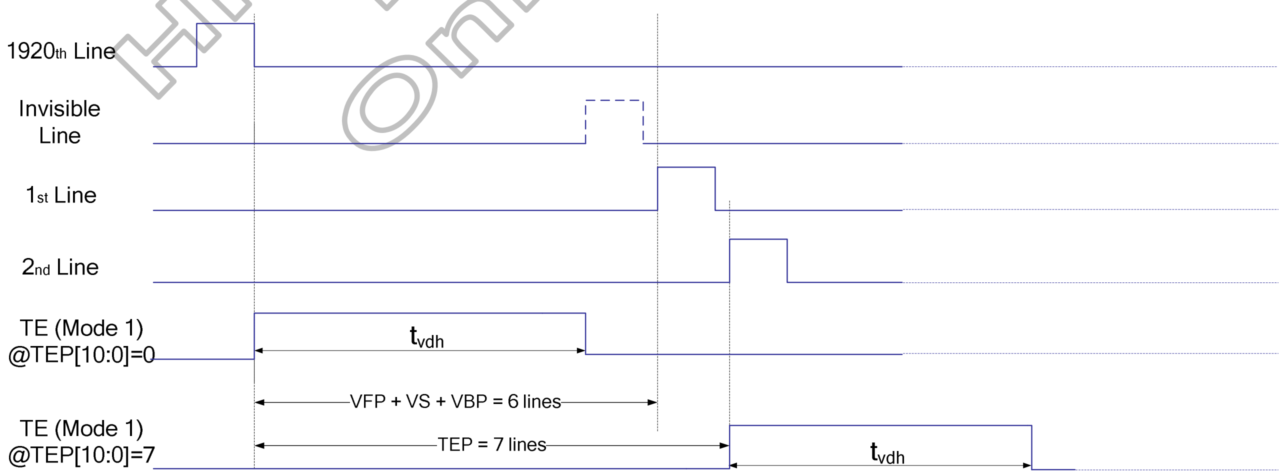


Figure 5.2: TE Delay Output

Mode 2, the Tearing Effect Output signal consists of V-Blanking and H-Blanking Information, there is one V-sync and N H-sync pulses per field.

N: If H_RES [2:0] = 3'b000 and V_RES[1:0] = 2'b00, the resolution is 1080 RGB X 1920, the N=1920.

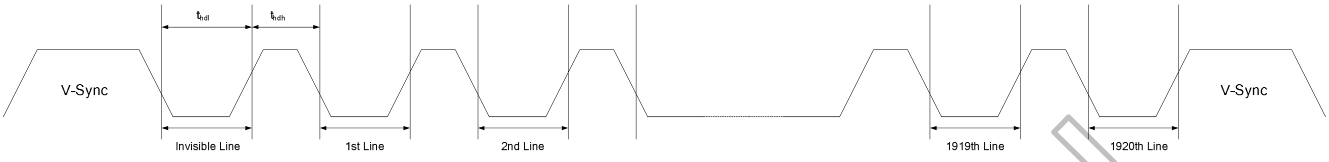


Figure 5.3: Tearing Effect Output signal mode 2

thdh= The LCD display is not updated

thdl= The LCD display is updated

Under Mode2, the H-sync pulses output amount will be defined by TESL[15:0] setting.

Ex: 1. TESL[15:0]=0, then TE signal will like TE mode 1.

TESL[15:0]=1, then TE signal will output 1920 H-sync.

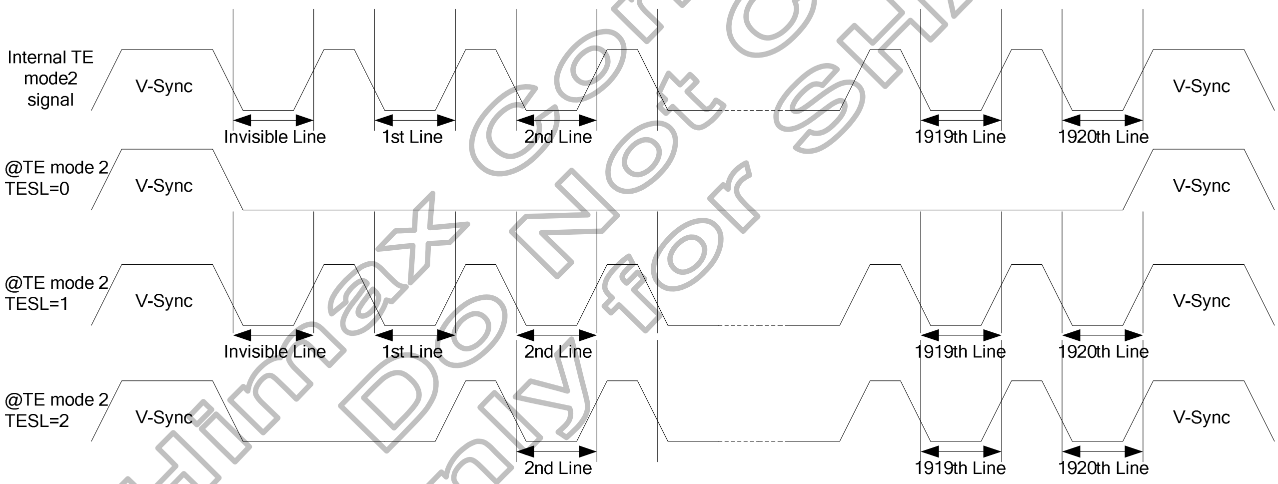


Figure 5.4: TE Output for TELINE setting

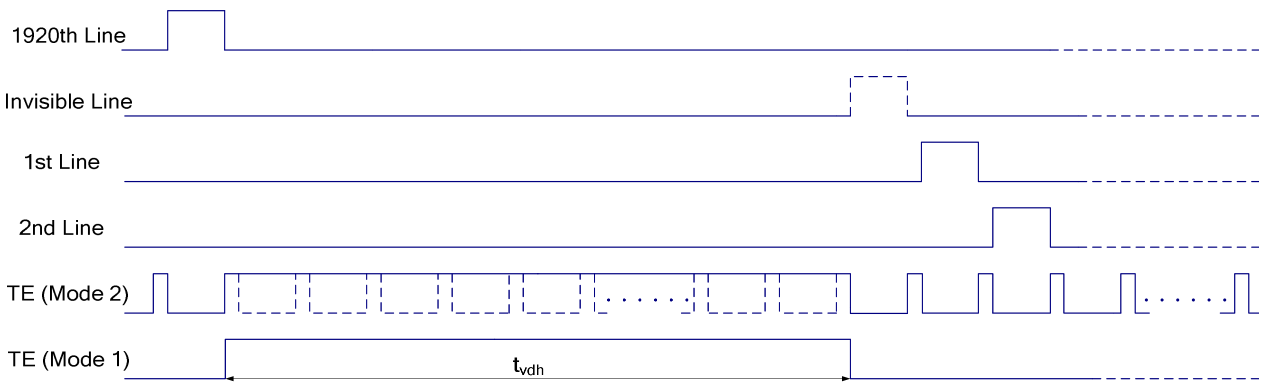


Figure 5.5: Tearing Effect Output signal

Note: During Sleep In Mode, the Tearing Output Pin is active Low

5.1.1 Tearing effect line timing

The Tearing Effect signal is described below:

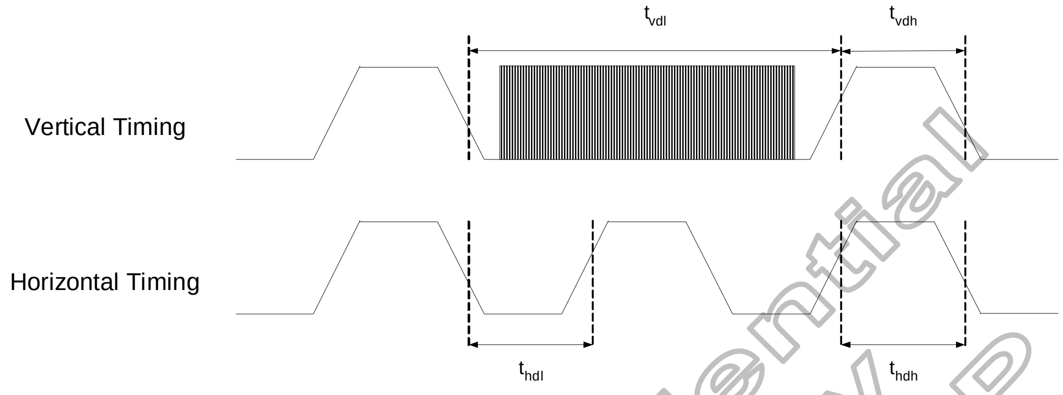


Figure 5.6: Tearing effect output line –tearing effect line timing

Idle Mode Off (Resolution 1080x1920 RGB, Frame Rate = 60 Hz)

Symbol	Parameter	Min.	Max.	Unit
tvdl	Vertical Timing Low Duration	15	-	ms
tvdh	Vertical Timing High Duration	1000	-	us
thdl	Horizontal Timing Low Duration	18	-	us
thdh	Horizontal Timing High Duration	0.13	500	us
tr	Rise time	-	15	ns
tf	Fall time	-	15	ns

Table 5.1: AC characteristics of tearing effect signal

The signal's rise and fall times (tf, tr) are stipulated to be equal to or less than 15ns.

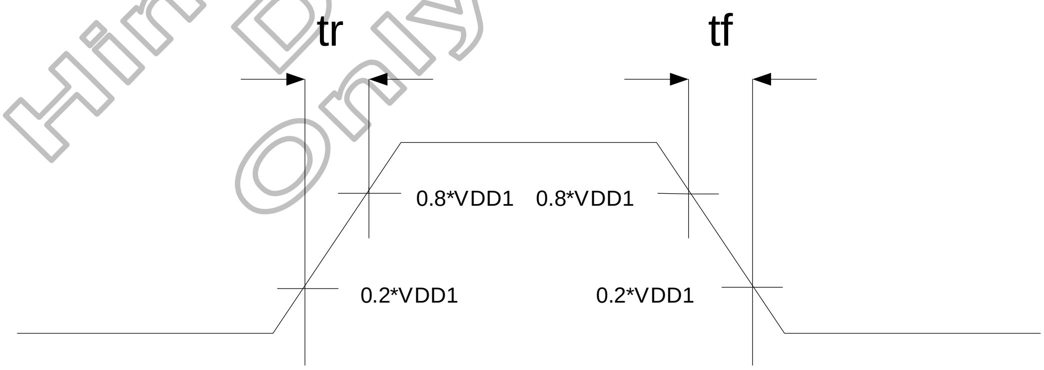


Figure 5.7: Tearing effect output line–definition of tf, tr

5.2 Oscillator

The HX8399-A can oscillate an internal R-C oscillator with an internal oscillation resistor (Rf). The oscillation frequency is changed according to the UADJ[3:0] internal register. Please refer to OSC control register. The default frequency is 73MHz(80MHz for 1200RGBx1920). The oscillation frequency tolerance is ±5%.

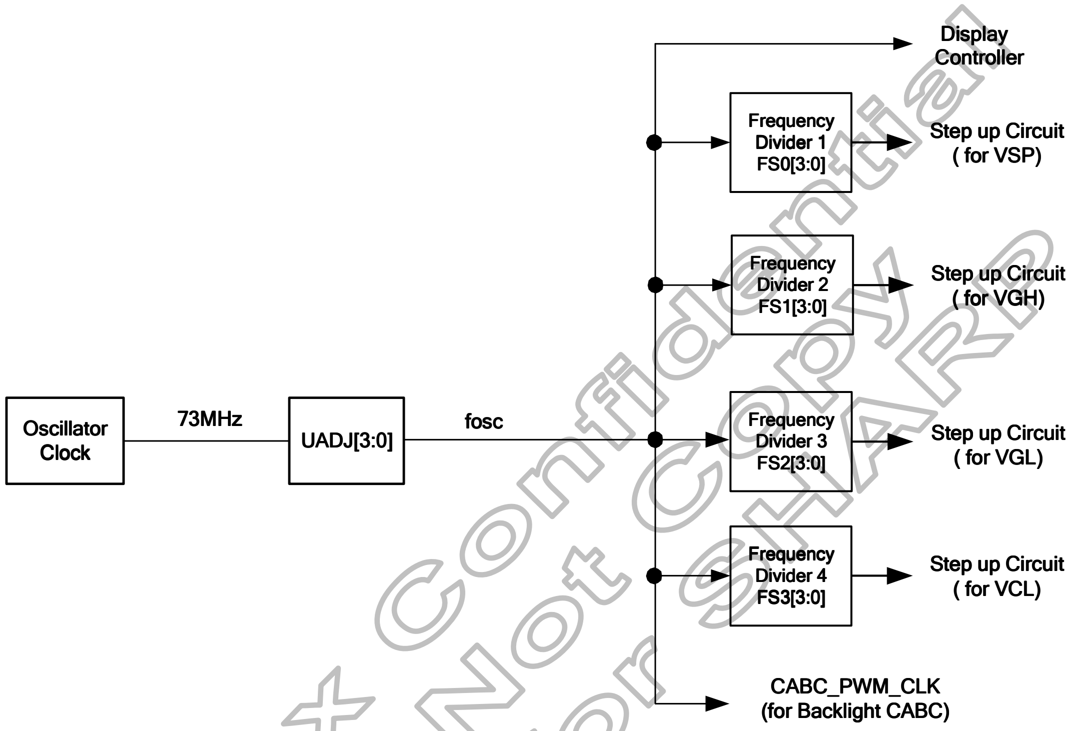


Figure 5.8: OSC aritecture

5.3 Source driver

The HX8399-A contains a 1202 channels of source driver which is used for driving the source line of TFT LCD panel. The source driver converts the input digital data into the analog voltage for 1202 channels and generates corresponding gray scale voltage output, which can realize a 16.7M colors display simultaneously. Since the output circuit of this source driver incorporates an operational amplifier, a positive and a negative voltage can be alternately outputted from each channel.

H_RES[2:0]	V_RES[1:0]	Resolution	Source channels
000	00	1080RGBX1920 dot	S1 ~ S540 , S661 ~ S1200
000	10	1080RGBX1440 dot	S1 ~ S540 , S661 ~ S1200
001	11	1024RGBX1280 dot	S1 ~ S512 , S689 ~ S1200
001	01	1024RGBX1600 dot	S1 ~ S512 , S689 ~ S1200
010	01	960RGBX1600 dot	S1 ~ S480 , S721 ~ S1200
010	10	960RGBX1440 dot	S1 ~ S480 , S721 ~ S1200
010	11	960RGBX1280 dot	S1 ~ S480 , S721 ~ S1200
011	01	900RGBX1600 dot	S1 ~ S450 , S751 ~ S1200
011	10	900RGBX1440 dot	S1 ~ S450 , S751 ~ S1200
100	00	1200RGBX1920 dot	S1 ~ S1200
101	11	720RGBx1280 dot	S1 ~ S360 , S841 ~ S1200
110	11	800RGBx1280 dot	S1 ~ S400 , S801 ~ S1200
111	-	Inhibited	-

Table 5.2: Source output for Panel resolution

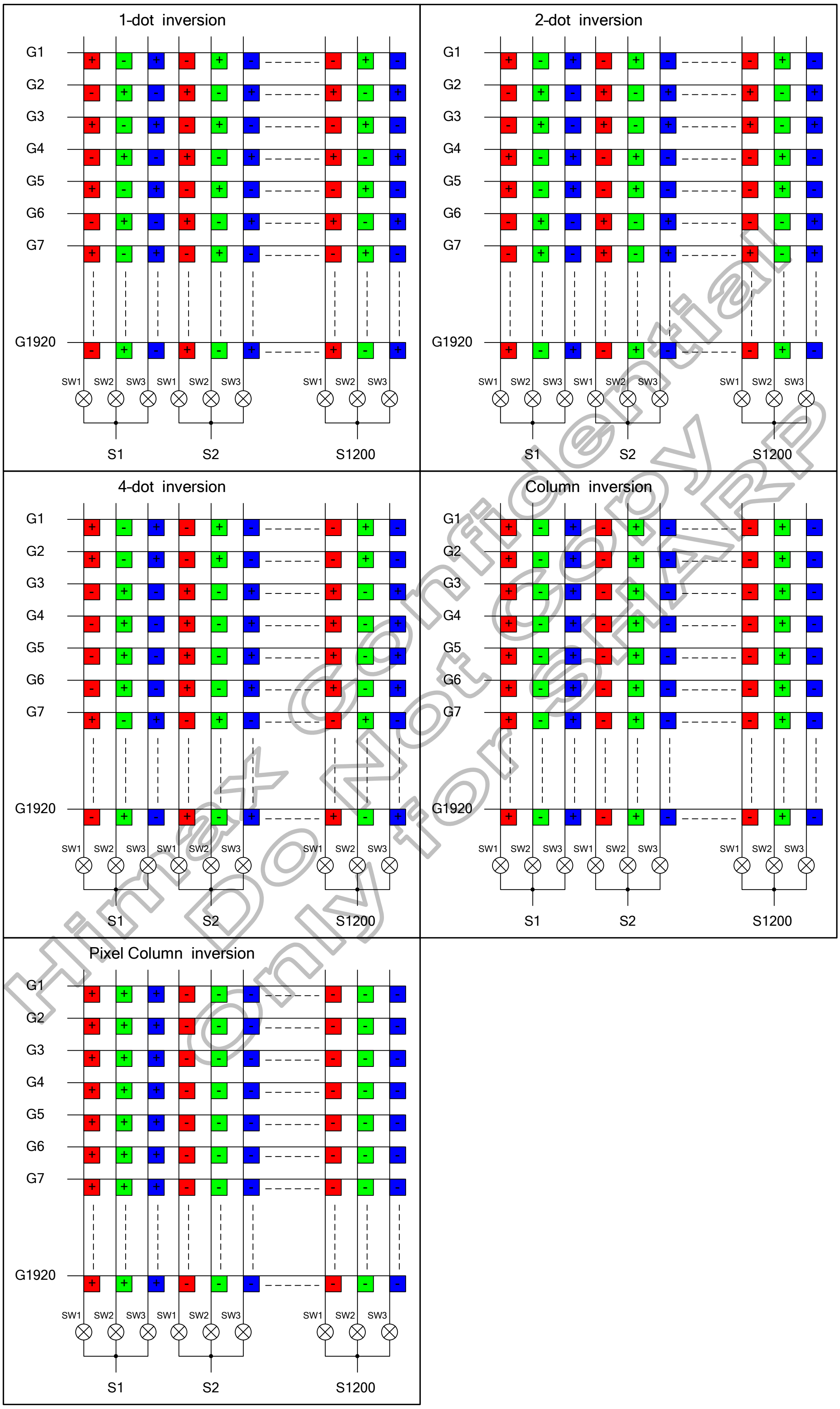


Figure 5.9: Inversion mode (MUX_SEL=0)

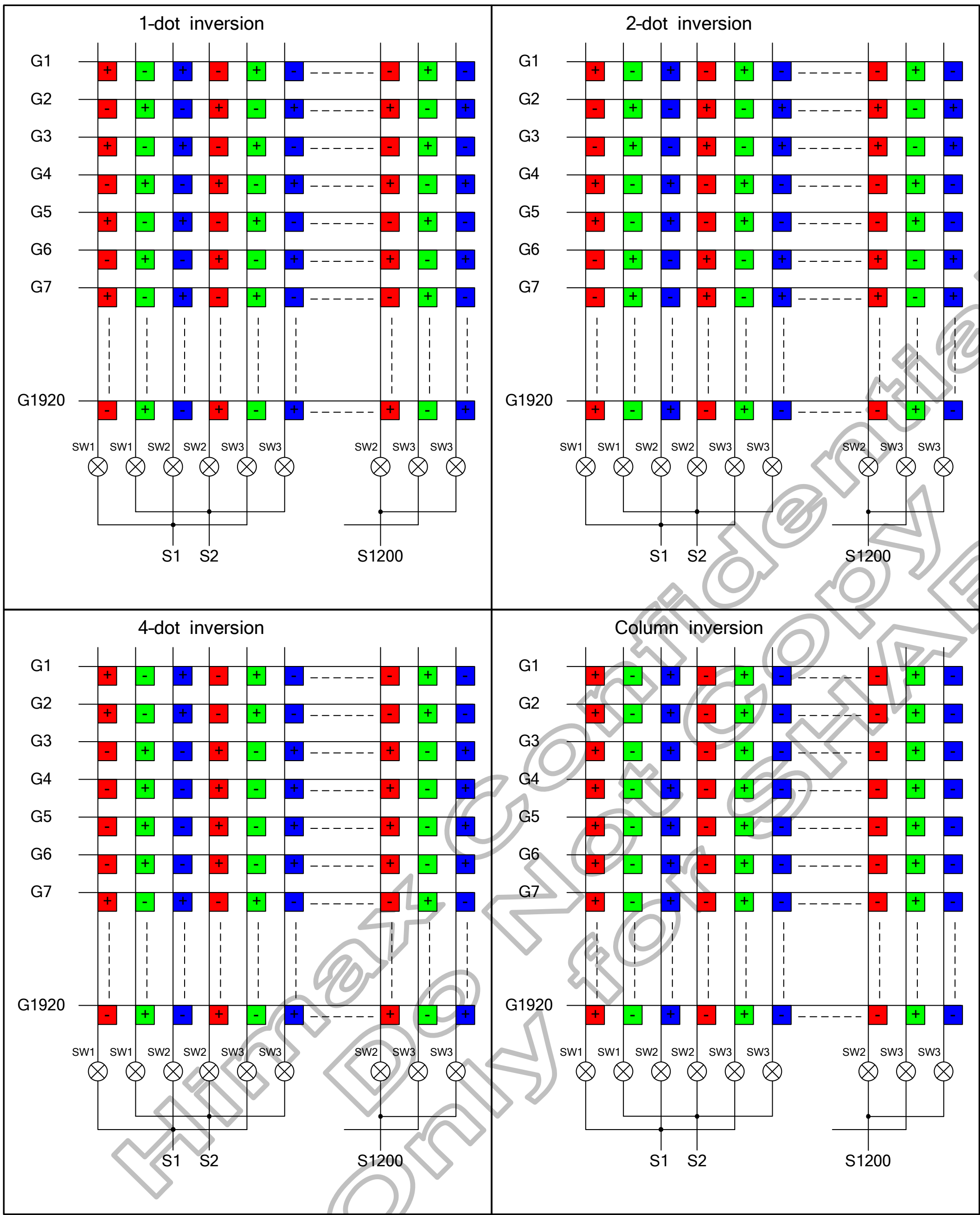


Figure 5.10: Inversion mode 2:6 (MUX_SEL=1)

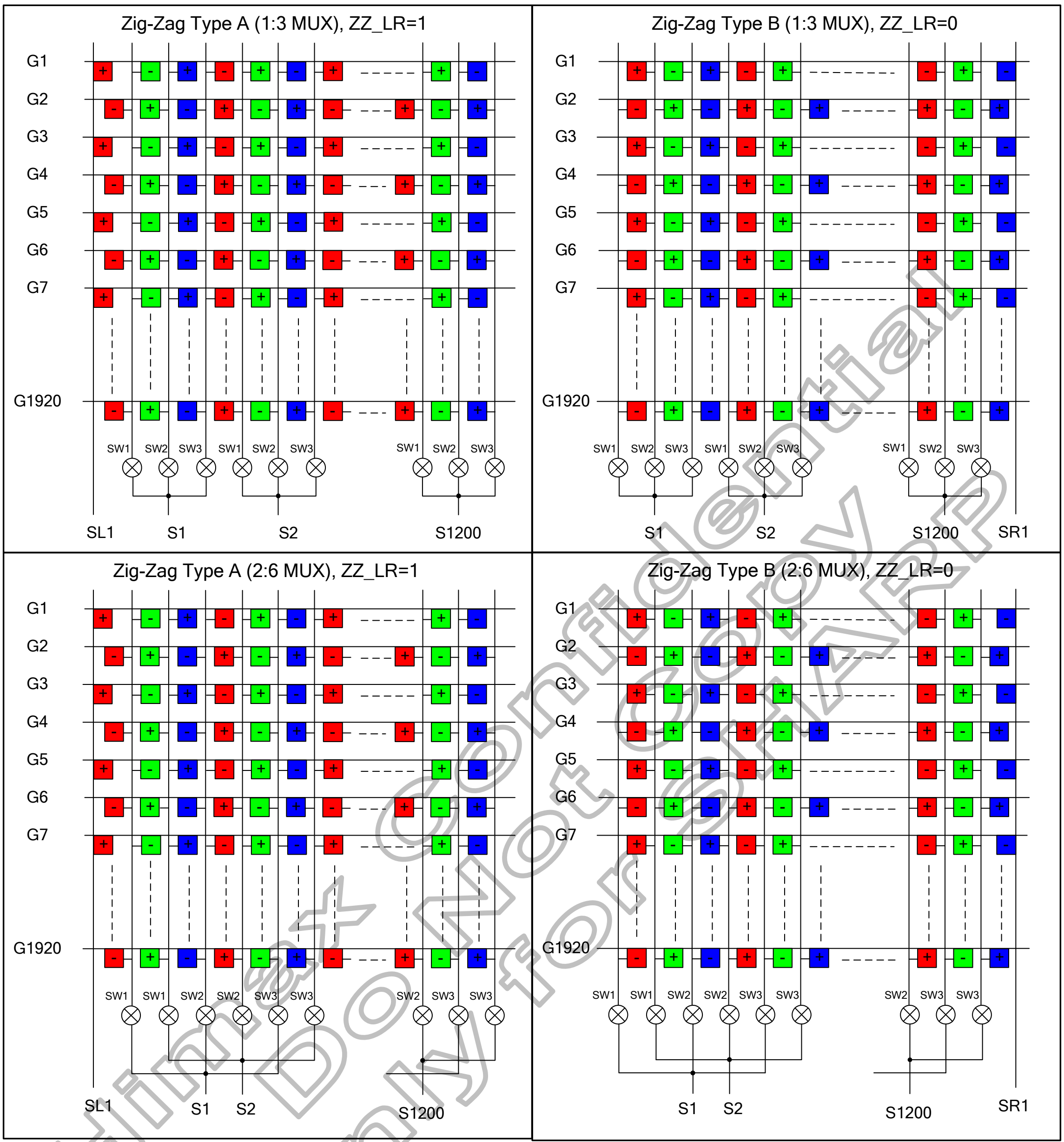


Figure 5.11: Zig-Zag Inversion mode(ZZ_EO=0)

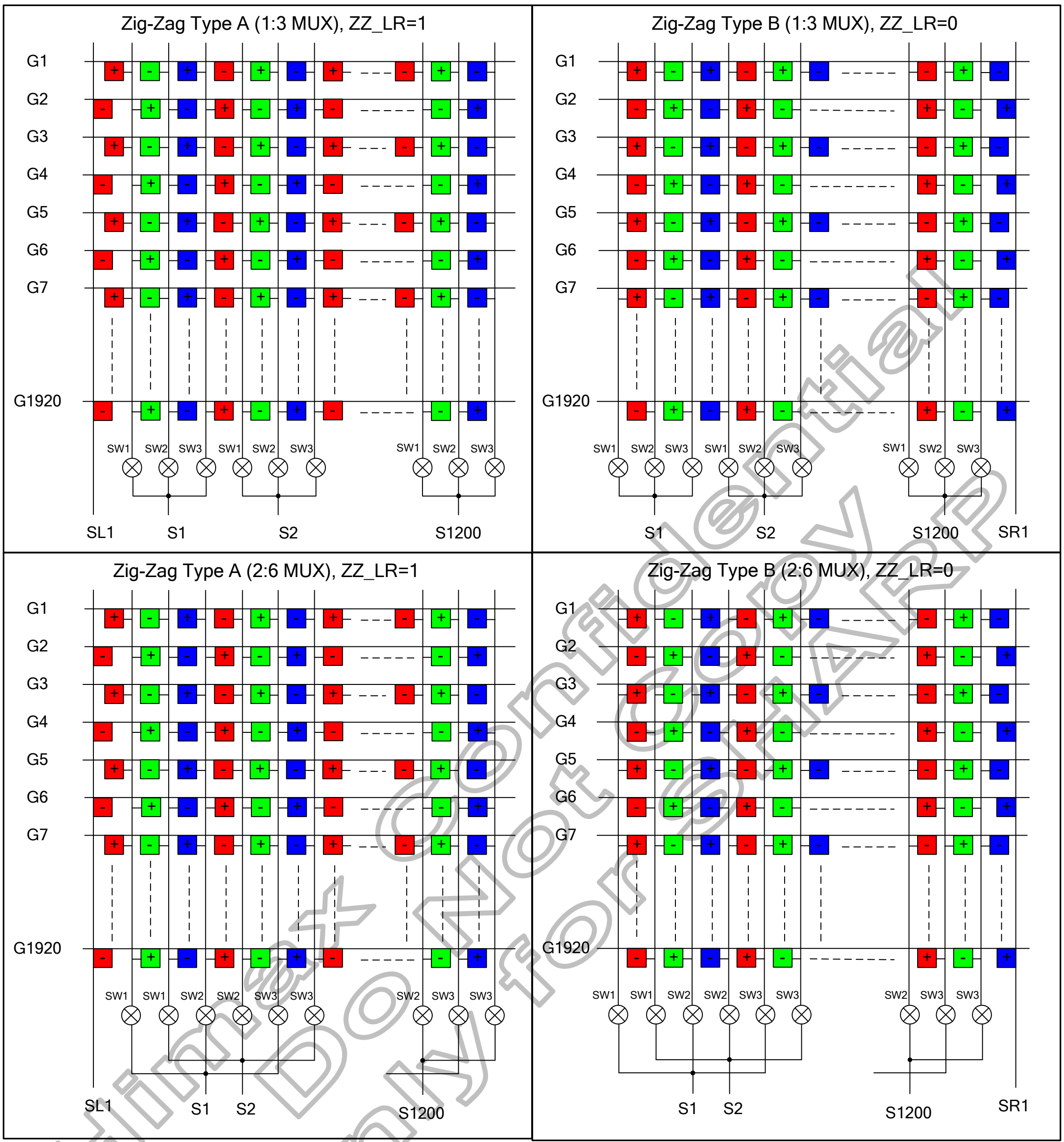


Figure 5.12: Zig-Zag Inversion mode(ZZ_EO=1)

5.4 LCD power generation scheme

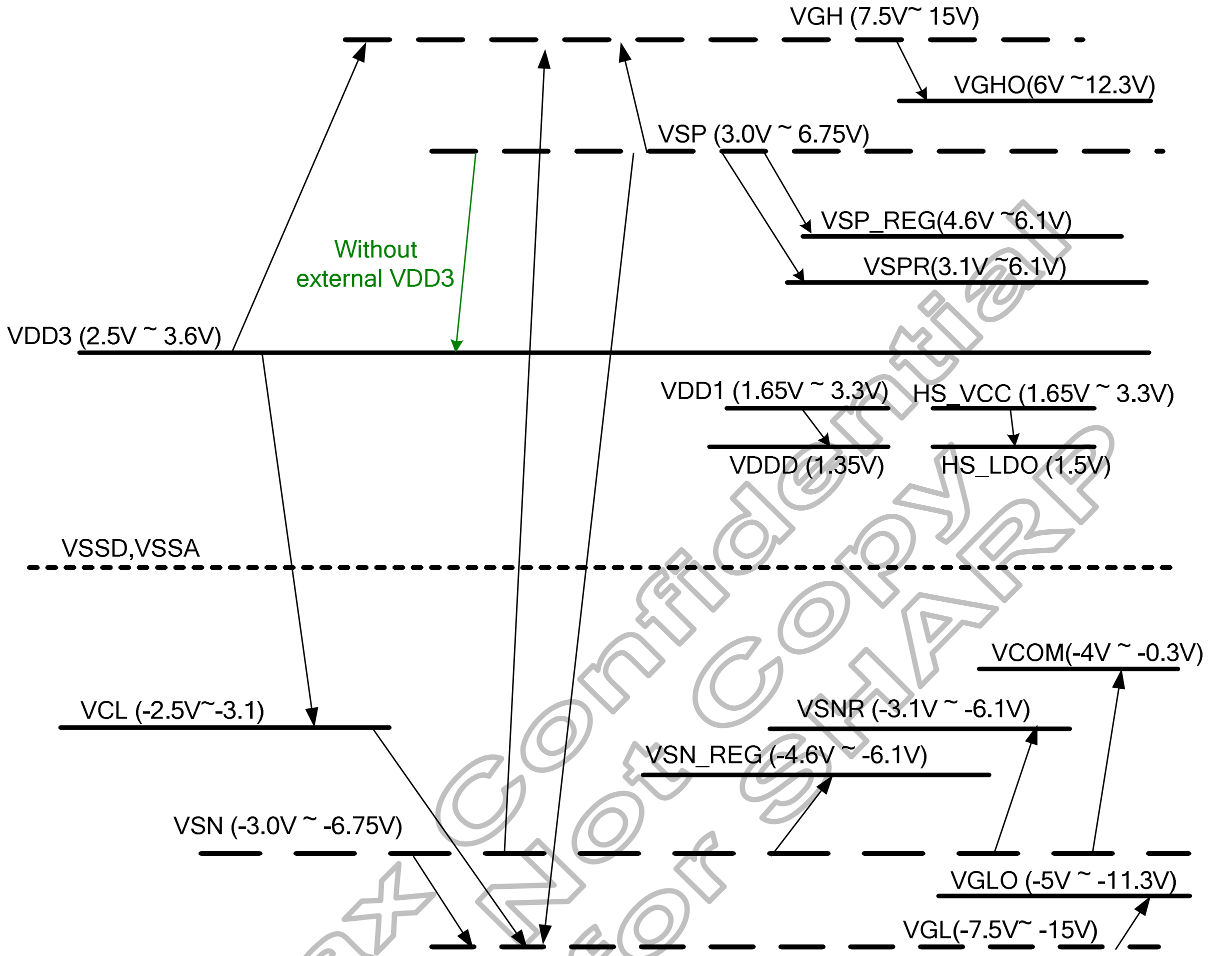


Figure 5.13: LCD power generation scheme

HX8399-A has an internal power supply circuit to drive LTPS LCD panel. Please set up each voltage output according to the LCD panel.

Name	Function	Set up value	Note
VSP_REG	DC/DC converter circuit output	4.6V ~ 6.1V	-
VSN_REG	DC/DC converter circuit output	-4.6V ~ -6.1V	-
VSPR	Reference voltage for gamma circuit	3.1V ~ 6.1V	-
VSNR	Reference voltage for gamma circuit	-3.1V ~ -6.1V	-
VDDD	Logic power supply	1.35V	-
VGH	DC/DC converter circuit output	7.5V ~ 15V	Depend on VSP and VSN
VGL	DC/DC converter circuit output	-7.5V ~ -15V	Depend on VSP and VSN
VGHO_L/R	Positive gate driver output voltage level	6V ~12.3V	-
VGLO_L/R	Negative gate driver output voltage level	-5V ~ -11.3V	-
VCL	DC/DC converter circuit output	-2.5V~-3.1 or –VDD3	-
VCOM	VCOM DC voltage	-4V ~ -0.3V	-
HS_LDO	Analog power for High speed interface circuit	1.5V	Depend on DSI I/F

Table 5.3: Voltage configuration

5.5 DC/DC converter circuit

5.5.1 Charge pump and step up circuit mode

HX8399-A supports various kinds of power generation mode, including PFM Type A, PFM Type B, PFM Type C and external HX5186 and external VSP & VSN and external VSP&VSN&VDD3. All power power mode can be set by hardware pins PCCS[2:0] as below:

PCCS2	PCCS1	PCCS0	Power source	Driving Mode
0	0	0	VDD3 / VDD1	PFM Type C
0	0	1	VDD3 /VDD1	HX5186-A/B/C
0	1	0	VSP /VDD1	PFM Type B(for VSN)
0	1	1	VSP / VSN / VDD1	External-1
1	0	1	VDD3 / VDD1	PFM Type A
1	1	1	VDD3 / VSP / VSN / VDD1	External-2
Others			Reserved	Reserved

Table 5.4: Power mde setting

5.5.2 Use HX5186-A/B/C

The HX5186-A is highly efficient switching voltage generator circuits that generate the high voltage level VSP/VSN required for source drivers. HX8399-A contains Charge Pump Controller for HX5186-A, including a comparator for VSP/VSN feedback control. HX5186-A can provide maximum efficiency and use minimum number of external components. The output voltage of the boost converter can be set from 3.0V to 6.75V (VSP) and -3.0V to -6.75V (VSN)

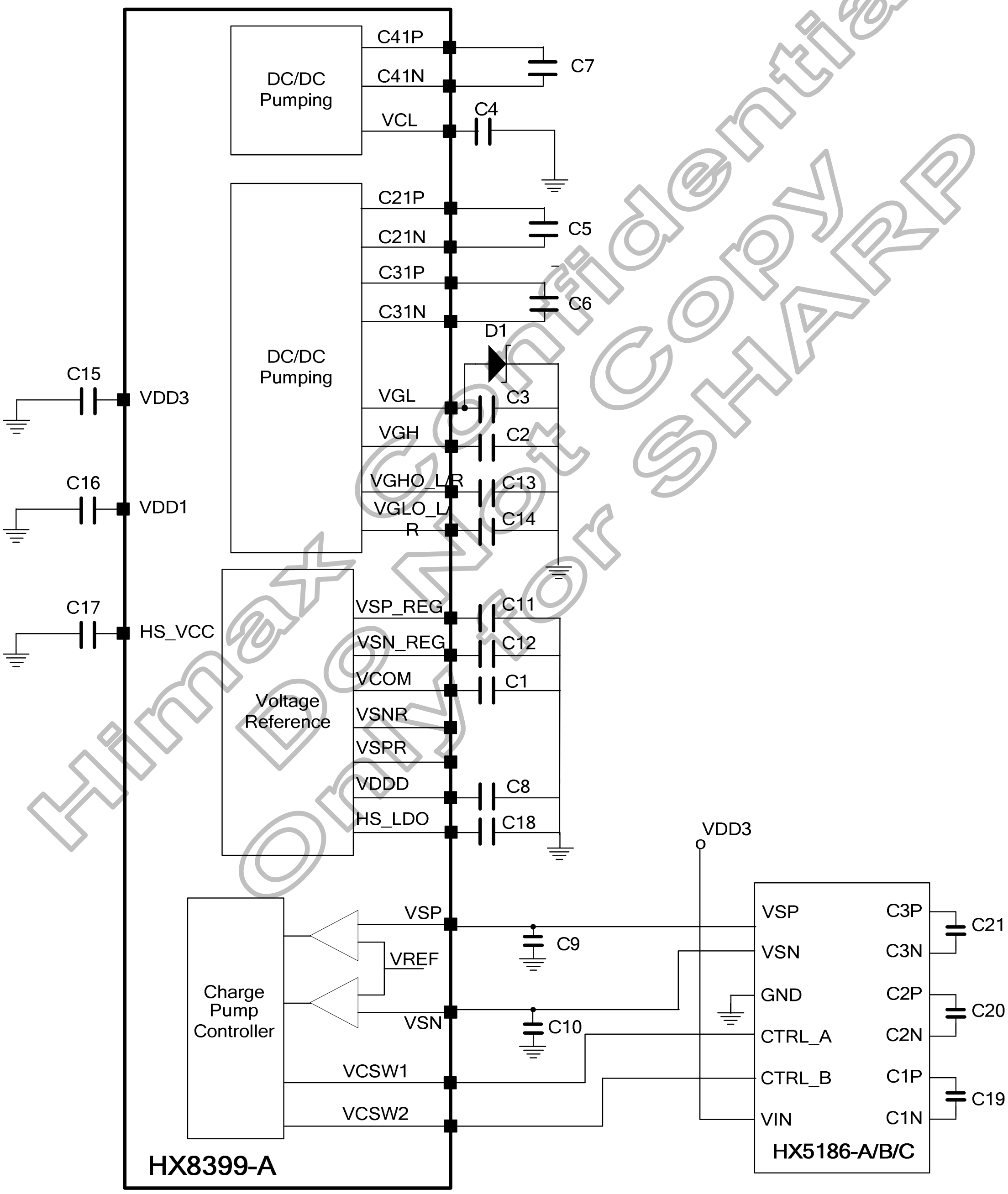


Figure 5.14: DC/DC converter circuit of HX5186-A/B/C

5.5.3 Use PFM DC/DC converter

The PFM DC-DC converter generates the high voltage level VSP/VSN required for source drivers. HX8399-A contains sub-circuits of the PFM boost converter, including a precision 1.8V reference voltage, comparator, PFM controlling logic, and the output buffer. The boost converter uses a external power transistor to provide maximum efficiency and to minimize the number of external components. The output voltage of the boost converter can be set from 3.0V to 6.75V (VSP) and -3.0 to -6.75V (VSN)

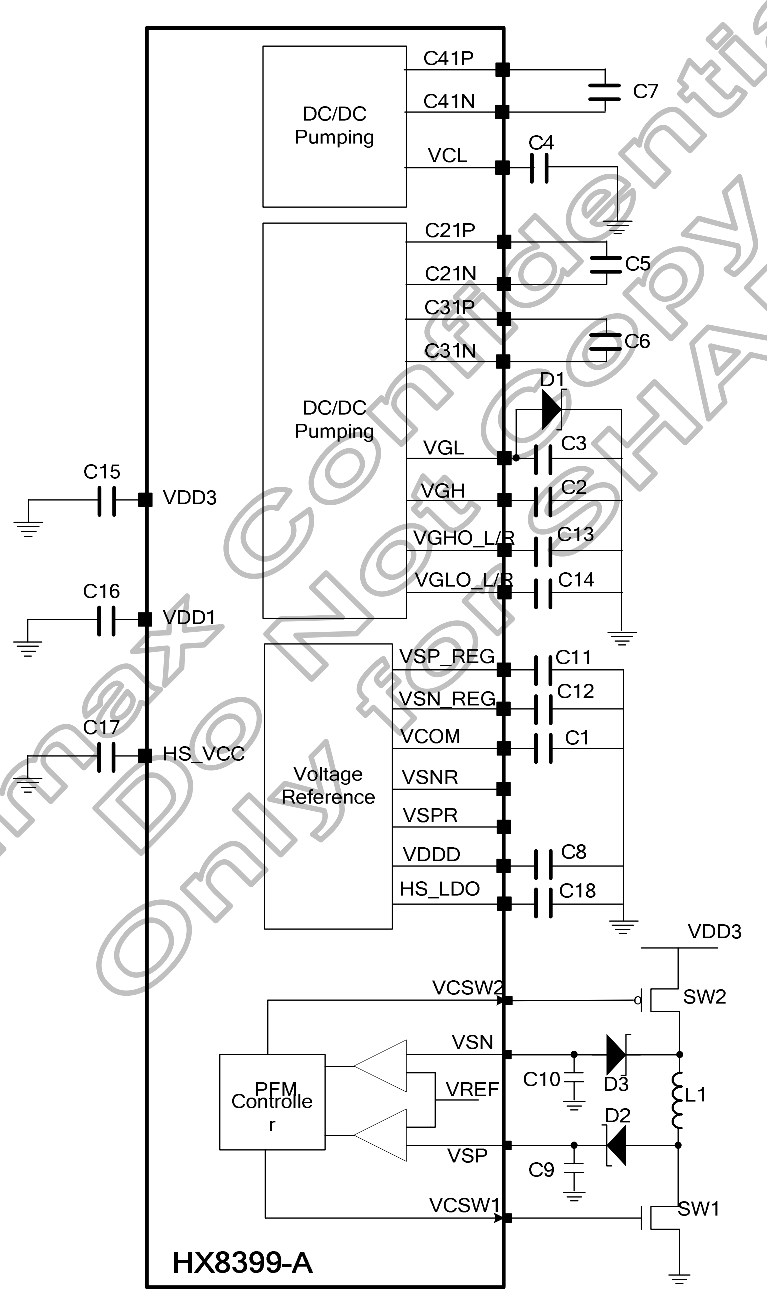


Figure 5.15: DC/DC converter circuit (PFM Type A)

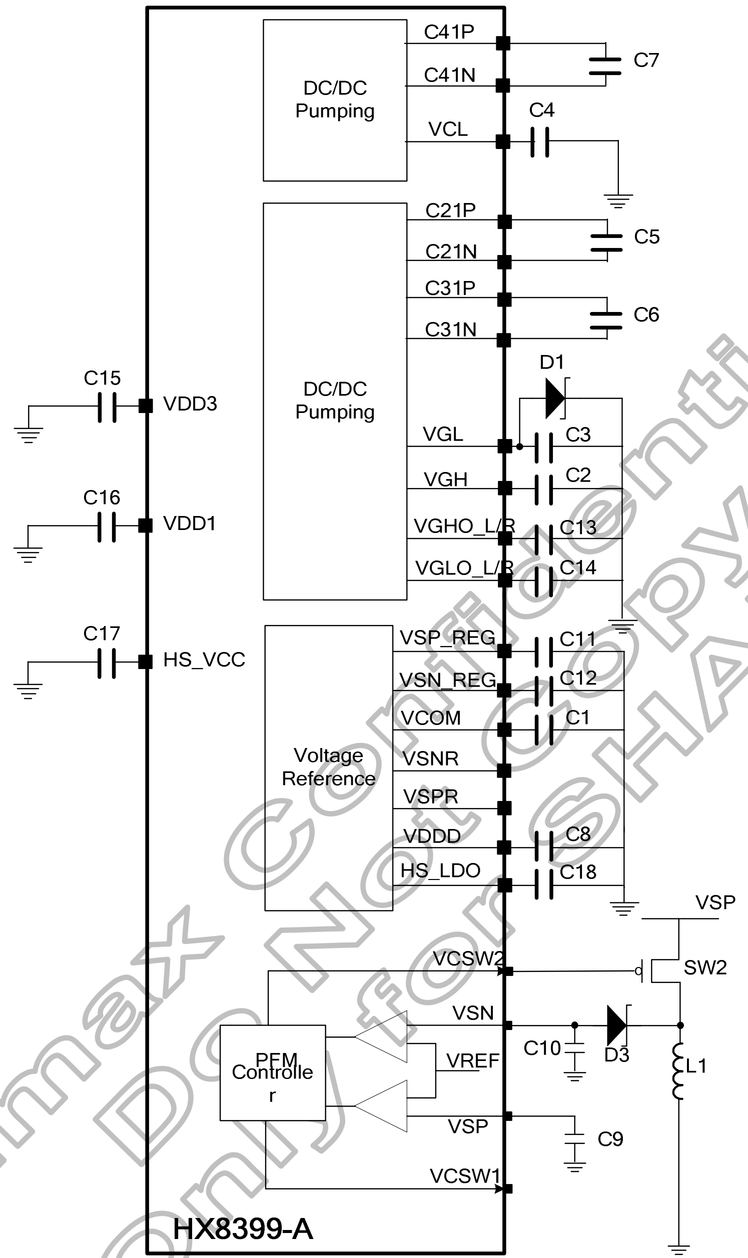


Figure 5.16: DC/DC converter circuit (PFM Type B)

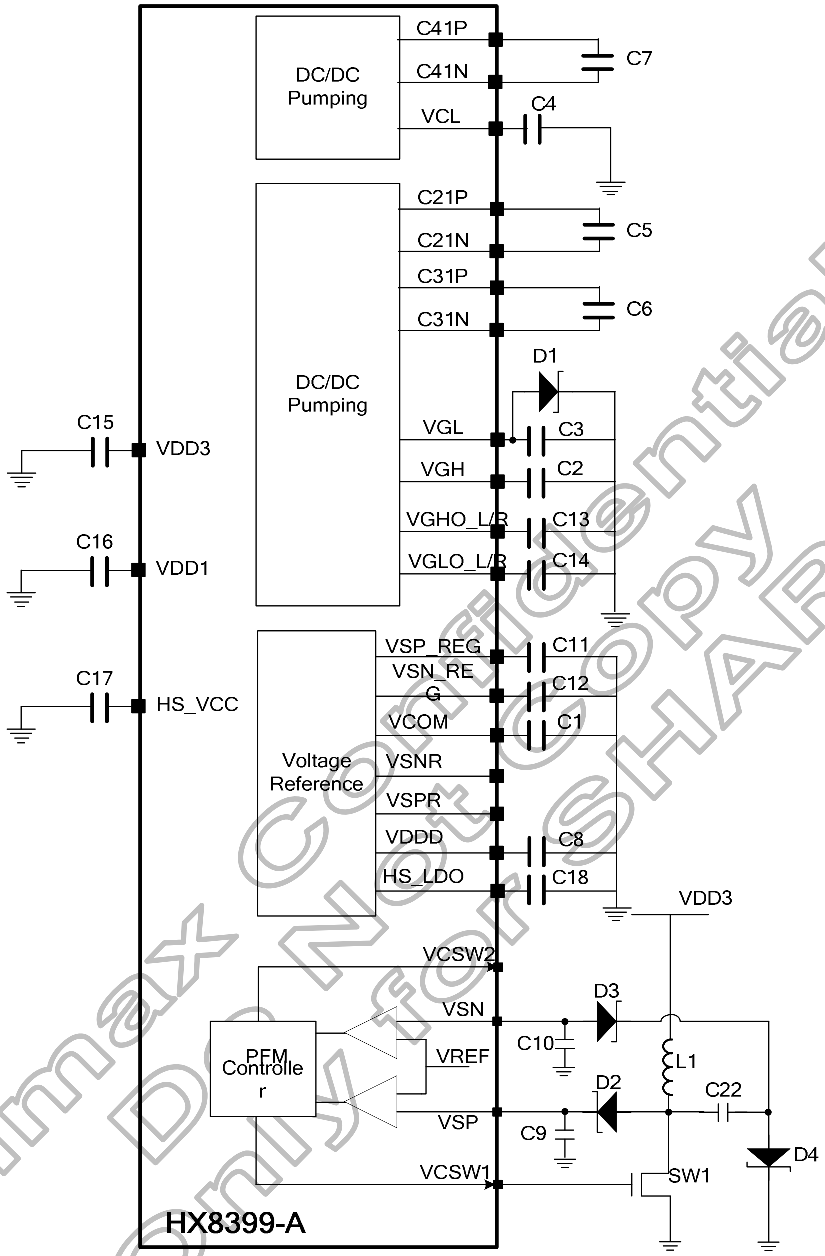


Figure 5.17: DC/DC converter circuit (PFM Type C)

5.5.4 Use external VSP and VSN circuit

VDD3 is generated from VSP by regulator.The input voltage range of VSP is from 4.8V ~ 6.75V. The input voltage range of VSN is from -4.8V ~ -6.75V.

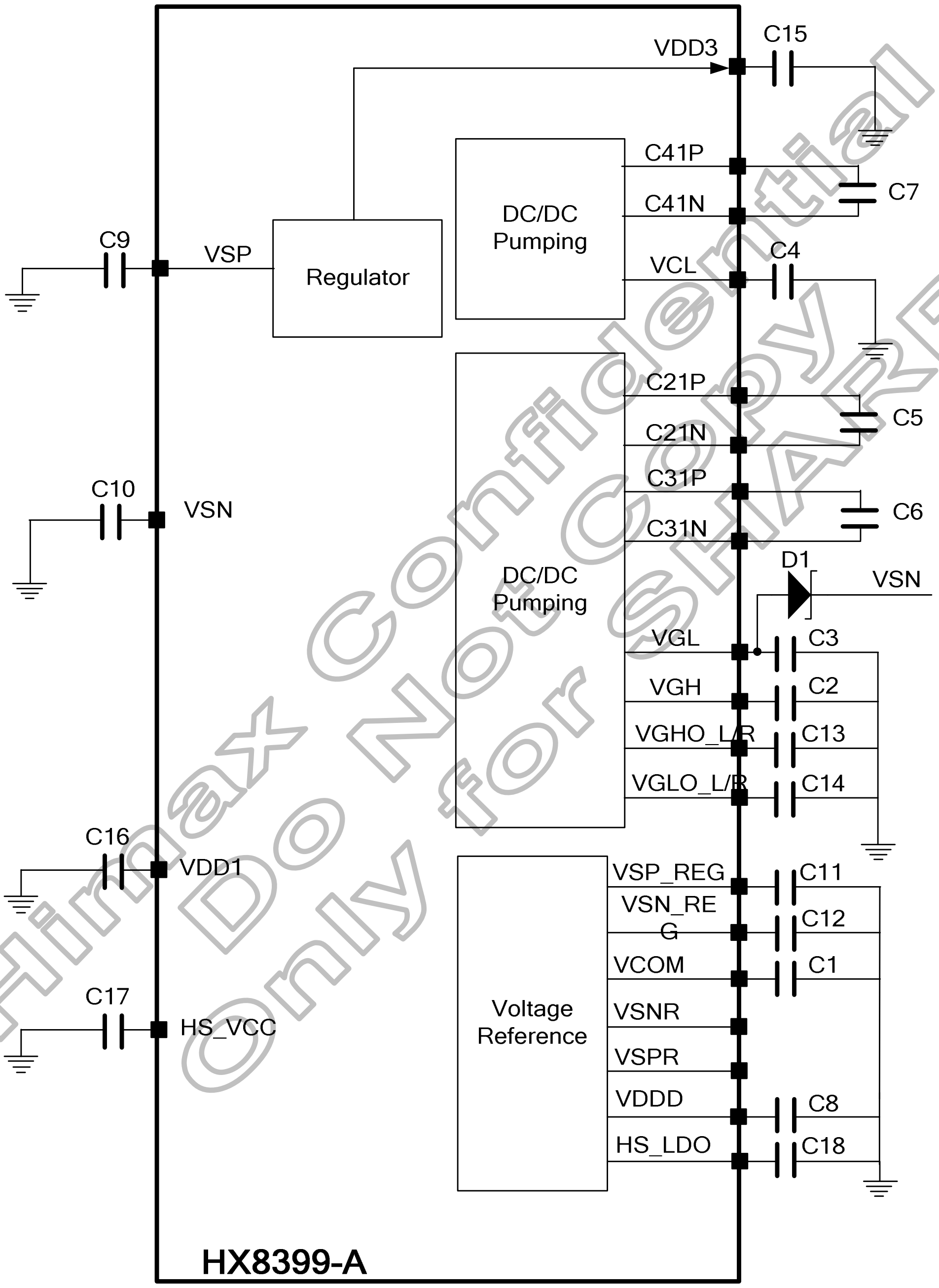


Figure 5.18: DC/DC converter circuit of external VSP/VSN

5.5.5 Use external VSP and VSN and VDD3 circuit

The input voltage range of VDD3 is from 2.5V ~ 3.6V.The input voltage range of VSP is from 4.8V ~ 6.75V. The input voltage range of VSN is from -4.8V ~ -6.75V.

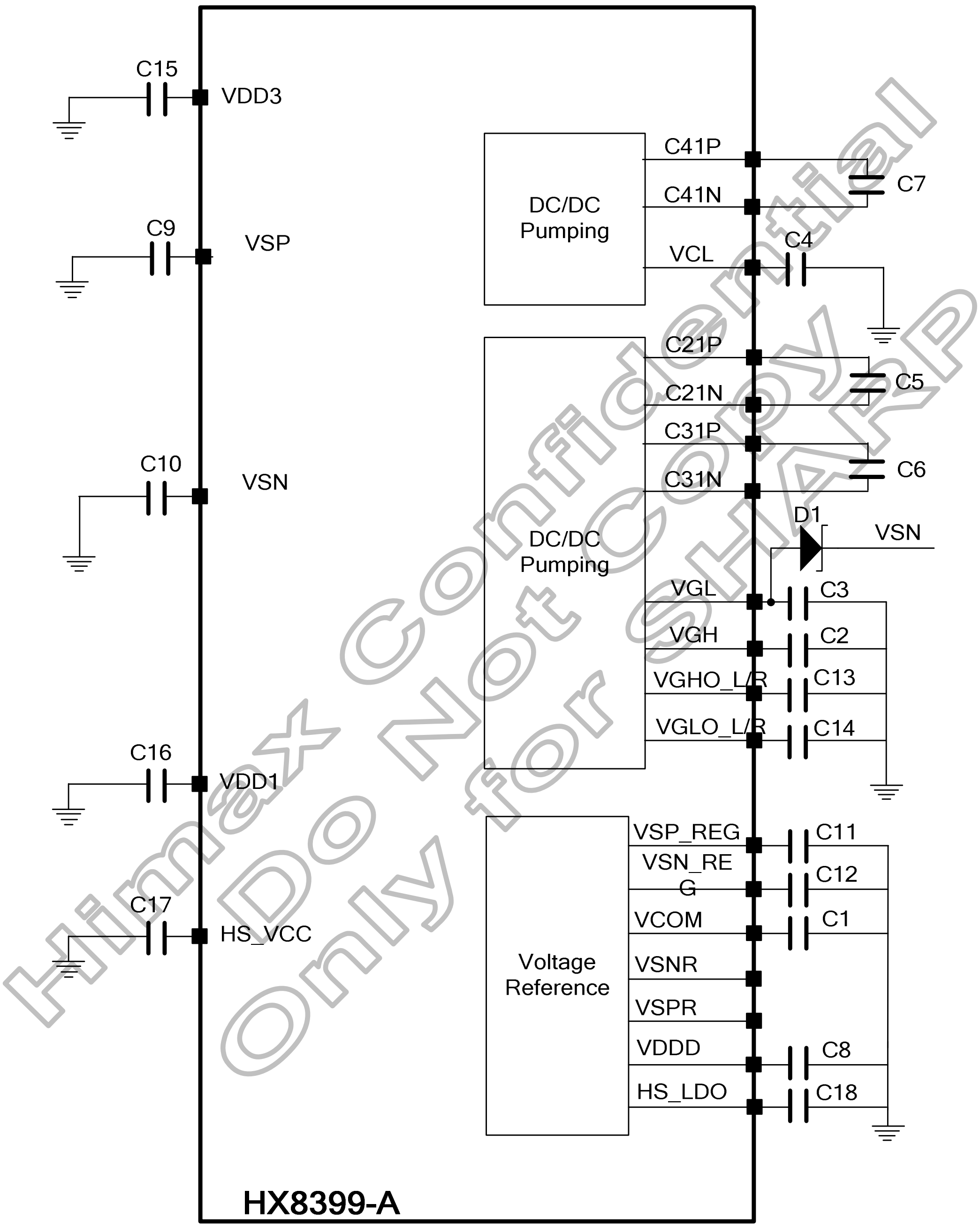


Figure 5.19: DC/DC converter circuit of external VDD3/VSP/VSN

5.6 Idle display

The HX8399-A supports an idle display mode. The grayscale level to be used is 0 and 255 with R7, G7, B7 decoding, and the other levels (1~254) are halted to reduce power consumption. In idle display mode, the Gamma-micro-adjustment registers are invalid and only the upper bits of RGB are used for display.

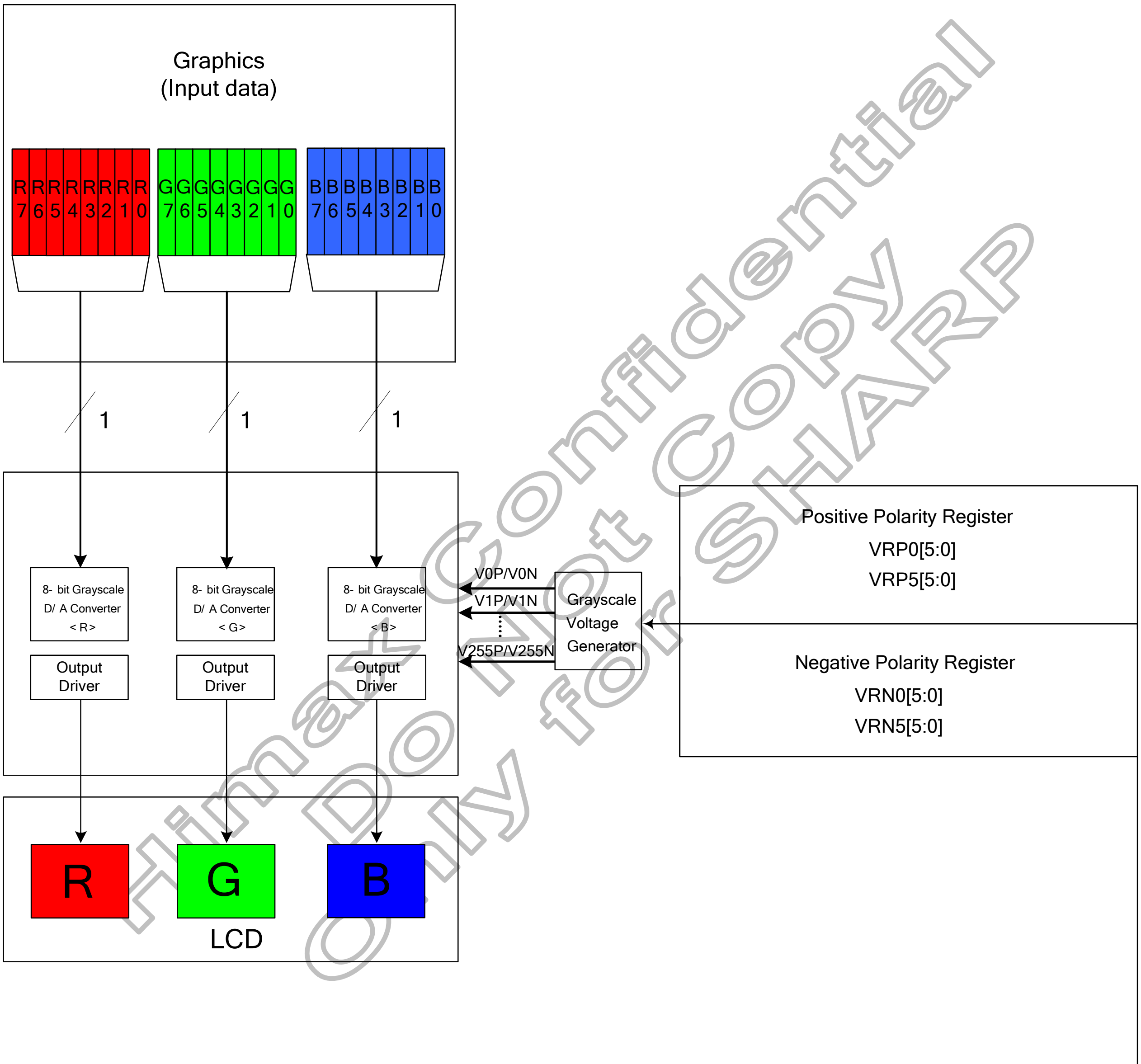
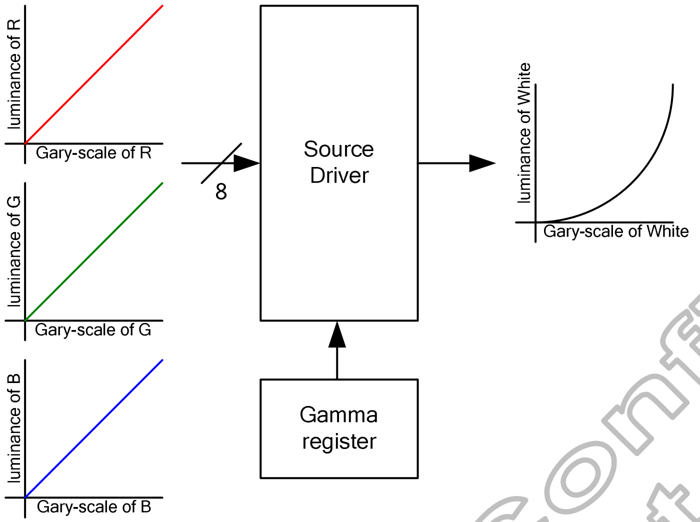


Figure 5.20: Idle mode grayscale control

5.7 Gamma characteristic correction function

The HX8399-A offers two kinds of Gamma adjustment ways to come to accord with LC characteristic, one kind is through Source Driver directly, another one is adjusted by the digital gamma correction. The Gamma adjustment way is selected by internal register DGC_EN bit.

A) Gamma adjustment of Source Driver



B) Gamma adjustment of Digital Gamma Correction

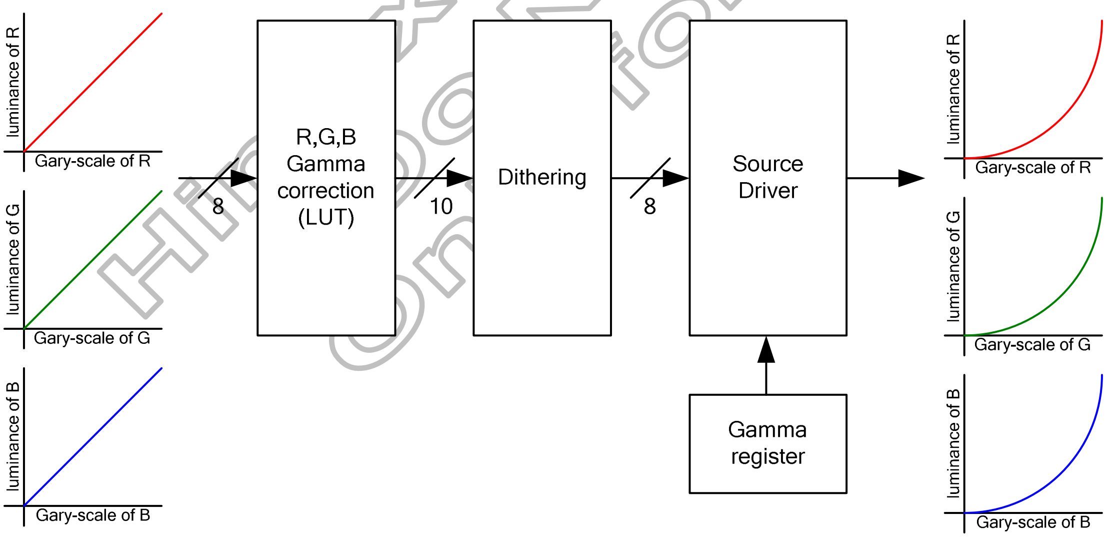


Figure 5.21: Gamma adjustments different of source driver with digital gamma correction

The HX8399-A incorporates gamma adjustment function for the 16,777,216-color display (256 grayscale for each R, G, B color). Gamma adjustment operation is implemented by deciding the16 grayscale levels firstly in gamma adjustment control registers to match the LCD panel. Then total 512 grayscale levels are generated in grayscale voltage generator. These registers are available for both polarities.

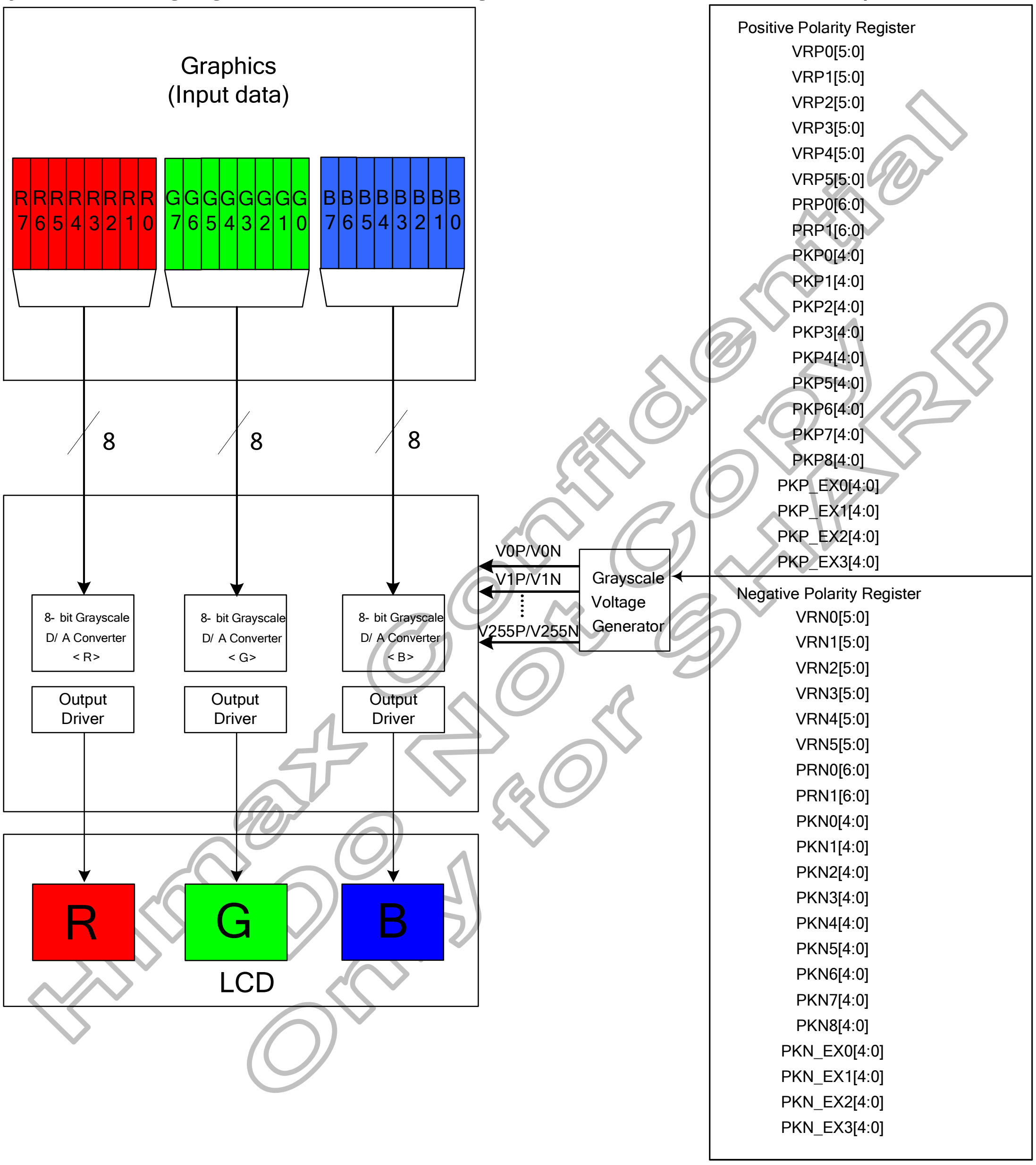


Figure 5.22: Grayscale control

5.7.1 Gamma-Characteristics adjustment register

The HX8399-A has register groups for specifying a series grayscale voltage that meets the Gamma-characteristics for the LCD panel used. These registers are divided into two groups, which correspond to the gradient, amplitude, and macro adjustment of the voltage for the grayscale characteristics. The polarity of each register can be specified independently.

(1) Offset adjustment registers

The offset adjustment variable registers are used to adjust the amplitude of the grayscale voltage. This function is implemented by controlling these selector in the top and bottom of the gamma resister stream for reference gamma voltage generation. These registers are available for both positive and negative polarities.

(2) Gamma center adjustment registers

The gamma center adjustment registers are used to adjust the reference gamma voltage in the middle level of grayscale without changing the dynamic range. This function is implemented by choosing one input of 88 to 1 selector in the gamma resister stream for reference gamma voltage generation. These registers are available for both positive and negative polarities.

(3) Gamma macro adjustment registers

The gamma macro adjustment registers can be used for fine adjustment of the reference gamma voltage. This function is implemented by controlling the 32-to-1 selectors (PKP/N0~8 ans PKP/N_EX0~3), each of which has 5-bit inputs and generates one reference voltage output.

Register Groups	Positive Polarity	Negative Polarity	Description
Center Adjustment	PRP0 6-0	PRN0 6-0	88-to-1 selector (voltage level of grayscale 52)
	PRP1 6-0	PRN1 6-0	88-to-1 selector (voltage level of grayscale 204)
Macro Adjustment	PKP0 4-0	PKN0 4-0	32-to-1 selector (voltage level of grayscale 12)
	PKP1 4-0	PKN1 4-0	32-to-1 selector (voltage level of grayscale 28)
	PKP2 4-0	PKN2 4-0	32-to-1 selector (voltage level of grayscale 76)
	PKP3 4-0	PKN3 4-0	32-to-1 selector (voltage level of grayscale 100)
	PKP4 4-0	PKN4 4-0	32-to-1 selector (voltage level of grayscale 132)
	PKP5 4-0	PKN5 4-0	32-to-1 selector (voltage level of grayscale 156)
	PKP6 4-0	PKN6 4-0	32-to-1 selector (voltage level of grayscale 180)
	PKP7 4-0	PKN7 4-0	32-to-1 selector (voltage level of grayscale 228)
	PKP8 4-0	PKN8 4-0	32-to-1 selector (voltage level of grayscale 243)
	PKP_EX0 4-0	PKN_EX0 4-0	32-to-1 selector (voltage level of grayscale 20)
	PKP_EX1 4-0	PKN_EX1 4-0	32-to-1 selector (voltage level of grayscale 40)
	PKP_EX2 4-0	PKN_EX2 4-0	32-to-1 selector (voltage level of grayscale 216)
	PKP_EX3 4-0	PKN_EX3 4-0	32-to-1 selector (voltage level of grayscale 236)
Offset Adjustment	VRP0 5-0	VRN0 5-0	64-to-1 selector (voltage level of grayscale 0)
	VRP1 5-0	VRN1 5-0	64-to-1 selector (voltage level of grayscale 4)
	VRP2 5-0	VRN2 5-0	64-to-1 selector (voltage level of grayscale 8)
	VRP3 5-0	VRN3 5-0	64-to-1 selector (voltage level of grayscale 247)
	VRP4 5-0	VRN4 5-0	64-to-1 selector (voltage level of grayscale 251)
	VRP5 5-0	VRN5 5-0	64-to-1 selector (voltage level of grayscale 255)

Table 5.5: Gamma-Adjustment registers

Gamma resister stream and selector

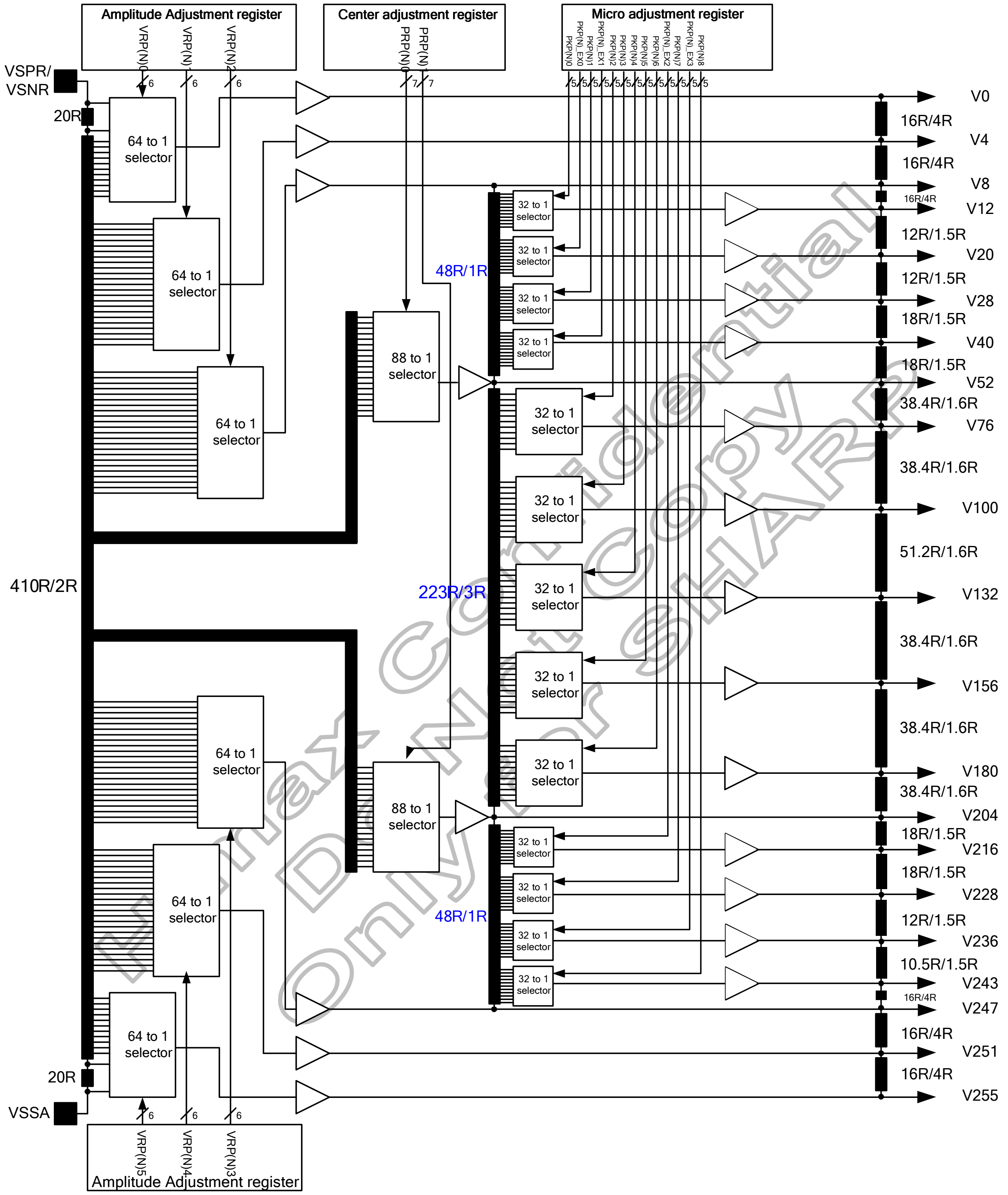


Figure 5.23: Gamma resister stream and gamma reference voltage

Variable resister

There are two types of variable resistors, one is for center adjustment and the other is for offset adjustment. The resistances are decided by setting values in the center adjustment, offset adjustment registers. Their relationships are shown below.

Value in Register VR(P/N)0 5-0	Resistance VR(P/N)0
000000	0R
000001	20R
000010	22R
000011	24R
• •	• •
011101	76R
011110	78R
011111	80R
100000	82R
100001	84R
100010	86R
• •	• •
111101	140R
111110	142R
111111	144R

Value in Register VR(P/N)1 5-0	Resistance VR(P/N)1
000000	0R
000001	2R
000010	4R
000011	6R
• •	• •
011101	58R
011110	60R
011111	62R
100000	64R
100001	66R
100010	68R
• •	• •
111101	122R
111110	124R
111111	126R

Value in Register VR(P/N)2 5-0	Resistance VR(P/N)2
000000	0R
000001	2R
000010	4R
000011	6R
• •	• •
011101	58R
011110	60R
011111	62R
100000	64R
100001	66R
100010	68R
• •	• •
111101	122R
111110	124R
111111	126R

Value in Register VR(P/N)3 5-0	Resistance VR(P/N)3
000000	0R
000001	2R
000010	4R
• •	• •
011101	58R
011110	60R
011111	62R
100000	64R
100001	66R
100010	68R
• •	• •
111100	120R
111101	122R
111110	124R
111111	126R

Value in Register VR(P/N)4 5-0	Resistance VR(P/N)4
000000	0R
000001	2R
000010	4R
• •	• •
011101	58R
011110	60R
011111	62R
100000	64R
100001	66R
100010	68R
• •	• •
111100	120R
111101	122R
111110	124R
111111	126R

Value in Register VR(P/N)5 5-0	Resistance VR(P/N)2
000000	0R
000001	2R
000010	4R
• •	• •
011101	58R
011110	60R
011111	62R
100000	64R
100001	66R
100010	68R
• •	• •
111100	120R
111101	122R
111110	124R
111111	144R

Table 5.6: Offset adjustment 0~5

Value in Register PR(P/N)0 6-0	Resistance PR(P/N)0
0000000	0R
0000001	2R
0000010	4R
• •	• •
1010101	170R
1010110	172R
1010111	174R

Value in Register PR(P/N)1 6-0	Resistance PR(P/N)1
0000000	0R
0000001	2R
0000010	4R
• •	• •
1010101	170R
1010110	172R
1010111	174R

Table 5.7: Center adjustment

The grayscale levels are determined by the following formulas:

Reference voltage	Macro adjustment value	VinP0 formula
VinP0	VRP0 5-0 = 000000	VSPR
	VRP0 5-0 = 000001	((450R - 20R) / 450R) * VSPR
	VRP0 5-0 = 000010	((450R - 22R) / 450R) * VSPR
	VRP0 5-0 = 000011	((450R - 24R) / 450R) * VSPR
	VRP0 5-0 = 000100	((450R - 26R) / 450R) * VSPR
	VRP0 5-0 = 000101	((450R - 28R) / 450R) * VSPR
	VRP0 5-0 = 000110	((450R - 30R) / 450R) * VSPR
	VRP0 5-0 = 000111	((450R - 32R) / 450R) * VSPR
	VRP0 5-0 = 001000	((450R - 34R) / 450R) * VSPR
	VRP0 5-0 = 001001	((450R - 36R) / 450R) * VSPR
	VRP0 5-0 = 001010	((450R - 38R) / 450R) * VSPR
	VRP0 5-0 = 001011	((450R - 40R) / 450R) * VSPR
	VRP0 5-0 = 001100	((450R - 42R) / 450R) * VSPR
	VRP0 5-0 = 001101	((450R - 44R) / 450R) * VSPR
	VRP0 5-0 = 001110	((450R - 46R) / 450R) * VSPR
	VRP0 5-0 = 001111	((450R - 48R) / 450R) * VSPR
	VRP0 5-0 = 010000	((450R - 50R) / 450R) * VSPR
	VRP0 5-0 = 010001	((450R - 52R) / 450R) * VSPR
	VRP0 5-0 = 010010	((450R - 54R) / 450R) * VSPR
	VRP0 5-0 = 010011	((450R - 56R) / 450R) * VSPR
	VRP0 5-0 = 010100	((450R - 58R) / 450R) * VSPR
	VRP0 5-0 = 010101	((450R - 60R) / 450R) * VSPR
	VRP0 5-0 = 010110	((450R - 62R) / 450R) * VSPR
	VRP0 5-0 = 010111	((450R - 64R) / 450R) * VSPR
	VRP0 5-0 = 011000	((450R - 66R) / 450R) * VSPR
	VRP0 5-0 = 011001	((450R - 68R) / 450R) * VSPR
	VRP0 5-0 = 011010	((450R - 70R) / 450R) * VSPR
	VRP0 5-0 = 011011	((450R - 72R) / 450R) * VSPR
	VRP0 5-0 = 011100	((450R - 74R) / 450R) * VSPR
	VRP0 5-0 = 011101	((450R - 76R) / 450R) * VSPR
	VRP0 5-0 = 011110	((450R - 78R) / 450R) * VSPR
	VRP0 5-0 = 011111	((450R - 80R) / 450R) * VSPR
	VRP0 5-0 = 100000	((450R - 82R) / 450R) * VSPR
	VRP0 5-0 = 100001	((450R - 84R) / 450R) * VSPR
	VRP0 5-0 = 100010	((450R - 86R) / 450R) * VSPR
	VRP0 5-0 = 100011	((450R - 88R) / 450R) * VSPR
	VRP0 5-0 = 100100	((450R - 90R) / 450R) * VSPR
	VRP0 5-0 = 100101	((450R - 92R) / 450R) * VSPR
	VRP0 5-0 = 100110	((450R - 94R) / 450R) * VSPR
	VRP0 5-0 = 100111	((450R - 96R) / 450R) * VSPR
	VRP0 5-0 = 101000	((450R - 98R) / 450R) * VSPR
	VRP0 5-0 = 101001	((450R - 100R) / 450R) * VSPR
	VRP0 5-0 = 101010	((450R - 102R) / 450R) * VSPR
	VRP0 5-0 = 101011	((450R - 104R) / 450R) * VSPR
	VRP0 5-0 = 101100	((450R - 106R) / 450R) * VSPR
	VRP0 5-0 = 101101	((450R - 108R) / 450R) * VSPR
	VRP0 5-0 = 101110	((450R - 110R) / 450R) * VSPR
	VRP0 5-0 = 101111	((450R - 112R) / 450R) * VSPR
	VRP0 5-0 = 110000	((450R - 114R) / 450R) * VSPR
	VRP0 5-0 = 110001	((450R - 116R) / 450R) * VSPR
	VRP0 5-0 = 110010	((450R - 118R) / 450R) * VSPR
	VRP0 5-0 = 110011	((450R - 120R) / 450R) * VSPR
	VRP0 5-0 = 110100	((450R - 122R) / 450R) * VSPR
	VRP0 5-0 = 110101	((450R - 124R) / 450R) * VSPR
	VRP0 5-0 = 110110	((450R - 126R) / 450R) * VSPR
	VRP0 5-0 = 110111	((450R - 128R) / 450R) * VSPR
	VRP0 5-0 = 111000	((450R - 130R) / 450R) * VSPR
	VRP0 5-0 = 111001	((450R - 132R) / 450R) * VSPR
	VRP0 5-0 = 111010	((450R - 134R) / 450R) * VSPR
	VRP0 5-0 = 111011	((450R - 136R) / 450R) * VSPR
	VRP0 5-0 = 111100	((450R - 138R) / 450R) * VSPR
	VRP0 5-0 = 111101	((450R - 140R) / 450R) * VSPR
	VRP0 5-0 = 111110	((450R - 142R) / 450R) * VSPR
	VRP0 5-0 = 111111	((450R - 144R) / 450R) * VSPR

Table 5.8: VinP0

Reference voltage	Macro adjustment value	VinP1 formula
VinP1	VRP1 5-0 = 000000	(430R / 450R) * VSPR
	VRP1 5-0 = 000001	((430R - 2R) / 450R) * VSPR
	VRP1 5-0 = 000010	((430R - 4R) / 450R) * VSPR
	VRP1 5-0 = 000011	((430R - 6R) / 450R) * VSPR
	VRP1 5-0 = 000100	((430R - 8R) / 450R) * VSPR
	VRP1 5-0 = 000101	((430R - 10R) / 450R) * VSPR
	VRP1 5-0 = 000110	((430R - 12R) / 450R) * VSPR
	VRP1 5-0 = 000111	((430R - 14R) / 450R) * VSPR
	VRP1 5-0 = 001000	((430R - 16R) / 450R) * VSPR
	VRP1 5-0 = 001001	((430R - 18R) / 450R) * VSPR
	VRP1 5-0 = 001010	((430R - 20R) / 450R) * VSPR
	VRP1 5-0 = 001011	((430R - 22R) / 450R) * VSPR
	VRP1 5-0 = 001100	((430R - 24R) / 450R) * VSPR
	VRP1 5-0 = 001101	((430R - 26R) / 450R) * VSPR
	VRP1 5-0 = 001110	((430R - 28R) / 450R) * VSPR
	VRP1 5-0 = 001111	((430R - 30R) / 450R) * VSPR
	VRP1 5-0 = 010000	((430R - 32R) / 450R) * VSPR
	VRP1 5-0 = 010001	((430R - 34R) / 450R) * VSPR
	VRP1 5-0 = 010010	((430R - 36R) / 450R) * VSPR
	VRP1 5-0 = 010011	((430R - 38R) / 450R) * VSPR
	VRP1 5-0 = 010100	((430R - 40R) / 450R) * VSPR
	VRP1 5-0 = 010101	((430R - 42R) / 450R) * VSPR
	VRP1 5-0 = 010110	((430R - 44R) / 450R) * VSPR
	VRP1 5-0 = 010111	((430R - 46R) / 450R) * VSPR
	VRP1 5-0 = 011000	((430R - 48R) / 450R) * VSPR
	VRP1 5-0 = 011001	((430R - 50R) / 450R) * VSPR
	VRP1 5-0 = 011010	((430R - 52R) / 450R) * VSPR
	VRP1 5-0 = 011011	((430R - 54R) / 450R) * VSPR
	VRP1 5-0 = 011100	((430R - 56R) / 450R) * VSPR
	VRP1 5-0 = 011101	((430R - 58R) / 450R) * VSPR
	VRP1 5-0 = 011110	((430R - 60R) / 450R) * VSPR
	VRP1 5-0 = 011111	((430R - 62R) / 450R) * VSPR
	VRP1 5-0 = 100000	((430R - 64R) / 450R) * VSPR
	VRP1 5-0 = 100001	((430R - 66R) / 450R) * VSPR
	VRP1 5-0 = 100010	((430R - 68R) / 450R) * VSPR
	VRP1 5-0 = 100011	((430R - 70R) / 450R) * VSPR
	VRP1 5-0 = 100100	((430R - 72R) / 450R) * VSPR
	VRP1 5-0 = 100101	((430R - 74R) / 450R) * VSPR
	VRP1 5-0 = 100110	((430R - 76R) / 450R) * VSPR
	VRP1 5-0 = 100111	((430R - 78R) / 450R) * VSPR
	VRP1 5-0 = 101000	((430R - 80R) / 450R) * VSPR
	VRP1 5-0 = 101001	((430R - 82R) / 450R) * VSPR
	VRP1 5-0 = 101010	((430R - 84R) / 450R) * VSPR
	VRP1 5-0 = 101011	((430R - 86R) / 450R) * VSPR
	VRP1 5-0 = 101100	((430R - 88R) / 450R) * VSPR
	VRP1 5-0 = 101101	((430R - 90R) / 450R) * VSPR
	VRP1 5-0 = 101110	((430R - 92R) / 450R) * VSPR
	VRP1 5-0 = 101111	((430R - 94R) / 450R) * VSPR
	VRP1 5-0 = 110000	((430R - 96R) / 450R) * VSPR
	VRP1 5-0 = 110001	((430R - 98R) / 450R) * VSPR
	VRP1 5-0 = 110010	((430R - 100R) / 450R) * VSPR
	VRP1 5-0 = 110011	((430R - 102R) / 450R) * VSPR
	VRP1 5-0 = 110100	((430R - 104R) / 450R) * VSPR
	VRP1 5-0 = 110101	((430R - 106R) / 450R) * VSPR
	VRP1 5-0 = 110110	((430R - 108R) / 450R) * VSPR
	VRP1 5-0 = 110111	((430R - 110R) / 450R) * VSPR
	VRP1 5-0 = 111000	((430R - 112R) / 450R) * VSPR
	VRP1 5-0 = 111001	((430R - 114R) / 450R) * VSPR
	VRP1 5-0 = 111010	((430R - 116R) / 450R) * VSPR
	VRP1 5-0 = 111011	((430R - 118R) / 450R) * VSPR
	VRP1 5-0 = 111100	((430R - 120R) / 450R) * VSPR
	VRP1 5-0 = 111101	((430R - 122R) / 450R) * VSPR
	VRP1 5-0 = 111110	((430R - 124R) / 450R) * VSPR
	VRP1 5-0 = 111111	((430R - 126R) / 450R) * VSPR

Table 5.9: VinP1

Reference voltage	Macro adjustment value	VinP2 formula
VinP2	VRP2 5-0 = 000000	(420R / 450R) * VSPR
	VRP2 5-0 = 000001	((420R - 2R) / 450R) * VSPR
	VRP2 5-0 = 000010	((420R - 4R) / 450R) * VSPR
	VRP2 5-0 = 000011	((420R - 6R) / 450R) * VSPR
	VRP2 5-0 = 000100	((420R - 8R) / 450R) * VSPR
	VRP2 5-0 = 000101	((420R - 10R) / 450R) * VSPR
	VRP2 5-0 = 000110	((420R - 12R) / 450R) * VSPR
	VRP2 5-0 = 000111	((420R - 14R) / 450R) * VSPR
	VRP2 5-0 = 001000	((420R - 16R) / 450R) * VSPR
	VRP2 5-0 = 001001	((420R - 18R) / 450R) * VSPR
	VRP2 5-0 = 001010	((420R - 20R) / 450R) * VSPR
	VRP2 5-0 = 001011	((420R - 22R) / 450R) * VSPR
	VRP2 5-0 = 001100	((420R - 24R) / 450R) * VSPR
	VRP2 5-0 = 001101	((420R - 26R) / 450R) * VSPR
	VRP2 5-0 = 001110	((420R - 28R) / 450R) * VSPR
	VRP2 5-0 = 001111	((420R - 30R) / 450R) * VSPR
	VRP2 5-0 = 010000	((420R - 32R) / 450R) * VSPR
	VRP2 5-0 = 010001	((420R - 34R) / 450R) * VSPR
	VRP2 5-0 = 010010	((420R - 36R) / 450R) * VSPR
	VRP2 5-0 = 010011	((420R - 38R) / 450R) * VSPR
	VRP2 5-0 = 010100	((420R - 40R) / 450R) * VSPR
	VRP2 5-0 = 010101	((420R - 42R) / 450R) * VSPR
	VRP2 5-0 = 010110	((420R - 44R) / 450R) * VSPR
	VRP2 5-0 = 010111	((420R - 46R) / 450R) * VSPR
	VRP2 5-0 = 011000	((420R - 48R) / 450R) * VSPR
	VRP2 5-0 = 011001	((420R - 50R) / 450R) * VSPR
	VRP2 5-0 = 011010	((420R - 52R) / 450R) * VSPR
	VRP2 5-0 = 011011	((420R - 54R) / 450R) * VSPR
	VRP2 5-0 = 011100	((420R - 56R) / 450R) * VSPR
	VRP2 5-0 = 011101	((420R - 58R) / 450R) * VSPR
	VRP2 5-0 = 011110	((420R - 60R) / 450R) * VSPR
	VRP2 5-0 = 011111	((420R - 62R) / 450R) * VSPR
	VRP2 5-0 = 100000	((420R - 64R) / 450R) * VSPR
	VRP2 5-0 = 100001	((420R - 66R) / 450R) * VSPR
	VRP2 5-0 = 100010	((420R - 68R) / 450R) * VSPR
	VRP2 5-0 = 100011	((420R - 70R) / 450R) * VSPR
	VRP2 5-0 = 100100	((420R - 72R) / 450R) * VSPR
	VRP2 5-0 = 100101	((420R - 74R) / 450R) * VSPR
	VRP2 5-0 = 100110	((420R - 76R) / 450R) * VSPR
	VRP2 5-0 = 100111	((420R - 78R) / 450R) * VSPR
	VRP2 5-0 = 101000	((420R - 80R) / 450R) * VSPR
	VRP2 5-0 = 101001	((420R - 82R) / 450R) * VSPR
	VRP2 5-0 = 101010	((420R - 84R) / 450R) * VSPR
	VRP2 5-0 = 101011	((420R - 86R) / 450R) * VSPR
	VRP2 5-0 = 101100	((420R - 88R) / 450R) * VSPR
	VRP2 5-0 = 101101	((420R - 90R) / 450R) * VSPR
	VRP2 5-0 = 101110	((420R - 92R) / 450R) * VSPR
	VRP2 5-0 = 101111	((420R - 94R) / 450R) * VSPR
	VRP2 5-0 = 110000	((420R - 96R) / 450R) * VSPR
	VRP2 5-0 = 110001	((420R - 98R) / 450R) * VSPR
	VRP2 5-0 = 110010	((420R - 100R) / 450R) * VSPR
	VRP2 5-0 = 110011	((420R - 102R) / 450R) * VSPR
	VRP2 5-0 = 110100	((420R - 104R) / 450R) * VSPR
	VRP2 5-0 = 110101	((420R - 106R) / 450R) * VSPR
	VRP2 5-0 = 110110	((420R - 108R) / 450R) * VSPR
	VRP2 5-0 = 110111	((420R - 110R) / 450R) * VSPR
	VRP2 5-0 = 111000	((420R - 112R) / 450R) * VSPR
	VRP2 5-0 = 111001	((420R - 114R) / 450R) * VSPR
	VRP2 5-0 = 111010	((420R - 116R) / 450R) * VSPR
	VRP2 5-0 = 111011	((420R - 118R) / 450R) * VSPR
	VRP2 5-0 = 111100	((420R - 120R) / 450R) * VSPR
	VRP2 5-0 = 111101	((420R - 122R) / 450R) * VSPR
	VRP2 5-0 = 111110	((420R - 124R) / 450R) * VSPR
	VRP2 5-0 = 111111	((420R - 126R) / 450R) * VSPR

Table 5.10: VinP2

Reference voltage	Macro adjustment value	VinP14 formula
VinP14	VRP3 5-0 = 000000	(156R / 450R) * VSPR
	VRP3 5-0 = 000001	((156R - 2R) / 450R) * VSPR
	VRP3 5-0 = 000010	((156R - 4R) / 450R) * VSPR
	VRP3 5-0 = 000011	((156R - 6R) / 450R) * VSPR
	VRP3 5-0 = 000100	((156R - 8R) / 450R) * VSPR
	VRP3 5-0 = 000101	((156R - 10R) / 450R) * VSPR
	VRP3 5-0 = 000110	((156R - 12R) / 450R) * VSPR
	VRP3 5-0 = 000111	((156R - 14R) / 450R) * VSPR
	VRP3 5-0 = 001000	((156R - 16R) / 450R) * VSPR
	VRP3 5-0 = 001001	((156R - 18R) / 450R) * VSPR
	VRP3 5-0 = 001010	((156R - 20R) / 450R) * VSPR
	VRP3 5-0 = 001011	((156R - 22R) / 450R) * VSPR
	VRP3 5-0 = 001100	((156R - 24R) / 450R) * VSPR
	VRP3 5-0 = 001101	((156R - 26R) / 450R) * VSPR
	VRP3 5-0 = 001110	((156R - 28R) / 450R) * VSPR
	VRP3 5-0 = 001111	((156R - 30R) / 450R) * VSPR
	VRP3 5-0 = 010000	((156R - 32R) / 450R) * VSPR
	VRP3 5-0 = 010001	((156R - 34R) / 450R) * VSPR
	VRP3 5-0 = 010010	((156R - 36R) / 450R) * VSPR
	VRP3 5-0 = 010011	((156R - 38R) / 450R) * VSPR
	VRP3 5-0 = 010100	((156R - 40R) / 450R) * VSPR
	VRP3 5-0 = 010101	((156R - 42R) / 450R) * VSPR
	VRP3 5-0 = 010110	((156R - 44R) / 450R) * VSPR
	VRP3 5-0 = 010111	((156R - 46R) / 450R) * VSPR
	VRP3 5-0 = 011000	((156R - 48R) / 450R) * VSPR
	VRP3 5-0 = 011001	((156R - 50R) / 450R) * VSPR
	VRP3 5-0 = 011010	((156R - 52R) / 450R) * VSPR
	VRP3 5-0 = 011011	((156R - 54R) / 450R) * VSPR
	VRP3 5-0 = 011100	((156R - 56R) / 450R) * VSPR
	VRP3 5-0 = 011101	((156R - 58R) / 450R) * VSPR
	VRP3 5-0 = 011110	((156R - 60R) / 450R) * VSPR
	VRP3 5-0 = 011111	((156R - 62R) / 450R) * VSPR
	VRP3 5-0 = 100000	((156R - 64R) / 450R) * VSPR
	VRP3 5-0 = 100001	((156R - 66R) / 450R) * VSPR
	VRP3 5-0 = 100010	((156R - 68R) / 450R) * VSPR
	VRP3 5-0 = 100011	((156R - 70R) / 450R) * VSPR
	VRP3 5-0 = 100100	((156R - 72R) / 450R) * VSPR
	VRP3 5-0 = 100101	((156R - 74R) / 450R) * VSPR
	VRP3 5-0 = 100110	((156R - 76R) / 450R) * VSPR
	VRP3 5-0 = 100111	((156R - 78R) / 450R) * VSPR
	VRP3 5-0 = 101000	((156R - 80R) / 450R) * VSPR
	VRP3 5-0 = 101001	((156R - 82R) / 450R) * VSPR
	VRP3 5-0 = 101010	((156R - 84R) / 450R) * VSPR
	VRP3 5-0 = 101011	((156R - 86R) / 450R) * VSPR
	VRP3 5-0 = 101100	((156R - 88R) / 450R) * VSPR
	VRP3 5-0 = 101101	((156R - 90R) / 450R) * VSPR
	VRP3 5-0 = 101110	((156R - 92R) / 450R) * VSPR
	VRP3 5-0 = 101111	((156R - 94R) / 450R) * VSPR
	VRP3 5-0 = 110000	((156R - 96R) / 450R) * VSPR
	VRP3 5-0 = 110001	((156R - 98R) / 450R) * VSPR
	VRP3 5-0 = 110010	((156R - 100R) / 450R) * VSPR
	VRP3 5-0 = 110011	((156R - 102R) / 450R) * VSPR
	VRP3 5-0 = 110100	((156R - 104R) / 450R) * VSPR
	VRP3 5-0 = 110101	((156R - 106R) / 450R) * VSPR
	VRP3 5-0 = 110110	((156R - 108R) / 450R) * VSPR
	VRP3 5-0 = 110111	((156R - 110R) / 450R) * VSPR
	VRP3 5-0 = 111000	((156R - 112R) / 450R) * VSPR
	VRP3 5-0 = 111001	((156R - 114R) / 450R) * VSPR
	VRP3 5-0 = 111010	((156R - 116R) / 450R) * VSPR
	VRP3 5-0 = 111011	((156R - 118R) / 450R) * VSPR
	VRP3 5-0 = 111100	((156R - 120R) / 450R) * VSPR
	VRP3 5-0 = 111101	((156R - 122R) / 450R) * VSPR
	VRP3 5-0 = 111110	((156R - 124R) / 450R) * VSPR
	VRP3 5-0 = 111111	((156R - 126R) / 450R) * VSPR

Table 5.11: VinP14

Reference voltage	Macro adjustment value	VinP15 formula
VinP15	VRP4 5-0 = 000000	(146R / 450R) * VSPR
	VRP4 5-0 = 000001	((146R - 2R) / 450R) * VSPR
	VRP4 5-0 = 000010	((146R - 4R) / 450R) * VSPR
	VRP4 5-0 = 000011	((146R - 6R) / 450R) * VSPR
	VRP4 5-0 = 000100	((146R - 8R) / 450R) * VSPR
	VRP4 5-0 = 000101	((146R - 10R) / 450R) * VSPR
	VRP4 5-0 = 000110	((146R - 12R) / 450R) * VSPR
	VRP4 5-0 = 000111	((146R - 14R) / 450R) * VSPR
	VRP4 5-0 = 001000	((146R - 16R) / 450R) * VSPR
	VRP4 5-0 = 001001	((146R - 18R) / 450R) * VSPR
	VRP4 5-0 = 001010	((146R - 20R) / 450R) * VSPR
	VRP4 5-0 = 001011	((146R - 22R) / 450R) * VSPR
	VRP4 5-0 = 001100	((146R - 24R) / 450R) * VSPR
	VRP4 5-0 = 001101	((146R - 26R) / 450R) * VSPR
	VRP4 5-0 = 001110	((146R - 28R) / 450R) * VSPR
	VRP4 5-0 = 001111	((146R - 30R) / 450R) * VSPR
	VRP4 5-0 = 010000	((146R - 32R) / 450R) * VSPR
	VRP4 5-0 = 010001	((146R - 34R) / 450R) * VSPR
	VRP4 5-0 = 010010	((146R - 36R) / 450R) * VSPR
	VRP4 5-0 = 010011	((146R - 38R) / 450R) * VSPR
	VRP4 5-0 = 010100	((146R - 40R) / 450R) * VSPR
	VRP4 5-0 = 010101	((146R - 42R) / 450R) * VSPR
	VRP4 5-0 = 010110	((146R - 44R) / 450R) * VSPR
	VRP4 5-0 = 010111	((146R - 46R) / 450R) * VSPR
	VRP4 5-0 = 011000	((146R - 48R) / 450R) * VSPR
	VRP4 5-0 = 011001	((146R - 50R) / 450R) * VSPR
	VRP4 5-0 = 011010	((146R - 52R) / 450R) * VSPR
	VRP4 5-0 = 011011	((146R - 54R) / 450R) * VSPR
	VRP4 5-0 = 011100	((146R - 56R) / 450R) * VSPR
	VRP4 5-0 = 011101	((146R - 58R) / 450R) * VSPR
	VRP4 5-0 = 011110	((146R - 60R) / 450R) * VSPR
	VRP4 5-0 = 011111	((146R - 62R) / 450R) * VSPR
	VRP4 5-0 = 100000	((146R - 64R) / 450R) * VSPR
	VRP4 5-0 = 100001	((146R - 66R) / 450R) * VSPR
	VRP4 5-0 = 100010	((146R - 68R) / 450R) * VSPR
	VRP4 5-0 = 100011	((146R - 70R) / 450R) * VSPR
	VRP4 5-0 = 100100	((146R - 72R) / 450R) * VSPR
	VRP4 5-0 = 100101	((146R - 74R) / 450R) * VSPR
	VRP4 5-0 = 100110	((146R - 76R) / 450R) * VSPR
	VRP4 5-0 = 100111	((146R - 78R) / 450R) * VSPR
	VRP4 5-0 = 101000	((146R - 80R) / 450R) * VSPR
	VRP4 5-0 = 101001	((146R - 82R) / 450R) * VSPR
	VRP4 5-0 = 101010	((146R - 84R) / 450R) * VSPR
	VRP4 5-0 = 101011	((146R - 86R) / 450R) * VSPR
	VRP4 5-0 = 101100	((146R - 88R) / 450R) * VSPR
	VRP4 5-0 = 101101	((146R - 90R) / 450R) * VSPR
	VRP4 5-0 = 101110	((146R - 92R) / 450R) * VSPR
	VRP4 5-0 = 101111	((146R - 94R) / 450R) * VSPR
	VRP4 5-0 = 110000	((146R - 96R) / 450R) * VSPR
	VRP4 5-0 = 110001	((146R - 98R) / 450R) * VSPR
	VRP4 5-0 = 110010	((146R - 100R) / 450R) * VSPR
	VRP4 5-0 = 110011	((146R - 102R) / 450R) * VSPR
	VRP4 5-0 = 110100	((146R - 104R) / 450R) * VSPR
	VRP4 5-0 = 110101	((146R - 106R) / 450R) * VSPR
	VRP4 5-0 = 110110	((146R - 108R) / 450R) * VSPR
	VRP4 5-0 = 110111	((146R - 110R) / 450R) * VSPR
	VRP4 5-0 = 111000	((146R - 112R) / 450R) * VSPR
	VRP4 5-0 = 111001	((146R - 114R) / 450R) * VSPR
	VRP4 5-0 = 111010	((146R - 116R) / 450R) * VSPR
	VRP4 5-0 = 111011	((146R - 118R) / 450R) * VSPR
	VRP4 5-0 = 111100	((146R - 120R) / 450R) * VSPR
	VRP4 5-0 = 111101	((146R - 122R) / 450R) * VSPR
	VRP4 5-0 = 111110	((146R - 124R) / 450R) * VSPR
	VRP4 5-0 = 111111	((146R - 126R) / 450R) * VSPR

Table 5.12: VinP15

Reference voltage	Macro adjustment value	VinP16 formula
VinP16	VRP5 5-0 = 000000	(144R / 450R) * VSPR
	VRP5 5-0 = 000001	((144R - 2R) / 450R) * VSPR
	VRP5 5-0 = 000010	((144R - 4R) / 450R) * VSPR
	VRP5 5-0 = 000011	((144R - 6R) / 450R) * VSPR
	VRP5 5-0 = 000100	((144R - 8R) / 450R) * VSPR
	VRP5 5-0 = 000101	((144R - 10R) / 450R) * VSPR
	VRP5 5-0 = 000110	((144R - 12R) / 450R) * VSPR
	VRP5 5-0 = 000111	((144R - 14R) / 450R) * VSPR
	VRP5 5-0 = 001000	((144R - 16R) / 450R) * VSPR
	VRP5 5-0 = 001001	((144R - 18R) / 450R) * VSPR
	VRP5 5-0 = 001010	((144R - 20R) / 450R) * VSPR
	VRP5 5-0 = 001011	((144R - 22R) / 450R) * VSPR
	VRP5 5-0 = 001100	((144R - 24R) / 450R) * VSPR
	VRP5 5-0 = 001101	((144R - 26R) / 450R) * VSPR
	VRP5 5-0 = 001110	((144R - 28R) / 450R) * VSPR
	VRP5 5-0 = 001111	((144R - 30R) / 450R) * VSPR
	VRP5 5-0 = 010000	((144R - 32R) / 450R) * VSPR
	VRP5 5-0 = 010001	((144R - 34R) / 450R) * VSPR
	VRP5 5-0 = 010010	((144R - 36R) / 450R) * VSPR
	VRP5 5-0 = 010011	((144R - 38R) / 450R) * VSPR
	VRP5 5-0 = 010100	((144R - 40R) / 450R) * VSPR
	VRP5 5-0 = 010101	((144R - 42R) / 450R) * VSPR
	VRP5 5-0 = 010110	((144R - 44R) / 450R) * VSPR
	VRP5 5-0 = 010111	((144R - 46R) / 450R) * VSPR
	VRP5 5-0 = 011000	((144R - 48R) / 450R) * VSPR
	VRP5 5-0 = 011001	((144R - 50R) / 450R) * VSPR
	VRP5 5-0 = 011010	((144R - 52R) / 450R) * VSPR
	VRP5 5-0 = 011011	((144R - 54R) / 450R) * VSPR
	VRP5 5-0 = 011100	((144R - 56R) / 450R) * VSPR
	VRP5 5-0 = 011101	((144R - 58R) / 450R) * VSPR
	VRP5 5-0 = 011110	((144R - 60R) / 450R) * VSPR
	VRP5 5-0 = 011111	((144R - 62R) / 450R) * VSPR
	VRP5 5-0 = 100000	((144R - 64R) / 450R) * VSPR
	VRP5 5-0 = 100001	((144R - 66R) / 450R) * VSPR
	VRP5 5-0 = 100010	((144R - 68R) / 450R) * VSPR
	VRP5 5-0 = 100011	((144R - 70R) / 450R) * VSPR
	VRP5 5-0 = 100100	((144R - 72R) / 450R) * VSPR
	VRP5 5-0 = 100101	((144R - 74R) / 450R) * VSPR
	VRP5 5-0 = 100110	((144R - 76R) / 450R) * VSPR
	VRP5 5-0 = 100111	((144R - 78R) / 450R) * VSPR
	VRP5 5-0 = 101000	((144R - 80R) / 450R) * VSPR
	VRP5 5-0 = 101001	((144R - 82R) / 450R) * VSPR
	VRP5 5-0 = 101010	((144R - 84R) / 450R) * VSPR
	VRP5 5-0 = 101011	((144R - 86R) / 450R) * VSPR
	VRP5 5-0 = 101100	((144R - 88R) / 450R) * VSPR
	VRP5 5-0 = 101101	((144R - 90R) / 450R) * VSPR
	VRP5 5-0 = 101110	((144R - 92R) / 450R) * VSPR
	VRP5 5-0 = 101111	((144R - 94R) / 450R) * VSPR
	VRP5 5-0 = 110000	((144R - 96R) / 450R) * VSPR
	VRP5 5-0 = 110001	((144R - 98R) / 450R) * VSPR
	VRP5 5-0 = 110010	((144R - 100R) / 450R) * VSPR
	VRP5 5-0 = 110011	((144R - 102R) / 450R) * VSPR
	VRP5 5-0 = 110100	((144R - 104R) / 450R) * VSPR
	VRP5 5-0 = 110101	((144R - 106R) / 450R) * VSPR
	VRP5 5-0 = 110110	((144R - 108R) / 450R) * VSPR
	VRP5 5-0 = 110111	((144R - 110R) / 450R) * VSPR
	VRP5 5-0 = 111000	((144R - 112R) / 450R) * VSPR
	VRP5 5-0 = 111001	((144R - 114R) / 450R) * VSPR
	VRP5 5-0 = 111010	((144R - 116R) / 450R) * VSPR
	VRP5 5-0 = 111011	((144R - 118R) / 450R) * VSPR
	VRP5 5-0 = 111100	((144R - 120R) / 450R) * VSPR
	VRP5 5-0 = 111101	((144R - 122R) / 450R) * VSPR
	VRP5 5-0 = 111110	((144R - 124R) / 450R) * VSPR
	VRP5 5-0 = 111111	VSSA

Table 5.13: VinP16

Reference voltage	Macro adjustment value	VinP5 formula
VinP5	PRP0 6-0 = 0000000	(350R / 450R) VSPR
	PRP0 6-0 = 0000001	((350R - 2R) / 450R) * VSPR
	PRP0 6-0 = 0000010	((350R - 4R) / 450R) * VSPR
	PRP0 6-0 = 0000011	((350R – 6R) / 450R) * VSPR
	PRP0 6-0 = 0000100	((350R – 8R) / 450R) * VSPR
	PRP0 6-0 = 0000101	((350R – 10R) / 450R) * VSPR
	PRP0 6-0 = 0000110	((350R – 12R) / 450R) * VSPR
	PRP0 6-0 = 0000111	((350R - 14R) / 450R) * VSPR
	PRP0 6-0 = 0001000	((350R – 16R) / 450R) * VSPR
	PRP0 6-0 = 0001001	((350R – 18R) / 450R) * VSPR
	PRP0 6-0 = 0001010	((350R – 20R) / 450R) * VSPR
	PRP0 6-0 = 0001011	((350R – 22R) / 450R) * VSPR
	PRP0 6-0 = 0001100	((350R – 24R) / 450R) * VSPR
	PRP0 6-0 = 0001101	((350R – 26R) / 450R) * VSPR
	PRP0 6-0 = 0001110	((350R – 28R) / 450R) * VSPR
	PRP0 6-0 = 0001111	((350R – 30R) / 450R) * VSPR
	PRP0 6-0 = 0010000	((350R – 32R) / 450R) * VSPR
	PRP0 6-0 = 0010001	((350R - 34R) / 450R) * VSPR
	PRP0 6-0 = 0010010	((350R – 36R) / 450R) * VSPR
	PRP0 6-0 = 0010011	((350R – 38R) / 450R) * VSPR
	PRP0 6-0 = 0010100	((350R – 40R) / 450R) * VSPR
	PRP0 6-0 = 0010101	((350R – 42R) / 450R) * VSPR
	PRP0 6-0 = 0010110	((350R – 44R) / 450R) * VSPR
	PRP0 6-0 = 0010111	((350R – 46R) / 450R) * VSPR
	PRP0 6-0 = 0011000	((350R – 48R) / 450R) * VSPR
	PRP0 6-0 = 0011001	((350R – 50R) / 450R) * VSPR
	PRP0 6-0 = 0011010	((350R – 52R) / 450R) * VSPR
	PRP0 6-0 = 0011011	((350R - 54R) / 450R) * VSPR
	PRP0 6-0 = 0011100	((350R – 56R) / 450R) * VSPR
	PRP0 6-0 = 0011101	((350R – 58R) / 450R) * VSPR
	PRP0 6-0 = 0011110	((350R – 60R) / 450R) * VSPR
	PRP0 6-0 = 0011111	((350R – 62R) / 450R) * VSPR
	PRP0 6-0 = 0100000	((350R - 64R) / 450R) * VSPR
	PRP0 6-0 = 0100001	((350R – 66R) / 450R) * VSPR
	PRP0 6-0 = 0100010	((350R – 68R) / 450R) * VSPR
	PRP0 6-0 = 0100011	((350R – 70R) / 450R) * VSPR
	PRP0 6-0 = 0100100	((350R – 72R) / 450R) * VSPR
	PRP0 6-0 = 0100101	((350R – 74R) / 450R) * VSPR
	PRP0 6-0 = 0100110	((350R – 76R) / 450R) * VSPR
	PRP0 6-0 = 0100111	((350R – 78R) / 450R) * VSPR
	PRP0 6-0 = 0101000	((350R – 80R) / 450R) * VSPR
	PRP0 6-0 = 0101001	((350R – 82R) / 450R) * VSPR
	PRP0 6-0 = 0101010	((350R - 84R) / 450R) * VSPR
	PRP0 6-0 = 0101011	((350R – 86R) / 450R) * VSPR
	PRP0 6-0 = 0101100	((350R – 88R) / 450R) * VSPR
	PRP0 6-0 = 0101101	((350R – 90R) / 450R) * VSPR
	PRP0 6-0 = 0101110	((350R – 92R) / 450R) * VSPR
	PRP0 6-0 = 0101111	((350R – 94R) / 450R) * VSPR
	PRP0 6-0 = 0110000	((350R – 96R) / 450R) * VSPR
	PRP0 6-0 = 0110001	((350R – 98R) / 450R) * VSPR
	PRP0 6-0 = 0110010	((350R – 100R) / 450R) * VSPR
	PRP0 6-0 = 0110011	((350R – 102R) / 450R) * VSPR
	PRP0 6-0 = 0110100	((350R – 104R) / 450R) * VSPR
	PRP0 6-0 = 0110101	((350R – 106R) / 450R) * VSPR
	PRP0 6-0 = 0110110	((350R – 108R) / 450R) * VSPR
	PRP0 6-0 = 0110111	((350R – 110R) / 450R) * VSPR
	PRP0 6-0 = 0111000	((350R – 112R) / 450R) * VSPR
	PRP0 6-0 = 0111001	((350R – 114R) / 450R) * VSPR
	PRP0 6-0 = 0111010	((350R – 116R) / 450R) * VSPR
	PRP0 6-0 = 0111011	((350R – 118R) / 450R) * VSPR
	PRP0 6-0 = 0111100	((350R – 120R) / 450R) * VSPR
	PRP0 6-0 = 0111101	((350R – 122R) / 450R) * VSPR
	PRP0 6-0 = 0111110	((350R - 124R) / 450R) * VSPR
	PRP0 6-0 = 0111111	((350R – 126R) / 450R) * VSPR
	PRP0 6-0 = 1000000	((350R – 128R) / 450R) * VSPR
	PRP0 6-0 = 1000001	((350R – 130R) / 450R) * VSPR
	PRP0 6-0 = 1000010	((350R - 132R) / 450R) * VSPR
	PRP0 6-0 = 1000011	((350R – 134R) / 450R) * VSPR
	PRP0 6-0 = 1000100	((350R – 136R) / 450R) * VSPR

	PRP0 6-0 = 1000101	((350R – 138R) / 450R) * VSPR
	PRP0 6-0 = 1000110	((350R – 140R) / 450R) * VSPR
	PRP0 6-0 = 1000111	((350R – 142R) / 450R) * VSPR
	PRP0 6-0 = 1001000	((350R – 144R) / 450R) * VSPR
	PRP0 6-0 = 1001001	((350R – 146R) / 450R) * VSPR
	PRP0 6-0 = 1001010	((350R – 148R) / 450R) * VSPR
	PRP0 6-0 = 1001011	((350R – 150R) / 450R) * VSPR
	PRP0 6-0 = 1001100	((350R - 152R) / 450R) * VSPR
	PRP0 6-0 = 1001101	((350R – 154R) / 450R) * VSPR
	PRP0 6-0 = 1001110	((350R – 156R) / 450R) * VSPR
	PRP0 6-0 = 1001111	((350R – 158R) / 450R) * VSPR
	PRP0 6-0 = 1010000	((350R – 160R) / 450R) * VSPR
	PRP0 6-0 = 1010001	((350R – 162R) / 450R) * VSPR
	PRP0 6-0 = 1010010	((350R – 164R) / 450R) * VSPR
	PRP0 6-0 = 1010011	((350R – 166R) / 450R) * VSPR
	PRP0 6-0 = 1010100	((350R – 168R) / 450R) * VSPR
	PRP0 6-0 = 1010101	((350R – 170R) / 450R) * VSPR
	PRP0 6-0 = 1010110	((350R – 172R) / 450R) * VSPR
	PRP0 6-0 = 1010111	((350R - 174R) / 450R) * VSPR
	PRP0 6-0 = 1011000	inhibit
	PRP0 6-0 = 1011001	inhibit
	PRP0 6-0 = 1011010	inhibit
	PRP0 6-0 = 1011011	inhibit
	PRP0 6-0 = 1011100	inhibit
	PRP0 6-0 = 1011101	inhibit
	PRP0 6-0 = 1011110	inhibit
	PRP0 6-0 = 1011111	inhibit
	PRP0 6-0 = 1100000	inhibit
	PRP0 6-0 = 1100001	inhibit
	PRP0 6-0 = 1100010	inhibit
	PRP0 6-0 = 1100011	inhibit
	PRP0 6-0 = 1100100	inhibit
	PRP0 6-0 = 1100101	inhibit
	PRP0 6-0 = 1100110	inhibit
	PRP0 6-0 = 1100111	inhibit
	PRP0 6-0 = 1101000	inhibit
	PRP0 6-0 = 1101001	inhibit
	PRP0 6-0 = 1101010	inhibit
	PRP0 6-0 = 1101011	inhibit
	PRP0 6-0 = 1101100	inhibit
	PRP0 6-0 = 1101101	inhibit
	PRP0 6-0 = 1101110	inhibit
	PRP0 6-0 = 1101111	inhibit
	PRP0 6-0 = 1110000	inhibit
	PRP0 6-0 = 1110001	inhibit
	PRP0 6-0 = 1110010	inhibit
	PRP0 6-0 = 1110011	inhibit
	PRP0 6-0 = 1110100	inhibit
	PRP0 6-0 = 1110101	inhibit
	PRP0 6-0 = 1110110	inhibit
	PRP0 6-0 = 1110111	inhibit
	PRP0 6-0 = 1111000	inhibit
	PRP0 6-0 = 1111001	inhibit
	PRP0 6-0 = 1111010	inhibit
	PRP0 6-0 = 1111011	inhibit
	PRP0 6-0 = 1111100	inhibit
	PRP0 6-0 = 1111101	inhibit
	PRP0 6-0 = 1111110	inhibit
	PRP0 6-0 = 1111111	inhibit

Table 5.14: VinP5

Reference voltage	Macro adjustment value	VinP11 formula
VinP11	PRP1 6-0 = 0000000	(274R / 450R) VSPR
	PRP1 6-0 = 0000001	((274R - 2R) / 450R) * VSPR
	PRP1 6-0 = 0000010	((274R - 4R) / 450R) * VSPR
	PRP1 6-0 = 0000011	((274R – 6R) / 450R) * VSPR
	PRP1 6-0 = 0000100	((274R – 8R) / 450R) * VSPR
	PRP1 6-0 = 0000101	((274R – 10R) / 450R) * VSPR
	PRP1 6-0 = 0000110	((274R – 12R) / 450R) * VSPR
	PRP1 6-0 = 0000111	((274R - 14R) / 450R) * VSPR
	PRP1 6-0 = 0001000	((274R – 16R) / 450R) * VSPR
	PRP1 6-0 = 0001001	((274R – 18R) / 450R) * VSPR
	PRP1 6-0 = 0001010	((274R – 20R) / 450R) * VSPR
	PRP1 6-0 = 0001011	((274R – 22R) / 450R) * VSPR
	PRP1 6-0 = 0001100	((274R – 24R) / 450R) * VSPR
	PRP1 6-0 = 0001101	((274R – 26R) / 450R) * VSPR
	PRP1 6-0 = 0001110	((274R – 28R) / 450R) * VSPR
	PRP1 6-0 = 0001111	((274R – 30R) / 450R) * VSPR
	PRP1 6-0 = 0010000	((274R – 32R) / 450R) * VSPR
	PRP1 6-0 = 0010001	((274R - 34R) / 450R) * VSPR
	PRP1 6-0 = 0010010	((274R – 36R) / 450R) * VSPR
	PRP1 6-0 = 0010011	((274R – 38R) / 450R) * VSPR
	PRP1 6-0 = 0010100	((274R – 40R) / 450R) * VSPR
	PRP1 6-0 = 0010101	((274R – 42R) / 450R) * VSPR
	PRP1 6-0 = 0010110	((274R – 44R) / 450R) * VSPR
	PRP1 6-0 = 0010111	((274R – 46R) / 450R) * VSPR
	PRP1 6-0 = 0011000	((274R – 48R) / 450R) * VSPR
	PRP1 6-0 = 0011001	((274R – 50R) / 450R) * VSPR
	PRP1 6-0 = 0011010	((274R – 52R) / 450R) * VSPR
	PRP1 6-0 = 0011011	((274R - 54R) / 450R) * VSPR
	PRP1 6-0 = 0011100	((274R – 56R) / 450R) * VSPR
	PRP1 6-0 = 0011101	((274R – 58R) / 450R) * VSPR
	PRP1 6-0 = 0011110	((274R – 60R) / 450R) * VSPR
	PRP1 6-0 = 0011111	((274R – 62R) / 450R) * VSPR
	PRP1 6-0 = 0100000	((274R - 64R) / 450R) * VSPR
	PRP1 6-0 = 0100001	((274R – 66R) / 450R) * VSPR
	PRP1 6-0 = 0100010	((274R – 68R) / 450R) * VSPR
	PRP1 6-0 = 0100011	((274R – 70R) / 450R) * VSPR
	PRP1 6-0 = 0100100	((274R – 72R) / 450R) * VSPR
	PRP1 6-0 = 0100101	((274R – 74R) / 450R) * VSPR
	PRP1 6-0 = 0100110	((274R – 76R) / 450R) * VSPR
	PRP1 6-0 = 0100111	((274R – 78R) / 450R) * VSPR
	PRP1 6-0 = 0101000	((274R – 80R) / 450R) * VSPR
	PRP1 6-0 = 0101001	((274R – 82R) / 450R) * VSPR
	PRP1 6-0 = 0101010	((274R - 84R) / 450R) * VSPR
	PRP1 6-0 = 0101011	((274R – 86R) / 450R) * VSPR
	PRP1 6-0 = 0101100	((274R – 88R) / 450R) * VSPR
	PRP1 6-0 = 0101101	((274R – 90R) / 450R) * VSPR
	PRP1 6-0 = 0101110	((274R – 92R) / 450R) * VSPR
	PRP1 6-0 = 0101111	((274R – 94R) / 450R) * VSPR
	PRP1 6-0 = 0110000	((274R – 96R) / 450R) * VSPR
	PRP1 6-0 = 0110001	((274R – 98R) / 450R) * VSPR
	PRP1 6-0 = 0110010	((274R – 100R) / 450R) * VSPR
	PRP1 6-0 = 0110011	((274R – 102R) / 450R) * VSPR
	PRP1 6-0 = 0110100	((274R – 104R) / 450R) * VSPR
	PRP1 6-0 = 0110101	((274R – 106R) / 450R) * VSPR
	PRP1 6-0 = 0110110	((274R – 108R) / 450R) * VSPR
	PRP1 6-0 = 0110111	((274R – 110R) / 450R) * VSPR
	PRP1 6-0 = 0111000	((274R – 112R) / 450R) * VSPR
	PRP1 6-0 = 0111001	((274R – 114R) / 450R) * VSPR
	PRP1 6-0 = 0111010	((274R – 116R) / 450R) * VSPR
	PRP1 6-0 = 0111011	((274R – 118R) / 450R) * VSPR
	PRP1 6-0 = 0111100	((274R – 120R) / 450R) * VSPR
	PRP1 6-0 = 0111101	((274R – 122R) / 450R) * VSPR
	PRP1 6-0 = 0111110	((274R - 124R) / 450R) * VSPR
	PRP1 6-0 = 0111111	((274R – 126R) / 450R) * VSPR
	PRP1 6-0 = 1000000	((274R – 128R) / 450R) * VSPR
	PRP1 6-0 = 1000001	((274R – 130R) / 450R) * VSPR
	PRP1 6-0 = 1000010	((274R - 132R) / 450R) * VSPR
	PRP1 6-0 = 1000011	((274R – 134R) / 450R) * VSPR
	PRP1 6-0 = 1000100	((274R – 136R) / 450R) * VSPR

	PRP1 6-0 = 1000101	((274R – 138R) / 450R) * VSPR
	PRP1 6-0 = 1000110	((274R – 140R) / 450R) * VSPR
	PRP1 6-0 = 1000111	((274R – 142R) / 450R) * VSPR
	PRP1 6-0 = 1001000	((274R – 144R) / 450R) * VSPR
	PRP1 6-0 = 1001001	((274R – 146R) / 450R) * VSPR
	PRP1 6-0 = 1001010	((274R – 148R) / 450R) * VSPR
	PRP1 6-0 = 1001011	((274R – 150R) / 450R) * VSPR
	PRP1 6-0 = 1001100	((274R - 152R) / 450R) * VSPR
	PRP1 6-0 = 1001101	((274R – 154R) / 450R) * VSPR
	PRP1 6-0 = 1001110	((274R – 156R) / 450R) * VSPR
	PRP1 6-0 = 1001111	((274R – 158R) / 450R) * VSPR
	PRP1 6-0 = 1010000	((274R – 160R) / 450R) * VSPR
	PRP1 6-0 = 1010001	((274R – 162R) / 450R) * VSPR
	PRP1 6-0 = 1010010	((274R – 164R) / 450R) * VSPR
	PRP1 6-0 = 1010011	((274R – 166R) / 450R) * VSPR
	PRP1 6-0 = 1010100	((274R – 168R) / 450R) * VSPR
	PRP1 6-0 = 1010101	((274R – 170R) / 450R) * VSPR
	PRP1 6-0 = 1010110	((274R – 172R) / 450R) * VSPR
	PRP1 6-0 = 1010111	((274R - 174R) / 450R) * VSPR
	PRP1 6-0 = 1011000	inhibit
	PRP1 6-0 = 1011001	inhibit
	PRP1 6-0 = 1011010	inhibit
	PRP1 6-0 = 1011011	inhibit
	PRP1 6-0 = 1011100	inhibit
	PRP1 6-0 = 1011101	inhibit
	PRP1 6-0 = 1011110	inhibit
	PRP1 6-0 = 1011111	inhibit
	PRP1 6-0 = 1100000	inhibit
	PRP1 6-0 = 1100001	inhibit
	PRP1 6-0 = 1100010	inhibit
	PRP1 6-0 = 1100011	inhibit
	PRP1 6-0 = 1100100	inhibit
	PRP1 6-0 = 1100101	inhibit
	PRP1 6-0 = 1100110	inhibit
	PRP1 6-0 = 1100111	inhibit
	PRP1 6-0 = 1101000	inhibit
	PRP1 6-0 = 1101001	inhibit
	PRP1 6-0 = 1101010	inhibit
	PRP1 6-0 = 1101011	inhibit
	PRP1 6-0 = 1101100	inhibit
	PRP1 6-0 = 1101101	inhibit
	PRP1 6-0 = 1101110	inhibit
	PRP1 6-0 = 1101111	inhibit
	PRP1 6-0 = 1110000	inhibit
	PRP1 6-0 = 1110001	inhibit
	PRP1 6-0 = 1110010	inhibit
	PRP1 6-0 = 1110011	inhibit
	PRP1 6-0 = 1110100	inhibit
	PRP1 6-0 = 1110101	inhibit
	PRP1 6-0 = 1110110	inhibit
	PRP1 6-0 = 1110111	inhibit
	PRP1 6-0 = 1111000	inhibit
	PRP1 6-0 = 1111001	inhibit
	PRP1 6-0 = 1111010	inhibit
	PRP1 6-0 = 1111011	inhibit
	PRP1 6-0 = 1111100	inhibit
	PRP1 6-0 = 1111101	inhibit
	PRP1 6-0 = 1111110	inhibit
	PRP1 6-0 = 1111111	inhibit

Table 5.15: VinP11

Reference voltage	Macro adjustment value	VinP3 formula
VinP3	PKP0 4-0 = 00000	(47R / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 00001	((47R – 1R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 00010	((47R – 2R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 00011	((47R – 3R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 00100	((47R – 4R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 00101	((47R – 5R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 00110	((47R – 6R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 00111	((47R – 7R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 01000	((47R – 8R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 01001	((47R – 9R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 01010	((47R - 10R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 01011	((47R - 11R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 01100	((47R - 12R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 01101	((47R - 13R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 01110	((47R - 14R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 01111	((47R - 15R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 10000	((47R - 16R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 10001	((47R - 17R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 10010	((47R - 18R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 10011	((47R - 19R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 10100	((47R - 20R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 10101	((47R - 21R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 10110	((47R - 22R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 10111	((47R - 23R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 11000	((47R - 24R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 11001	((47R - 25R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 11010	((47R - 26R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 11011	((47R - 27R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 11100	((47R - 28R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 11101	((47R - 29R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 11110	((47R - 30R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP0 4-0 = 11111	((47R - 31R) / 48R) * (VinP2 - VinP5) + VinP5

Table 5.16: VinP3

Reference voltage	Macro adjustment value	VinP4 formula
VinP4	PKP1 4-0 = 00000	(32R / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 00001	((32R - 1R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 00010	((32R - 2R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 00011	((32R - 3R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 00100	((32R - 4R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 00101	((32R - 5R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 00110	((32R - 6R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 00111	((32R - 7R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 01000	((32R - 8R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 01001	((32R - 9R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 01010	((32R - 10R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 01011	((32R - 11R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 01100	((32R - 12R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 01101	((32R - 13R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 01110	((32R - 14R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 01111	((32R - 15R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 10000	((32R - 16R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 10001	((32R - 17R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 10010	((32R - 18R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 10011	((32R - 19R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 10100	((32R - 20R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 10101	((32R - 21R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 10110	((32R - 22R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 10111	((32R - 23R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 11000	((32R - 24R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 11001	((32R - 25R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 11010	((32R - 26R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 11011	((32R - 27R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 11100	((32R - 28R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 11101	((32R - 29R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 11110	((32R - 30R) / 48R) * (VinP2 - VinP5) + VinP5
	PKP1 4-0 = 11111	((32R - 31R) / 48R) * (VinP2 - VinP5) + VinP5

Table 5.17: VinP4

Reference voltage	Macro adjustment value	VinP6 formula
VinP6	PKP2 4-0 = 00000	(220R / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 00001	((220R - 3R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 00010	((220R - 6R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 00011	((220R - 9R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 00100	((220R - 12R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 00101	((220R - 15R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 00110	((220R - 18R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 00111	((220R - 21R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 01000	((220R - 24R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 01001	((220R - 27R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 01010	((220R - 30R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 01011	((220R - 33R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 01100	((220R - 36R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 01101	((220R - 39R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 01110	((220R - 42R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 01111	((220R - 45R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 10000	((220R - 48R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 10001	((220R - 51R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 10010	((220R - 54R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 10011	((220R - 57R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 10100	((220R - 60R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 10101	((220R - 63R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 10110	((220R - 66R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 10111	((220R - 69R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 11000	((220R - 72R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 11001	((220R - 75R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 11010	((220R - 78R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 11011	((220R - 81R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 11100	((220R - 84R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 11101	((220R - 87R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 11110	((220R - 90R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP2 4-0 = 11111	((220R - 93R) / 223R) * (VinP5 - VinP11) + VinP11

Table 5.18: VinP6

Reference voltage	Macro adjustment value	VinP7 formula
VinP7	PKP3 4-0 = 00000	(193R / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 00001	((193R - 3R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 00010	((193R - 6R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 00011	((193R - 9R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 00100	((193R - 12R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 00101	((193R - 15R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 00110	((193R - 18R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 00111	((193R - 21R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 01000	((193R - 24R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 01001	((193R - 27R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 01010	((193R - 30R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 01011	((193R - 33R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 01100	((193R - 36R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 01101	((193R - 39R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 01110	((193R - 42R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 01111	((193R - 45R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 10000	((193R - 48R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 10001	((193R - 51R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 10010	((193R - 54R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 10011	((193R - 57R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 10100	((193R - 60R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 10101	((193R - 63R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 10110	((193R - 66R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 10111	((193R - 69R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 11000	((193R - 72R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 11001	((193R - 75R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 11010	((193R - 78R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 11011	((193R - 81R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 11100	((193R - 84R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 11101	((193R - 87R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 11110	((193R - 90R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP3 4-0 = 11111	((193R - 93R) / 223R) * (VinP5 - VinP11) + VinP11

Table 5.19: VinP7

Reference voltage	Macro adjustment value	VinP8 formula
VinP8	PKP4 4-0 = 00000	(158R / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 00001	((158R - 3R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 00010	((158R - 6R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 00011	((158R - 9R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 00100	((158R - 12R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 00101	((158R - 15R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 00110	((158R - 18R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 00111	((158R - 21R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 01000	((158R - 24R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 01001	((158R - 27R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 01010	((158R - 30R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 01011	((158R - 33R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 01100	((158R - 36R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 01101	((158R - 39R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 01110	((158R - 42R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 01111	((158R - 45R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 10000	((158R - 48R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 10001	((158R - 51R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 10010	((158R - 54R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 10011	((158R - 57R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 10100	((158R - 60R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 10101	((158R - 63R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 10110	((158R - 66R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 10111	((158R - 69R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 11000	((158R - 72R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 11001	((158R - 75R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 11010	((158R - 78R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 11011	((158R - 81R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 11100	((158R - 84R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 11101	((158R - 87R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 11110	((158R - 90R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP4 4-0 = 11111	((158R - 93R) / 223R) * (VinP5 - VinP11) + VinP11

Table 5.20: VinP8

Reference voltage	Macro adjustment value	VinP9 formula
VinP9	PKP5 4-0 = 00000	(123R / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 00001	((123R - 3R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 00010	((123R - 6R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 00011	((123R - 9R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 00100	((123R - 12R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 00101	((123R - 15R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 00110	((123R - 18R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 00111	((123R - 21R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 01000	((123R - 24R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 01001	((123R - 27R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 01010	((123R - 30R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 01011	((123R - 33R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 01100	((123R - 36R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 01101	((123R - 39R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 01110	((123R - 42R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 01111	((123R - 45R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 10000	((123R - 48R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 10001	((123R - 51R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 10010	((123R - 54R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 10011	((123R - 57R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 10100	((123R - 60R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 10101	((123R - 63R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 10110	((123R - 66R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 10111	((123R - 69R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 11000	((123R - 72R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 11001	((123R - 75R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 11010	((123R - 78R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 11011	((123R - 81R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 11100	((123R - 84R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 11101	((123R - 87R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 11110	((123R - 90R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP5 4-0 = 11111	((123R - 93R) / 223R) * (VinP5 - VinP11) + VinP11

Table 5.21: VinP9

Reference voltage	Macro adjustment value	VinP10 formula
VinP10	PKP6 4-0 = 00000	(96R / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 00001	((96R - 3R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 00010	((96R - 6R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 00011	((96R - 9R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 00100	((96R - 12R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 00101	((96R - 15R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 00110	((96R - 18R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 00111	((96R - 21R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 01000	((96R - 24R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 01001	((96R - 27R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 01010	((96R - 30R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 01011	((96R - 33R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 01100	((96R - 36R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 01101	((96R - 39R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 01110	((96R - 42R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 01111	((96R - 45R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 10000	((96R - 48R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 10001	((96R - 51R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 10010	((96R - 54R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 10011	((96R - 57R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 10100	((96R - 60R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 10101	((96R - 63R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 10110	((96R - 66R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 10111	((96R - 69R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 11000	((96R - 72R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 11001	((96R - 75R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 11010	((96R - 78R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 11011	((96R - 81R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 11100	((96R - 84R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 11101	((96R - 87R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 11110	((96R - 90R) / 223R) * (VinP5 - VinP11) + VinP11
	PKP6 4-0 = 11111	((96R - 93R) / 223R) * (VinP5 - VinP11) + VinP11

Table 5.22: VinP10

Reference voltage	Macro adjustment value	VinP12 formula
VinP12	PKP7 4-0 = 00000	(47R / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 00001	((47R - 1R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 00010	((47R - 2R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 00011	((47R - 3R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 00100	((47R - 4R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 00101	((47R - 5R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 00110	((47R - 6R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 00111	((47R - 7R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 01000	((47R - 8R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 01001	((47R - 9R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 01010	((47R - 10R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 01011	((47R - 11R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 01100	((47R - 12R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 01101	((47R - 13R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 01110	((47R - 14R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 01111	((47R - 15R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 10000	((47R - 16R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 10001	((47R - 17R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 10010	((47R - 18R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 10011	((47R - 19R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 10100	((47R - 20R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 10101	((47R - 21R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 10110	((47R - 22R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 10111	((47R - 23R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 11000	((47R - 24R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 11001	((47R - 25R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 11010	((47R - 26R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 11011	((47R - 27R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 11100	((47R - 28R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 11101	((47R - 29R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 11110	((47R - 30R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP7 4-0 = 11111	((47R - 31R) / 48R) * (VinP11 - VinP14) + VinP14

Table 5.23: VinP12

Reference voltage	Macro adjustment value	VinP13 formula
VinP13	PKP8 4-0 = 00000	(32R / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 00001	((32R - 1R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 00010	((32R - 2R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 00011	((32R - 3R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 00100	((32R - 4R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 00101	((32R - 5R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 00110	((32R - 6R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 00111	((32R - 7R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 01000	((32R - 8R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 01001	((32R - 9R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 01010	((32R - 10R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 01011	((32R - 11R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 01100	((32R - 12R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 01101	((32R - 13R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 01110	((32R - 14R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 01111	((32R - 15R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 10000	((32R - 16R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 10001	((32R - 17R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 10010	((32R - 18R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 10011	((32R - 19R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 10100	((32R - 20R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 10101	((32R - 21R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 10110	((32R - 22R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 10111	((32R - 23R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 11000	((32R - 24R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 11001	((32R - 25R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 11010	((32R - 26R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 11011	((32R - 27R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 11100	((32R - 28R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 11101	((32R - 29R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 11110	((32R - 30R) / 48R) * (VinP11 - VinP14) + VinP14
	PKP8 4-0 = 11111	((32R - 31R) / 48R) * (VinP11 - VinP14) + VinP14

Table 5.24: VinP13

Reference voltage	Macro adjustment value	VinP_EX1 formula
VinP_EX0	PKP_EX0 4-0 = 00000	(39R / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 00001	((39R – 1R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 00010	((39R – 2R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 00011	((39R – 3R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 00100	((39R – 4R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 00101	((39R – 5R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 00110	((39R – 6R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 00111	((39R – 7R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 01000	((39R – 8R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 01001	((39R – 9R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 01010	((39R – 10R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 01011	((39R – 11R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 01100	((39R – 12R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 01101	((39R – 13R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 01110	((39R – 14R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 01111	((39R – 15R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 10000	((39R – 16R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 10001	((39R – 17R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 10010	((39R – 18R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 10011	((39R – 19R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 10100	((39R – 20R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 10101	((39R – 21R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 10110	((39R – 22R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 10111	((39R – 23R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 11000	((39R – 24R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 11001	((39R – 25R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 11010	((39R – 26R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 11011	((39R – 27R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 11100	((39R – 28R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 11101	((39R – 29R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 11110	((39R – 30R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX0 4-0 = 11111	((39R – 31R) / 48R) * (VinP2 – VinP5) + VinP5

Table 5.25: VinP_EX0

Reference voltage	Macro adjustment value	VinP_EX2 formula
VinP_EX1	PKP_EX1 4-0 = 00000	(32R / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 00001	((32R – 1R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 00010	((32R – 2R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 00011	((32R – 3R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 00100	((32R – 4R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 00101	((32R – 5R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 00110	((32R – 6R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 00111	((32R – 7R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 01000	((32R – 8R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 01001	((32R – 9R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 01010	((32R – 10R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 01011	((32R – 11R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 01100	((32R – 12R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 01101	((32R – 13R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 01110	((32R – 14R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 01111	((32R – 15R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 10000	((32R – 16R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 10001	((32R – 17R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 10010	((32R – 18R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 10011	((32R – 19R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 10100	((32R – 20R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 10101	((32R – 21R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 10110	((32R – 22R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 10111	((32R – 23R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 11000	((32R – 24R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 11001	((32R – 25R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 11010	((32R – 26R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 11011	((32R – 27R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 11100	((32R – 28R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 11101	((32R – 29R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 11110	((32R – 30R) / 48R) * (VinP2 – VinP5) + VinP5
	PKP_EX1 4-0 = 11111	((32R – 31R) / 48R) * (VinP2 – VinP5) + VinP5

Table 5.26: VinP_EX1

Reference voltage	Macro adjustment value	VinP_EX3 formula
VinP_EX2	PKP_EX2 4-0 = 00000	(47R / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 00001	((47R – 1R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 00010	((47R – 2R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 00011	((47R – 3R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 00100	((47R – 4R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 00101	((47R – 5R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 00110	((47R – 6R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 00111	((47R – 7R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 01000	((47R – 8R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 01001	((47R – 9R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 01010	((47R – 10R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 01011	((47R – 11R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 01100	((47R – 12R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 01101	((47R – 13R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 01110	((47R – 14R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 01111	((47R – 15R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 10000	((47R – 16R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 10001	((47R – 17R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 10010	((47R – 18R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 10011	((47R – 19R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 10100	((47R – 20R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 10101	((47R – 21R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 10110	((47R – 22R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 10111	((47R – 23R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 11000	((47R – 24R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 11001	((47R – 25R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 11010	((47R – 26R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 11011	((47R – 27R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 11100	((47R – 28R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 11101	((47R – 29R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 11110	((47R – 30R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX2 4-0 = 11111	((47R – 31R) / 48R) * (VinP11 – VinP14) + VinP14

Table 5.27: VinP_EX2

Reference voltage	Macro adjustment value	VinP_EX4 formula
VinP_EX3	PKP_EX3 4-0 = 00000	(39R / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 00001	((39R – 1R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 00010	((39R – 2R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 00011	((39R – 3R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 00100	((39R – 4R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 00101	((39R – 5R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 00110	((39R – 6R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 00111	((39R – 7R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 01000	((39R – 8R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 01001	((39R – 9R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 01010	((39R – 10R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 01011	((39R – 11R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 01100	((39R – 12R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 01101	((39R – 13R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 01110	((39R – 14R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 01111	((39R – 15R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 10000	((39R – 16R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 10001	((39R – 17R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 10010	((39R – 18R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 10011	((39R – 19R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 10100	((39R – 20R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 10101	((39R – 21R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 10110	((39R – 22R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 10111	((39R – 23R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 11000	((39R – 24R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 11001	((39R – 25R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 11010	((39R – 26R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 11011	((39R – 27R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 11100	((39R – 28R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 11101	((39R – 29R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 11110	((39R – 30R) / 48R) * (VinP11 – VinP14) + VinP14
	PKP_EX3 4-0 = 11111	((39R – 31R) / 48R) * (VinP11 – VinP14) + VinP14

Table 5.28: VinP_EX3

Reference voltage	Macro adjustment value	VinN0 formula
VinN0	VRN0 5-0 = 000000	VSNR
	VRN0 5-0 = 000001	((450R - 20R) / 450R) * VSNR
	VRN0 5-0 = 000010	((450R - 22R) / 450R) * VSNR
	VRN0 5-0 = 000011	((450R - 24R) / 450R) * VSNR
	VRN0 5-0 = 000100	((450R - 26R) / 450R) * VSNR
	VRN0 5-0 = 000101	((450R - 28R) / 450R) * VSNR
	VRN0 5-0 = 000110	((450R - 30R) / 450R) * VSNR
	VRN0 5-0 = 000111	((450R - 32R) / 450R) * VSNR
	VRN0 5-0 = 001000	((450R - 34R) / 450R) * VSNR
	VRN0 5-0 = 001001	((450R - 36R) / 450R) * VSNR
	VRN0 5-0 = 001010	((450R - 38R) / 450R) * VSNR
	VRN0 5-0 = 001011	((450R - 40R) / 450R) * VSNR
	VRN0 5-0 = 001100	((450R - 42R) / 450R) * VSNR
	VRN0 5-0 = 001101	((450R - 44R) / 450R) * VSNR
	VRN0 5-0 = 001110	((450R - 46R) / 450R) * VSNR
	VRN0 5-0 = 001111	((450R - 48R) / 450R) * VSNR
	VRN0 5-0 = 010000	((450R - 50R) / 450R) * VSNR
	VRN0 5-0 = 010001	((450R - 52R) / 450R) * VSNR
	VRN0 5-0 = 010010	((450R - 54R) / 450R) * VSNR
	VRN0 5-0 = 010011	((450R - 56R) / 450R) * VSNR
	VRN0 5-0 = 010100	((450R - 58R) / 450R) * VSNR
	VRN0 5-0 = 010101	((450R - 60R) / 450R) * VSNR
	VRN0 5-0 = 010110	((450R - 62R) / 450R) * VSNR
	VRN0 5-0 = 010111	((450R - 64R) / 450R) * VSNR
	VRN0 5-0 = 011000	((450R - 66R) / 450R) * VSNR
	VRN0 5-0 = 011001	((450R - 68R) / 450R) * VSNR
	VRN0 5-0 = 011010	((450R - 70R) / 450R) * VSNR
	VRN0 5-0 = 011011	((450R - 72R) / 450R) * VSNR
	VRN0 5-0 = 011100	((450R - 74R) / 450R) * VSNR
	VRN0 5-0 = 011101	((450R - 76R) / 450R) * VSNR
	VRN0 5-0 = 011110	((450R - 78R) / 450R) * VSNR
	VRN0 5-0 = 011111	((450R - 80R) / 450R) * VSNR
	VRN0 5-0 = 100000	((450R - 82R) / 450R) * VSNR
	VRN0 5-0 = 100001	((450R - 84R) / 450R) * VSNR
	VRN0 5-0 = 100010	((450R - 86R) / 450R) * VSNR
	VRN0 5-0 = 100011	((450R - 88R) / 450R) * VSNR
	VRN0 5-0 = 100100	((450R - 90R) / 450R) * VSNR
	VRN0 5-0 = 100101	((450R - 92R) / 450R) * VSNR
	VRN0 5-0 = 100110	((450R - 94R) / 450R) * VSNR
	VRN0 5-0 = 100111	((450R - 96R) / 450R) * VSNR
	VRN0 5-0 = 101000	((450R - 98R) / 450R) * VSNR
	VRN0 5-0 = 101001	((450R - 100R) / 450R) * VSNR
	VRN0 5-0 = 101010	((450R - 102R) / 450R) * VSNR
	VRN0 5-0 = 101011	((450R - 104R) / 450R) * VSNR
	VRN0 5-0 = 101100	((450R - 106R) / 450R) * VSNR
	VRN0 5-0 = 101101	((450R - 108R) / 450R) * VSNR
	VRN0 5-0 = 101110	((450R - 110R) / 450R) * VSNR
	VRN0 5-0 = 101111	((450R - 112R) / 450R) * VSNR
	VRN0 5-0 = 110000	((450R - 114R) / 450R) * VSNR
	VRN0 5-0 = 110001	((450R - 116R) / 450R) * VSNR
	VRN0 5-0 = 110010	((450R - 118R) / 450R) * VSNR
	VRN0 5-0 = 110011	((450R - 120R) / 450R) * VSNR
	VRN0 5-0 = 110100	((450R - 122R) / 450R) * VSNR
	VRN0 5-0 = 110101	((450R - 124R) / 450R) * VSNR
	VRN0 5-0 = 110110	((450R - 126R) / 450R) * VSNR
	VRN0 5-0 = 110111	((450R - 128R) / 450R) * VSNR
	VRN0 5-0 = 111000	((450R - 130R) / 450R) * VSNR
	VRN0 5-0 = 111001	((450R - 132R) / 450R) * VSNR
	VRN0 5-0 = 111010	((450R - 134R) / 450R) * VSNR
	VRN0 5-0 = 111011	((450R - 136R) / 450R) * VSNR
	VRN0 5-0 = 111100	((450R - 138R) / 450R) * VSNR
	VRN0 5-0 = 111101	((450R - 140R) / 450R) * VSNR
	VRN0 5-0 = 111110	((450R - 142R) / 450R) * VSNR
	VRN0 5-0 = 111111	((450R - 144R) / 450R) * VSNR

Table 5.29: VinN0

Reference voltage	Macro adjustment value	VinN1 formula
VinN1	VRN1 5-0 = 000000	(430R / 450R) * VSNR
	VRN1 5-0 = 000001	((430R - 2R) / 450R) * VSNR
	VRN1 5-0 = 000010	((430R - 4R) / 450R) * VSNR
	VRN1 5-0 = 000011	((430R - 6R) / 450R) * VSNR
	VRN1 5-0 = 000100	((430R - 8R) / 450R) * VSNR
	VRN1 5-0 = 000101	((430R - 10R) / 450R) * VSNR
	VRN1 5-0 = 000110	((430R - 12R) / 450R) * VSNR
	VRN1 5-0 = 000111	((430R - 14R) / 450R) * VSNR
	VRN1 5-0 = 001000	((430R - 16R) / 450R) * VSNR
	VRN1 5-0 = 001001	((430R - 18R) / 450R) * VSNR
	VRN1 5-0 = 001010	((430R - 20R) / 450R) * VSNR
	VRN1 5-0 = 001011	((430R - 22R) / 450R) * VSNR
	VRN1 5-0 = 001100	((430R - 24R) / 450R) * VSNR
	VRN1 5-0 = 001101	((430R - 26R) / 450R) * VSNR
	VRN1 5-0 = 001110	((430R - 28R) / 450R) * VSNR
	VRN1 5-0 = 001111	((430R - 30R) / 450R) * VSNR
	VRN1 5-0 = 010000	((430R - 32R) / 450R) * VSNR
	VRN1 5-0 = 010001	((430R - 34R) / 450R) * VSNR
	VRN1 5-0 = 010010	((430R - 36R) / 450R) * VSNR
	VRN1 5-0 = 010011	((430R - 38R) / 450R) * VSNR
	VRN1 5-0 = 010100	((430R - 40R) / 450R) * VSNR
	VRN1 5-0 = 010101	((430R - 42R) / 450R) * VSNR
	VRN1 5-0 = 010110	((430R - 44R) / 450R) * VSNR
	VRN1 5-0 = 010111	((430R - 46R) / 450R) * VSNR
	VRN1 5-0 = 011000	((430R - 48R) / 450R) * VSNR
	VRN1 5-0 = 011001	((430R - 50R) / 450R) * VSNR
	VRN1 5-0 = 011010	((430R - 52R) / 450R) * VSNR
	VRN1 5-0 = 011011	((430R - 54R) / 450R) * VSNR
	VRN1 5-0 = 011100	((430R - 56R) / 450R) * VSNR
	VRN1 5-0 = 011101	((430R - 58R) / 450R) * VSNR
	VRN1 5-0 = 011110	((430R - 60R) / 450R) * VSNR
	VRN1 5-0 = 011111	((430R - 62R) / 450R) * VSNR
	VRN1 5-0 = 100000	((430R - 64R) / 450R) * VSNR
	VRN1 5-0 = 100001	((430R - 66R) / 450R) * VSNR
	VRN1 5-0 = 100010	((430R - 68R) / 450R) * VSNR
	VRN1 5-0 = 100011	((430R - 70R) / 450R) * VSNR
	VRN1 5-0 = 100100	((430R - 72R) / 450R) * VSNR
	VRN1 5-0 = 100101	((430R - 74R) / 450R) * VSNR
	VRN1 5-0 = 100110	((430R - 76R) / 450R) * VSNR
	VRN1 5-0 = 100111	((430R - 78R) / 450R) * VSNR
	VRN1 5-0 = 101000	((430R - 80R) / 450R) * VSNR
	VRN1 5-0 = 101001	((430R - 82R) / 450R) * VSNR
	VRN1 5-0 = 101010	((430R - 84R) / 450R) * VSNR
	VRN1 5-0 = 101011	((430R - 86R) / 450R) * VSNR
	VRN1 5-0 = 101100	((430R - 88R) / 450R) * VSNR
	VRN1 5-0 = 101101	((430R - 90R) / 450R) * VSNR
	VRN1 5-0 = 101110	((430R - 92R) / 450R) * VSNR
	VRN1 5-0 = 101111	((430R - 94R) / 450R) * VSNR
	VRN1 5-0 = 110000	((430R - 96R) / 450R) * VSNR
	VRN1 5-0 = 110001	((430R - 98R) / 450R) * VSNR
	VRN1 5-0 = 110010	((430R - 100R) / 450R) * VSNR
	VRN1 5-0 = 110011	((430R - 102R) / 450R) * VSNR
	VRN1 5-0 = 110100	((430R - 104R) / 450R) * VSNR
	VRN1 5-0 = 110101	((430R - 106R) / 450R) * VSNR
	VRN1 5-0 = 110110	((430R - 108R) / 450R) * VSNR
	VRN1 5-0 = 110111	((430R - 110R) / 450R) * VSNR
	VRN1 5-0 = 111000	((430R - 112R) / 450R) * VSNR
	VRN1 5-0 = 111001	((430R - 114R) / 450R) * VSNR
	VRN1 5-0 = 111010	((430R - 116R) / 450R) * VSNR
	VRN1 5-0 = 111011	((430R - 118R) / 450R) * VSNR
	VRN1 5-0 = 111100	((430R - 120R) / 450R) * VSNR
	VRN1 5-0 = 111101	((430R - 122R) / 450R) * VSNR
	VRN1 5-0 = 111110	((430R - 124R) / 450R) * VSNR
	VRN1 5-0 = 111111	((430R - 126R) / 450R) * VSNR

Table 5.30: VinN1

Reference voltage	Macro adjustment value	VinN2 formula
VinN2	VRN2 5-0 = 000000	(420R / 450R) * VSNR
	VRN2 5-0 = 000001	((420R - 2R) / 450R) * VSNR
	VRN2 5-0 = 000010	((420R - 4R) / 450R) * VSNR
	VRN2 5-0 = 000011	((420R - 6R) / 450R) * VSNR
	VRN2 5-0 = 000100	((420R - 8R) / 450R) * VSNR
	VRN2 5-0 = 000101	((420R - 10R) / 450R) * VSNR
	VRN2 5-0 = 000110	((420R - 12R) / 450R) * VSNR
	VRN2 5-0 = 000111	((420R - 14R) / 450R) * VSNR
	VRN2 5-0 = 001000	((420R - 16R) / 450R) * VSNR
	VRN2 5-0 = 001001	((420R - 18R) / 450R) * VSNR
	VRN2 5-0 = 001010	((420R - 20R) / 450R) * VSNR
	VRN2 5-0 = 001011	((420R - 22R) / 450R) * VSNR
	VRN2 5-0 = 001100	((420R - 24R) / 450R) * VSNR
	VRN2 5-0 = 001101	((420R - 26R) / 450R) * VSNR
	VRN2 5-0 = 001110	((420R - 28R) / 450R) * VSNR
	VRN2 5-0 = 001111	((420R - 30R) / 450R) * VSNR
	VRN2 5-0 = 010000	((420R - 32R) / 450R) * VSNR
	VRN2 5-0 = 010001	((420R - 34R) / 450R) * VSNR
	VRN2 5-0 = 010010	((420R - 36R) / 450R) * VSNR
	VRN2 5-0 = 010011	((420R - 38R) / 450R) * VSNR
	VRN2 5-0 = 010100	((420R - 40R) / 450R) * VSNR
	VRN2 5-0 = 010101	((420R - 42R) / 450R) * VSNR
	VRN2 5-0 = 010110	((420R - 44R) / 450R) * VSNR
	VRN2 5-0 = 010111	((420R - 46R) / 450R) * VSNR
	VRN2 5-0 = 011000	((420R - 48R) / 450R) * VSNR
	VRN2 5-0 = 011001	((420R - 50R) / 450R) * VSNR
	VRN2 5-0 = 011010	((420R - 52R) / 450R) * VSNR
	VRN2 5-0 = 011011	((420R - 54R) / 450R) * VSNR
	VRN2 5-0 = 011100	((420R - 56R) / 450R) * VSNR
	VRN2 5-0 = 011101	((420R - 58R) / 450R) * VSNR
	VRN2 5-0 = 011110	((420R - 60R) / 450R) * VSNR
	VRN2 5-0 = 011111	((420R - 62R) / 450R) * VSNR
	VRN2 5-0 = 100000	((420R - 64R) / 450R) * VSNR
	VRN2 5-0 = 100001	((420R - 66R) / 450R) * VSNR
	VRN2 5-0 = 100010	((420R - 68R) / 450R) * VSNR
	VRN2 5-0 = 100011	((420R - 70R) / 450R) * VSNR
	VRN2 5-0 = 100100	((420R - 72R) / 450R) * VSNR
	VRN2 5-0 = 100101	((420R - 74R) / 450R) * VSNR
	VRN2 5-0 = 100110	((420R - 76R) / 450R) * VSNR
	VRN2 5-0 = 100111	((420R - 78R) / 450R) * VSNR
	VRN2 5-0 = 101000	((420R - 80R) / 450R) * VSNR
	VRN2 5-0 = 101001	((420R - 82R) / 450R) * VSNR
	VRN2 5-0 = 101010	((420R - 84R) / 450R) * VSNR
	VRN2 5-0 = 101011	((420R - 86R) / 450R) * VSNR
	VRN2 5-0 = 101100	((420R - 88R) / 450R) * VSNR
	VRN2 5-0 = 101101	((420R - 90R) / 450R) * VSNR
	VRN2 5-0 = 101110	((420R - 92R) / 450R) * VSNR
	VRN2 5-0 = 101111	((420R - 94R) / 450R) * VSNR
	VRN2 5-0 = 110000	((420R - 96R) / 450R) * VSNR
	VRN2 5-0 = 110001	((420R - 98R) / 450R) * VSNR
	VRN2 5-0 = 110010	((420R - 100R) / 450R) * VSNR
	VRN2 5-0 = 110011	((420R - 102R) / 450R) * VSNR
	VRN2 5-0 = 110100	((420R - 104R) / 450R) * VSNR
	VRN2 5-0 = 110101	((420R - 106R) / 450R) * VSNR
	VRN2 5-0 = 110110	((420R - 108R) / 450R) * VSNR
	VRN2 5-0 = 110111	((420R - 110R) / 450R) * VSNR
	VRN2 5-0 = 111000	((420R - 112R) / 450R) * VSNR
	VRN2 5-0 = 111001	((420R - 114R) / 450R) * VSNR
	VRN2 5-0 = 111010	((420R - 116R) / 450R) * VSNR
	VRN2 5-0 = 111011	((420R - 118R) / 450R) * VSNR
	VRN2 5-0 = 111100	((420R - 120R) / 450R) * VSNR
	VRN2 5-0 = 111101	((420R - 122R) / 450R) * VSNR
	VRN2 5-0 = 111110	((420R - 124R) / 450R) * VSNR
	VRN2 5-0 = 111111	((420R - 126R) / 450R) * VSNR

Table 5.31: VinN2

Reference voltage	Macro adjustment value	VinN14 formula
VinN14	VRN3 5-0 = 000000	(156R / 450R) * VSNR
	VRN3 5-0 = 000001	((156R - 2R) / 450R) * VSNR
	VRN3 5-0 = 000010	((156R - 4R) / 450R) * VSNR
	VRN3 5-0 = 000011	((156R - 6R) / 450R) * VSNR
	VRN3 5-0 = 000100	((156R - 8R) / 450R) * VSNR
	VRN3 5-0 = 000101	((156R - 10R) / 450R) * VSNR
	VRN3 5-0 = 000110	((156R - 12R) / 450R) * VSNR
	VRN3 5-0 = 000111	((156R - 14R) / 450R) * VSNR
	VRN3 5-0 = 001000	((156R - 16R) / 450R) * VSNR
	VRN3 5-0 = 001001	((156R - 18R) / 450R) * VSNR
	VRN3 5-0 = 001010	((156R - 20R) / 450R) * VSNR
	VRN3 5-0 = 001011	((156R - 22R) / 450R) * VSNR
	VRN3 5-0 = 001100	((156R - 24R) / 450R) * VSNR
	VRN3 5-0 = 001101	((156R - 26R) / 450R) * VSNR
	VRN3 5-0 = 001110	((156R - 28R) / 450R) * VSNR
	VRN3 5-0 = 001111	((156R - 30R) / 450R) * VSNR
	VRN3 5-0 = 010000	((156R - 32R) / 450R) * VSNR
	VRN3 5-0 = 010001	((156R - 34R) / 450R) * VSNR
	VRN3 5-0 = 010010	((156R - 36R) / 450R) * VSNR
	VRN3 5-0 = 010011	((156R - 38R) / 450R) * VSNR
	VRN3 5-0 = 010100	((156R - 40R) / 450R) * VSNR
	VRN3 5-0 = 010101	((156R - 42R) / 450R) * VSNR
	VRN3 5-0 = 010110	((156R - 44R) / 450R) * VSNR
	VRN3 5-0 = 010111	((156R - 46R) / 450R) * VSNR
	VRN3 5-0 = 011000	((156R - 48R) / 450R) * VSNR
	VRN3 5-0 = 011001	((156R - 50R) / 450R) * VSNR
	VRN3 5-0 = 011010	((156R - 52R) / 450R) * VSNR
	VRN3 5-0 = 011011	((156R - 54R) / 450R) * VSNR
	VRN3 5-0 = 011100	((156R - 56R) / 450R) * VSNR
	VRN3 5-0 = 011101	((156R - 58R) / 450R) * VSNR
	VRN3 5-0 = 011110	((156R - 60R) / 450R) * VSNR
	VRN3 5-0 = 011111	((156R - 62R) / 450R) * VSNR
	VRN3 5-0 = 100000	((156R - 64R) / 450R) * VSNR
	VRN3 5-0 = 100001	((156R - 66R) / 450R) * VSNR
	VRN3 5-0 = 100010	((156R - 68R) / 450R) * VSNR
	VRN3 5-0 = 100011	((156R - 70R) / 450R) * VSNR
	VRN3 5-0 = 100100	((156R - 72R) / 450R) * VSNR
	VRN3 5-0 = 100101	((156R - 74R) / 450R) * VSNR
	VRN3 5-0 = 100110	((156R - 76R) / 450R) * VSNR
	VRN3 5-0 = 100111	((156R - 78R) / 450R) * VSNR
	VRN3 5-0 = 101000	((156R - 80R) / 450R) * VSNR
	VRN3 5-0 = 101001	((156R - 82R) / 450R) * VSNR
	VRN3 5-0 = 101010	((156R - 84R) / 450R) * VSNR
	VRN3 5-0 = 101011	((156R - 86R) / 450R) * VSNR
	VRN3 5-0 = 101100	((156R - 88R) / 450R) * VSNR
	VRN3 5-0 = 101101	((156R - 90R) / 450R) * VSNR
	VRN3 5-0 = 101110	((156R - 92R) / 450R) * VSNR
	VRN3 5-0 = 101111	((156R - 94R) / 450R) * VSNR
	VRN3 5-0 = 110000	((156R - 96R) / 450R) * VSNR
	VRN3 5-0 = 110001	((156R - 98R) / 450R) * VSNR
	VRN3 5-0 = 110010	((156R - 100R) / 450R) * VSNR
	VRN3 5-0 = 110011	((156R - 102R) / 450R) * VSNR
	VRN3 5-0 = 110100	((156R - 104R) / 450R) * VSNR
	VRN3 5-0 = 110101	((156R - 106R) / 450R) * VSNR
	VRN3 5-0 = 110110	((156R - 108R) / 450R) * VSNR
	VRN3 5-0 = 110111	((156R - 110R) / 450R) * VSNR
	VRN3 5-0 = 111000	((156R - 112R) / 450R) * VSNR
	VRN3 5-0 = 111001	((156R - 114R) / 450R) * VSNR
	VRN3 5-0 = 111010	((156R - 116R) / 450R) * VSNR
	VRN3 5-0 = 111011	((156R - 118R) / 450R) * VSNR
	VRN3 5-0 = 111100	((156R - 120R) / 450R) * VSNR
	VRN3 5-0 = 111101	((156R - 122R) / 450R) * VSNR
	VRN3 5-0 = 111110	((156R - 124R) / 450R) * VSNR
	VRN3 5-0 = 111111	((156R - 126R) / 450R) * VSNR

Table 5.32: VinN14

Reference voltage	Macro adjustment value	VinN15 formula
VinN15	VRN4 5-0 = 000000	(146R / 450R) * VSNR
	VRN4 5-0 = 000001	((146R - 2R) / 450R) * VSNR
	VRN4 5-0 = 000010	((146R - 4R) / 450R) * VSNR
	VRN4 5-0 = 000011	((146R - 6R) / 450R) * VSNR
	VRN4 5-0 = 000100	((146R - 8R) / 450R) * VSNR
	VRN4 5-0 = 000101	((146R - 10R) / 450R) * VSNR
	VRN4 5-0 = 000110	((146R - 12R) / 450R) * VSNR
	VRN4 5-0 = 000111	((146R - 14R) / 450R) * VSNR
	VRN4 5-0 = 001000	((146R - 16R) / 450R) * VSNR
	VRN4 5-0 = 001001	((146R - 18R) / 450R) * VSNR
	VRN4 5-0 = 001010	((146R - 20R) / 450R) * VSNR
	VRN4 5-0 = 001011	((146R - 22R) / 450R) * VSNR
	VRN4 5-0 = 001100	((146R - 24R) / 450R) * VSNR
	VRN4 5-0 = 001101	((146R - 26R) / 450R) * VSNR
	VRN4 5-0 = 001110	((146R - 28R) / 450R) * VSNR
	VRN4 5-0 = 001111	((146R - 30R) / 450R) * VSNR
	VRN4 5-0 = 010000	((146R - 32R) / 450R) * VSNR
	VRN4 5-0 = 010001	((146R - 34R) / 450R) * VSNR
	VRN4 5-0 = 010010	((146R - 36R) / 450R) * VSNR
	VRN4 5-0 = 010011	((146R - 38R) / 450R) * VSNR
	VRN4 5-0 = 010100	((146R - 40R) / 450R) * VSNR
	VRN4 5-0 = 010101	((146R - 42R) / 450R) * VSNR
	VRN4 5-0 = 010110	((146R - 44R) / 450R) * VSNR
	VRN4 5-0 = 010111	((146R - 46R) / 450R) * VSNR
	VRN4 5-0 = 011000	((146R - 48R) / 450R) * VSNR
	VRN4 5-0 = 011001	((146R - 50R) / 450R) * VSNR
	VRN4 5-0 = 011010	((146R - 52R) / 450R) * VSNR
	VRN4 5-0 = 011011	((146R - 54R) / 450R) * VSNR
	VRN4 5-0 = 011100	((146R - 56R) / 450R) * VSNR
	VRN4 5-0 = 011101	((146R - 58R) / 450R) * VSNR
	VRN4 5-0 = 011110	((146R - 60R) / 450R) * VSNR
	VRN4 5-0 = 011111	((146R - 62R) / 450R) * VSNR
	VRN4 5-0 = 100000	((146R - 64R) / 450R) * VSNR
	VRN4 5-0 = 100001	((146R - 66R) / 450R) * VSNR
	VRN4 5-0 = 100010	((146R - 68R) / 450R) * VSNR
	VRN4 5-0 = 100011	((146R - 70R) / 450R) * VSNR
	VRN4 5-0 = 100100	((146R - 72R) / 450R) * VSNR
	VRN4 5-0 = 100101	((146R - 74R) / 450R) * VSNR
	VRN4 5-0 = 100110	((146R - 76R) / 450R) * VSNR
	VRN4 5-0 = 100111	((146R - 78R) / 450R) * VSNR
	VRN4 5-0 = 101000	((146R - 80R) / 450R) * VSNR
	VRN4 5-0 = 101001	((146R - 82R) / 450R) * VSNR
	VRN4 5-0 = 101010	((146R - 84R) / 450R) * VSNR
	VRN4 5-0 = 101011	((146R - 86R) / 450R) * VSNR
	VRN4 5-0 = 101100	((146R - 88R) / 450R) * VSNR
	VRN4 5-0 = 101101	((146R - 90R) / 450R) * VSNR
	VRN4 5-0 = 101110	((146R - 92R) / 450R) * VSNR
	VRN4 5-0 = 101111	((146R - 94R) / 450R) * VSNR
	VRN4 5-0 = 110000	((146R - 96R) / 450R) * VSNR
	VRN4 5-0 = 110001	((146R - 98R) / 450R) * VSNR
	VRN4 5-0 = 110010	((146R - 100R) / 450R) * VSNR
	VRN4 5-0 = 110011	((146R - 102R) / 450R) * VSNR
	VRN4 5-0 = 110100	((146R - 104R) / 450R) * VSNR
	VRN4 5-0 = 110101	((146R - 106R) / 450R) * VSNR
	VRN4 5-0 = 110110	((146R - 108R) / 450R) * VSNR
	VRN4 5-0 = 110111	((146R - 110R) / 450R) * VSNR
	VRN4 5-0 = 111000	((146R - 112R) / 450R) * VSNR
	VRN4 5-0 = 111001	((146R - 114R) / 450R) * VSNR
	VRN4 5-0 = 111010	((146R - 116R) / 450R) * VSNR
	VRN4 5-0 = 111011	((146R - 118R) / 450R) * VSNR
	VRN4 5-0 = 111100	((146R - 120R) / 450R) * VSNR
	VRN4 5-0 = 111101	((146R - 122R) / 450R) * VSNR
	VRN4 5-0 = 111110	((146R - 124R) / 450R) * VSNR
	VRN4 5-0 = 111111	((146R - 126R) / 450R) * VSNR

Table 5.33: VinN15

Reference voltage	Macro adjustment value	VinN16 formula
VinN16	VRN5 5-0 = 000000	(144R / 450R) * VSNR
	VRN5 5-0 = 000001	((144R - 2R) / 450R) * VSNR
	VRN5 5-0 = 000010	((144R - 4R) / 450R) * VSNR
	VRN5 5-0 = 000011	((144R - 6R) / 450R) * VSNR
	VRN5 5-0 = 000100	((144R - 8R) / 450R) * VSNR
	VRN5 5-0 = 000101	((144R - 10R) / 450R) * VSNR
	VRN5 5-0 = 000110	((144R - 12R) / 450R) * VSNR
	VRN5 5-0 = 000111	((144R - 14R) / 450R) * VSNR
	VRN5 5-0 = 001000	((144R - 16R) / 450R) * VSNR
	VRN5 5-0 = 001001	((144R - 18R) / 450R) * VSNR
	VRN5 5-0 = 001010	((144R - 20R) / 450R) * VSNR
	VRN5 5-0 = 001011	((144R - 22R) / 450R) * VSNR
	VRN5 5-0 = 001100	((144R - 24R) / 450R) * VSNR
	VRN5 5-0 = 001101	((144R - 26R) / 450R) * VSNR
	VRN5 5-0 = 001110	((144R - 28R) / 450R) * VSNR
	VRN5 5-0 = 001111	((144R - 30R) / 450R) * VSNR
	VRN5 5-0 = 010000	((144R - 32R) / 450R) * VSNR
	VRN5 5-0 = 010001	((144R - 34R) / 450R) * VSNR
	VRN5 5-0 = 010010	((144R - 36R) / 450R) * VSNR
	VRN5 5-0 = 010011	((144R - 38R) / 450R) * VSNR
	VRN5 5-0 = 010100	((144R - 40R) / 450R) * VSNR
	VRN5 5-0 = 010101	((144R - 42R) / 450R) * VSNR
	VRN5 5-0 = 010110	((144R - 44R) / 450R) * VSNR
	VRN5 5-0 = 010111	((144R - 46R) / 450R) * VSNR
	VRN5 5-0 = 011000	((144R - 48R) / 450R) * VSNR
	VRN5 5-0 = 011001	((144R - 50R) / 450R) * VSNR
	VRN5 5-0 = 011010	((144R - 52R) / 450R) * VSNR
	VRN5 5-0 = 011011	((144R - 54R) / 450R) * VSNR
	VRN5 5-0 = 011100	((144R - 56R) / 450R) * VSNR
	VRN5 5-0 = 011101	((144R - 58R) / 450R) * VSNR
	VRN5 5-0 = 011110	((144R - 60R) / 450R) * VSNR
	VRN5 5-0 = 011111	((144R - 62R) / 450R) * VSNR
	VRN5 5-0 = 100000	((144R - 64R) / 450R) * VSNR
	VRN5 5-0 = 100001	((144R - 66R) / 450R) * VSNR
	VRN5 5-0 = 100010	((144R - 68R) / 450R) * VSNR
	VRN5 5-0 = 100011	((144R - 70R) / 450R) * VSNR
	VRN5 5-0 = 100100	((144R - 72R) / 450R) * VSNR
	VRN5 5-0 = 100101	((144R - 74R) / 450R) * VSNR
	VRN5 5-0 = 100110	((144R - 76R) / 450R) * VSNR
	VRN5 5-0 = 100111	((144R - 78R) / 450R) * VSNR
	VRN5 5-0 = 101000	((144R - 80R) / 450R) * VSNR
	VRN5 5-0 = 101001	((144R - 82R) / 450R) * VSNR
	VRN5 5-0 = 101010	((144R - 84R) / 450R) * VSNR
	VRN5 5-0 = 101011	((144R - 86R) / 450R) * VSNR
	VRN5 5-0 = 101100	((144R - 88R) / 450R) * VSNR
	VRN5 5-0 = 101101	((144R - 90R) / 450R) * VSNR
	VRN5 5-0 = 101110	((144R - 92R) / 450R) * VSNR
	VRN5 5-0 = 101111	((144R - 94R) / 450R) * VSNR
	VRN5 5-0 = 110000	((144R - 96R) / 450R) * VSNR
	VRN5 5-0 = 110001	((144R - 98R) / 450R) * VSNR
	VRN5 5-0 = 110010	((144R - 100R) / 450R) * VSNR
	VRN5 5-0 = 110011	((144R - 102R) / 450R) * VSNR
	VRN5 5-0 = 110100	((144R - 104R) / 450R) * VSNR
	VRN5 5-0 = 110101	((144R - 106R) / 450R) * VSNR
	VRN5 5-0 = 110110	((144R - 108R) / 450R) * VSNR
	VRN5 5-0 = 110111	((144R - 110R) / 450R) * VSNR
	VRN5 5-0 = 111000	((144R - 112R) / 450R) * VSNR
	VRN5 5-0 = 111001	((144R - 114R) / 450R) * VSNR
	VRN5 5-0 = 111010	((144R - 116R) / 450R) * VSNR
	VRN5 5-0 = 111011	((144R - 118R) / 450R) * VSNR
	VRN5 5-0 = 111100	((144R - 120R) / 450R) * VSNR
	VRN5 5-0 = 111101	((144R - 122R) / 450R) * VSNR
	VRN5 5-0 = 111110	((144R - 124R) / 450R) * VSNR
	VRN5 5-0 = 111111	VSSA

Table 5.34: VinN16

Reference voltage	Macro adjustment value	VinN5 formula
VinN5	PRN0 6-0 = 0000000	(350R / 450R) VSNR
	PRN0 6-0 = 0000001	((350R - 2R) / 450R) * VSNR
	PRN0 6-0 = 0000010	((350R - 4R) / 450R) * VSNR
	PRN0 6-0 = 0000011	((350R – 6R) / 450R) * VSNR
	PRN0 6-0 = 0000100	((350R – 8R) / 450R) * VSNR
	PRN0 6-0 = 0000101	((350R – 10R) / 450R) * VSNR
	PRN0 6-0 = 0000110	((350R – 12R) / 450R) * VSNR
	PRN0 6-0 = 0000111	((350R - 14R) / 450R) * VSNR
	PRN0 6-0 = 0001000	((350R – 16R) / 450R) * VSNR
	PRN0 6-0 = 0001001	((350R – 18R) / 450R) * VSNR
	PRN0 6-0 = 0001010	((350R – 20R) / 450R) * VSNR
	PRN0 6-0 = 0001011	((350R – 22R) / 450R) * VSNR
	PRN0 6-0 = 0001100	((350R – 24R) / 450R) * VSNR
	PRN0 6-0 = 0001101	((350R – 26R) / 450R) * VSNR
	PRN0 6-0 = 0001110	((350R – 28R) / 450R) * VSNR
	PRN0 6-0 = 0001111	((350R – 30R) / 450R) * VSNR
	PRN0 6-0 = 0010000	((350R – 32R) / 450R) * VSNR
	PRN0 6-0 = 0010001	((350R - 34R) / 450R) * VSNR
	PRN0 6-0 = 0010010	((350R – 36R) / 450R) * VSNR
	PRN0 6-0 = 0010011	((350R – 38R) / 450R) * VSNR
	PRN0 6-0 = 0010100	((350R – 40R) / 450R) * VSNR
	PRN0 6-0 = 0010101	((350R – 42R) / 450R) * VSNR
	PRN0 6-0 = 0010110	((350R – 44R) / 450R) * VSNR
	PRN0 6-0 = 0010111	((350R – 46R) / 450R) * VSNR
	PRN0 6-0 = 0011000	((350R – 48R) / 450R) * VSNR
	PRN0 6-0 = 0011001	((350R – 50R) / 450R) * VSNR
	PRN0 6-0 = 0011010	((350R – 52R) / 450R) * VSNR
	PRN0 6-0 = 0011011	((350R - 54R) / 450R) * VSNR
	PRN0 6-0 = 0011100	((350R – 56R) / 450R) * VSNR
	PRN0 6-0 = 0011101	((350R – 58R) / 450R) * VSNR
	PRN0 6-0 = 0011110	((350R – 60R) / 450R) * VSNR
	PRN0 6-0 = 0011111	((350R – 62R) / 450R) * VSNR
	PRN0 6-0 = 0100000	((350R - 64R) / 450R) * VSNR
	PRN0 6-0 = 0100001	((350R – 66R) / 450R) * VSNR
	PRN0 6-0 = 0100010	((350R – 68R) / 450R) * VSNR
	PRN0 6-0 = 0100011	((350R – 70R) / 450R) * VSNR
	PRN0 6-0 = 0100100	((350R – 72R) / 450R) * VSNR
	PRN0 6-0 = 0100101	((350R – 74R) / 450R) * VSNR
	PRN0 6-0 = 0100110	((350R – 76R) / 450R) * VSNR
	PRN0 6-0 = 0100111	((350R – 78R) / 450R) * VSNR
	PRN0 6-0 = 0101000	((350R – 80R) / 450R) * VSNR
	PRN0 6-0 = 0101001	((350R – 82R) / 450R) * VSNR
	PRN0 6-0 = 0101010	((350R - 84R) / 450R) * VSNR
	PRN0 6-0 = 0101011	((350R – 86R) / 450R) * VSNR
	PRN0 6-0 = 0101100	((350R – 88R) / 450R) * VSNR
	PRN0 6-0 = 0101101	((350R – 90R) / 450R) * VSNR
	PRN0 6-0 = 0101110	((350R – 92R) / 450R) * VSNR
	PRN0 6-0 = 0101111	((350R – 94R) / 450R) * VSNR
	PRN0 6-0 = 0110000	((350R – 96R) / 450R) * VSNR
	PRN0 6-0 = 0110001	((350R – 98R) / 450R) * VSNR
	PRN0 6-0 = 0110010	((350R – 100R) / 450R) * VSNR
	PRN0 6-0 = 0110011	((350R – 102R) / 450R) * VSNR
	PRN0 6-0 = 0110100	((350R – 104R) / 450R) * VSNR
	PRN0 6-0 = 0110101	((350R – 106R) / 450R) * VSNR
	PRN0 6-0 = 0110110	((350R – 108R) / 450R) * VSNR
	PRN0 6-0 = 0110111	((350R – 110R) / 450R) * VSNR
	PRN0 6-0 = 0111000	((350R – 112R) / 450R) * VSNR
	PRN0 6-0 = 0111001	((350R – 114R) / 450R) * VSNR
	PRN0 6-0 = 0111010	((350R – 116R) / 450R) * VSNR
	PRN0 6-0 = 0111011	((350R – 118R) / 450R) * VSNR
	PRN0 6-0 = 0111100	((350R – 120R) / 450R) * VSNR
	PRN0 6-0 = 0111101	((350R – 122R) / 450R) * VSNR
	PRN0 6-0 = 0111110	((350R - 124R) / 450R) * VSNR
	PRN0 6-0 = 0111111	((350R – 126R) / 450R) * VSNR
	PRN0 6-0 = 1000000	((350R – 128R) / 450R) * VSNR
	PRN0 6-0 = 1000001	((350R – 130R) / 450R) * VSNR
	PRN0 6-0 = 1000010	((350R - 132R) / 450R) * VSNR
	PRN0 6-0 = 1000011	((350R – 134R) / 450R) * VSNR
	PRN0 6-0 = 1000100	((350R – 136R) / 450R) * VSNR

	PRN0 6-0 = 1000101	((350R – 138R) / 450R) * VSNR
	PRN0 6-0 = 1000110	((350R – 140R) / 450R) * VSNR
	PRN0 6-0 = 1000111	((350R – 142R) / 450R) * VSNR
	PRN0 6-0 = 1001000	((350R – 144R) / 450R) * VSNR
	PRN0 6-0 = 1001001	((350R – 146R) / 450R) * VSNR
	PRN0 6-0 = 1001010	((350R – 148R) / 450R) * VSNR
	PRN0 6-0 = 1001011	((350R – 150R) / 450R) * VSNR
	PRN0 6-0 = 1001100	((350R - 152R) / 450R) * VSNR
	PRN0 6-0 = 1001101	((350R – 154R) / 450R) * VSNR
	PRN0 6-0 = 1001110	((350R – 156R) / 450R) * VSNR
	PRN0 6-0 = 1001111	((350R – 158R) / 450R) * VSNR
	PRN0 6-0 = 1010000	((350R – 160R) / 450R) * VSNR
	PRN0 6-0 = 1010001	((350R – 162R) / 450R) * VSNR
	PRN0 6-0 = 1010010	((350R – 164R) / 450R) * VSNR
	PRN0 6-0 = 1010011	((350R – 166R) / 450R) * VSNR
	PRN0 6-0 = 1010100	((350R – 168R) / 450R) * VSNR
	PRN0 6-0 = 1010101	((350R – 170R) / 450R) * VSNR
	PRN0 6-0 = 1010110	((350R – 172R) / 450R) * VSNR
	PRN0 6-0 = 1010111	((350R - 174R) / 450R) * VSNR
	PRN0 6-0 = 1011000	inhibit
	PRN0 6-0 = 1011001	inhibit
	PRN0 6-0 = 1011010	inhibit
	PRN0 6-0 = 1011011	inhibit
	PRN0 6-0 = 1011100	inhibit
	PRN0 6-0 = 1011101	inhibit
	PRN0 6-0 = 1011110	inhibit
	PRN0 6-0 = 1011111	inhibit
	PRN0 6-0 = 1100000	inhibit
	PRN0 6-0 = 1100001	inhibit
	PRN0 6-0 = 1100010	inhibit
	PRN0 6-0 = 1100011	inhibit
	PRN0 6-0 = 1100100	inhibit
	PRN0 6-0 = 1100101	inhibit
	PRN0 6-0 = 1100110	inhibit
	PRN0 6-0 = 1100111	inhibit
	PRN0 6-0 = 1101000	inhibit
	PRN0 6-0 = 1101001	inhibit
	PRN0 6-0 = 1101010	inhibit
	PRN0 6-0 = 1101011	inhibit
	PRN0 6-0 = 1101100	inhibit
	PRN0 6-0 = 1101101	inhibit
	PRN0 6-0 = 1101110	inhibit
	PRN0 6-0 = 1101111	inhibit
	PRN0 6-0 = 1110000	inhibit
	PRN0 6-0 = 1110001	inhibit
	PRN0 6-0 = 1110010	inhibit
	PRN0 6-0 = 1110011	inhibit
	PRN0 6-0 = 1110100	inhibit
	PRN0 6-0 = 1110101	inhibit
	PRN0 6-0 = 1110110	inhibit
	PRN0 6-0 = 1110111	inhibit
	PRN0 6-0 = 1111000	inhibit
	PRN0 6-0 = 1111001	inhibit
	PRN0 6-0 = 1111010	inhibit
	PRN0 6-0 = 1111011	inhibit
	PRN0 6-0 = 1111100	inhibit
	PRN0 6-0 = 1111101	inhibit
	PRN0 6-0 = 1111110	inhibit
	PRN0 6-0 = 1111111	inhibit

Table 5.35: VinN5

Reference voltage	Macro adjustment value	VinN11 formula
VinN11	PRN1 6-0 = 0000000	(274R / 450R) VSNR
	PRN1 6-0 = 0000001	((274R - 2R) / 450R) * VSNR
	PRN1 6-0 = 0000010	((274R - 4R) / 450R) * VSNR
	PRN1 6-0 = 0000011	((274R – 6R) / 450R) * VSNR
	PRN1 6-0 = 0000100	((274R – 8R) / 450R) * VSNR
	PRN1 6-0 = 0000101	((274R – 10R) / 450R) * VSNR
	PRN1 6-0 = 0000110	((274R – 12R) / 450R) * VSNR
	PRN1 6-0 = 0000111	((274R - 14R) / 450R) * VSNR
	PRN1 6-0 = 0001000	((274R – 16R) / 450R) * VSNR
	PRN1 6-0 = 0001001	((274R – 18R) / 450R) * VSNR
	PRN1 6-0 = 0001010	((274R – 20R) / 450R) * VSNR
	PRN1 6-0 = 0001011	((274R – 22R) / 450R) * VSNR
	PRN1 6-0 = 0001100	((274R – 24R) / 450R) * VSNR
	PRN1 6-0 = 0001101	((274R – 26R) / 450R) * VSNR
	PRN1 6-0 = 0001110	((274R – 28R) / 450R) * VSNR
	PRN1 6-0 = 0001111	((274R – 30R) / 450R) * VSNR
	PRN1 6-0 = 0010000	((274R – 32R) / 450R) * VSNR
	PRN1 6-0 = 0010001	((274R - 34R) / 450R) * VSNR
	PRN1 6-0 = 0010010	((274R – 36R) / 450R) * VSNR
	PRN1 6-0 = 0010011	((274R – 38R) / 450R) * VSNR
	PRN1 6-0 = 0010100	((274R – 40R) / 450R) * VSNR
	PRN1 6-0 = 0010101	((274R – 42R) / 450R) * VSNR
	PRN1 6-0 = 0010110	((274R – 44R) / 450R) * VSNR
	PRN1 6-0 = 0010111	((274R – 46R) / 450R) * VSNR
	PRN1 6-0 = 0011000	((274R – 48R) / 450R) * VSNR
	PRN1 6-0 = 0011001	((274R – 50R) / 450R) * VSNR
	PRN1 6-0 = 0011010	((274R – 52R) / 450R) * VSNR
	PRN1 6-0 = 0011011	((274R - 54R) / 450R) * VSNR
	PRN1 6-0 = 0011100	((274R – 56R) / 450R) * VSNR
	PRN1 6-0 = 0011101	((274R – 58R) / 450R) * VSNR
	PRN1 6-0 = 0011110	((274R – 60R) / 450R) * VSNR
	PRN1 6-0 = 0011111	((274R – 62R) / 450R) * VSNR
	PRN1 6-0 = 0100000	((274R - 64R) / 450R) * VSNR
	PRN1 6-0 = 0100001	((274R – 66R) / 450R) * VSNR
	PRN1 6-0 = 0100010	((274R – 68R) / 450R) * VSNR
	PRN1 6-0 = 0100011	((274R – 70R) / 450R) * VSNR
	PRN1 6-0 = 0100100	((274R – 72R) / 450R) * VSNR
	PRN1 6-0 = 0100101	((274R – 74R) / 450R) * VSNR
	PRN1 6-0 = 0100110	((274R – 76R) / 450R) * VSNR
	PRN1 6-0 = 0100111	((274R – 78R) / 450R) * VSNR
	PRN1 6-0 = 0101000	((274R – 80R) / 450R) * VSNR
	PRN1 6-0 = 0101001	((274R – 82R) / 450R) * VSNR
	PRN1 6-0 = 0101010	((274R - 84R) / 450R) * VSNR
	PRN1 6-0 = 0101011	((274R – 86R) / 450R) * VSNR
	PRN1 6-0 = 0101100	((274R – 88R) / 450R) * VSNR
	PRN1 6-0 = 0101101	((274R – 90R) / 450R) * VSNR
	PRN1 6-0 = 0101110	((274R – 92R) / 450R) * VSNR
	PRN1 6-0 = 0101111	((274R – 94R) / 450R) * VSNR
	PRN1 6-0 = 0110000	((274R – 96R) / 450R) * VSNR
	PRN1 6-0 = 0110001	((274R – 98R) / 450R) * VSNR
	PRN1 6-0 = 0110010	((274R – 100R) / 450R) * VSNR
	PRN1 6-0 = 0110011	((274R – 102R) / 450R) * VSNR
	PRN1 6-0 = 0110100	((274R – 104R) / 450R) * VSNR
	PRN1 6-0 = 0110101	((274R – 106R) / 450R) * VSNR
	PRN1 6-0 = 0110110	((274R – 108R) / 450R) * VSNR
	PRN1 6-0 = 0110111	((274R – 110R) / 450R) * VSNR
	PRN1 6-0 = 0111000	((274R – 112R) / 450R) * VSNR
	PRN1 6-0 = 0111001	((274R – 114R) / 450R) * VSNR
	PRN1 6-0 = 0111010	((274R – 116R) / 450R) * VSNR
	PRN1 6-0 = 0111011	((274R – 118R) / 450R) * VSNR
	PRN1 6-0 = 0111100	((274R – 120R) / 450R) * VSNR
	PRN1 6-0 = 0111101	((274R – 122R) / 450R) * VSNR
	PRN1 6-0 = 0111110	((274R - 124R) / 450R) * VSNR
	PRN1 6-0 = 0111111	((274R – 126R) / 450R) * VSNR
	PRN1 6-0 = 1000000	((274R – 128R) / 450R) * VSNR
	PRN1 6-0 = 1000001	((274R – 130R) / 450R) * VSNR
	PRN1 6-0 = 1000010	((274R - 132R) / 450R) * VSNR
	PRN1 6-0 = 1000011	((274R – 134R) / 450R) * VSNR
	PRN1 6-0 = 1000100	((274R – 136R) / 450R) * VSNR

	PRN1 6-0 = 1000101	((274R – 138R) / 450R) * VSNR
	PRN1 6-0 = 1000110	((274R – 140R) / 450R) * VSNR
	PRN1 6-0 = 1000111	((274R – 142R) / 450R) * VSNR
	PRN1 6-0 = 1001000	((274R – 144R) / 450R) * VSNR
	PRN1 6-0 = 1001001	((274R – 146R) / 450R) * VSNR
	PRN1 6-0 = 1001010	((274R – 148R) / 450R) * VSNR
	PRN1 6-0 = 1001011	((274R – 150R) / 450R) * VSNR
	PRN1 6-0 = 1001100	((274R - 152R) / 450R) * VSNR
	PRN1 6-0 = 1001101	((274R – 154R) / 450R) * VSNR
	PRN1 6-0 = 1001110	((274R – 156R) / 450R) * VSNR
	PRN1 6-0 = 1001111	((274R – 158R) / 450R) * VSNR
	PRN1 6-0 = 1010000	((274R – 160R) / 450R) * VSNR
	PRN1 6-0 = 1010001	((274R – 162R) / 450R) * VSNR
	PRN1 6-0 = 1010010	((274R – 164R) / 450R) * VSNR
	PRN1 6-0 = 1010011	((274R – 166R) / 450R) * VSNR
	PRN1 6-0 = 1010100	((274R – 168R) / 450R) * VSNR
	PRN1 6-0 = 1010101	((274R – 170R) / 450R) * VSNR
	PRN1 6-0 = 1010110	((274R – 172R) / 450R) * VSNR
	PRN1 6-0 = 1010111	((274R - 174R) / 450R) * VSNR
	PRN1 6-0 = 1011000	inhibit
	PRN1 6-0 = 1011001	inhibit
	PRN1 6-0 = 1011010	inhibit
	PRN1 6-0 = 1011011	inhibit
	PRN1 6-0 = 1011100	inhibit
	PRN1 6-0 = 1011101	inhibit
	PRN1 6-0 = 1011110	inhibit
	PRN1 6-0 = 1011111	inhibit
	PRN1 6-0 = 1100000	inhibit
	PRN1 6-0 = 1100001	inhibit
	PRN1 6-0 = 1100010	inhibit
	PRN1 6-0 = 1100011	inhibit
	PRN1 6-0 = 1100100	inhibit
	PRN1 6-0 = 1100101	inhibit
	PRN1 6-0 = 1100110	inhibit
	PRN1 6-0 = 1100111	inhibit
	PRN1 6-0 = 1101000	inhibit
	PRN1 6-0 = 1101001	inhibit
	PRN1 6-0 = 1101010	inhibit
	PRN1 6-0 = 1101011	inhibit
	PRN1 6-0 = 1101100	inhibit
	PRN1 6-0 = 1101101	inhibit
	PRN1 6-0 = 1101110	inhibit
	PRN1 6-0 = 1101111	inhibit
	PRN1 6-0 = 1110000	inhibit
	PRN1 6-0 = 1110001	inhibit
	PRN1 6-0 = 1110010	inhibit
	PRN1 6-0 = 1110011	inhibit
	PRN1 6-0 = 1110100	inhibit
	PRN1 6-0 = 1110101	inhibit
	PRN1 6-0 = 1110110	inhibit
	PRN1 6-0 = 1110111	inhibit
	PRN1 6-0 = 1111000	inhibit
	PRN1 6-0 = 1111001	inhibit
	PRN1 6-0 = 1111010	inhibit
	PRN1 6-0 = 1111011	inhibit
	PRN1 6-0 = 1111100	inhibit
	PRN1 6-0 = 1111101	inhibit
	PRN1 6-0 = 1111110	inhibit
	PRN1 6-0 = 1111111	inhibit

Table 5.36: VinN11

Reference voltage	Macro adjustment value	VinN3 formula
VinN3	PKN0 4-0 = 00000	(47R / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 00001	((47R – 1R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 00010	((47R – 2R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 00011	((47R – 3R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 00100	((47R – 4R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 00101	((47R – 5R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 00110	((47R – 6R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 00111	((47R – 7R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 01000	((47R – 8R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 01001	((47R – 9R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 01010	((47R - 10R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 01011	((47R - 11R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 01100	((47R - 12R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 01101	((47R - 13R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 01110	((47R - 14R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 01111	((47R - 15R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 10000	((47R - 16R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 10001	((47R - 17R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 10010	((47R - 18R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 10011	((47R - 19R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 10100	((47R - 20R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 10101	((47R - 21R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 10110	((47R - 22R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 10111	((47R - 23R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 11000	((47R - 24R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 11001	((47R - 25R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 11010	((47R - 26R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 11011	((47R - 27R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 11100	((47R - 28R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 11101	((47R - 29R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 11110	((47R - 30R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN0 4-0 = 11111	((47R - 31R) / 48R) * (VinN2 - VinN5) + VinN5

Table 5.37: VinN3

Reference voltage	Macro adjustment value	VinN4 formula
VinN4	PKN1 4-0 = 00000	(32R / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 00001	((32R - 1R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 00010	((32R - 2R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 00011	((32R - 3R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 00100	((32R - 4R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 00101	((32R - 5R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 00110	((32R - 6R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 00111	((32R - 7R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 01000	((32R - 8R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 01001	((32R - 9R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 01010	((32R - 10R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 01011	((32R - 11R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 01100	((32R - 12R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 01101	((32R - 13R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 01110	((32R - 14R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 01111	((32R - 15R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 10000	((32R - 16R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 10001	((32R - 17R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 10010	((32R - 18R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 10011	((32R - 19R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 10100	((32R - 20R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 10101	((32R - 21R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 10110	((32R - 22R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 10111	((32R - 23R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 11000	((32R - 24R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 11001	((32R - 25R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 11010	((32R - 26R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 11011	((32R - 27R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 11100	((32R - 28R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 11101	((32R - 29R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 11110	((32R - 30R) / 48R) * (VinN2 - VinN5) + VinN5
	PKN1 4-0 = 11111	((32R - 31R) / 48R) * (VinN2 - VinN5) + VinN5

Table 5.38: VinN4

Reference voltage	Macro adjustment value	VinN6 formula
VinN6	PKN2 4-0 = 00000	(220R / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 00001	((220R - 3R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 00010	((220R - 6R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 00011	((220R - 9R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 00100	((220R - 12R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 00101	((220R - 15R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 00110	((220R - 18R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 00111	((220R - 21R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 01000	((220R - 24R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 01001	((220R - 27R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 01010	((220R - 30R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 01011	((220R - 33R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 01100	((220R - 36R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 01101	((220R - 39R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 01110	((220R - 42R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 01111	((220R - 45R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 10000	((220R - 48R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 10001	((220R - 51R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 10010	((220R - 54R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 10011	((220R - 57R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 10100	((220R - 60R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 10101	((220R - 63R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 10110	((220R - 66R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 10111	((220R - 69R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 11000	((220R - 72R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 11001	((220R - 75R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 11010	((220R - 78R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 11011	((220R - 81R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 11100	((220R - 84R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 11101	((220R - 87R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 11110	((220R - 90R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN2 4-0 = 11111	((220R - 93R) / 223R) * (VinN5 - VinN11) + VinN11

Table 5.39: VinN6

Reference voltage	Macro adjustment value	VinN7 formula
VinN7	PKN3 4-0 = 00000	(193R / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 00001	((193R - 3R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 00010	((193R - 6R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 00011	((193R - 9R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 00100	((193R - 12R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 00101	((193R - 15R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 00110	((193R - 18R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 00111	((193R - 21R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 01000	((193R - 24R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 01001	((193R - 27R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 01010	((193R - 30R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 01011	((193R - 33R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 01100	((193R - 36R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 01101	((193R - 39R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 01110	((193R - 42R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 01111	((193R - 45R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 10000	((193R - 48R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 10001	((193R - 51R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 10010	((193R - 54R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 10011	((193R - 57R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 10100	((193R - 60R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 10101	((193R - 63R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 10110	((193R - 66R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 10111	((193R - 69R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 11000	((193R - 72R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 11001	((193R - 75R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 11010	((193R - 78R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 11011	((193R - 81R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 11100	((193R - 84R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 11101	((193R - 87R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 11110	((193R - 90R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN3 4-0 = 11111	((193R - 93R) / 223R) * (VinN5 - VinN11) + VinN11

Table 5.40: VinN7

Reference voltage	Macro adjustment value	VinN8 formula
VinN8	PKN4 4-0 = 00000	(158R / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 00001	((158R - 3R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 00010	((158R - 6R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 00011	((158R - 9R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 00100	((158R - 12R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 00101	((158R - 15R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 00110	((158R - 18R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 00111	((158R - 21R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 01000	((158R - 24R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 01001	((158R - 27R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 01010	((158R - 30R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 01011	((158R - 33R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 01100	((158R - 36R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 01101	((158R - 39R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 01110	((158R - 42R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 01111	((158R - 45R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 10000	((158R - 48R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 10001	((158R - 51R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 10010	((158R - 54R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 10011	((158R - 57R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 10100	((158R - 60R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 10101	((158R - 63R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 10110	((158R - 66R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 10111	((158R - 69R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 11000	((158R - 72R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 11001	((158R - 75R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 11010	((158R - 78R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 11011	((158R - 81R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 11100	((158R - 84R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 11101	((158R - 87R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 11110	((158R - 90R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN4 4-0 = 11111	((158R - 93R) / 223R) * (VinN5 - VinN11) + VinN11

Table 5.41: VinN8

Reference voltage	Macro adjustment value	VinN9 formula
VinN9	PKN5 4-0 = 00000	(123R / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 00001	((123R - 3R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 00010	((123R - 6R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 00011	((123R - 9R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 00100	((123R - 12R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 00101	((123R - 15R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 00110	((123R - 18R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 00111	((123R - 21R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 01000	((123R - 24R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 01001	((123R - 27R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 01010	((123R - 30R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 01011	((123R - 33R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 01100	((123R - 36R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 01101	((123R - 39R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 01110	((123R - 42R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 01111	((123R - 45R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 10000	((123R - 48R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 10001	((123R - 51R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 10010	((123R - 54R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 10011	((123R - 57R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 10100	((123R - 60R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 10101	((123R - 63R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 10110	((123R - 66R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 10111	((123R - 69R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 11000	((123R - 72R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 11001	((123R - 75R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 11010	((123R - 78R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 11011	((123R - 81R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 11100	((123R - 84R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 11101	((123R - 87R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 11110	((123R - 90R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN5 4-0 = 11111	((123R - 93R) / 223R) * (VinN5 - VinN11) + VinN11

Table 5.42: VinN9

Reference voltage	Macro adjustment value	VinN10 formula
VinN10	PKN6 4-0 = 00000	(96R / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 00001	((96R - 3R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 00010	((96R - 6R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 00011	((96R - 9R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 00100	((96R - 12R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 00101	((96R - 15R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 00110	((96R - 18R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 00111	((96R - 21R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 01000	((96R - 24R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 01001	((96R - 27R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 01010	((96R - 30R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 01011	((96R - 33R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 01100	((96R - 36R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 01101	((96R - 39R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 01110	((96R - 42R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 01111	((96R - 45R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 10000	((96R - 48R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 10001	((96R - 51R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 10010	((96R - 54R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 10011	((96R - 57R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 10100	((96R - 60R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 10101	((96R - 63R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 10110	((96R - 66R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 10111	((96R - 69R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 11000	((96R - 72R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 11001	((96R - 75R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 11010	((96R - 78R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 11011	((96R - 81R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 11100	((96R - 84R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 11101	((96R - 87R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 11110	((96R - 90R) / 223R) * (VinN5 - VinN11) + VinN11
	PKN6 4-0 = 11111	((96R - 93R) / 223R) * (VinN5 - VinN11) + VinN11

Table 5.43: VinN10

Reference voltage	Macro adjustment value	VinN12 formula
VinN12	PKN7 4-0 = 00000	(47R / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 00001	((47R - 1R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 00010	((47R - 2R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 00011	((47R - 3R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 00100	((47R - 4R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 00101	((47R - 5R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 00110	((47R - 6R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 00111	((47R - 7R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 01000	((47R - 8R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 01001	((47R - 9R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 01010	((47R - 10R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 01011	((47R - 11R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 01100	((47R - 12R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 01101	((47R - 13R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 01110	((47R - 14R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 01111	((47R - 15R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 10000	((47R - 16R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 10001	((47R - 17R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 10010	((47R - 18R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 10011	((47R - 19R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 10100	((47R - 20R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 10101	((47R - 21R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 10110	((47R - 22R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 10111	((47R - 23R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 11000	((47R - 24R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 11001	((47R - 25R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 11010	((47R - 26R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 11011	((47R - 27R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 11100	((47R - 28R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 11101	((47R - 29R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 11110	((47R - 30R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN7 4-0 = 11111	((47R - 31R) / 48R) * (VinN11 - VinN14) + VinN14

Table 5.44: VinN12

Reference voltage	Macro adjustment value	VinN13 formula
VinN13	PKN8 4-0 = 00000	(32R / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 00001	((32R - 1R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 00010	((32R - 2R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 00011	((32R - 3R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 00100	((32R - 4R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 00101	((32R - 5R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 00110	((32R - 6R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 00111	((32R - 7R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 01000	((32R - 8R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 01001	((32R - 9R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 01010	((32R - 10R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 01011	((32R - 11R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 01100	((32R - 12R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 01101	((32R - 13R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 01110	((32R - 14R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 01111	((32R - 15R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 10000	((32R - 16R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 10001	((32R - 17R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 10010	((32R - 18R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 10011	((32R - 19R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 10100	((32R - 20R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 10101	((32R - 21R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 10110	((32R - 22R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 10111	((32R - 23R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 11000	((32R - 24R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 11001	((32R - 25R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 11010	((32R - 26R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 11011	((32R - 27R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 11100	((32R - 28R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 11101	((32R - 29R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 11110	((32R - 30R) / 48R) * (VinN11 - VinN14) + VinN14
	PKN8 4-0 = 11111	((32R - 31R) / 48R) * (VinN11 - VinN14) + VinN14

Table 5.45: VinN13

Reference voltage	Macro adjustment value	VinP_EX1 formula
VinN_EX0	PKN_EX0 4-0 = 00000	(39R / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 00001	((39R – 1R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 00010	((39R – 2R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 00011	((39R – 3R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 00100	((39R – 4R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 00101	((39R – 5R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 00110	((39R – 6R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 00111	((39R – 7R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 01000	((39R – 8R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 01001	((39R – 9R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 01010	((39R – 10R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 01011	((39R – 11R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 01100	((39R – 12R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 01101	((39R – 13R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 01110	((39R – 14R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 01111	((39R – 15R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 10000	((39R – 16R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 10001	((39R – 17R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 10010	((39R – 18R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 10011	((39R – 19R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 10100	((39R – 20R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 10101	((39R – 21R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 10110	((39R – 22R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 10111	((39R – 23R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 11000	((39R – 24R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 11001	((39R – 25R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 11010	((39R – 26R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 11011	((39R – 27R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 11100	((39R – 28R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 11101	((39R – 29R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 11110	((39R – 30R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX0 4-0 = 11111	((39R – 31R) / 48R) * (VinN2 – VinN5) + VinN5

Table 5.46: VinN_EX0

Reference voltage	Macro adjustment value	VinN_EX2 formula
VinN_EX1	PKN_EX1 4-0 = 00000	(32R / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 00001	((32R – 1R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 00010	((32R – 2R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 00011	((32R – 3R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 00100	((32R – 4R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 00101	((32R – 5R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 00110	((32R – 6R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 00111	((32R – 7R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 01000	((32R – 8R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 01001	((32R – 9R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 01010	((32R – 10R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 01011	((32R – 11R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 01100	((32R – 12R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 01101	((32R – 13R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 01110	((32R – 14R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 01111	((32R – 15R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 10000	((32R – 16R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 10001	((32R – 17R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 10010	((32R – 18R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 10011	((32R – 19R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 10100	((32R – 20R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 10101	((32R – 21R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 10110	((32R – 22R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 10111	((32R – 23R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 11000	((32R – 24R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 11001	((32R – 25R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 11010	((32R – 26R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 11011	((32R – 27R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 11100	((32R – 28R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 11101	((32R – 29R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 11110	((32R – 30R) / 48R) * (VinN2 – VinN5) + VinN5
	PKN_EX1 4-0 = 11111	((32R – 31R) / 48R) * (VinN2 – VinN5) + VinN5

Table 5.47: VinN_EX1

Reference voltage	Macro adjustment value	VinN_EX3 formula
VinN_EX2	PKN_EX2 4-0 = 00000	(47R / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 00001	((47R – 1R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 00010	((47R – 2R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 00011	((47R – 3R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 00100	((47R – 4R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 00101	((47R – 5R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 00110	((47R – 6R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 00111	((47R – 7R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 01000	((47R – 8R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 01001	((47R – 9R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 01010	((47R – 10R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 01011	((47R – 11R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 01100	((47R – 12R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 01101	((47R – 13R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 01110	((47R – 14R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 01111	((47R – 15R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 10000	((47R – 16R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 10001	((47R – 17R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 10010	((47R – 18R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 10011	((47R – 19R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 10100	((47R – 20R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 10101	((47R – 21R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 10110	((47R – 22R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 10111	((47R – 23R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 11000	((47R – 24R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 11001	((47R – 25R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 11010	((47R – 26R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 11011	((47R – 27R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 11100	((47R – 28R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 11101	((47R – 29R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 11110	((47R – 30R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX2 4-0 = 11111	((47R – 31R) / 48R) * (VinN11 – VinN14) + VinN14

Table 5.48: VinN_EX2

Reference voltage	Macro adjustment value	VinN_EX4 formula
VinN_EX3	PKN_EX3 4-0 = 00000	(39R / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 00001	((39R – 1R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 00010	((39R – 2R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 00011	((39R – 3R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 00100	((39R – 4R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 00101	((39R – 5R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 00110	((39R – 6R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 00111	((39R – 7R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 01000	((39R – 8R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 01001	((39R – 9R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 01010	((39R – 10R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 01011	((39R – 11R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 01100	((39R – 12R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 01101	((39R – 13R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 01110	((39R – 14R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 01111	((39R – 15R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 10000	((39R – 16R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 10001	((39R – 17R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 10010	((39R – 18R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 10011	((39R – 19R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 10100	((39R – 20R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 10101	((39R – 21R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 10110	((39R – 22R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 10111	((39R – 23R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 11000	((39R – 24R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 11001	((39R – 25R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 11010	((39R – 26R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 11011	((39R – 27R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 11100	((39R – 28R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 11101	((39R – 29R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 11110	((39R – 30R) / 48R) * (VinN11 – VinN14) + VinN14
	PKN_EX3 4-0 = 11111	((39R – 31R) / 48R) * (VinN11 – VinN14) + VinN14

Table 5.49: VinN_EX3

Grayscale voltage	Formula
V0	VinP/N0
V1	VinP/N0 - (VinP/N0 - VinP/N1)*(4R/16R)
V2	VinP/N0 - (VinP/N0 - VinP/N1)*(8R/16R)
V3	VinP/N0 - (VinP/N0 - VinP/N1)*(12R/16R)
V4	VinP/N1
V5	VinP/N1 - (VinP/N1 - VinP/N2)*(4R/16R)
V6	VinP/N1 - (VinP/N1 - VinP/N2)*(8R/16R)
V7	VinP/N1 - (VinP/N1 - VinP/N2)*(12R/16R)
V8	VinP/N2
V9	VinP/N2 - (VinP/N2 - VinP/N3)*(4R/16R)
V10	VinP/N2 - (VinP/N2 - VinP/N3)*(8R/16R)
V11	VinP/N2 - (VinP/N2 - VinP/N3)*(12R/16R)
V12	VinP/N3
V13	VinP/N3 - (VinP/N3 - VinP/N_EX0)*(1.5R/12R)
V14	VinP/N3 - (VinP/N3 - VinP/N_EX0)*(3R/12R)
V15	VinP/N3 - (VinP/N3 - VinP/N_EX0)*(4.5R/12R)
V16	VinP/N3 - (VinP/N3 - VinP/N_EX0)*(6R/12R)
V17	VinP/N3 - (VinP/N3 - VinP/N_EX0)*(7.5R/12R)
V18	VinP/N3 - (VinP/N3 - VinP/N_EX0)*(9R/12R)
V19	VinP/N3 - (VinP/N3 - VinP/N_EX0)*(10.5R/12R)
V20	VinP/N_EX0
V21	VinP/N_EX0 - (VinP/N_EX0 - VinP/N4)*(1.5R/12R)
V22	VinP/N_EX0 - (VinP/N_EX0 - VinP/N4)*(3R/12R)
V23	VinP/N_EX0 - (VinP/N_EX0 - VinP/N4)*(4.5R/12R)
V24	VinP/N_EX0 - (VinP/N_EX0 - VinP/N4)*(6R/12R)
V25	VinP/N_EX0 - (VinP/N_EX0 - VinP/N4)*(7.5R/12R)
V26	VinP/N_EX0 - (VinP/N_EX0 - VinP/N4)*(9R/12R)
V27	VinP/N_EX0 - (VinP/N_EX0 - VinP/N4)*(10.5R/12R)
V28	VinP/N4
V29	VinP/N4 - (VinP/N4 - VinP/N_EX1)*(1.5R/18R)
V30	VinP/N4 - (VinP/N4 - VinP/N_EX1)*(3R/18R)
V31	VinP/N4 - (VinP/N4 - VinP/N_EX1)*(4.5R/18R)
V32	VinP/N4 - (VinP/N4 - VinP/N_EX1)*(6R/18R)
V33	VinP/N4 - (VinP/N4 - VinP/N_EX1)*(7.5R/18R)
V34	VinP/N4 - (VinP/N4 - VinP/N_EX1)*(9R/18R)
V35	VinP/N4 - (VinP/N4 - VinP/N_EX1)*(10.5R/18R)
V36	VinP/N4 - (VinP/N4 - VinP/N_EX1)*(12R/18R)
V37	VinP/N4 - (VinP/N4 - VinP/N_EX1)*(13.5R/18R)
V38	VinP/N4 - (VinP/N4 - VinP/N_EX1)*(15R/18R)
V39	VinP/N4 - (VinP/N4 - VinP/N_EX1)*(16.5R/18R)
V40	VinP/N_EX1
V41	VinP/N_EX1 - (VinP/N_EX1 - VinP/N5)*(1.5R/18R)
V42	VinP/N_EX1 - (VinP/N_EX1 - VinP/N5)*(3R/18R)
V43	VinP/N_EX1 - (VinP/N_EX1 - VinP/N5*(4.5R/18R)

Grayscale voltage	Formula
V44	VinP/N_EX1 - (VinP/N_EX1 - VinP/N5)*(6R/18R)
V45	VinP/N_EX1 - (VinP/N_EX1 - VinP/N5)*(7.5R/18R)
V46	VinP/N_EX1 - (VinP/N_EX1 - VinP/N5)*(9R/18R)
V47	VinP/N_EX1 - (VinP/N_EX1 - VinP/N5)*(10.5R/18R)
V48	VinP/N_EX1 - (VinP/N_EX1 - VinP/N5)*(12R/18R)
V49	VinP/N_EX1 - (VinP/N_EX1 - VinP/N5)*(13.5R/18R)
V50	VinP/N_EX1 - (VinP/N_EX1 - VinP/N5)*(15R/18R)
V51	VinP/N_EX1 - (VinP/N_EX1 - VinP/N5)*(16.5R/18R)
V52	VinP/N5
V53	VinP/N5 - (VinP/N5 - VinP/N6)*(1.6R/38.4R)
V54	VinP/N5 - (VinP/N5 - VinP/N6)*(3.2R/38.4R)
V55	VinP/N5 - (VinP/N5 - VinP/N6)*(4.8R/38.4R)
V56	VinP/N5 - (VinP/N5 - VinP/N6)*(6.4R/38.4R)
V57	VinP/N5 - (VinP/N5 - VinP/N6)*(8R/38.4R)
V58	VinP/N5 - (VinP/N5 - VinP/N6)*(9.6R/38.4R)
V59	VinP/N5 - (VinP/N5 - VinP/N6)*(11.2R/38.4R)
V60	VinP/N5 - (VinP/N5 - VinP/N6)*(12.8R/38.4R)
V61	VinP/N5 - (VinP/N5 - VinP/N6)*(14.4R/38.4R)
V62	VinP/N5 - (VinP/N5 - VinP/N6)*(16R/38.4R)
V63	VinP/N5 - (VinP/N5 - VinP/N6)*(17.6R/38.4R)
V64	VinP/N5 - (VinP/N5 - VinP/N6)*(19.2R/38.4R)
V65	VinP/N5 - (VinP/N5 - VinP/N6)*(20.8R/38.4R)
V66	VinP/N5 - (VinP/N5 - VinP/N6)*(22.4R/38.4R)
V67	VinP/N5 - (VinP/N5 - VinP/N6)*(24R/38.4R)
V68	VinP/N5 - (VinP/N5 - VinP/N6)*(25.6R/38.4R)
V69	VinP/N5 - (VinP/N5 - VinP/N6)*(27.2R/38.4R)
V70	VinP/N5 - (VinP/N5 - VinP/N6)*(28.8R/38.4R)
V71	VinP/N5 - (VinP/N5 - VinP/N6)*(30.4R/38.4R)
V72	VinP/N5 - (VinP/N5 - VinP/N6)*(32R/38.4R)
V73	VinP/N5 - (VinP/N5 - VinP/N6)*(33.6R/38.4R)
V74	VinP/N5 - (VinP/N5 - VinP/N6)*(35.2R/38.4R)
V75	VinP/N5 - (VinP/N5 - VinP/N6)*(36.8R/38.4R)
V76	VinP/N6
V77	VinP/N6 - (VinP/N6 - VinP/N7)* (1.6R/38.4R)
V78	VinP/N6 - (VinP/N6 - VinP/N7)* (3.2R/38.4R)
V79	VinP/N6 - (VinP/N6 - VinP/N7)* (4.8R/38.4R)
V80	VinP/N6 - (VinP/N6 - VinP/N7)* (6.4R/38.4R)
V81	VinP/N6 - (VinP/N6 - VinP/N7)* (8R/38.4R)
V82	VinP/N6 - (VinP/N6 - VinP/N7)* (9.6R/38.4R)
V83	VinP/N6 - (VinP/N6 - VinP/N7)* (11.2R/38.4R)
V84	VinP/N6 - (VinP/N6 - VinP/N7)* (12.8R/38.4R)
V85	VinP/N6 - (VinP/N6 - VinP/N7)* (14.4R/38.4R)
V86	VinP/N6 - (VinP/N6 - VinP/N7)* (16R/38.4R)
V87	VinP/N6 - (VinP/N6 - VinP/N7)* (17.6R/38.4R)

Grayscale voltage	Formula
V88	VinP/N6 - (VinP/N6 - VinP/N7)* (19.2R/38.4R)
V89	VinP/N6 - (VinP/N6 - VinP/N7)* (20.8R/38.4R)
V90	VinP/N6 - (VinP/N6 - VinP/N7)* (22.4R/38.4R)
V91	VinP/N6 - (VinP/N6 - VinP/N7)* (24R/38.4R)
V92	VinP/N6 - (VinP/N6 - VinP/N7)* (25.6R/38.4R)
V93	VinP/N6 - (VinP/N6 - VinP/N7)* (27.2R/38.4R)
V94	VinP/N6 - (VinP/N6 - VinP/N7)* (28.8R/38.4R)
V95	VinP/N6 - (VinP/N6 - VinP/N7)* (30.4R/38.4R)
V96	VinP/N6 - (VinP/N6 - VinP/N7)* (32R/38.4R)
V97	VinP/N6 - (VinP/N6 - VinP/N7)* (33.6R/38.4R)
V98	VinP/N6 - (VinP/N6 - VinP/N7)* (35.2R/38.4R)
V99	VinP/N6 - (VinP/N6 - VinP/N7)* (36.8R/38.4R)
V100	VinP/N7
V101	VinP/N7 - (VinP/N7 - VinP/N8)*(1.6R/51.2R)
V102	VinP/N7 - (VinP/N7 - VinP/N8)*(3.2R/51.2R)
V103	VinP/N7 - (VinP/N7 - VinP/N8)*(4.8R/51.2R)
V104	VinP/N7 - (VinP/N7 - VinP/N8)*(6.4R/51.2R)
V105	VinP/N7 - (VinP/N7 - VinP/N8)*(8R/51.2R)
V106	VinP/N7 - (VinP/N7 - VinP/N8)*(9.6R/51.2R)
V107	VinP/N7 - (VinP/N7 - VinP/N8)*(11.2R/51.2R)
V108	VinP/N7 - (VinP/N7 - VinP/N8)*(12.8R/51.2R)
V109	VinP/N7 - (VinP/N7 - VinP/N8)*(14.4R/51.2R)
V110	VinP/N7 - (VinP/N7 - VinP/N8)*(16R/51.2R)
V111	VinP/N7 - (VinP/N7 - VinP/N8)*(17.6R/51.2R)
V112	VinP/N7 - (VinP/N7 - VinP/N8)*(19.2R/51.2R)
V113	VinP/N7 - (VinP/N7 - VinP/N8)*(20.8R/51.2R)
V114	VinP/N7 - (VinP/N7 - VinP/N8)*(22.4R/51.2R)
V115	VinP/N7 - (VinP/N7 - VinP/N8)*(24R/51.2R)
V116	VinP/N7 - (VinP/N7 - VinP/N8)*(25.6R/51.2R)
V117	VinP/N7 - (VinP/N7 - VinP/N8)*(27.2R/51.2R)
V118	VinP/N7 - (VinP/N7 - VinP/N8)*(28.8R/51.2R)
V119	VinP/N7 - (VinP/N7 - VinP/N8)*(30.4R/51.2R)
V120	VinP/N7 - (VinP/N7 - VinP/N8)*(32R/51.2R)
V121	VinP/N7 - (VinP/N7 - VinP/N8)*(33.6R/51.2R)
V122	VinP/N7 - (VinP/N7 - VinP/N8)*(35.2R/51.2R)
V123	VinP/N7 - (VinP/N7 - VinP/N8)*(36.8R/51.2R)
V124	VinP/N7 - (VinP/N7 - VinP/N8)*(38.4R/51.2R)
V125	VinP/N7 - (VinP/N7 - VinP/N8)*(40R/51.2R)
V126	VinP/N7 - (VinP/N7 - VinP/N8)*(41.6R/51.2R)
V127	VinP/N7 - (VinP/N7 - VinP/N8)*(43.2R/51.2R)
V128	VinP/N7 - (VinP/N7 - VinP/N8)*(44.8R/51.2R)
V129	VinP/N7 - (VinP/N7 - VinP/N8)*(46.4R/51.2R)
V130	VinP/N7 - (VinP/N7 - VinP/N8)*(48R/51.2R)
V131	VinP/N7 - (VinP/N7 - VinP/N8)*(49.6R/51.2R)

Grayscale voltage	Formula
V132	VinP/N8
V133	VinP/N8 - (VinP/N8 - VinP/N9)*(1.6R/38.4R)
V134	VinP/N8 - (VinP/N8 - VinP/N9)*(3.2R/38.4R)
V135	VinP/N8 - (VinP/N8 - VinP/N9)*(4.8R/38.4R)
V136	VinP/N8 - (VinP/N8 - VinP/N9)*(6.4R/38.4R)
V137	VinP/N8 - (VinP/N8 - VinP/N9)*(8R/38.4R)
V138	VinP/N8 - (VinP/N8 - VinP/N9)*(9.6R/38.4R)
V139	VinP/N8 - (VinP/N8 - VinP/N9)*(11.2R/38.4R)
V140	VinP/N8 - (VinP/N8 - VinP/N9)*(12.8R/38.4R)
V141	VinP/N8 - (VinP/N8 - VinP/N9)*(14.4R/38.4R)
V142	VinP/N8 - (VinP/N8 - VinP/N9)*(16R/38.4R)
V143	VinP/N8 - (VinP/N8 - VinP/N9)*(17.6R/38.4R)
V144	VinP/N8 - (VinP/N8 - VinP/N9)*(19.2R/38.4R)
V145	VinP/N8 - (VinP/N8 - VinP/N9)*(20.8R/38.4R)
V146	VinP/N8 - (VinP/N8 - VinP/N9)*(22.4R/38.4R)
V147	VinP/N8 - (VinP/N8 - VinP/N9)*(24R/38.4R)
V148	VinP/N8 - (VinP/N8 - VinP/N9)*(25.6R/38.4R)
V149	VinP/N8 - (VinP/N8 - VinP/N9)*(27.2R/38.4R)
V150	VinP/N8 - (VinP/N8 - VinP/N9)*(28.8R/38.4R)
V151	VinP/N8 - (VinP/N8 - VinP/N9)*(30.4R/38.4R)
V152	VinP/N8 - (VinP/N8 - VinP/N9)*(32R/38.4R)
V153	VinP/N8 - (VinP/N8 - VinP/N9)*(33.6R/38.4R)
V154	VinP/N8 - (VinP/N8 - VinP/N9)*(35.2R/38.4R)
V155	VinP/N8 - (VinP/N8 - VinP/N9)*(36.8R/38.4R)
V156	VinP/N9
V157	VinP/N9 - (VinP/N9 - VinP/N10)*(1.6R/38.4R)
V158	VinP/N9 - (VinP/N9 - VinP/N10)*(3.2R/38.4R)
V159	VinP/N9 - (VinP/N9 - VinP/N10)*(4.8R/38.4R)
V160	VinP/N9 - (VinP/N9 - VinP/N10)*(6.4R/38.4R)
V161	VinP/N9 - (VinP/N9 - VinP/N10)*(8R/38.4R)
V162	VinP/N9 - (VinP/N9 - VinP/N10)*(9.6R/38.4R)
V163	VinP/N9 - (VinP/N9 - VinP/N10)*(11.2R/38.4R)
V164	VinP/N9 - (VinP/N9 - VinP/N10)*(12.8R/38.4R)
V165	VinP/N9 - (VinP/N9 - VinP/N10)*(14.4R/38.4R)
V166	VinP/N9 - (VinP/N9 - VinP/N10)*(16R/38.4R)
V167	VinP/N9 - (VinP/N9 - VinP/N10)*(17.6R/38.4R)
V168	VinP/N9 - (VinP/N9 - VinP/N10)*(19.2R/38.4R)
V169	VinP/N9 - (VinP/N9 - VinP/N10)*(20.8R/38.4R)
V170	VinP/N9 - (VinP/N9 - VinP/N10)*(22.4R/38.4R)
V171	VinP/N9 - (VinP/N9 - VinP/N10)*(24R/38.4R)
V172	VinP/N9 - (VinP/N9 - VinP/N10)*(25.6R/38.4R)
V173	VinP/N9 - (VinP/N9 - VinP/N10)*(27.2R/38.4R)
V174	VinP/N9 - (VinP/N9 - VinP/N10)*(28.8R/38.4R)
V175	VinP/N9 - (VinP/N9 - VinP/N10)*(30.4R/38.4R)

Grayscale voltage	Formula
V176	VinP/N9 - (VinP/N9 - VinP/N10)*(32R/38.4R)
V177	VinP/N9 - (VinP/N9 - VinP/N10)*(33.6R/38.4R)
V178	VinP/N9 - (VinP/N9 - VinP/N10)*(35.2R/38.4R)
V179	VinP/N9 - (VinP/N9 - VinP/N10)*(36.8R/38.4R)
V180	VinP/N10
V181	VinP/N10 - (VinP/N10 - VinP/N11)*(1.6R/38.4R)
V182	VinP/N10 - (VinP/N10 - VinP/N11)*(3.2R/38.4R)
V183	VinP/N10 - (VinP/N10 - VinP/N11)*(4.8R/38.4R)
V184	VinP/N10 - (VinP/N10 - VinP/N11)*(6.4R/38.4R)
V185	VinP/N10 - (VinP/N10 - VinP/N11)*(8R/38.4R)
V186	VinP/N10 - (VinP/N10 - VinP/N11)*(9.6R/38.4R)
V187	VinP/N10 - (VinP/N10 - VinP/N11)*(11.2R/38.4R)
V188	VinP/N10 - (VinP/N10 - VinP/N11)*(12.8R/38.4R)
V189	VinP/N10 - (VinP/N10 - VinP/N11)*(14.4R/38.4R)
V190	VinP/N10 - (VinP/N10 - VinP/N11)*(16R/38.4R)
V191	VinP/N10 - (VinP/N10 - VinP/N11)*(17.6R/38.4R)
V192	VinP/N10 - (VinP/N10 - VinP/N11)*(19.2R/38.4R)
V193	VinP/N10 - (VinP/N10 - VinP/N11)*(20.8R/38.4R)
V194	VinP/N10 - (VinP/N10 - VinP/N11)*(22.4R/38.4R)
V195	VinP/N10 - (VinP/N10 - VinP/N11)*(24R/38.4R)
V196	VinP/N10 - (VinP/N10 - VinP/N11)*(25.6R/38.4R)
V197	VinP/N10 - (VinP/N10 - VinP/N11)*(27.2R/38.4R)
V198	VinP/N10 - (VinP/N10 - VinP/N11)*(28.8R/38.4R)
V199	VinP/N10 - (VinP/N10 - VinP/N11)*(30.4R/38.4R)
V200	VinP/N10 - (VinP/N10 - VinP/N11)*(32R/38.4R)
V201	VinP/N10 - (VinP/N10 - VinP/N11)*(33.6R/38.4R)
V202	VinP/N10 - (VinP/N10 - VinP/N11)*(35.2R/38.4R)
V203	VinP/N10 - (VinP/N10 - VinP/N11)*(36.8R/38.4R)
V204	VinP/N11
V205	VinP/N11 - (VinP/N11 - VinP/N_EX2)*(1.5R/18R)
V206	VinP/N11 - (VinP/N11 - VinP/N_EX2)*(3R/18R)
V207	VinP/N11 - (VinP/N11 - VinP/N_EX2)*(4.5R/18R)
V208	VinP/N11 - (VinP/N11 - VinP/N_EX2)*(6R/18R)
V209	VinP/N11 - (VinP/N11 - VinP/N_EX2)*(7.5R/18R)
V210	VinP/N11 - (VinP/N11 - VinP/N_EX2)*(9R/18R)
V211	VinP/N11 - (VinP/N11 - VinP/N_EX2)*(10.5R/18R)
V212	VinP/N11 - (VinP/N11 - VinP/N_EX2)*(12R/18R)
V213	VinP/N11 - (VinP/N11 - VinP/N_EX2)*(13.5R/18R)
V214	VinP/N11 - (VinP/N11 - VinP/N_EX2)*(15R/18R)
V215	VinP/N11 - (VinP/N11 - VinP/N_EX2)*(16.5R/18R)

Grayscale voltage	Formula
V216	VinP/N_EX2
V217	VinP/N_EX2 - (VinP/N_EX2 - VinP/N12)*(1.5R/18R)
V218	VinP/N_EX2 - (VinP/N_EX2 - VinP/N12)*(3R/18R)
V219	VinP/N_EX2 - (VinP/N_EX2 - VinP/N12*(4.5R/18R)
V220	VinP/N_EX2 - (VinP/N_EX2 - VinP/N12)*(6R/18R)
V221	VinP/N_EX2 - (VinP/N_EX2 - VinP/N12)*(7.5R/18R)
V222	VinP/N_EX2 - (VinP/N_EX2 - VinP/N12)*(9R/18R)
V223	VinP/N_EX2 - (VinP/N_EX2 - VinP/N12)*(10.5R/18R)
V224	VinP/N_EX2 - (VinP/N_EX2 - VinP/N12)*(12R/18R)
V225	VinP/N_EX2 - (VinP/N_EX2 - VinP/N12)*(13.5R/18R)
V226	VinP/N_EX2 - (VinP/N_EX2 - VinP/N12)*(15R/18R)
V227	VinP/N_EX2 - (VinP/N_EX2 - VinP/N12)*(16.5R/18R)
V228	VinP/N12
V229	VinP/N12 - (VinP/N12 - VinP/N_EX3)*(1.5R/12R)
V230	VinP/N12 - (VinP/N12 - VinP/N_EX3)*(3R/12R)
V231	VinP/N12 - (VinP/N12 - VinP/N_EX3)*(4.5R/12R)
V232	VinP/N12 - (VinP/N12 - VinP/N_EX3)*(6R/12R)
V233	VinP/N12 - (VinP/N12 - VinP/N_EX3)*(7.5R/12R)
V234	VinP/N12 - (VinP/N12 - VinP/N_EX3)*(9R/12R)
V235	VinP/N12 - (VinP/N12 - VinP/N_EX3)*(10.5R/12R)
V236	VinP/N_EX3
V237	VinP/N_EX3 - (VinP/N_EX3 - VinP/N13)*(1.5R/10.5R)
V238	VinP/N_EX3 - (VinP/N_EX3 - VinP/N13)*(3R10.5R)
V239	VinP/N_EX3 - (VinP/N_EX3 - VinP/N13)*(4.5R/10.5R)
V240	VinP/N_EX3 - (VinP/N_EX3 - VinP/N13)*(6R/10.5R)
V241	VinP/N_EX3 - (VinP/N_EX3 - VinP/N13)*(7.5R/10.5R)
V242	VinP/N_EX3 - (VinP/N_EX3 - VinP/N13)*(9R/10.5R)
V243	VinP/N13
V244	VinP/N13 - (VinP/N13 - VinP/N14)*(4R/16R)
V245	VinP/N13 - (VinP/N13 - VinP/N14)*(8R/16R)
V246	VinP/N13 - (VinP/N13 - VinP/N14)*(12R/16R)
V247	VinP/N14
V248	VinP/N14 - (VinP/N14 - VinP/N15)*(4R/16R)
V249	VinP/N14 - (VinP/N14 - VinP/N15)*(8R/16R)
V250	VinP/N14 - (VinP/N14 - VinP/N15)*(12R/16R)
V251	VinP/N15
V252	VinP/N15 - (VinP/N15 - VinP/N16)*(4R/16R)
V253	VinP/N15 - (VinP/N15 - VinP/N16)*(8R/16R)
V254	VinP/N15 - (VinP/N15 - VinP/N16)*(12R/16R)
V255	VinP/N16

Table 5.50: Voltage calculation formula of 256-grayscale voltage (positive/negative polarity)

5.7.2 Gray voltage generator for digital gamma correction

The HX8399-A digital gamma correction can reach the independent GAMMA curve of RGB. HX8399-A utilizes DGC_LUT (Digital Gamma Correction Look Up Table) to change input data from 8-bit into 10-bit and sends 10-bit data to Dithering circuit, and then drive Source Driver via Dithering circuit. The following of the block diagram of the function.

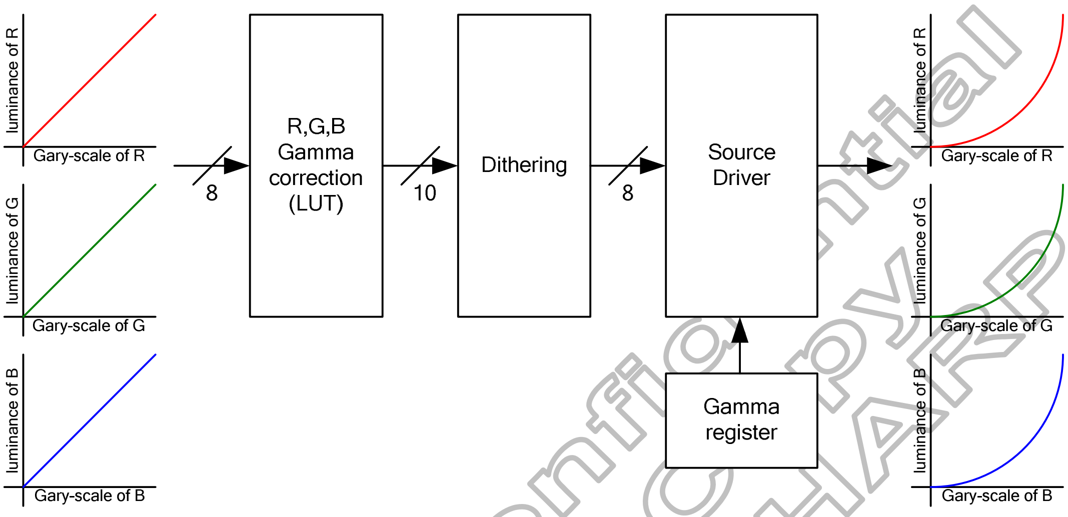


Figure 5.24: Block diagram of digital gamma correction

There are 126 bytes DGC LUT to set R, G, B gamma independently. When DGC_EN=1, R, G, B gamma will mapping V0, V8, V16, ..., V240, V248, V255 voltage to the LUT register setting gray level voltage.

LUT	D7	D6	D5	D4	D3	D2	D1	D0	Default
1st	R009	R008	R007	R006	R005	R004	R003	R002	00h
2nd	R019	R018	R017	R016	R015	R014	R013	R012	08h
3rd	R029	R028	R027	R026	R025	R024	R023	R022	10h
:	:	:	:	:	:	:	:	:	:
32th	R319	R318	R317	R316	R315	R314	R313	R312	F8h
33th	R329	R328	R327	R326	R325	R324	R323	R322	FFh
34th	R001	R000	R011	R010	R021	R020	R031	R030	00h
35th	R041	R040	R051	R050	R061	R060	R071	R070	00h
:	:	:	:	:	:	:	:	:	:
41th	R281	R280	R291	R290	R301	R300	R311	R310	00h
42th	R321	R320	0	0	0	0	0	0	00h
43th	G009	G008	G007	G006	G005	G004	G003	G002	00h
44th	G019	G018	G017	G016	G015	G014	G013	G012	08h
45th	G029	G028	G027	G026	G025	G024	G023	G022	10h
:	:	:	:	:	:	:	:	:	:
74th	G319	G318	G317	G316	G315	G314	G313	G312	F8h
75th	G329	G328	G327	G326	G325	G324	G323	G322	FFh
76th	G001	G000	G011	G010	G021	G020	G031	G030	00h
77th	G041	G040	G051	G050	G061	G060	G071	G070	00h
:	:	:	:	:	:	:	:	:	:
83th	G281	G280	G291	G290	G301	G300	G311	G310	00h
84th	G321	G320	0	0	0	0	0	0	00h
85th	B009	B008	B007	B006	B005	B004	B003	B002	00h
86th	B019	B018	B017	B016	B015	B014	B013	B012	08h
87th	B029	B028	B027	B026	B025	B024	B023	B022	10h
:	:	:	:	:	:	:	:	:	:

:	:	:	:	:	:	:	:	:	:
116th	B319	B318	B317	B316	B315	B314	B313	B312	F8h
117th	B329	B328	B327	B326	B325	B324	B323	B322	FFh
118th	B001	B000	B011	B010	B021	B020	B031	B030	00h
119th	B041	B040	B051	B050	B061	B060	B071	B070	00h
: :	: :	: :	: :	: :	: :	: :	: :	: :	: :
125th	B281	B280	B291	B290	B301	B300	B311	B310	00h
126th	B321	B320	0	0	0	0	0	0	00h

Table 5.51: DGC Look-up table

5.8 Characteristics of I/O

5.8.1 Output or bi-directional (I/O) pins

Output or bi-directional pins	After power on	After hardware reset	After software reset
TE	Low	Low	Low
TE1	Low	Low	Low
SDO	High-Z (Inactive)	High-Z (Inactive)	High-Z (Inactive)
CABC_PWM_OUT	Low	Low	Low

Table 5.52: Characteristics of output or bi-directional (I/O) pins

5.8.2 Input pins

Input pins	During power on process	After power on	After hardware reset	After software reset	During power off process
RESX	Input valid	Input valid	Input valid	Input valid	Input valid
CSX	Input valid	Input valid	Input valid	Input valid	Input valid
DCX	Input valid	Input valid	Input valid	Input valid	Input valid
SCL	Input valid	Input valid	Input valid	Input valid	Input valid
DB7~DB0 SDI_SDA	Input valid	Input valid	Input valid	Input valid	Input valid
HSYNC	Input valid	Input valid	Input valid	Input valid	Input valid
VSYNC	Input valid	Input valid	Input valid	Input valid	Input valid
PCLK	Input valid	Input valid	Input valid	Input valid	Input valid
DE	Input valid	Input valid	Input valid	Input valid	Input valid
OSC, IM2, IM1, IM0,	Input valid	Input valid	Input valid	Input valid	Input valid
TEST2, TEST1, TEST0	Low	Low	Low	Low	Low

Table 5.53: Characteristics of input pins

5.9 Sleep Out –command and self-diagnostic functions of the display module

5.9.1 Register loading detection

Sleep Out-command (See "Sleep Out (11h)") is a trigger for an internal function of the display module, which indicates, if the display module loading function of factory default values from OTP (or similar device) to registers of the display controller is working properly. There are compared factory values of the OTP and register values of the display controller by the display controller. If those both values (OTP and register values) are same, there is inverted (=increased by 1) a bit, which is defined in command "Read Display Self-Diagnostic Result (0Fh)" (=RDDSDR) (The used bit of this command is D7). If those both values are not same, this bit (D7) is not inverted (=increased by 1).
The flow chart for this internal function is following:

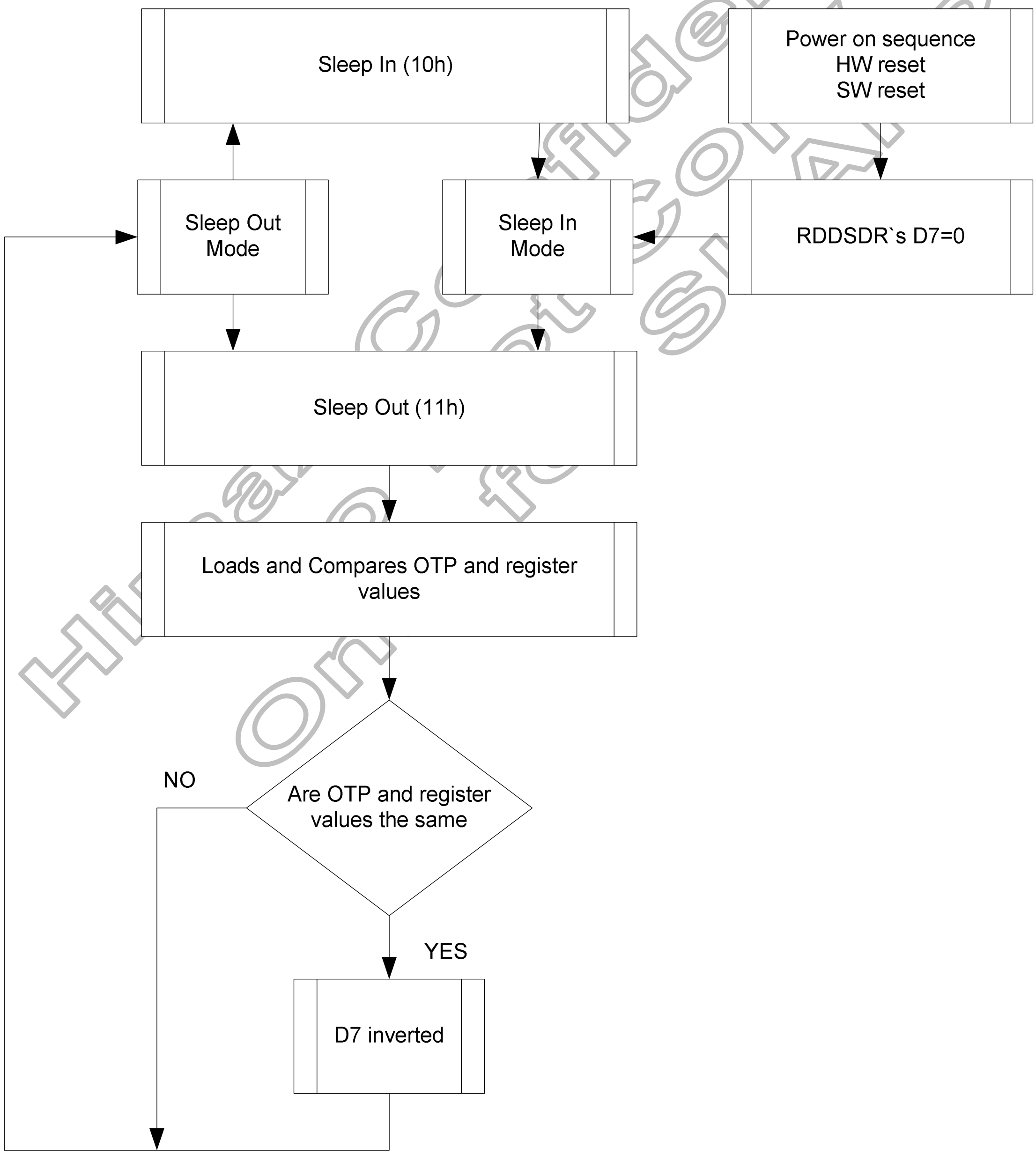
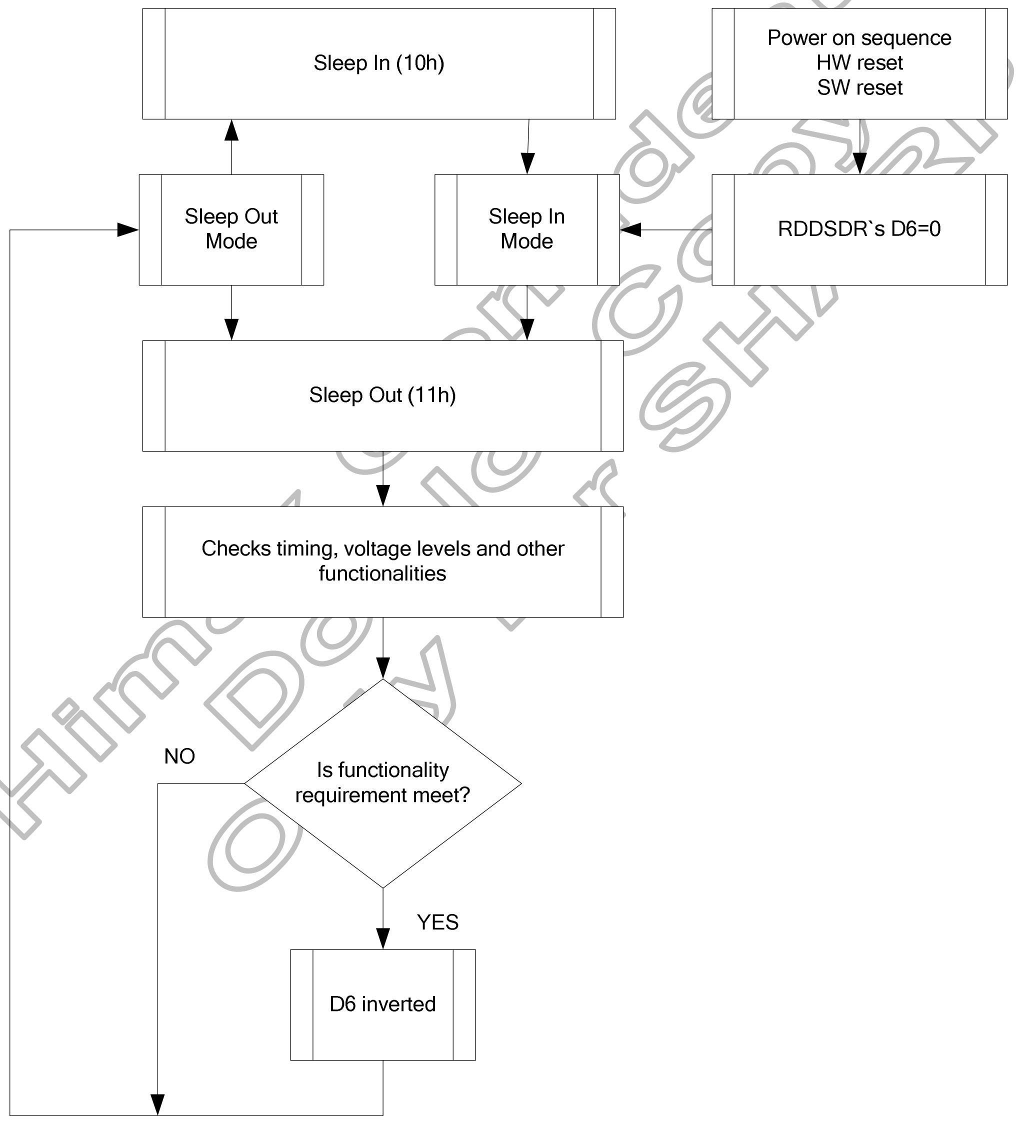


Figure 5.25: Sleep out flow chart–command and self-diagnostic functions

5.9.2 Functionality detection

Sleep Out-command (See "Sleep Out (11h)") is a trigger for an internal function of the display module, which indicates, if the display module is still running and meets functionality requirements.
The internal function (=the display controller) is comparing, if the display module still meets functionality requirements (e.g. booster voltage levels, timings, etc.). If functionality requirement is met, 1 bit will be inverted (=increased by 1), which is defined in command "Read Display Self- Diagnostic Result (0Fh)" (=RDDSDR) (The used bit of this command is D6). If functionality requirement is not the same, this bit (D6) is not inverted (=increased by 1). The flow chart for this internal function is shown as below.



Note: There is needed 120msec after Sleep Out -command, when there is changing from Sleep In–mode toSleep Out -mode, before there is possible to check if Customer's functionality requirements are met and a value of RDDSDR's D6 is valid. Otherwise, there is 5msec delay for D6's value, when Sleep Out –command is sent in Sleep Out -mode.

Figure 5.26: Sleep out flow chart internal function detection

5.10 Power on/off sequence

The Power supply On/Off, Sleep In/Out and Display On/Off sequence is illustrated below.

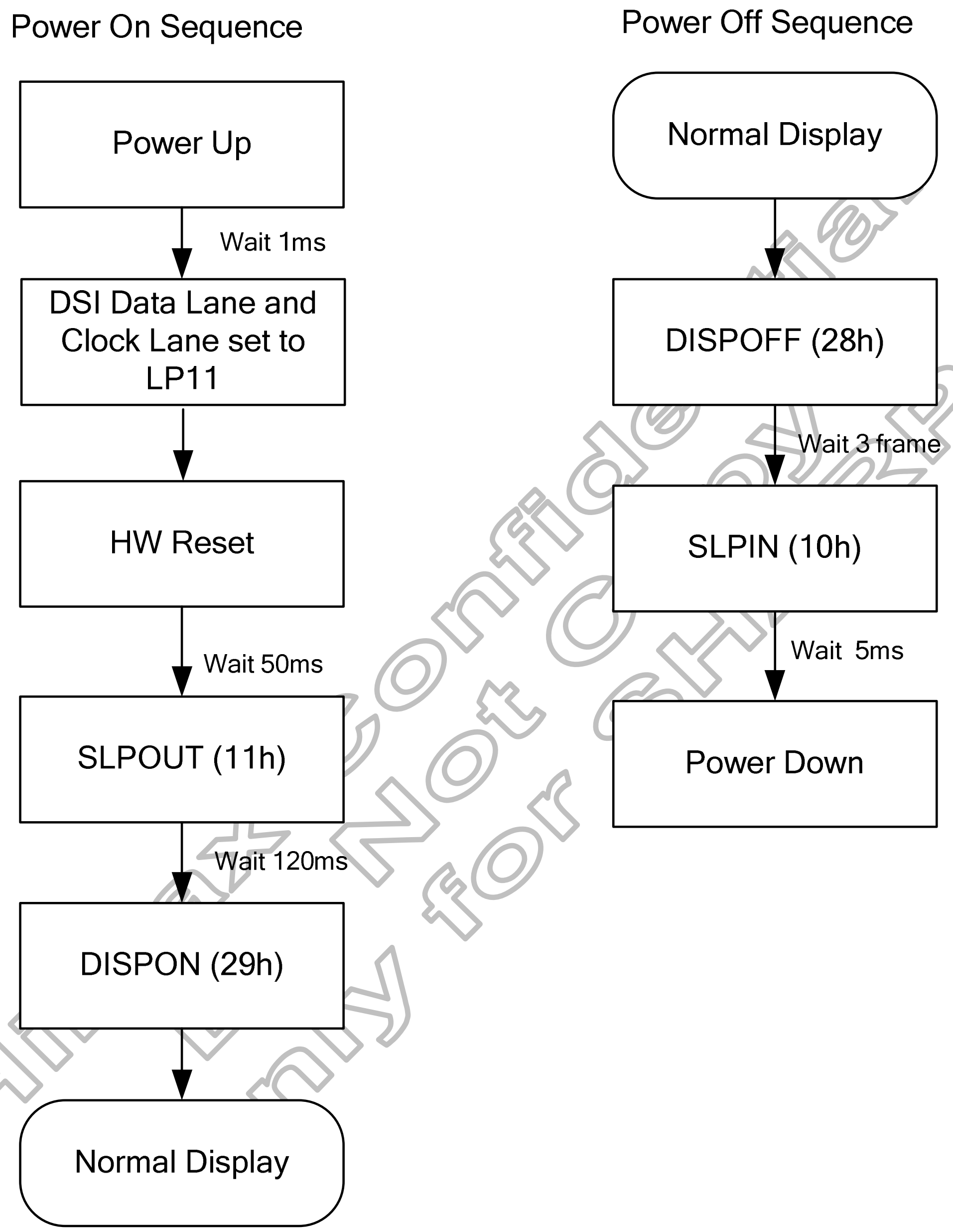


Figure 5.27: Power on/off sequence

5.10.1 VDD3/VDD1 iput power(PCCS[2:0]=000, 001, 101)

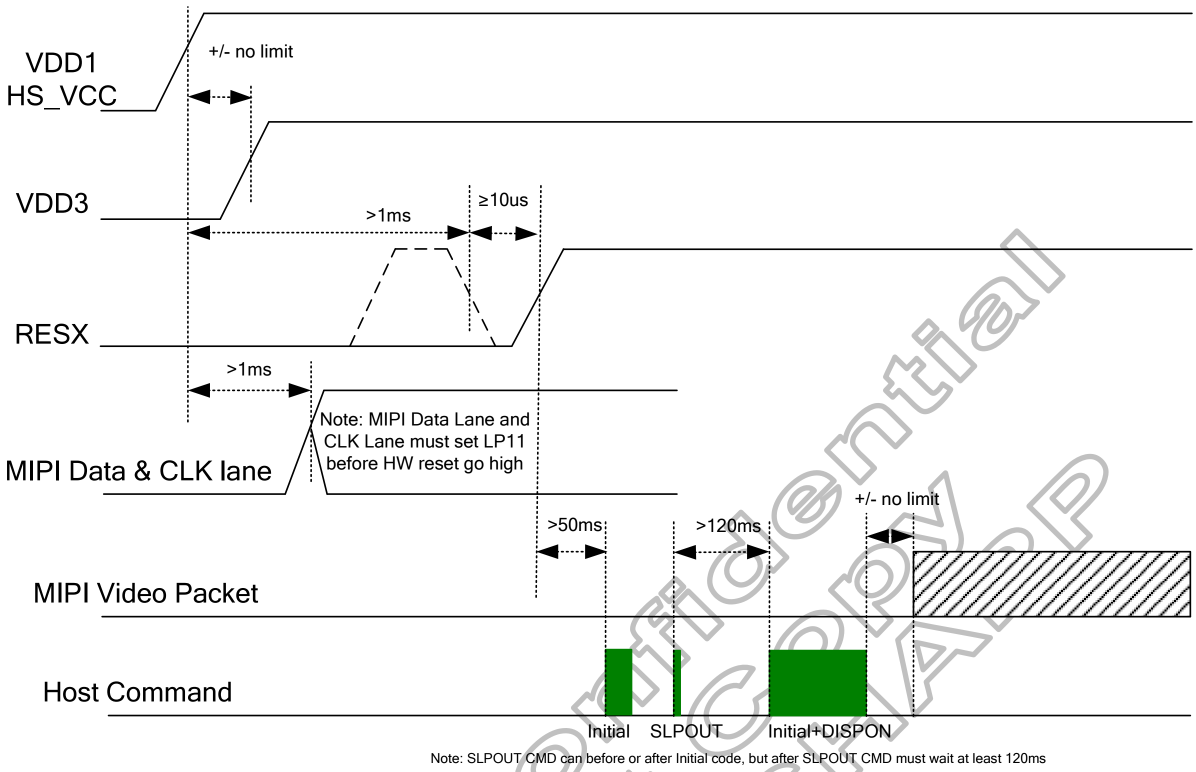


Figure 5.28: VDD3/VDD1 input power on sequence

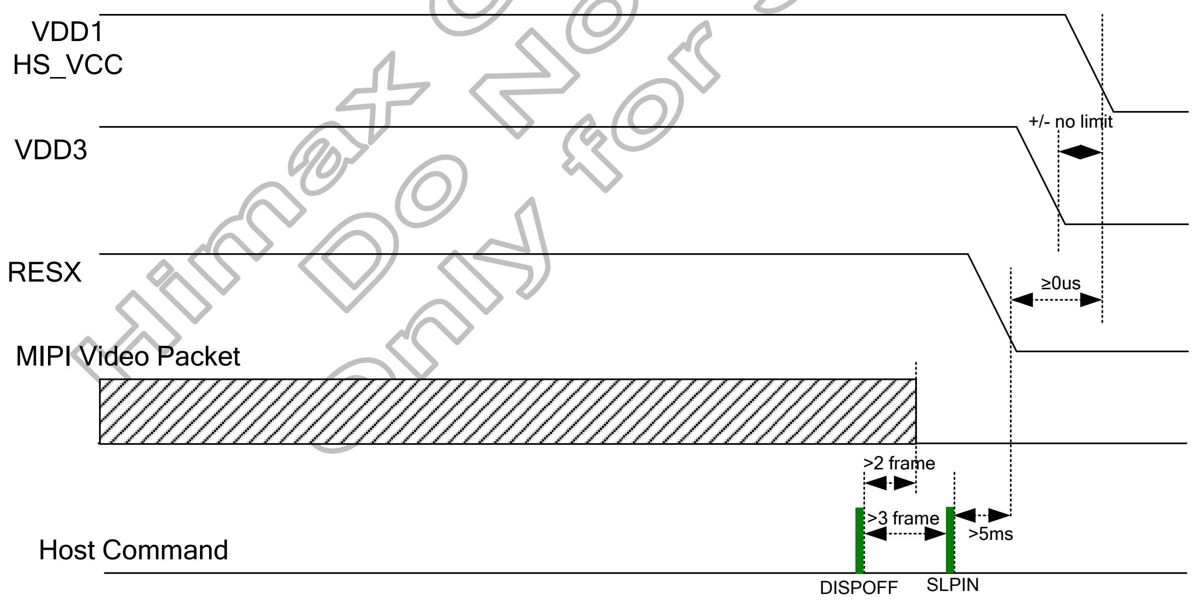


Figure 5.29: VDD3/VDD1 input power off sequence

5.10.2 VSP/VDD1 iput power(PCCS[2:0]=010)

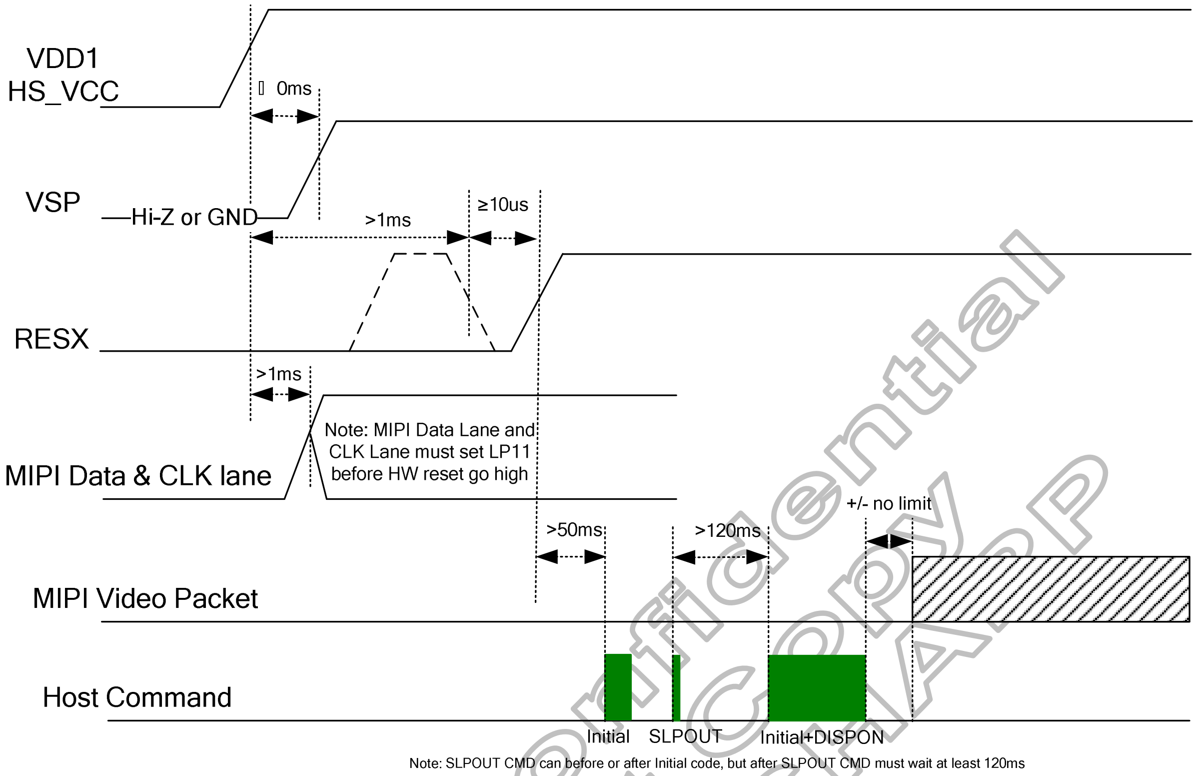


Figure 5.30: VSP/VDD1 input power on sequence

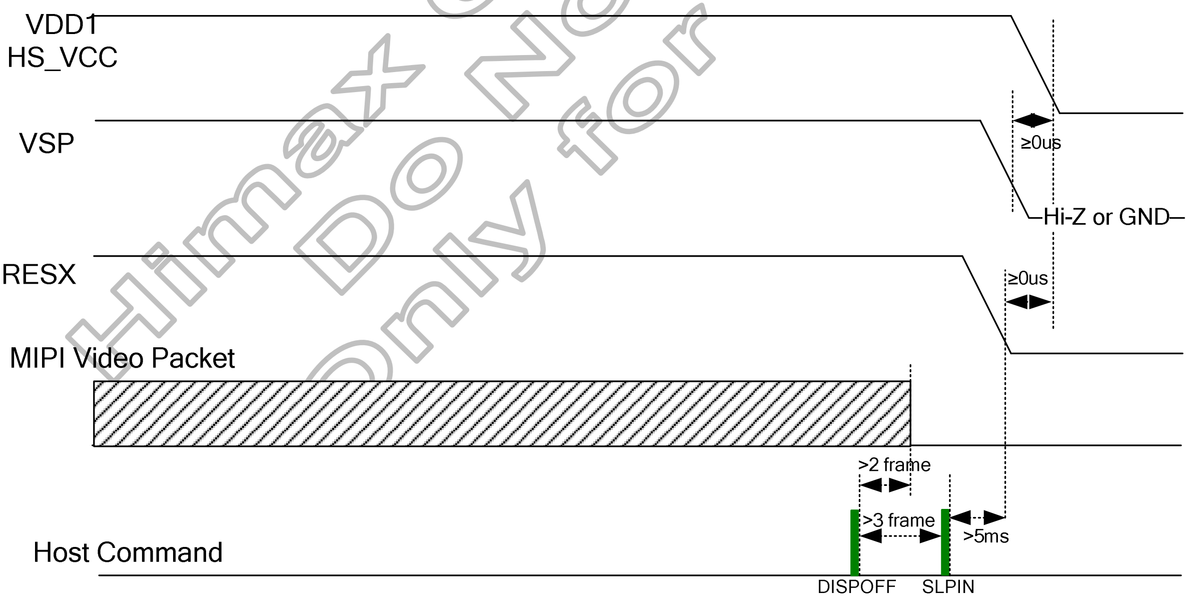


Figure 5.31: VSP/VDD1 input power off sequence

5.10.3 VSP/VSN/VDD1 iput power(PCCS[2:0]=011)

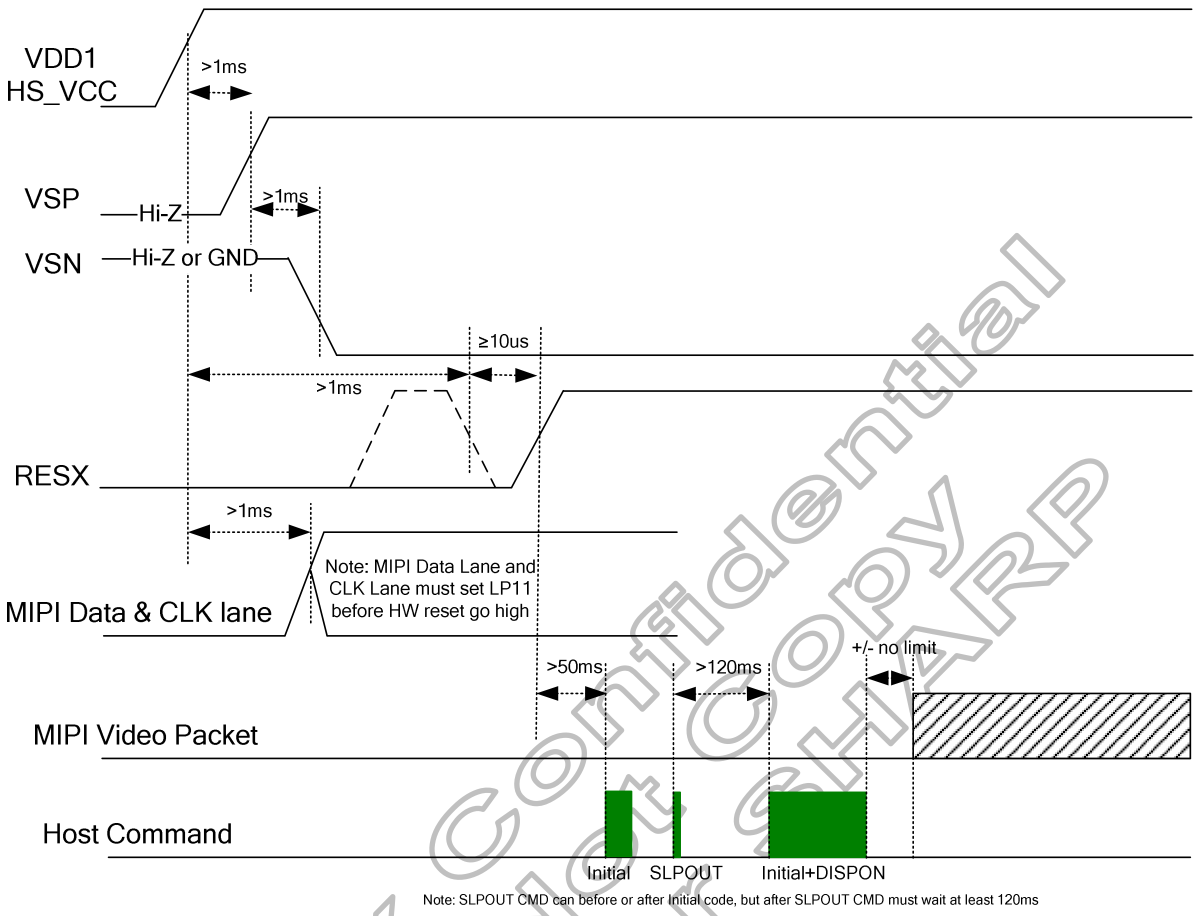


Figure 5.32: VSP/VSN/VDD1 input power on sequence

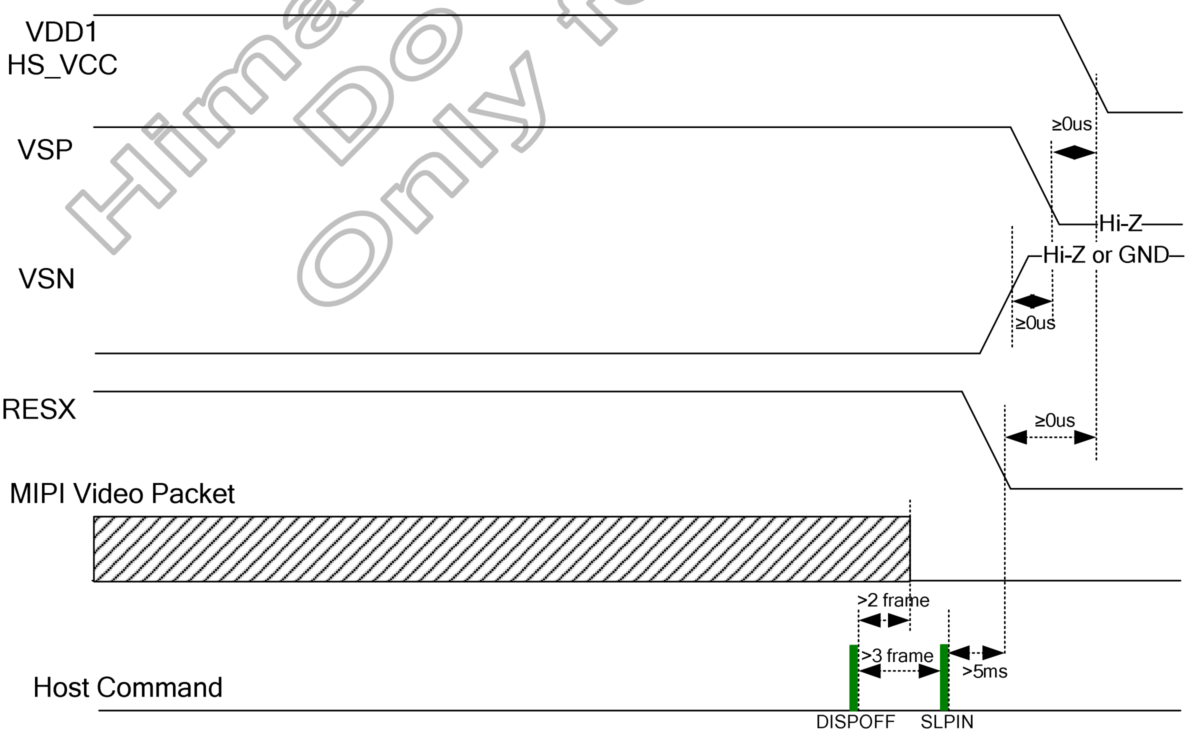


Figure 5.33: VSP/VSN/VDD1 input power off sequence

5.10.4 VDD3/VSP/VSN/VDD1 iput power(PCCS[2:0]=111)

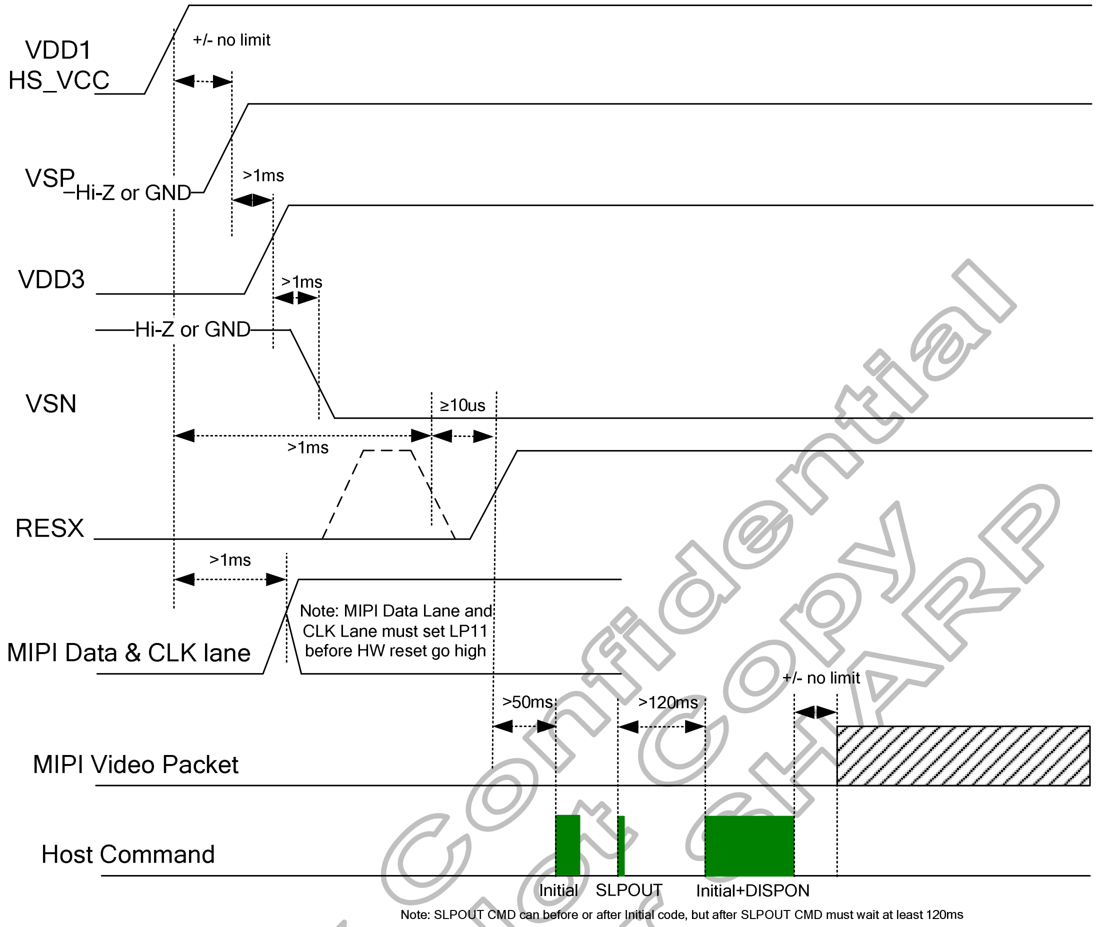


Figure 5.34: VDD3/VSP/VSN/VDD1 input power on sequence

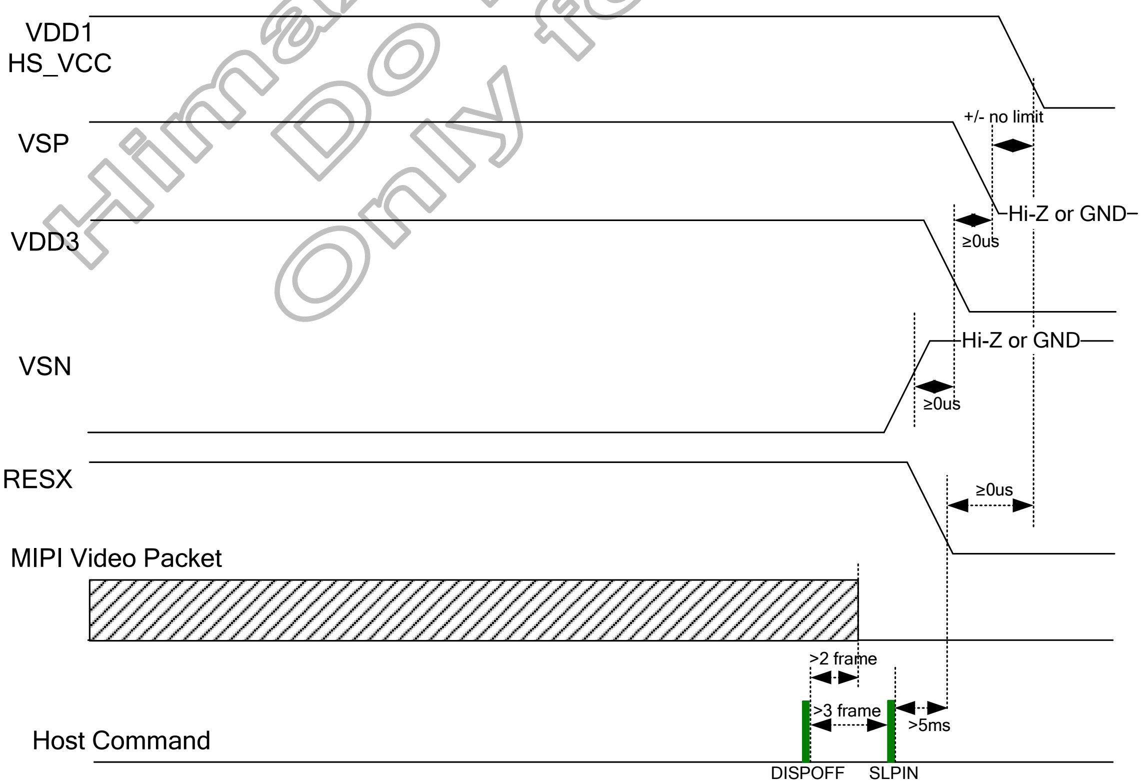
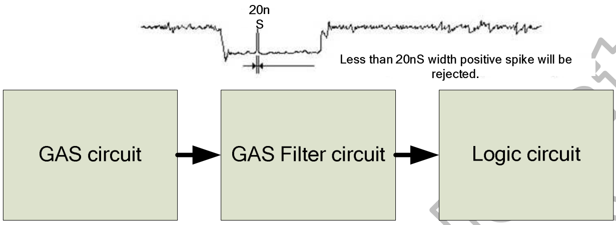


Figure 5.35: VDD3/VSP/VSN/VDD1 input power off sequence

5.11 Uncontrolled power off

The uncontrolled power off means a situation when e.g. there is removed a battery without the controlled power off sequence. There will not be any damages for the display module or the display module will not cause any damages for the host or lines of the interface. At an uncontrolled power off the display will go blank and there will not be any visible effects within 1 second on the display (blank display) and remains blank until “Power On Sequence” powers it up.

Note: HX8399-A is support the noise reject filter (20ns) to reject spike or noise.



5.12 Content adaptive brightness control (CABC) function

The general block diagram of the CABC and the brightness control is illustrated below:

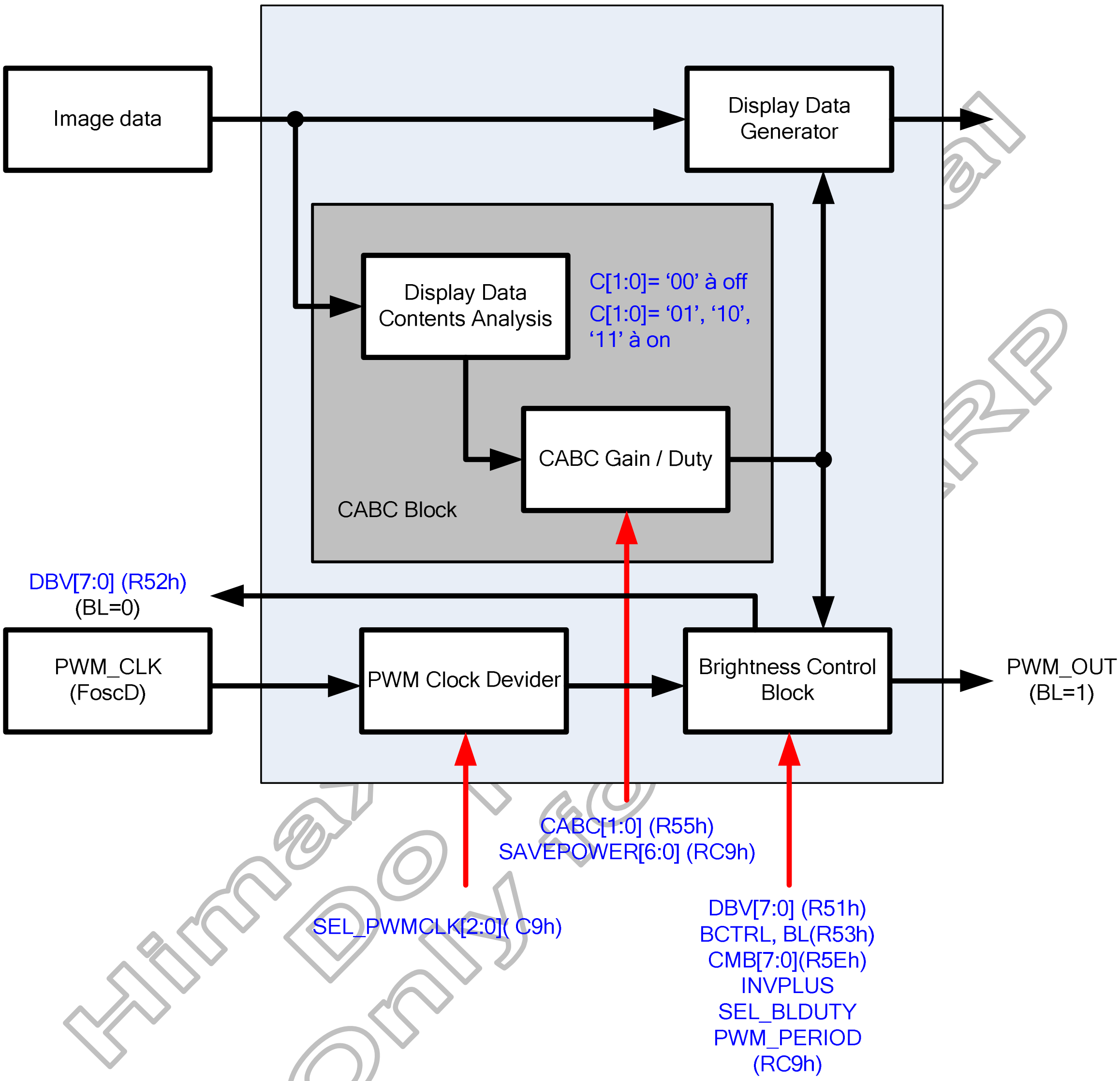
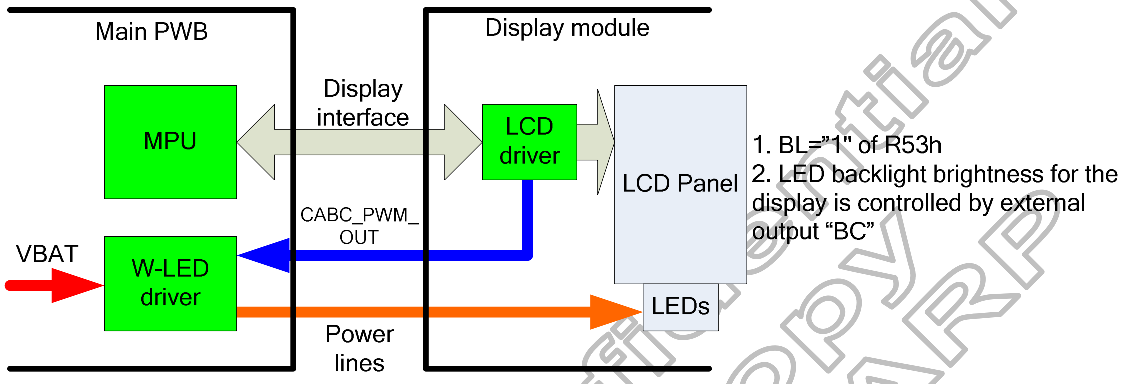


Figure 5.36: CABC block diagram

5.12.1 Module architectures

HX8399-A can support two module architectures for CABC operation. The BL bit setting of R53h can be used to select used display module architecture. White LED driver circuit for display backlight is located on the main PWB, not in the display module both in architecture I and II.

- Architecture I



- Architecture II

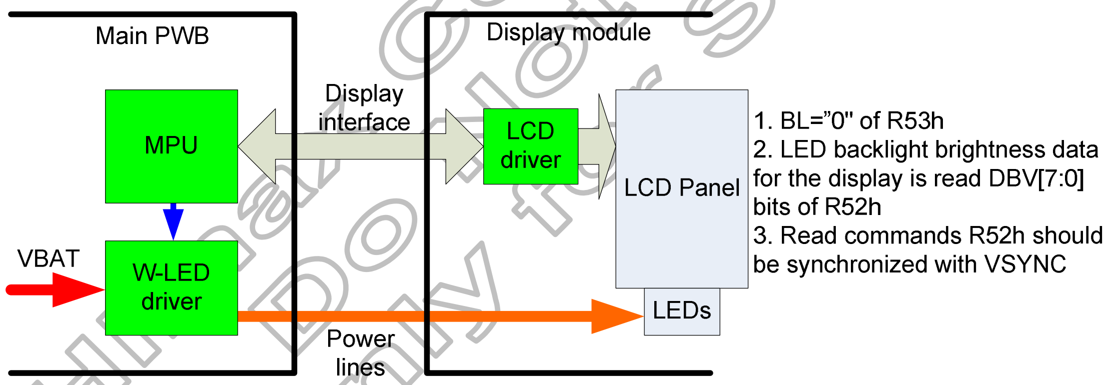


Figure 5.37: Module architecture

5.12.2 CABC block

There are DBG0~8[6:0] register bits in CABC block to define the “CABC gain”/ “CABC duty” table. Every DBGx[6:0] has 33 gain/duty value setting.

After one-frame display data content analysis, LSI will generate one CABC gain / CABC duty value calculated from DBG0~8[6:0] register bits setting (by using interpolated method) for display data generating and for backlight PWM pulse generating.

Please note that the CABC gain / CABC duty value calculated by the LSI is one of the 33 gain/duty value setting in DBGxx[6:0].

Please note that : Duty (valid level period (LED on) / one complete period)=1/ gain.

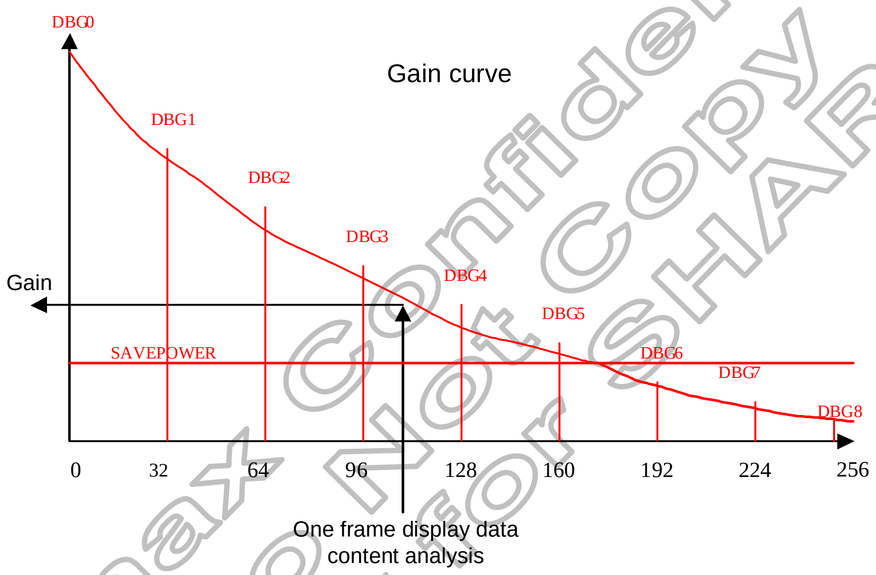


Figure 5.38: CABC gain / CABC duty generation

For power saving of backlight module, there are SAVEPOWER[6:0] bits to define the “minimum gain”/ “maximum duty” of CABC block output. If the CABC gain / duty after one-frame display data contents analysis is smaller(gain) / larger(duty) than SAVEPOWER[6:0] bits setting, the CABC block will output CABC gain / duty equal to SAVEPOWER[6:0] and ignore the result of display data contents analysis.

5.12.3 Brightness control block

There is an external output signal from brightness block, CABC_PWM_OUT, to control the LED driver IC in order to control display brightness.

There are resister bits, DBV[7:0] of R51h, for display brightness of manual brightness setting. The CABC_PWM_OUT duty is calculated as (DBV[7:0])/255 x CABC duty (generated after one-frame display data content analysis).

For ex: CABC_PWM_OUT period=2.95 ms, and DBV[7:0](R51h)=‘228DEC’ and CABC duty is 74%. Then CABC_PWM_OUT duty=(228) / 255 x 74.42%≡66.54%. Correspond to the CABC_PWM_OUT period=2.95 ms, the high-level of CABC_PWM_OUT (high effective) = 1.96ms, and the low-level of CABC_PWM_OUT =0.99ms.

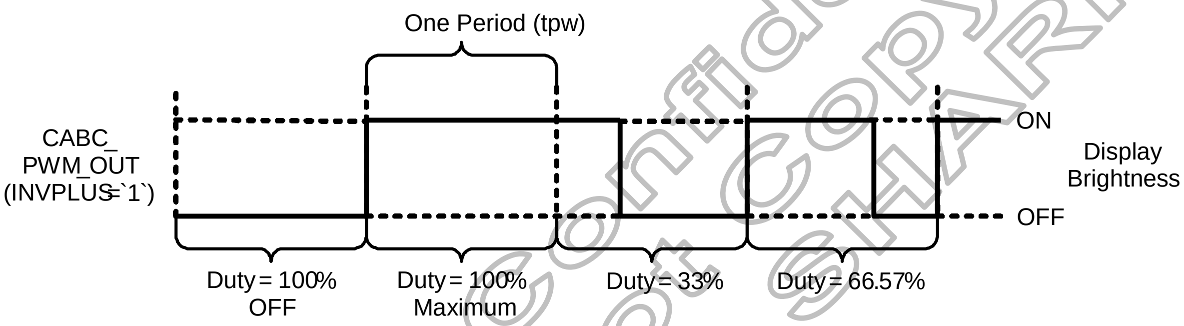


Figure 5.39: CABC_PWM_OUT output duty

Symbol	Parameter	Min.	Max.	Unit	Description
tpw	Pulse width	0.0333	8.33	ms	-

Table 5.54: CABC timing table

Note1: The signal rise and fall times (tf, tr) are stipulated to be equal to or less than 15ns.
Note2: The pulse width range by setting CABC related registers is locate between 0.0333ms to 8.33ms.

When Architecture II module is used (BL=’0’) with the example below, the CABC_PWM_OUT is always output low and the DBV[7:0](R51h) will be read a value as 169DEC ((169)/255≡ 66.27%).

5.12.4 Minimum brightness setting of CABC function

CABC function is automatically reduced backlight brightness based on image contents. In the case of the combination with the CABC or manual brightness setting, display brightness is too dark. It must affect to image quality degradation. CABC minimum brightness setting (CMB[7:0] bits of R5Eh) is to avoid too much brightness reduction.

When CABC is active, CABC can not reduce the display brightness to less than CABC minimum brightness setting. Image processing function is worked as normal, even if the brightness can not be changed.

This function does not affect to the other function, manual brightness setting. Manual brightness can be set the display brightness to less than CABC minimum brightness. Smooth transition and dimming function can be worked as normal.

When display brightness is turned off (BCTRL='0' of R53h), CABC minimum brightness setting is ignored. "CMB[7:0], Read CABC minimum brightness (R5Fh)" always read the setting value of "CMB[7:0], Write CABC minimum brightness (R5Eh)"

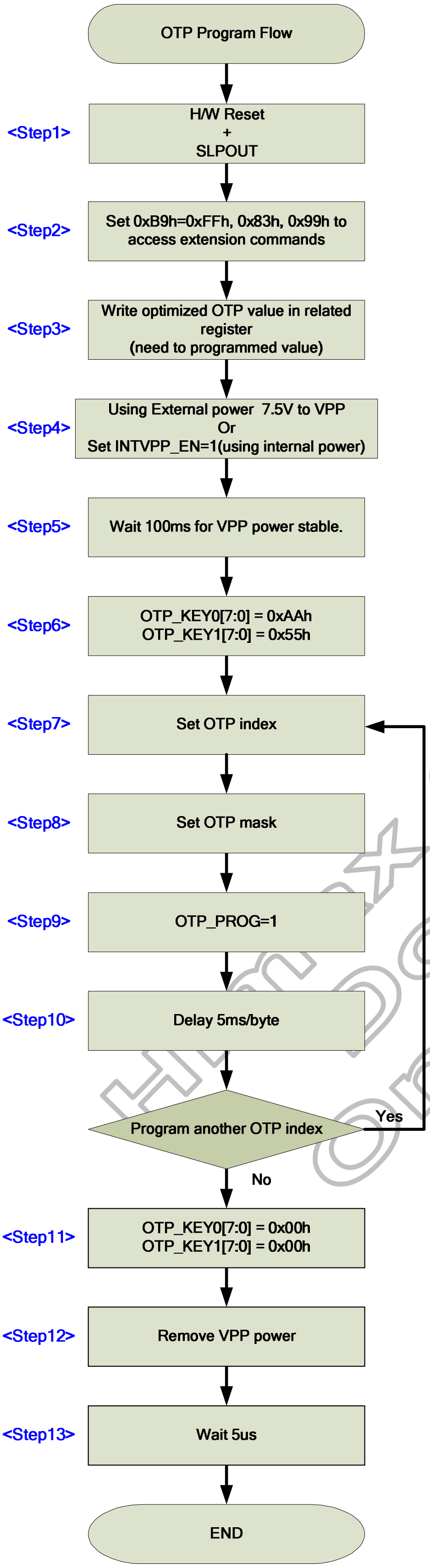
5.13 OTP programing

5.13.1 OTP table

OTP_INDEX (HEX)	B7	B6	B5	B4	B3	B2	B1	B0
00	ID1_1[7:0]							
01	ID2_1[7:0]							
02	ID3_1[7:0]							
03	ID4_1[7:0]							
04	ID1_2[7:0]							
05	ID2_2[7:0]							
06	ID3_2[7:0]							
07	ID4_2[7:0]							
08	ID1_3[7:0]							
09	ID2_3[7:0]							
0A	ID3_3[7:0]							
0B	ID4_3[7:0]							
0C	NVALID_ID1	NVALID_ID2	NVALID_ID3	-	-	-	-	-
0D	VCMC_F1[7:0]							
0E	VCMC_B1[7:0]							
0F	VCMC_F2[7:0]							
10	VCMC_B2[7:0]							
11	VCMC_F3[7:0]							
12	VCMC_B3[7:0]							
13	-	-	VCMC_B38	VCMC_B38	VCMC_B28	VCMC_B28	VCMC_B18	VCMC_B18
14	NVALID_VCMC1	NVALID_VCMC2	NVALID_VCMC3	-	-	-	-	-
15	NVALID_PANEL	-	-	-	SS_PANEL	GS_PANEL	REV_PANEL	BGR_PANEL
16	NVALID_I2C_SA	I2C_SA[6:0]						

Table 5.55: OTP table

5.13.2 OTP programming flow



OTP_KEY0[7:0] OTP_KEY1[7:0]	Description	Note
OTP_KEY0[7:0] = 0xAAh OTP_KEY1[7:0] = 0x55h	Enter OTP program mode	
OTP_KEY0[7:0] = 0x00h OTP_KEY1[7:0] = 0x00h	Leave OTP program mode	
Other value	Invalid	1. If HX8399-A operate on OTP program mode, then keep on OTP program mode. 2. If HX8399-A operate on non-OTP program mode, then keep on non-OTP program mode.

Figure 5.40: OTP programming sequence

5.13.3 Programming sequence

Step	Operation
1	Power on and reset the module.
2	SLPOUT and set 0xB9h = 0xFFh, 0x83h, 0x99h to access the extension commands.
3	Write optimized values to related registers.
4	Set INTVPP_EN=1 for OTP programming state for using internal power mode. Or using the external power 7.5V to VPP.
5	Wait 100ms for VPP power stable.
6	Set OTP_KEY0[7:0]=0xAAh and OTP_KEY1[7:0]=0x55h to enter OTP program mode.
7	Specify OTP_index, please refer to the OTP table.
8	Set OTP_Mask=0x00h, programming the entire bit of one parameter.
9	Set OTP_PROG=1, Internal register begin write to OTP according to OTP_index.
10	Wait 5 ms/byte for programming time (**Note 1**)
11	Complete programming one parameter to OTP. If continue to programming other parameter, return to step (7). Otherwise, Set OTP_KEY0[7:0]=0x00h and OTP_KEY1[7:0]=0x00h to leave OTP program mode.
12	Remove the external power on VPP pin.
13	Wait 5us

Note1: When do the OTP programming process, it must be added 5ms/byte delay time after setting OTP_PROG=1.
Note2: If user want to program ID1~ID4, only need program OTP Index 0x00h.State machine will program ID1~ID4 and valid bit automically.
Note3: If user want to program VCMC_F and VCMC_B, only need program OTP Index 0x0Dh.State machine will program VCMC_F and VCMC_B and valid bit automically.

Table 5.56: OTP Programming sequence

5.13.4 OTP Programming example of VCOM setting VCMC

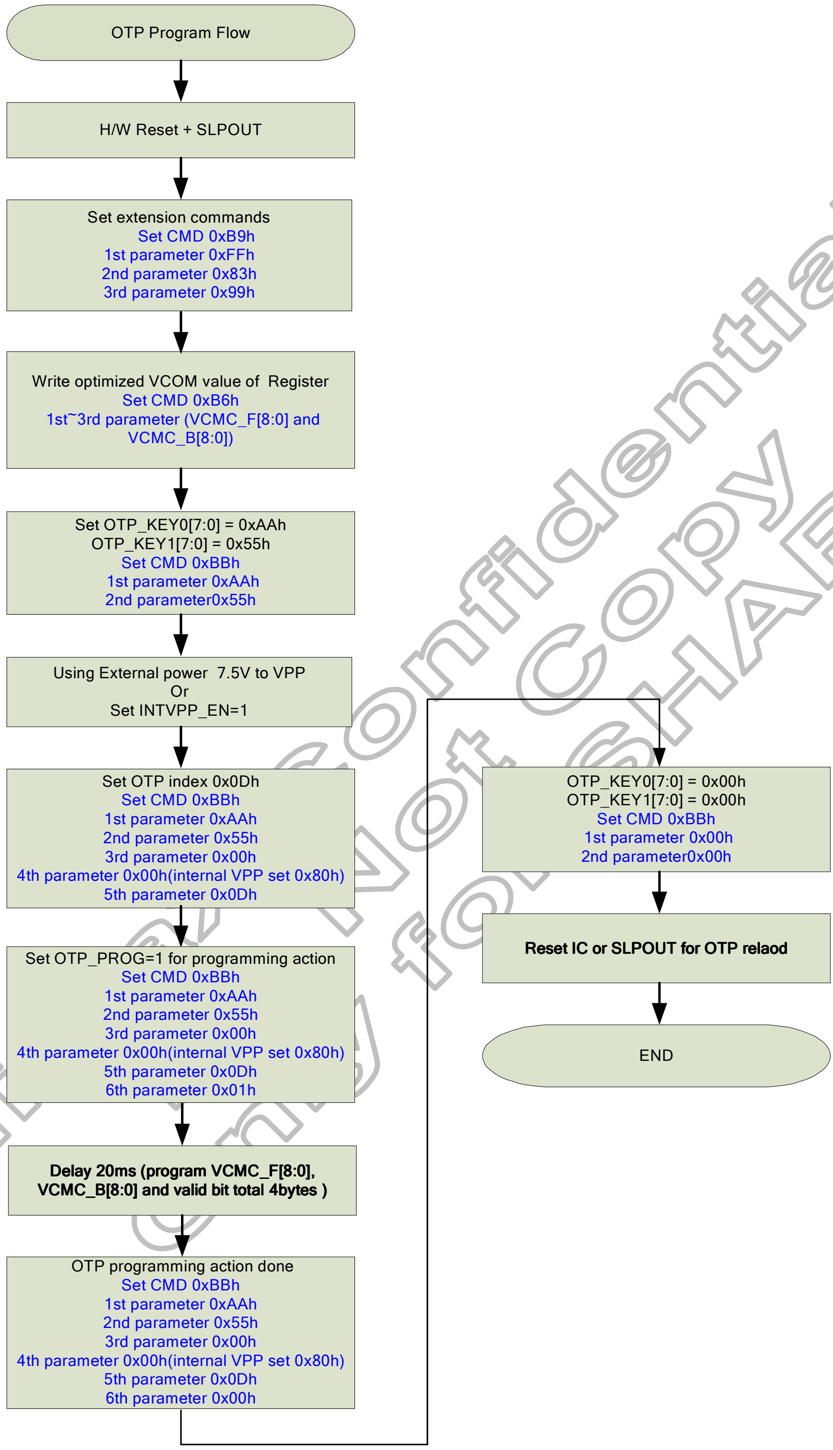


Figure 5.41: OTP programming sequence example 1.

5.13.5 OTP Programming example of ID1, ID2, ID3 and ID4

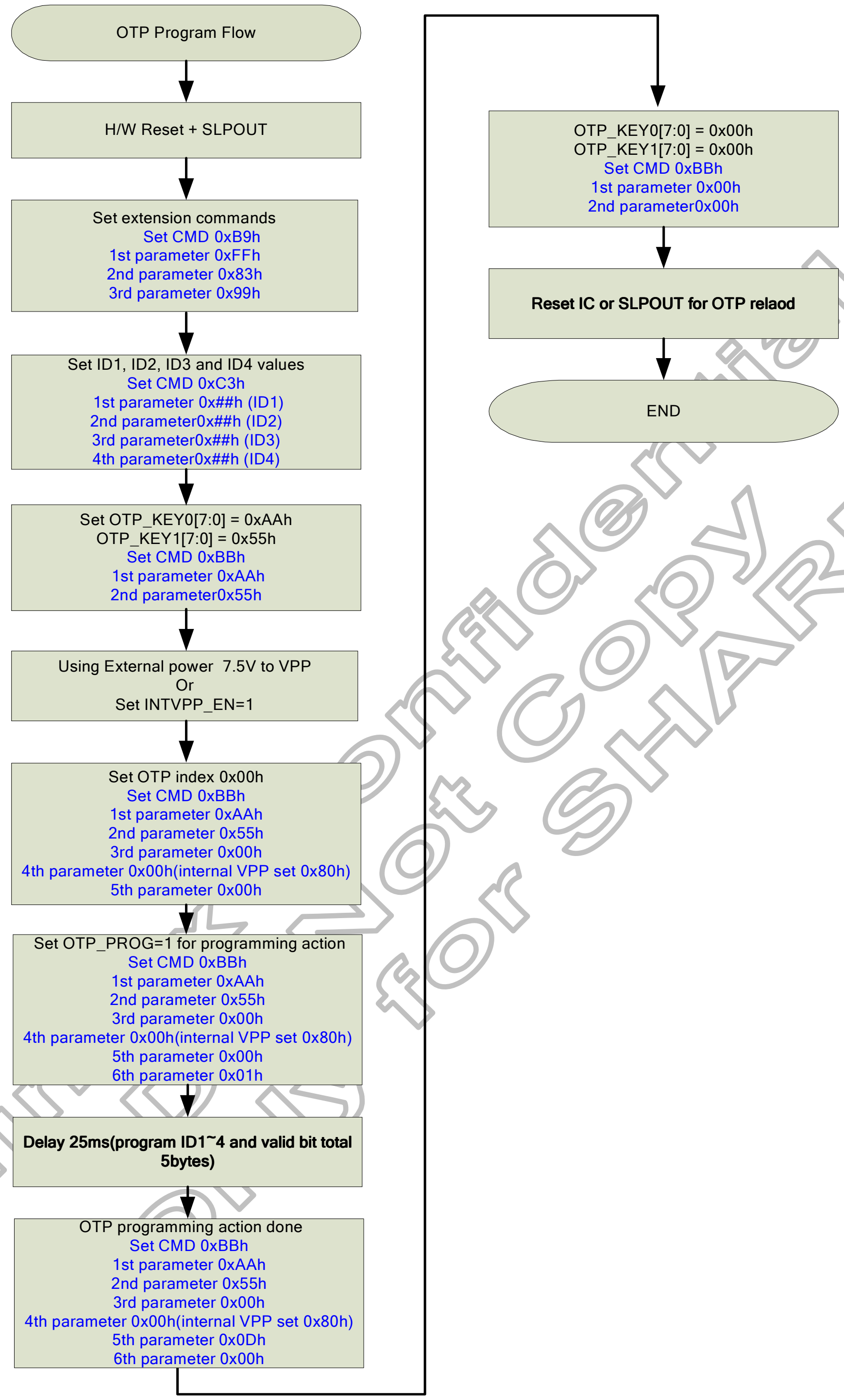


Figure 5.42: OTP programming sequence example 2.

5.13.6 OTP read example of 0x00h (ID1)

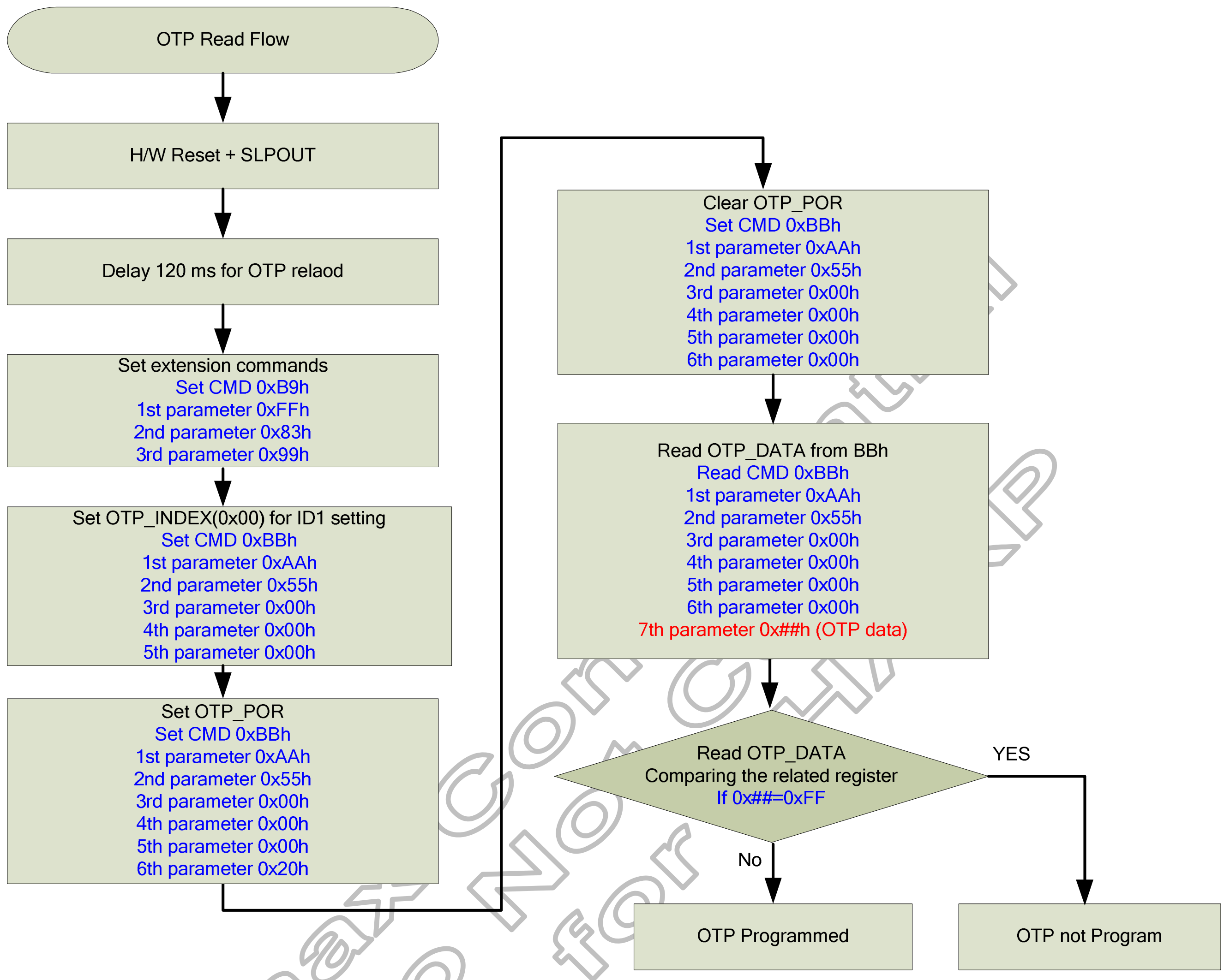


Figure 5.43: OTP read sequence flow of index 0x00h.

6. Command

6.1 Command list

6.1.1 Standard command

(Hex)	Operation code	DCX	D7	D6	D5	D4	D3	D2	D1	D0	Function	Default (Hex)
00	NOP	0	0	0	0	0	0	0	0	0	No Operation	-
01	SWRESET	0	0	0	0	0	0	0	0	1	Software Reset	-
04	RDDIDIF	0	0	0	0	0	0	1	0	0	Read Display Identification Information	-
		1	ID1[7:0]									83h
		1	ID2[7:0]									99h
		1	ID3[7:0]									0Ah
05	RDNUMPE	0	0	0	0	0	0	1	0	1	Read Number of DSI Parity Error	-
		1	P[7:0]									-
06	RDRED	0	0	0	0	0	0	1	1	0	Read Red Colour	-
		1	R7	R6	R5	R4	R3	R2	R1	R0		-
07	RDGREEN	0	0	0	0	0	0	1	1	1	Read Green Colour	-
		1	G7	G6	G5	G4	G3	G2	G1	G0		-
08	RDBLUE	0	0	0	0	0	1	0	0	0	Read Blue Colour	-
		1	B7	B6	B5	B4	B3	B2	B1	B0		-
09	RDDST	0	0	0	0	0	1	0	0	1	Read display status	-
		1	D31	D30	D29	0	0	D26	D25	D24		00h
		1	D23	D22	D21	D20	D19	0	D17	D16		71h
		1	0	0	D13	D12	D11	D10	D9	D8		00h
		1	D7	D6	D5	D4	D3	D2	D1	D0		00h
0A	RDDPM	0	0	0	0	0	1	0	1	0	Read display power mode	-
		1	D7	D6	0	D4	D3	D2	0	0		08h
0B	RDDMADCTL	0	0	0	0	0	1	0	1	1	Read display MADCTL	-
		1	D7	D6	-	-	D3	D2	D1	D0		00h
0C	RDDCOLMOD	0	0	0	0	0	1	1	0	0	Read display pixel format	-
		1	0	D6	D5	D4	0	0	0	0		70h
0D	RDDIM	0	0	0	0	0	1	1	0	1	Read display image mode	-
		1	0	0	D5	D4	D3	D2	D1	D0		00h
0E	RDDSM	0	0	0	0	0	1	1	1	0	Read display signal mode	-
		1	D7	D6	D5	D4	D3	D2	0	D0		00h
0F	RDDSDR	0	0	0	0	0	1	1	1	1	Read display self-diagnostic result	-
		1	D7	D6	D5	D4	0	0	0	0		00h
10	SLPIN	0	0	0	0	1	0	0	0	0	Sleep In	-
11	SLPOUT	0	0	0	0	1	0	0	0	1	Sleep Out	-
13	NORON	0	0	0	0	1	0	0	1	1	Normal display mode on	-
20	INVOFF	0	0	0	1	0	0	0	0	0	Display inversion off	-
21	INVON	0	0	0	1	0	0	0	0	1	Display inversion on	-
22	ALLPOFF	0	0	0	1	0	0	0	1	0	All pixel off (black)	-
23	ALLPON	0	0	0	1	0	0	0	1	1	All pixel on (white)	-
26	GAMSET	0	0	0	1	0	0	1	1	0	Gamma set	-
		1	GC[7:0]									01h
28	DISPOFF	0	0	0	1	0	1	0	0	0	Display off	-
29	DISPON	0	0	0	1	0	1	0	0	1	Display on	-
34	TEOFF	0	0	0	1	1	0	1	0	0	Tearing Effect Line OFF	-
35	TEON	0	0	0	1	1	0	1	0	1	Tearing Effect Line ON	-
		1	X	X	X	X	X	X	X	M		00h
36	MADCTL	0	0	0	1	1	0	1	1	0	Memory access Control	-
		1	D7	D6	X	X	D3	D2	D1	D0		00h
38	IDMOFF	0	0	0	1	1	1	0	0	0	Idle mode off	-
39	IDMON	0	0	0	1	1	1	0	0	1	Idle mode on	-
3A	COLMOD	0	0	0	1	1	1	0	1	0	-	-
		1	X	D6	D5	D4	X	X	X	X	-	70h

(Hex)	Operation Code	DCX	D7	D6	D5	D4	D3	D2	D1	D0	Function	Default (Hex)
44	TESL	0	0	1	0	0	0	1	0	0	Tearing Effect Scan Line number	-
		1	TELINE[15:8]									00h
		1	TELINE[7:0]									00h
45	GETSCAN	0	0	1	0	0	0	1	0	1	Reture the current scanline SLN[15:0]	-
		1	SLN[15:8]									00h
		1	SLN[7:0]									00h
51	WRDISBV	0	0	1	0	1	0	0	0	1	Write Display Brightness	-
		1	DBV[7:0]									00h
52	RDDISBV	0	0	1	0	1	0	0	1	0	Read Display Brightness Value	-
		1	DBV[7:0]									00h
53	WRCTRLD	0	0	1	0	1	0	0	1	1	Write CTRL Display	-
		1	X	X	BCTRL	X	DD	BL	X	X		00h
54	RDCTRLD	0	0	1	0	1	0	1	0	0	Read Control Value Display	-
		1	0	0	BCTRL	0	DD	BL	0	0		00h
55	WRCABC	0	0	1	0	1	0	1	0	1	Write Adaptive Brightness Control	-
		1	X	X	X	X	X	X	CABC[1:0]			00h
56	RDCABC	0	0	1	0	1	0	1	1	0	Read Adaptive Brightness Control Content	-
		1	0	0	0	0	0	0	C1	C0		00h
5E	WRCABCMB	0	0	1	0	1	1	1	1	0	Write CABC minimum brightness	-
		1	CMB[7:0]									00h
5F	RDCABCMB	0	0	1	0	1	1	1	1	1	Read CABC minimum brightness	-
		1	CMB[7:0]									00h
68	RDABCSDR	0	0	1	1	0	1	0	0	0	Read Automatic Brightness Control Self-Diagnostic Result	-
		1	D[7:6]		0	0	0	0	0	0		00h
A1	Read_DDB_start	0	1	0	1	0	0	0	0	1	Read the DDB from the provided location.	-
		1	xx	xx	xx	xx	xx	xx	xx	xx		-
		1	xx	xx	xx	xx	xx	xx	xx	xx		-
		1	xx	xx	xx	xx	xx	xx	xx	xx		-
A8	Read_DDB_continue	0	1	0	1	0	1	0	0	0	Continue reading the DDB from the last read location.	-
		1	xx	xx	xx	xx	xx	xx	xx	xx		-
		1	xx	xx	xx	xx	xx	xx	xx	xx		-
		1	xx	xx	xx	xx	xx	xx	xx	xx		-
DA	RDID1	0	1	1	0	1	1	0	1	0	Read ID1	-
		1	ID1[7:0]									83h
DB	RDID2	0	1	1	0	1	1	0	1	1	Read ID2	-
		1	ID2[7:0]									99h
DC	RDID3	0	1	1	0	1	1	1	0	0	Read ID3	-
		1	ID3[7:0]									0Ah

Note: (1) Undefined commands are treated as NOP (00h) command.
(2) B0h to D8h and E0h to FFh are for factory use of display supplier.

6.1.2 User define command list table

User define command list is available only set "SETEXC" command.

(Hex)	Operation Code	DCX	D7	D6	D5	D4	D3	D2	D1	D0	Default	Function
B0	SETSEQUENCE	0	1	0	1	1	0	0	0	0	-	Set sequecnce
		1	-	AUTO_OPT[6:0]							00h	
		1	-	-	-	-	-	-	-	OSC_EN	00h	
		1	-	VSN_EN	VSP_EN	VGL_EN	VGH_EN	VCL_EN	-	STB	7Dh	
		1	-	-	-	-	GON	DTE	D[1:0]		0Ch	
B1	SETPOWER	0	1	0	1	1	0	0	0	1	-	Set power control
		1	-	-	-	-	-	-	-	DSTB	00h	
		1	VSP_FBOFF	APF_EN	DD_TU	GASVCI_OPT[1:0]		APP[2:0]			64h	
		1	-	VCI_LDOS[1:0]		VRHP[4:0]					32h	
		1	-	DT[1:0]		VRHN[4:0]					32h	
		1	VSN_REGS[3:0]				VSP_REGS[3:0]				44h	
		1	-	-	-	DUTY_ADJ_DLY[1:0]		XDK[2:0]			09h	
		1	FS0[3:0]				FS1[3:0]				22h	
		1	FS2[3:0]				FS3[3:0]				22h	
		1	VCL[2:0]			BTP[4:0]					71h	
		1	CLK_OPT[2:0]			BTN[4:0]					F1h	
		1	VGHS[7:0]								E4h	
		1	VGLS[7:0]								E4h	
		1	DCDIVS[3:0]				DCDIV[3:0]				06h	
		1	DCPS[3:0]				DCP[3:0]				FBh	
		1	DCNS[3:0]				DCN[3:0]				FCh	
		1	DTPS[3:0]				DTP[3:0]				05h	
		1	DTNS[3:0]				DTN[3:0]				07h	
		1	MAXDUTY[3:0]				MINDUTY[3:0]				E3h	
		1	RPRIORITY[1:0]		STEP[3:0]				EN_VSN_CLAMP	EN_DUTY_ADJ	06h	
B2	SETDISP	0	1	0	1	1	0	0	1	0	-	Set Display
		1	ZZ_LR	ZZ_EO	-	-	-	NW[2:0]			80h	
		1	MUX_SEL	-	-	-	TGS[3:0]				00h	
		1	-	-	-	H_RES[2:0]			V_RES[1:0]		00h	
		1	NL[7:0]								7Fh	
		1	BP [7:0]								0Bh	
		1	FP [7:0]								11h	
		1	RTN[7:0]								23h	
		1	RTN_GAS[7:0]								4Dh	
		1	-	INIT_SD_SEL	INIT_VCOM_SEL	-	INIT_SET[3:0]				01h	
		1	-	END_SD_SEL	END_VCOM_SEL	-	END_SET[3:0]				01h	

(Hex)	Operation Code	DCX	D7	D6	D5	D4	D3	D2	D1	D0	Default	Function
B4	SETCYC	0	1	0	1	1	0	1	0	0	-	Set Cycle
		1	GEN_ON[7:0]								00h	
		1	GEN_OFF[7:0]								FFh	
		1	SPON[7:0]								03h	
		1	SPOFF[7:0]								38h	
		1	CON[7:0]								05h	
		1	COFF[7:0]								36h	
		1	CON1[7:0]								05h	
		1	COFF1[7:0]								36h	
		1	N_t1[7:0]								00h	
		1	N_t2[7:0]								09h	
		1	N_t3[7:0]								01h	
		1	N_t4[7:0]								01h	
		1	N_t5[7:0]								00h	
		1	N_t6[7:0]								10h	
		1	N_t7[7:0]								01h	
		1	N_t8[7:0]								02h	
		1	N_t9[7:0]								02h	
		1	SAP[3:0]				EQT[3:0]				71h	
		1	N_t10[7:0]								00h	
		1	SON[7:0]								00h	
		1	SOFF[7:0]								FFh	
		1	SPON_MPU[7:0]								03h	
		1	SPOFF_MPU[7:0]								38h	
		1	CON_MPU[7:0]								05h	
		1	COFF_MPU[7:0]								36h	
		1	CON1_MPU[7:0]								05h	
		1	COFF1_MPU[7:0]								36h	
		1	N_t1_MPU[7:0]								00h	
		1	N_t2_MPU[7:0]								09h	
		1	N_t3_MPU[7:0]								01h	
		1	N_t4_MPU[7:0]								01h	
		1	N_t5_MPU[7:0]								00h	
		1	N_t6_MPU[7:0]								10h	
		1	N_t7_MPU[7:0]								01h	
		1	N_t8_MPU[7:0]								02h	
		1	N_t9_MPU[7:0]								02h	
		1	-	-	-	-	EQT_MPU[3:0]				01h	
		1	N_t10_MPU[7:0]								00h	
		1	SON_MPU[7:0]								00h	
		1	SOFF_MPU[7:0]								FFh	
B6	SETVCOM	0	1	0	1	1	0	1	1	0	-	Set VCOM Voltage
		1	VCMC_F[7:0]								46h	
		1	VCMC_B[7:0]								46h	
		1	VCOM_TIMES[2:0]			-	-	-	VCMC_B8	VCMC_F8	E0h	
B7	SETTE	0	1	0	1	1	0	1	1	1	-	Set TE function
		1	TEI[3:0]				-	TEP[10:8]			00h	
		1	TEP[7:0]								00h	
B9	SETEXTC	0	1	0	1	1	1	0	0	1	-	Set extended command
		1	EXTC1[7:0]								00h	
		1	EXTC2[7:0]								00h	
		1	EXTC3[7:0]								00h	
BA	SETRMIPI	0	1	0	1	1	1	0	1	0	-	Set MIPI Control
		1	-	DSISETUP0[6:0]							03h	
		1	DSISETUP1[7:0]								82h	

(Hex)	Operation Code	DCX	D7	D6	D5	D4	D3	D2	D1	D0	Default	Function
BB	SETOTP	0	1	0	1	1	1	0	1	1	-	Set OTP
		1	OTP_KEY0[7:0]								00h	
		1	OTP_KEY1[7:0]								00h	
		1	OTP_MASK[7:0]								00h	
		1	INTVPP_EN	-	-	-	-	-	-	-	00h	
		1	OTP_INDEX[7:0]								00h	
		1	OTP_LOAD_DISABLE	OTP_TEST	OTP_POR	OTP_PWE	OTP_PTM[1:0]		OTP_PWR_SEL	OTP_PROG	00h	
		1	OTP_DATA[7:0]								-	
BD	SETBANK	0	1	0	1	1	1	1	0	1	-	Set Register Bank
		1	-	-	-	-	-	-	BANK_INDEX[1:0]		00h	
C1	SETDGC	0	1	1	0	0	0	0	0	1	-	Set DGC
		1	-	-	-	-	-	-	-	DGC_EN	00h	Bank0
		1	R_GAMMA0[9:2]									
		1	R_GAMMA1[9:2]									
		1	:									
		1	R_GAMMA32[9:2]									
		1	R_GAMMA0[1:0]		R_GAMMA1[1:0]		R_GAMMA2[1:0]		R_GAMMA3[1:0]			
		1	:									
		1	R_GAMMA32[1:0]		-	-	-	-	-	-		
		1	G_GAMMA0[9:2]									Bank1
		1	G_GAMMA1[9:2]									
		1	:									
		1	G_GAMMA32[9:2]									
		1	G_GAMMA0[1:0]		G_GAMMA1[1:0]		G_GAMMA2[1:0]		G_GAMMA3[1:0]			
		1	:									
		1	G_GAMMA32[1:0]		-	-	-	-	-	-		
		1	B_GAMMA0[9:2]									Bank2
		1	B_GAMMA1[9:2]									
		1	:									
		1	B_GAMMA32[9:2]									
		1	B_GAMMA0[1:0]		B_GAMMA1[1:0]		B_GAMMA2[1:0]		B_GAMMA3[1:0]			
		1	:									
		1	B_GAMMA32[1:0]		-	-	-	-	-	-		
C3	SETID	0	1	1	0	0	0	0	1	1	-	Set ID
		1	ID1[7:0]								83h	
		1	ID2[7:0]								99h	
		1	ID3[7:0]								0Ah	
		1	ID4[7:0]								00h	
		1	ID_TIMES[2:0]			-	-	-	-	-	E0h	
C4	SETDDB	0	1	1	0	0	0	1	0	0	-	Set DDB
		1	DDB1[7:0]								00h	
		1	DDB2[7:0]								00h	
		1	DDB3[7:0]								00h	
		1	DDB4[7:0]								00h	
C9	SETCABC	0	1	1	0	0	1	0	0	1	-	Set CABC Control
		1	PWM_PERIOD[16]	SEL_PWMCLK[2:0]			SEL_GAIN[1:0]		INVPULS	SEL_BLDUTY	1Fh	
		1	PWM_PERIOD[15:8]								00h	
		1	PWM_PERIOD[7:0]								2Eh	
		1	CABC_FSYNC	DIM_FRAME[6:0]							1Eh	
		1	CABC_DD	-	-	-	CABC_FLM[3:0]				01h	
		1	CABC_STEP[7:0]								1Eh	
		1	-	SAVEPOWER[6:0]							00h	
CB	SETCLOCK	0	1	1	0	0	1	0	1	1	-	Set Oscillator
		1	-	-	-	-	UADJ[3:0]				07h	
CC	SETPANEL	0	1	1	0	0	1	1	0	0	-	Set Panel
		1	-	-	-	-	SS_PANEL	GS_PANEL	REV_PANEL	BGR_PANEL	02h	

(Hex)	Operation Code	DCX	D7	D6	D5	D4	D3	D2	D1	D0	Default	Function
D0	SET_SLR_MODE	0	1	1	0	1	0	0	0	0	-	Set SLR mode
		1	reg_amb_val[7:0]								80h	
		1	reg_CTRL_EHLV[7:0]								40h	
		1	-	-	-	-	slr_sel_opt	slr_demo_on	slr_bps_en	reg_slr_en	00h	
D3	SETGIP0	0	1	1	0	1	0	0	1	1	-	Set GIP Option0
		1	-	-	GIP_EQ_MODE[1:0]		EQ_DELAY_HSYNC[3:0]				00h	
		1	EQ_DELAY[7:0]								00h	
		1	-	-	-	-	-	-	OVERLAP_OPT	CKV_VGH_OPT	00h	
		1	-	-	GTO[5:0]						00h	
		1	GNO[7:0]								00h	
		1	USER_GIP_GATE[7:0]								08h	
		1	USER_GIP_GATE1[7:0]								08h	
		1	SHR0_3[3:0]				SHR0_2[3:0]				32h	
		1	SHR0_1[3:0]				SHR0[11:8]				10h	
		1	SHR0[7:0]								00h	
		1	-	-	-	-	SHR0_GS[11:8]				00h	
		1	SHR0_GS[7:0]								00h	
		1	SHR1_3[3:0]				SHR1_2[3:0]				32h	
		1	SHR1_1[3:0]				SHR1[11:8]				13h	
		1	SHR1[7:0]								C0h	
		1	-	-	-	-	SHR1_GS[11:8]				00h	
		1	SHR1_GS[7:0]								00h	
		1	SHR2_3[3:0]				SHR2_2[3:0]				32h	
		1	SHR2_1[3:0]				SHR2[11:8]				10h	
		1	SHR2[7:0]								08h	
		1	-	-	-	-	SHR2_GS[11:8]				00h	
		1	SHR2_GS[7:0]								00h	
		1	SHP[3:0]				SCP[3:0]				23h	
		1	CHR0[7:0]								04h	
		1	CHR0_GS[7:0]								00h	
		1	CHP0[3:0]				CCP0[3:0]				27h	
		1	CHR1[7:0]								04h	
		1	CHR1_GS[7:0]								00h	
		1	CHP1[3:0]				CCP1[3:0]				27h	
		1	vbp_setting[7:0]								00h	
		1	-	-	-	-	vbp_self_learning	-	-	-	00h	

(Hex)	Operation Code	DCX	D7	D6	D5	D4	D3	D2	D1	D0	Default	Function
D5	SETGIP1	0	1	1	0	1	0	1	0	1	-	Set GIP Option1
		1	-	CGTS_L_INV[1]	COS1_L[5:0]						00h	
		1	-	CGTS_R_INV[1]	COS1_R[5:0]						00h	
		1	-	CGTS_L_INV[2]	COS2_L[5:0]						00h	
		1	-	CGTS_R_INV[2]	COS2_R[5:0]						00h	
		1	-	CGTS_L_INV[3]	COS3_L[5:0]						00h	
		1	-	CGTS_R_INV[3]	COS3_R[5:0]						00h	
		1	-	CGTS_L_INV[4]	COS4_L[5:0]						00h	
		1	-	CGTS_R_INV[4]	COS4_R[5:0]						00h	
		1	-	CGTS_L_INV[5]	COS5_L[5:0]						00h	
		1	-	CGTS_R_INV[5]	COS5_R[5:0]						00h	
		1	-	CGTS_L_INV[6]	COS6_L[5:0]						00h	
		1	-	CGTS_R_INV[6]	COS6_R[5:0]						00h	
		1	-	CGTS_L_INV[7]	COS7_L[5:0]						00h	
		1	-	CGTS_R_INV[7]	COS7_R[5:0]						00h	
		1	-	CGTS_L_INV[8]	COS8_L[5:0]						00h	
		1	-	CGTS_R_INV[8]	COS8_R[5:0]						00h	
		1	-	CGTS_L_INV[9]	COS9_L[5:0]						00h	
		1	-	CGTS_R_INV[9]	COS9_R[5:0]						00h	
		1	-	CGTS_L_INV[10]	COS10_L[5:0]						00h	
		1	-	CGTS_R_INV[10]	COS10_R[5:0]						00h	
		1	-	CGTS_L_INV[11]	COS11_L[5:0]						00h	
		1	-	CGTS_R_INV[11]	COS11_R[5:0]						00h	
		1	-	CGTS_L_INV[12]	COS12_L[5:0]						00h	
		1	-	CGTS_R_INV[12]	COS12_R[5:0]						00h	
		1	-	CGTS_L_INV[13]	COS13_L[5:0]						00h	
		1	-	CGTS_R_INV[13]	COS13_R[5:0]						00h	
		1	-	CGTS_L_INV[14]	COS14_L[5:0]						00h	
		1	-	CGTS_R_INV[14]	COS14_R[5:0]						00h	
		1	-	CGTS_L_INV[15]	COS15_L[5:0]						00h	
		1	-	CGTS_R_INV[15]	COS15_R[5:0]						00h	
		1	-	CGTS_L_INV[16]	COS16_L[5:0]						00h	
		1	-	CGTS_R_INV[16]	COS16_R[5:0]						00h	
D6	SETGIP2	0	1	1	0	1	0	1	1	0	-	Set GIP Option2
		1	-	CGOUT_L_HIZ_IN[1]	COS1_L_GS[5:0]						00h	
		1	-	CGOUT_R_HIZ_IN[1]	COS1_R_GS[5:0]						00h	
		1	-	CGOUT_L_HIZ_IN[2]	COS2_L_GS[5:0]						00h	
		1	-	CGOUT_R_HIZ_IN[2]	COS2_R_GS[5:0]						00h	

		1	-	CGOUT_L_HIZ_IN[3]	COS3_L_GS[5:0]						00h	
		1	-	CGOUT_R_HIZ_IN[3]	COS3_R_GS[5:0]						00h	
		1	-	CGOUT_L_HIZ_IN[4]	COS4_L_GS[5:0]						00h	
		1	-	CGOUT_R_HIZ_IN[4]	COS4_R_GS[5:0]						00h	
		1	-	CGOUT_L_HIZ_IN[5]	COS5_L_GS[5:0]						00h	
		1	-	CGOUT_R_HIZ_IN[5]	COS5_R_GS[5:0]						00h	
		1	-	CGOUT_L_HIZ_IN[6]	COS6_L_GS[5:0]						00h	
		1	-	CGOUT_R_HIZ_IN[6]	COS6_R_GS[5:0]						00h	
		1	-	CGOUT_L_HIZ_IN[7]	COS7_L_GS[5:0]						00h	
		1	-	CGOUT_R_HIZ_IN[7]	COS7_R_GS[5:0]						00h	
		1	-	CGOUT_L_HIZ_IN[8]	COS8_L_GS[5:0]						00h	
		1	-	CGOUT_R_HIZ_IN[8]	COS8_R_GS[5:0]						00h	
		1	-	CGOUT_L_HIZ_IN[9]	COS9_L_GS[5:0]						00h	
		1	-	CGOUT_R_HIZ_IN[9]	COS9_R_GS[5:0]						00h	
		1	-	CGOUT_L_HIZ_IN[10]	COS10_L_GS[5:0]						00h	
		1	-	CGOUT_R_HIZ_IN[10]	COS10_R_GS[5:0]						00h	
		1	-	CGOUT_L_HIZ_IN[11]	COS11_L_GS[5:0]						00h	
		1	-	CGOUT_R_HIZ_IN[11]	COS11_R_GS[5:0]						00h	
		1	-	CGOUT_L_HIZ_IN[12]	COS12_L_GS[5:0]						00h	
		1	-	CGOUT_R_HIZ_IN[12]	COS12_R_GS[5:0]						00h	
		1	-	CGOUT_L_HIZ_IN[13]	COS13_L_GS[5:0]						00h	
		1	-	CGOUT_R_HIZ_IN[13]	COS13_R_GS[5:0]						00h	
		1	-	CGOUT_L_HIZ_IN[14]	COS14_L_GS[5:0]						00h	
		1	-	CGOUT_R_HIZ_IN[14]	COS14_R_GS[5:0]						00h	
		1	-	CGOUT_L_HIZ_IN[15]	COS15_L_GS[5:0]						00h	
		1	-	CGOUT_R_HIZ_IN[15]	COS15_R_GS[5:0]						00h	
		1	-	CGOUT_L_HIZ_IN[16]	COS16_L_GS[5:0]						00h	
		1	-	CGOUT_R_HIZ_IN[16]	COS16_R_GS[5:0]						00h	
D8	SETGIP_3	0	1	1	0	1	1	0	0	0	-	Set GIP

		1	INIT_0_SEL_CG OUT1_L[1:0]	INIT_0_SEL_CG OUT2_L[1:0]	INIT_0_SEL_CG OUT3_L[1:0]	INIT_0_SEL_CG OUT4_L[1:0]	00h	Option3
		1	INIT_0_SEL_CG OUT5_L[1:0]	INIT_0_SEL_CG OUT6_L[1:0]	INIT_0_SEL_CG OUT7_L[1:0]	INIT_0_SEL_CG OUT8_L[1:0]	00h	
		1	INIT_0_SEL_CG OUT9_L[1:0]	INIT_0_SEL_CG OUT10_L[1:0]	INIT_0_SEL_CG OUT11_L[1:0]	INIT_0_SEL_CG OUT12_L[1:0]	00h	
		1	INIT_0_SEL_CG OUT13_L[1:0]	INIT_0_SEL_CG OUT14_L[1:0]	INIT_0_SEL_CG OUT15_L[1:0]	INIT_0_SEL_CG OUT16_L[1:0]	00h	
		1	INIT_0_SEL_CG OUT1_R[1:0]	INIT_0_SEL_CG OUT2_R[1:0]	INIT_0_SEL_CG OUT3_R[1:0]	INIT_0_SEL_CG OUT4_R[1:0]	00h	
		1	INIT_0_SEL_CG OUT5_R[1:0]	INIT_0_SEL_CG OUT6_R[1:0]	INIT_0_SEL_CG OUT7_R[1:0]	INIT_0_SEL_CG OUT8_R[1:0]	00h	
		1	INIT_0_SEL_CG OUT9_R[1:0]	INIT_0_SEL_CG OUT10_R[1:0]	INIT_0_SEL_CG OUT11_R[1:0]	INIT_0_SEL_CG OUT12_R[1:0]	00h	
		1	INIT_0_SEL_CG OUT13_R[1:0]	INIT_0_SEL_CG OUT14_R[1:0]	INIT_0_SEL_CG OUT15_R[1:0]	INIT_0_SEL_CG OUT16_R[1:0]	00h	
		1	INIT_1_SEL_CG OUT1_L[1:0]	INIT_1_SEL_CG OUT2_L[1:0]	INIT_1_SEL_CG OUT3_L[1:0]	INIT_1_SEL_CG OUT4_L[1:0]	00h	
		1	INIT_1_SEL_CG OUT5_L[1:0]	INIT_1_SEL_CG OUT6_L[1:0]	INIT_1_SEL_CG OUT7_L[1:0]	INIT_1_SEL_CG OUT8_L[1:0]	00h	
		1	INIT_1_SEL_CG OUT9_L[1:0]	INIT_1_SEL_CG OUT10_L[1:0]	INIT_1_SEL_CG OUT11_L[1:0]	INIT_1_SEL_CG OUT12_L[1:0]	00h	
		1	INIT_1_SEL_CG OUT13_L[1:0]	INIT_1_SEL_CG OUT14_L[1:0]	INIT_1_SEL_CG OUT15_L[1:0]	INIT_1_SEL_CG OUT16_L[1:0]	00h	
		1	INIT_1_SEL_CG OUT1_R[1:0]	INIT_1_SEL_CG OUT2_R[1:0]	INIT_1_SEL_CG OUT3_R[1:0]	INIT_1_SEL_CG OUT4_R[1:0]	00h	
		1	INIT_1_SEL_CG OUT5_R[1:0]	INIT_1_SEL_CG OUT6_R[1:0]	INIT_1_SEL_CG OUT7_R[1:0]	INIT_1_SEL_CG OUT8_R[1:0]	00h	
		1	INIT_1_SEL_CG OUT9_R[1:0]	INIT_1_SEL_CG OUT10_R[1:0]	INIT_1_SEL_CG OUT11_R[1:0]	INIT_1_SEL_CG OUT12_R[1:0]	00h	
		1	INIT_1_SEL_CG OUT13_R[1:0]	INIT_1_SEL_CG OUT14_R[1:0]	INIT_1_SEL_CG OUT15_R[1:0]	INIT_1_SEL_CG OUT16_R[1:0]	00h	
		1	END_0_SEL_CG OUT1_L[1:0]	END_0_SEL_CG OUT2_L[1:0]	END_0_SEL_CG OUT3_L[1:0]	END_0_SEL_CG OUT4_L[1:0]	00h	
		1	END_0_SEL_CG OUT5_L[1:0]	END_0_SEL_CG OUT6_L[1:0]	END_0_SEL_CG OUT7_L[1:0]	END_0_SEL_CG OUT8_L[1:0]	00h	
		1	END_0_SEL_CG OUT9_L[1:0]	END_0_SEL_CG OUT10_L[1:0]	END_0_SEL_CG OUT11_L[1:0]	END_0_SEL_CG OUT12_L[1:0]	00h	
		1	END_0_SEL_CG OUT13_L[1:0]	END_0_SEL_CG OUT14_L[1:0]	END_0_SEL_CG OUT15_L[1:0]	END_0_SEL_CG OUT16_L[1:0]	00h	
		1	END_0_SEL_CG OUT1_R[1:0]	END_0_SEL_CG OUT2_R[1:0]	END_0_SEL_CG OUT3_R[1:0]	END_0_SEL_CG OUT4_R[1:0]	00h	
		1	END_0_SEL_CG OUT5_R[1:0]	END_0_SEL_CG OUT6_R[1:0]	END_0_SEL_CG OUT7_R[1:0]	END_0_SEL_CG OUT8_R[1:0]	00h	
		1	END_0_SEL_CG OUT9_R[1:0]	END_0_SEL_CG OUT10_R[1:0]	END_0_SEL_CG OUT11_R[1:0]	END_0_SEL_CG OUT12_R[1:0]	00h	
		1	END_0_SEL_CG OUT13_R[1:0]	END_0_SEL_CG OUT14_R[1:0]	END_0_SEL_CG OUT15_R[1:0]	END_0_SEL_CG OUT16_R[1:0]	00h	
		1	END_1_SEL_CG OUT1_L[1:0]	END_1_SEL_CG OUT2_L[1:0]	END_1_SEL_CG OUT3_L[1:0]	END_1_SEL_CG OUT4_L[1:0]	00h	
		1	END_1_SEL_CG OUT5_L[1:0]	END_1_SEL_CG OUT6_L[1:0]	END_1_SEL_CG OUT7_L[1:0]	END_1_SEL_CG OUT8_L[1:0]	00h	
		1	END_1_SEL_CG OUT9_L[1:0]	END_1_SEL_CG OUT10_L[1:0]	END_1_SEL_CG OUT11_L[1:0]	END_1_SEL_CG OUT12_L[1:0]	00h	
		1	END_1_SEL_CG OUT13_L[1:0]	END_1_SEL_CG OUT14_L[1:0]	END_1_SEL_CG OUT15_L[1:0]	END_1_SEL_CG OUT16_L[1:0]	00h	
		1	END_1_SEL_CG OUT1_R[1:0]	END_1_SEL_CG OUT2_R[1:0]	END_1_SEL_CG OUT3_R[1:0]	END_1_SEL_CG OUT4_R[1:0]	00h	
		1	END_1SEL_CG OUT5_R[1:0]	END_1_SEL_CG OUT6_R[1:0]	END_1_SEL_CG OUT7_R[1:0]	END_1_SEL_CG OUT8_R[1:0]	00h	
		1	END_1_SEL_CG OUT9_R[1:0]	END_1_SEL_CG OUT10_R[1:0]	END_1_SEL_CG OUT11_R[1:0]	END_1_SEL_CG OUT12_R[1:0]	00h	
		1	END_1_SEL_CG OUT13_R[1:0]	END_1_SEL_CG OUT14_R[1:0]	END_1_SEL_CG OUT15_R[1:0]	END_1_SEL_CG OUT16_R[1:0]	00h	
		1	END_2_SEL_CG OUT1_L[1:0]	END_2_SEL_CG OUT2_L[1:0]	END_2_SEL_CG OUT3_L[1:0]	END_2_SEL_CG OUT4_L[1:0]	00h	
		1	END_2_SEL_CG OUT5_L[1:0]	END_2_SEL_CG OUT6_L[1:0]	END_2_SEL_CG OUT7_L[1:0]	END_2_SEL_CG OUT8_L[1:0]	00h	
		1	END_2_SEL_CG OUT9_L[1:0]	END_2_SEL_CG OUT10_L[1:0]	END_2_SEL_CG OUT11_L[1:0]	END_2_SEL_CG OUT12_L[1:0]	00h	
		1	END_2_SEL_CG OUT13_L[1:0]	END_2_SEL_CG OUT14_L[1:0]	END_2_SEL_CG OUT15_L[1:0]	END_2_SEL_CG OUT16_L[1:0]	00h	
		1	END_2_SEL_CG OUT1_R[1:0]	END_2_SEL_CG OUT2_R[1:0]	END_2_SEL_CG OUT3_R[1:0]	END_2_SEL_CG OUT4_R[1:0]	00h	

		1	END_2_SEL_CG OUT5_R[1:0]	END_2_SEL_CG OUT6_R[1:0]	END_2_SEL_CG OUT7_R[1:0]	END_2_SEL_CG OUT8_R[1:0]	00h	
		1	END_2_SEL_CG OUT9_R[1:0]	END_2_SEL_CG OUT10_R[1:0]	END_2_SEL_CG OUT11_R[1:0]	END_2_SEL_CG OUT12_R[1:0]	00h	
		1	END_2_SEL_CG OUT13_R[1:0]	END_2_SEL_CG OUT14_R[1:0]	END_2_SEL_CG OUT15_R[1:0]	END_2_SEL_CG OUT16_R[1:0]	00h	
		1	GAS_0_SEL_CG OUT1_L[1:0]	GAS_0_SEL_CG OUT2_L[1:0]	GAS_0_SEL_CG OUT3_L[1:0]	GAS_0_SEL_CG OUT4_L[1:0]	00h	
		1	GAS_0_SEL_CG OUT5_L[1:0]	GAS_0_SEL_CG OUT6_L[1:0]	GAS_0_SEL_CG OUT7_L[1:0]	GAS_0_SEL_CG OUT8_L[1:0]	00h	
		1	GAS_0_SEL_CG OUT9_L[1:0]	GAS_0_SEL_CG OUT10_L[1:0]	GAS_0_SEL_CG OUT11_L[1:0]	GAS_0_SEL_CG OUT12_L[1:0]	00h	
		1	GAS_0_SEL_CG OUT13_L[1:0]	GAS_0_SEL_CG OUT14_L[1:0]	GAS_0_SEL_CG OUT15_L[1:0]	GAS_0_SEL_CG OUT16_L[1:0]	00h	
		1	GAS_0_SEL_CG OUT1_R[1:0]	GAS_0_SEL_CG OUT2_R[1:0]	GAS_0_SEL_CG OUT3_R[1:0]	GAS_0_SEL_CG OUT4_R[1:0]	00h	
		1	GAS_0_SEL_CG OUT5_R[1:0]	GAS_0_SEL_CG OUT6_R[1:0]	GAS_0_SEL_CG OUT7_R[1:0]	GAS_0_SEL_CG OUT8_R[1:0]	00h	
		1	GAS_0_SEL_CG OUT9_R[1:0]	GAS_0_SEL_CG OUT10_R[1:0]	GAS_0_SEL_CG OUT11_R[1:0]	GAS_0_SEL_CG OUT12_R[1:0]	00h	
		1	GAS_0_SEL_CG OUT13_R[1:0]	GAS_0_SEL_CG OUT14_R[1:0]	GAS_0_SEL_CG OUT15_R[1:0]	GAS_0_SEL_CG OUT16_R[1:0]	00h	

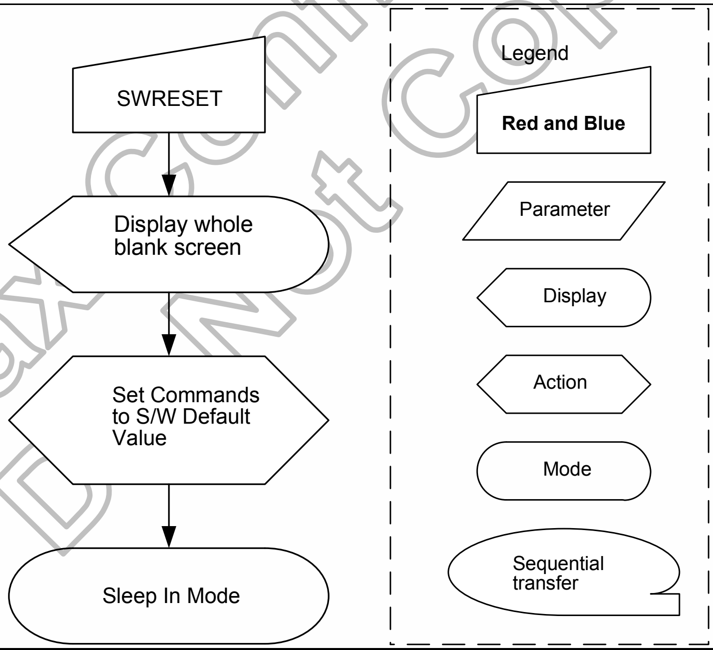
(Hex)	Operation Code	DCX	D7	D6	D5	D4	D3	D2	D1	D0	Default	Function
D9	SETGPO	0	1	1	0	1	1	0	0	1	-	Set GPO
		1	-	-	-	-	TE_GPO[3:0]				00h	
		1	-	-	-	-	TE1_GPO[3:0]				01h	
		1	-	-	-	-	CABC_GPO[3:0]				02h	
		1	-	-	-	-	SDO_GPO[3:0]				07h	
E0	SETGAMMA	0	1	1	1	0	0	0	0	0	-	Set Gamma
		1	-	-	VRP0[5:0]						04h	
		1	-	-	VRP1[5:0]						0Ch	
		1	-	-	VRP2[5:0]						0Dh	
		1	-	-	VRP3[5:0]						0Ah	
		1	-	-	VRP4[5:0]						15h	
		1	-	-	VRP5[5:0]						21h	
		1	-	PRP0[6:0]							0Dh	
		1	-	PRP1[6:0]							19h	
		1	-	-	-	PKP0[4:0]					0Ah	
		1	-	-	-	PKP1[4:0]					0Ch	
		1	-	-	-	PKP2[4:0]					0Fh	
		1	-	-	-	PKP3[4:0]					13h	
		1	-	-	-	PKP4[4:0]					1Ah	
		1	-	-	-	PKP5[4:0]					14h	
		1	-	-	-	PKP6[4:0]					15h	
		1	-	-	-	PKP7[4:0]					0Dh	
		1	-	-	-	PKP8[4:0]					13h	
		1	-	-	-	PKP_EX0[4:0]					0Ah	
		1	-	-	-	PKP_EX1[4:0]					16h	
		1	-	-	-	PKP_EX2[4:0]					06h	
		1	-	-	-	PKP_EX3[4:0]					0Fh	
		1	-	-	VRN0[5:0]						04h	
		1	-	-	VRN1[5:0]						0Ch	
		1	-	-	VRN2[5:0]						0Dh	
		1	-	-	VRN3[5:0]						0Ah	
		1	-	-	VRN4[5:0]						15h	
		1	-	-	VRN5[5:0]						21h	
		1	-	PRN0[6:0]							0Dh	
		1	-	PRN1[6:0]							19h	
		1	-	-	-	PKN0[4:0]					06h	
		1	-	-	-	PKN1[4:0]					0Ch	
		1	-	-	-	PKN2[4:0]					0Fh	
		1	-	-	-	PKN3[4:0]					13h	
		1	-	-	-	PKN4[4:0]					16h	
		1	-	-	-	PKN5[4:0]					14h	
		1	-	-	-	PKN6[4:0]					15h	
		1	-	-	-	PKN7[4:0]					0Dh	
		1	-	-	-	PKN8[4:0]					13h	
		1	-	-	-	PKN_EX0[4:0]					07h	
		1	-	-	-	PKN_EX1[4:0]					16h	
		1	-	-	-	PKN_EX2[4:0]					06h	
		1	-	-	-	PKN_EX3[4:0]					11h	
E4	SETCHEMODE	0	1	1	1	0	0	1	0	0	-	Set color enhancement mode
		1	-	-	-	-	-	-	-	DYN_CEH_EN	01h	
		1	HUE_MODE[1:0]		SE_MODE[1:0]		BE_MODE[1:0]		CE_MODE[1:0]		00h	
E8	SETI2C_SA	0	1	1	1	0	1	0	0	0	-	Set I2C slave address
		1	-	I2C_SA[6:0]							00h	
FD	SETCNCD/ GETCNCD	0	1	1	1	1	1	1	0	1	-	Set/Get Continue Command
		1	WR_CMD_CN[7:0]								-	
FE	SET SPIREAD INDEX	0	1	1	1	1	1	1	1	0	-	SET SPI read Command Address
		1	CMD_ADD[7:0]								-	
FF	GETSPIREAD	0	1	1	1	1	1	1	1	1	-	Read SPI Command Data
		1	CMD_DATA1[7:0]								-	
		1	:								-	
		1	CMD_DATAN[7:0]								-	

6.2 Command description

6.2.1 NOP (00h)

00H	NOP (No Operation)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	0	0	0	0	0	00
Parameter	NO PARAMETER									
Description	This command is an empty command; it does not have any effect on the display module.									
Restriction	-									
Register Availability	Status: Normal Mode On, Idle Mode Off, Sleep Out — Availability: Yes Status: Normal Mode On, Idle Mode On, Sleep Out — Availability: Yes Status: Sleep In or Booster Off — Availability: Yes									
Default	N/A									
Flow Chart	-									

6.2.2 Software reset (01h)

01H	SWRESET (Software Reset)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	0	0	0	0	1	01
Parameter	NO PARAMETER									
Description	When the Software Reset command is written, it causes a software reset. It resets the commands and parameters to their S/W Reset default values. (See default tables in each command description.)									
Restriction	It will be necessary to wait 5msec before sending new command following software reset. The display module loads all display supplier's factory default values to the registers during this 5msec. If Software Reset is applied during Sleep Out mode, it will be necessary to wait 120msec before sending Sleep out command. Software Reset Command cannot be sent during Sleep Out sequence. The host processor needs continuing to send PCLK, HSYNC, VSYNC and DE signals to HX8399-A for two frames after this command is sent.									
Register Availability	Status / Availability: Normal Mode On, Idle Mode Off, Sleep Out: Yes Normal Mode On, Idle Mode On, Sleep Out: Yes Sleep In or Booster Off: Yes									
Default	N/A									
Flow Chart										

6.2.3 Read Display Identification Information (04h)

04H	RDDIDIF (Read Display Identification Information)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	0	0	1	0	0	04
1st parameter	1	ID1[7:0]								83
2nd parameter	1	ID2[7:0]								99
3rd parameter	1	ID3[7:0]								0A

Description

This read byte returns 24-bit display identification information. The 1st Parameter identifies the LCD module's manufacturer. It is specified by display supplier and for xx is defined as xxHEX. The 2nd Parameter has 2 purposes. Bit7 (MSB) defines the type of panel. 0=Driver (STN B/W), 1=Module (Colour). Bits 6..0 are used to track the LCD module/driver version. It is defined by display supplier and it changes each time a revision is made to the display, material or construction specifications. See Table:

ID Byte Value V[7:0]	Version	Changes
80h	--	--
81h	--	--
82h	--	--
83h	--	--
84h	--	--
85h	--	--

The 3rd parameter identifies the LCD module/driver. It is specified by display supplier and for this LCD project module is defined as xxHEX.

Restriction

-

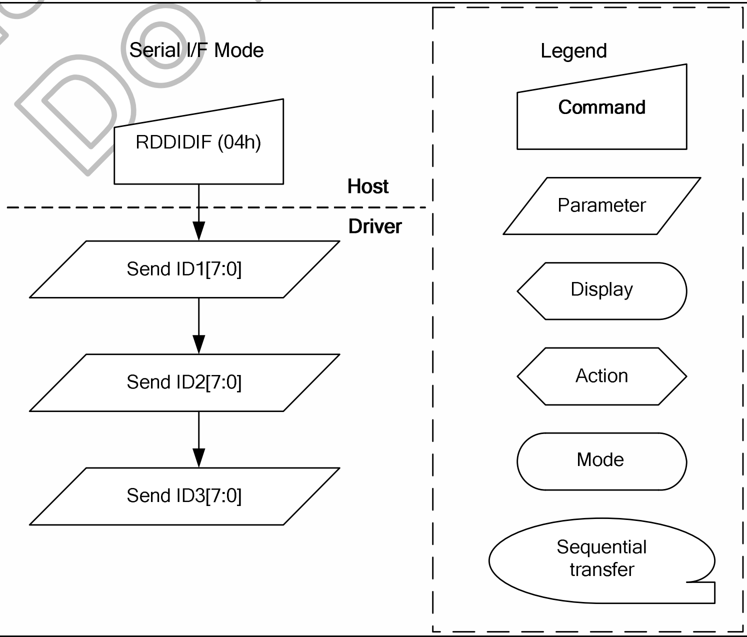
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

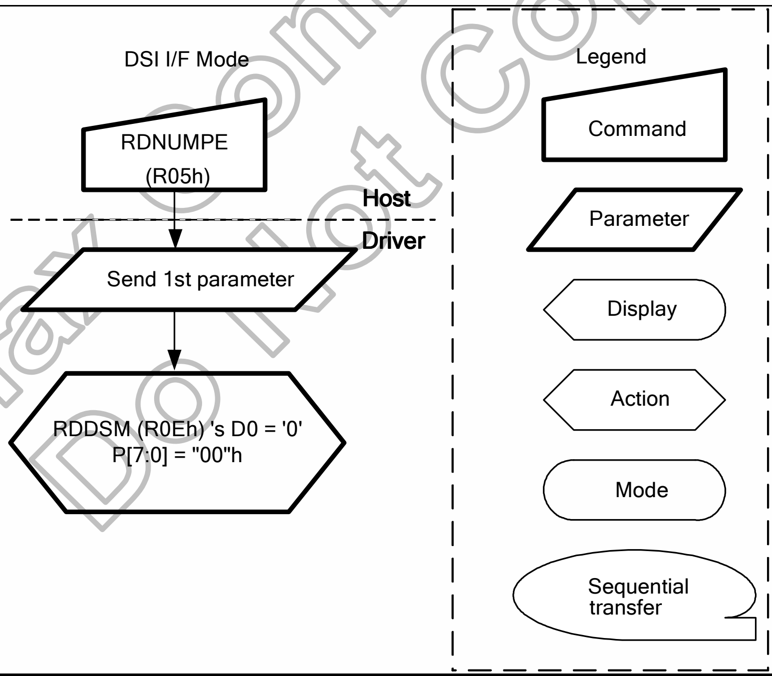
Default

Status	Default Value
Power On Sequence	OTP Value
S/W Reset	OTP Value
H/W Reset	OTP Value

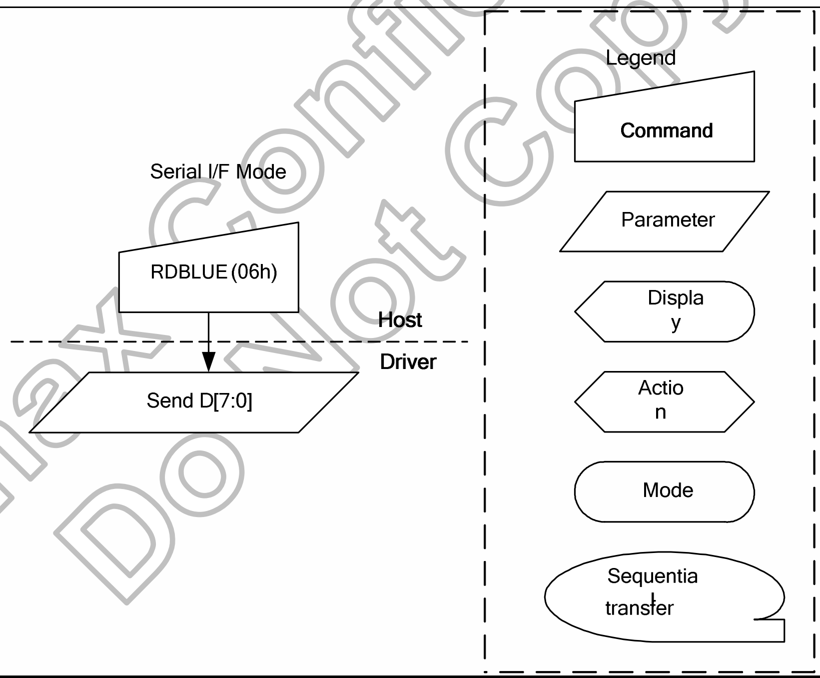
Flow Chart



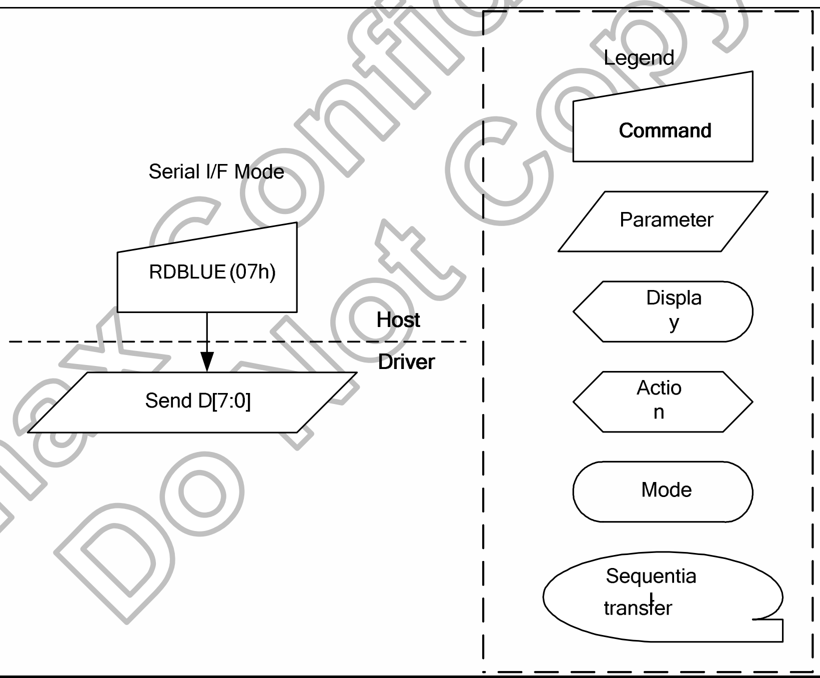
6.2.4 RDNUMPE: Read number of the parity errors (05h)

<table>
<tr><th rowspan="2">05H</th><th colspan="10">RDNUMPE (Read Number of the Parity Errors)</th></tr>
<tr><th>DCX</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>05</td></tr>
<tr><td>1st parameter</td><td>1</td><td>P7</td><td>P6</td><td>P5</td><td>P4</td><td>P3</td><td>P2</td><td>P1</td><td>P0</td><td>00</td></tr>
<tr><td>Description</td><td colspan="10">The first parameter is telling a number of the errors on DSI. The more detailed description of the bits is below.
P[6..0] bits are telling a number of the errors.
P[7] is set to ‘1’ if there is overflow with P[6..0] bits.
P[7..0] bits are set to ‘0’s (as well as RDDSM(0Eh)’s D0 is set ‘0’ at the same time) after there is sent the second parameter information (The read function is completed).</td></tr>
<tr><td>Restriction</td><td colspan="10">SETEXTC turn on to enable this command</td></tr>
<tr><td>Register Availability</td><td colspan="10"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="10"><table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>S/W Reset</td><td>00h</td></tr><tr><td>H/W Reset</td><td>00h</td></tr></table></td></tr>
<tr><td>Flow Chart</td><td colspan="10">
</td></tr>
</table>

6.2.5 Get_red_channel (06h)

<table>
<tr><th rowspan="2">06H</th><th colspan="10">RDRED (Read Red Colour)</th></tr>
<tr><th>DCX</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>06</td></tr>
<tr><td>1st parameter</td><td>1</td><td>R7</td><td>R6</td><td>R5</td><td>R4</td><td>R3</td><td>R2</td><td>R1</td><td>R0</td><td>00</td></tr>
<tr><td>Description</td><td colspan="10">The first parameter is telling red colour value of the first pixel of the frame when there is used DPI I/F.
16 bit format: R5 is MSB and R1 is LSB. R7, R6 and R0 are set to ‘0’.
18 bit format: R5 is MSB and R0 is LSB. R7 and R6 are set to ‘0’.
24 bit format: R7 is MSB and R0 is LSB.</td></tr>
<tr><td>Restriction</td><td colspan="10">-</td></tr>
<tr><td>Register Availability</td><td colspan="10"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="10"><table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>S/W Reset</td><td>00h</td></tr><tr><td>H/W Reset</td><td>00h</td></tr></table></td></tr>
<tr><td>Flow Chart</td><td colspan="10">
</td></tr>
</table>

6.2.6 Get_green_channel (07h)

<table>
<tr><th rowspan="2">07H</th><th colspan="10">RDGREEN (Read Green Colour)</th></tr>
<tr><th>DCX</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>07</td></tr>
<tr><td>1st parameter</td><td>1</td><td>G7</td><td>G6</td><td>G5</td><td>G4</td><td>G3</td><td>G2</td><td>G1</td><td>G0</td><td>00</td></tr>
<tr><td>Description</td><td colspan="10">The first parameter is telling green colour value of the first pixel of the frame when there is used DPI I/F.
16 and 18 bit formats: G5 is MSB and G0 is LSB. G7 and G6 are set to ‘0’.
24 bit format: G7 is MSB and G0 is LSB.</td></tr>
<tr><td>Restriction</td><td colspan="10">-</td></tr>
<tr><td>Register Availability</td><td colspan="10"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="10"><table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>S/W Reset</td><td>00h</td></tr><tr><td>H/W Reset</td><td>00h</td></tr></table></td></tr>
<tr><td>Flow Chart</td><td colspan="10"></td></tr>
</table>

6.2.7 Get_blue_channel (08h)

08H	RDBLUE (Read Blue Colour)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	0	1	0	0	0	08
1st parameter	1	B7	B6	B5	B4	B3	B2	B1	B0	00
Description	The first parameter is telling blue colour value of the first pixel of the frame when there is used DPI I/F. 16 bit format: B5 is MSB and B1 is LSB. B7, B6 and B0 are set to ‘0’. 18 bit format: B5 is MSB and B0 is LSB. B7 and B6 are set to ‘0’. 24 bit format: B7 is MSB and B0 is LSB.									
Restriction	-									

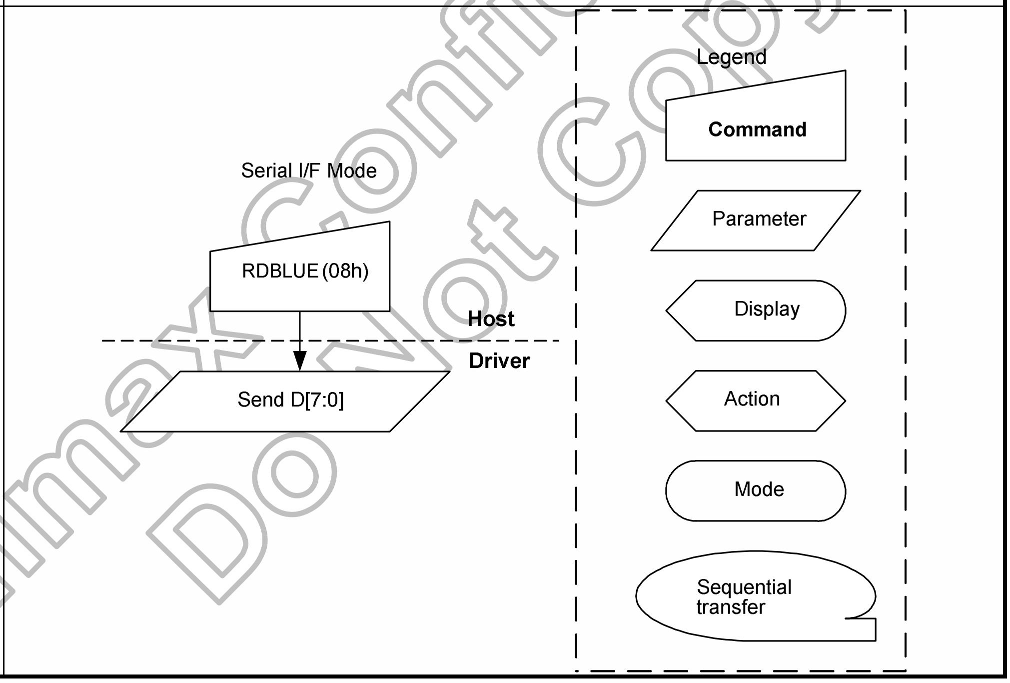
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



6.2.8 Read Display Status (09h)

09H	RDDST (Read Display Status)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	0	1	0	0	1	09
1st parameter	1	D[31:24]								00
2nd parameter	1	D[23:16]								71
3rd parameter	1	D[15:8]								00
4th parameter	1	D[7:0]								00

Description

This command indicates the current status of the display as described in the table below:

Bit	Description	Comment
D31	Booster Voltage Status	-
D30	Page Address Order	-
D29	Column Address Order	-
D28	Page/Column Order	Set to '0'
D27	Line Address Order	Set to '0'
D26	RGB/BGR Order	-
D25	Display Data Latch Order	-
D24	Flip Horizontal	-
D23	Flip Vertical	-
D22	Interface Colour Pixel Format Definition	-
D21		
D20		
D19	Idle Mode On/Off	-
D18	Partial Mode On/Off	Set to '0'
D17	Sleep In/Out	-
D16	Display Normal Mode On/Off	-
D15	Vertical Scrolling Status	Set to '0'
D14	Horizontal Scrolling Status	Set to '0'
D13	Inversion Status	-
D12	All Pixels On	-
D11	All Pixels Off	-
D10	Display On/Off	-
D9	Tearing Effect Line On/Off	-
D8	Gamma Curve Selection	-
D7		
D6		
D5	Tearing Effect Output Line Mode	-
D4	Horizontal Sync. (HSYNC, DPI I/F)	-
D3	Vertical Sync. (VSYNC, DPI I/F)	-
D2	Pixel Clock (DCK, DPI I/F)	-
D1	Data Enable (ENABLE, DPI I/F)	-
D0	Error on DSI	-

Bit Values are explained overleaf.

Bit D31 – Booster Voltage Status
'0' = Booster Off or has a fault.
'1' = Booster On and working OK (Meets display supplier's optical requirements).

Bit D30 – Page Address Order
'0' = Top to Bottom (When MADCTL B7='0').
'1' = Bottom to Top (When MADCTL B7='1').

Bit D29 – Column Address Order
'0' = Left to Right (When MADCTL B6='0').
'1' = Right to Left (When MADCTL B6='1').

Bit D28 – Page/Column Order
'0' = Normal (When MADCTL B5='0').
'1' = Rotation (When MADCTL B5='1').
This bit is not applicable for this project, so it is set to '0'

Bit D27 – Line Address Order
‘0’ = LCD Refresh Top to Bottom (When MADCTL B4=’0’).
‘1’ = LCD Refresh Bottom to Top (When MADCTL B4=’1’).
This bit is not applicable for this project, so it is set to ‘0’
Bit D26 – RGB/BGR Order
‘0’ = RGB (When MADCTL B3=’0’).
‘1’ = BGR (When MADCTL B3=’1’).
Bit D25 – Display Data Latch Order
‘0’ = LCD Refresh Left to Right (When MADCTL B2=’0’).
‘1’ = LCD Refresh Right to Left (When MADCTL B2=’1’).
Bit D24 – Flip Horizontal
‘0’ = Normal (When MADCTL B1=’0’).
‘1’ = Flipped (When MADCTL B1=’1’).
Bit D23 – Flip Vertical
‘0’ = Normal (When MADCTL B0=’0’).
‘1’ = Flipped (When MADCTL B0=’1’).
Bits D22, D21, D20 – Interface Colour Pixel Format Definition

Interface Format	D22	D21	D20
Not Defined	0	0	0
Not Defined	0	0	1
Not Defined	0	1	0
Not Defined	0	1	1
Not Defined	1	0	0
16 Bit/Pixel	1	0	1
18 Bit/Pixel	1	1	0
24 Bit/Pixel	1	1	1

Bit D19 – Idle Mode On/Off
‘0’ = Idle Mode Off.
‘1’ = Idle Mode On.
Bit D18 – Partial Mode On/Off
‘0’ = Partial Mode Off.
‘1’ = Partial Mode On.
This bit is not applicable for this project, so it is set to ‘0’
Bit D17 – Sleep In/Out
‘0’ = Sleep In Mode.
‘1’ = Sleep Out Mode.
Bit D16 – Display Normal Mode On/Off
‘0’ = Display Normal Mode Off.
‘1’ = Display Normal Mode On.
Bit D15 – Vertical Scrolling On/Off
‘0’ = Vertical Scrolling is Off.
‘1’ = Vertical Scrolling is On.
This bit is not applicable for this project, so it is set to ‘0’
Bit D14 – Horizontal Scrolling Status
This bit is not applicable for this project, so it is set to ‘0’
Bit D13 – Inversion On/Off
‘0’ = Inversion is Off.
‘1’ = Inversion is On.
Bit D12 – All Pixels On.
‘0’ = Nornal mode.
‘1’ = All Pixels On.
Bit D11 – All Pixels Off.
0’ = Nornal mode.
‘1’ = All Pixels Off.
Bit D10 – Display On/Off
‘0’ = Display is Off.
‘1’ = Display is On.
Bit D9 – Tearing Effect Line On/Off
‘0’ =Tearing Effect Line Off.
‘1’ = Tearing Effect On.
Bits D8, D7, D6 – Gamma Curve Selection

Gamma Curve Selected	D8	D7	D6	Gamma Set (26h) Parameter
Gamma Curve 1	0	0	0	GC0
Gamma Curve 2	0	0	1	Setting Inhibit
Gamma Curve 3	0	1	0	Setting Inhibit
Gamma Curve 4	0	1	1	Setting Inhibit
Not Defined	1	0	0	Not Defined
Not Defined	1	0	1	Not Defined
Not Defined	1	1	0	Not Defined
Not Defined	1	1	1	Not Defined

Bit D5 – Tearing Effect Line Output Mode.
'0' = Mode 1, V-Blanking only.
'1' = Mode 2, both H-Blanking and V-Blanking.
Bit D4 – Horizontal Sync. (HSYNC) On/Off, Note
'0' = Horizontal Sync. Line is Off ("Low").
'1' = Horizontal Sync. Line is On ("High").
Bit D3 – Vertical Sync. (VSYNC) On/Off, Note
'0' = Vertical Sync. Line is Off ("Low").
'1' = Vertical Sync. Line is On ("High").
Bit D2 – Pixel Clock (PCLK) On/Off, Note
'0' = Pixel Clock line is Off ("Low").
'1' = Pixel Clock line is On ("High").
Bit D1 – Data Enable (DE) On/Off, Note
'0' = Data Enable line is Off ("Low").
'1' = Data Enable line is On ("High").
Bit D0 –Error on DSI
'0' = No Error.
'1' = Error.
Note: This bit indicates current status of the line when this command has been sent.

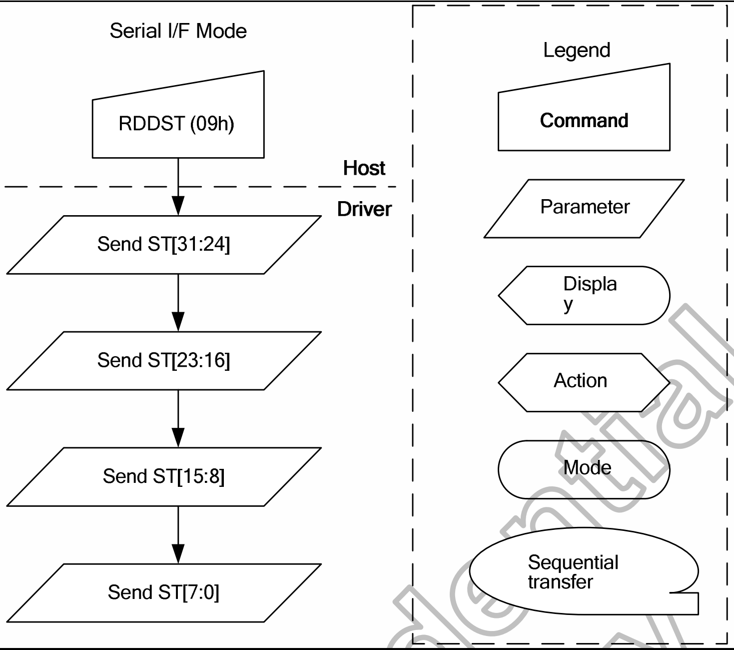
Restriction: -

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	Refer to Description
S/W Reset	Refer to Description
H/W Reset	Refer to Description

Flow Chart	Serial I/F Mode RDDST (09h) Host Driver Send ST[31:24] Send ST[23:16] Send ST[15:8] Send ST[7:0] Legend Command Parameter Displa y Action Mode Sequential transfer

6.2.9 Get_power_mode (0Ah)

0AH	RDDPM (Read Display Power Mode)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	0	1	0	1	0	0A
1st parameter	1	D7	D6	0	D4	D3	D2	0	0	08

Description

This command indicates the current status of the display as described in the table below:

Bit	Description	Comment
D7	Booster Voltage Status	-
D6	Idle Mode On/Off	-
D5	Partial Mode On/Off	Set to '0'
D4	Sleep In/Out	-
D3	Display Normal Mode On/Off	-
D2	Display On/Off	-
D1	Not Defined	Set to '0'
D0	Not Defined	Set to '0'

Bit D7 – Booster Voltage Status
'0' = Booster Off or has a fault.
'1' = Booster On and working OK (Meets display supplier's optical requirements).
Bit D6 – Idle Mode On/Off
'0' = Idle Mode Off.
'1' = Idle Mode On.
Bit D5 – Partial Mode On/Off
'0' = Partial Mode Off.
'1' = Partial Mode On.
This bit is not applicable for this project, so it is set to '0'
Bit D4 – Sleep In/Out
'0' = Sleep In Mode.
'1' = Sleep Out Mode.
Bit D3 – Display Normal Mode On/Off
'0' = Display Normal Mode Off.
'1' = Display Normal Mode On.
Bit D2 – Display On/Off
'0' = Display is Off.
'1' = Display is On.

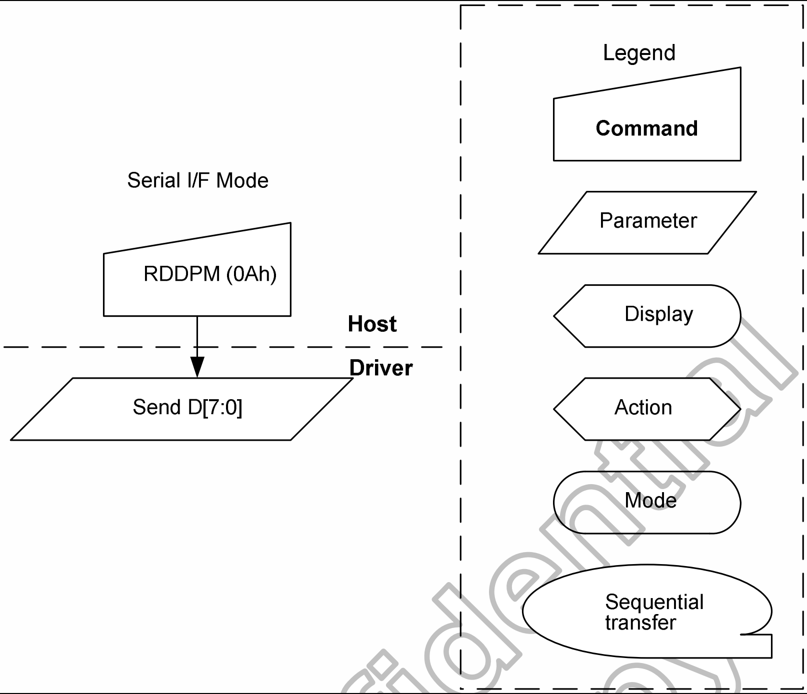
Restrictions: -

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	08h
S/W Reset	08h
H/W Reset	08h

Flow Chart	Serial I/F Mode
RDDPM (0Ah)
Host
Driver
Send D[7:0]
Legend
Command
Parameter
Display
Action
Mode
Sequential transfer

6.2.10 Read display MADCTL (0Bh)

0BH	RDDMADCTL (Read Display MADCTL)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	0	1	0	1	1	0B
1st parameter	1	D7	D6	0	0	D3	D2	D1	D0	00

Description

This command indicates the current status of the display as described in the table below:

Bit	Description	Comment
D7	Page Address Order	-
D6	Column Address Order	-
D5	Page/Column Order	Set to '0'
D4	Line Address Order	Set to '0'
D3	RGB/BGR Order	-
D2	Display Data Latch Order	-
D1	Flip Horizontal	-
D0	Flip Vertical	-

Bit D7 – Page Address Order
'0' = Top to Bottom (When MADCTL B7='0').
'1' = Bottom to Top (When MADCTL B7='1').
Bit D6 – Column Address Order
'0' = Left to Right (When MADCTL B6='0').
'1' = Right to Left (When MADCTL B6='1').
Bit D5 – Page/Column Order
'0' = Normal (When MADCTL B5='0').
'1' = Rotation (When MADCTL B5='1').
This bit is not applicable for this project, so it is set to '0'
Bit D4 – Line Address Order
'0' = LCD Refresh Top to Bottom (When MADCTL B4='0').
'1' = LCD Refresh Bottom to Top (When MADCTL B4='1').
This bit is not applicable for this project, so it is set to '0'
Bit D3 – RGB/BGR Order
'0' = RGB (When MADCTL B3='0').
'1' = BGR (When MADCTL B3='1').
Bit D2 – Display Data Latch Order
'0' = LCD Refresh Left to Right (When MADCTL B2='0').
'1' = LCD Refresh Right to Left (When MADCTL B2='1').
Bit D1 – Flip Horizontal
'0' = Normal (When MADCTL B1='0').
'1' = Flipped (When MADCTL B1='1').
Bit D0 – Flip Vertical
'0' = Normal (When MADCTL B0='0').

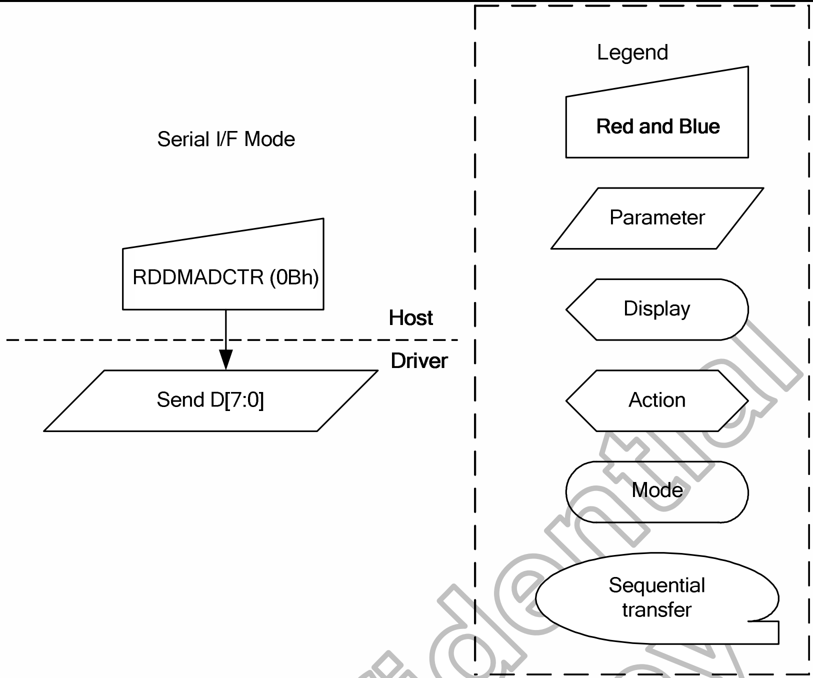
Restrictions: -

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	No Change
H/W Reset	00h

Flow Chart	

Serial I/F Mode
RDDMADCTR (0Bh)
Host
Driver
Send D[7:0]
Legend
Red and Blue
Parameter
Display
Action
Mode
Sequential transfer

6.2.11 Get_pixel_format (0Ch)

0CH	RDDCOLMOD (Read Display COLMOD)								
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	0	1	1	0	0	0C
1st parameter	1	0	D6	D5	D4	0	0	0	0	70

Description

This command indicates the current status of the display as described in the table below:

Bit	Description	Comment
D7	Reserved	Set to '0'
D6	DPI Interface Pixel format	-
D5		-
D4		-
D3	Reserved	Set to '0'
D2	Reserved	Set to '0'
D1	Reserved	Set to '0'
D0	Reserved	Set to '0'

Bits D6, D5, D4 – DPI Interface Colour Pixel Format Definition
Bits D2, D1, D0 – DBI Interface Colour Pixel Format Definition. These bits are not applicable for this project, so it is set to "0".

Interface Colour Format	D6	D5	D4
Not Defined	0	0	0
Not Defined	0	0	1
Not Defined	0	1	0
Not Defined	0	1	1
Not Defined	1	0	0
16 bit/pixel	1	0	1
18 bit/pixel	1	1	0
24 bit/pixel	1	1	1

If the setting is not used then the corresponding bits in the parameter returned from

Restrictions

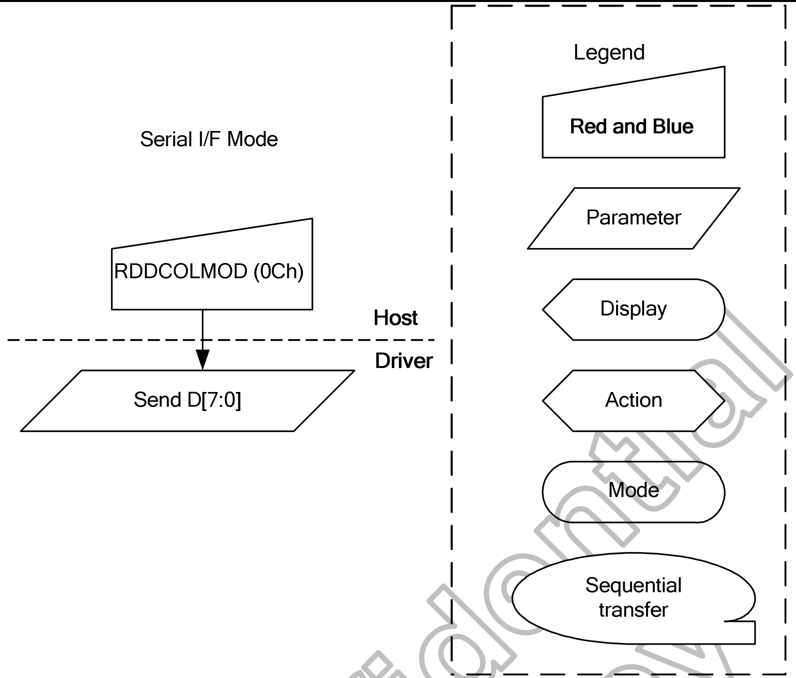
-

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	70h
S/W Reset	No Change
H/W Reset	70h

Flow Chart	 Serial I/F Mode RDDCOLMOD (0Ch) Host Driver Send D[7:0] Legend Red and Blue Parameter Display Action Mode Sequential transfer

6.2.12 Get_display_mode (0Dh)

0DH	RDDIM (Read Display Image Mode)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	0	1	1	0	1	0D
1st parameter	1	0	0	D5	D4	D3	D2	D1	D0	00

Description

This command indicates the current status of the display as described in the table below:

Bit D7 – Vertical Scrolling On/Off
This bit is not applicable for this project, so it is set to '0'

Bit D6 – Horizontal Scrolling Status
This bit is not applicable for this project, so it is set to '0'

Bit D5 – Inversion On/Off
'0' = Inversion is Off.
'1' = Inversion is On.

Bit D4 – All Pixels On
'0' = Normal Display
'1' = White Display

Bit D3 – All Pixels Off
'0' = Normal Display
'1' = Black Display

Bits D2, D1, D0 – Gamma Curve Selection

Gamma Curve Selected	D2	D1	D0	Gamma Set (26h) Parameter
Gamma Curve1	0	0	0	GC0
Gamma Curve2	0	0	1	Setting Inhibit
Gamma Curve3	0	1	0	Setting Inhibit
Gamma Curve4	0	1	1	Setting Inhibit
Not Defined	1	0	0	Not Defined
Not Defined	1	0	1	Not Defined
Not Defined	1	1	0	Not Defined
Not Defined	1	1	1	Not Defined

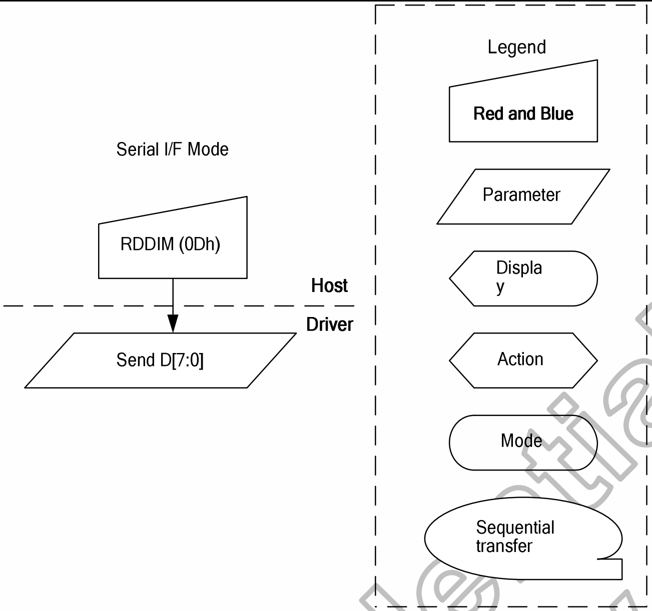
Restrictions: -

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart	

Serial I/F Mode
RDDIM (0Dh)
Host
Driver
Send D[7:0]
Legend
Red and Blue
Parameter
Displa
y
Action
Mode
Sequential
transfer

6.2.13 Get_signal_mode (0Eh)

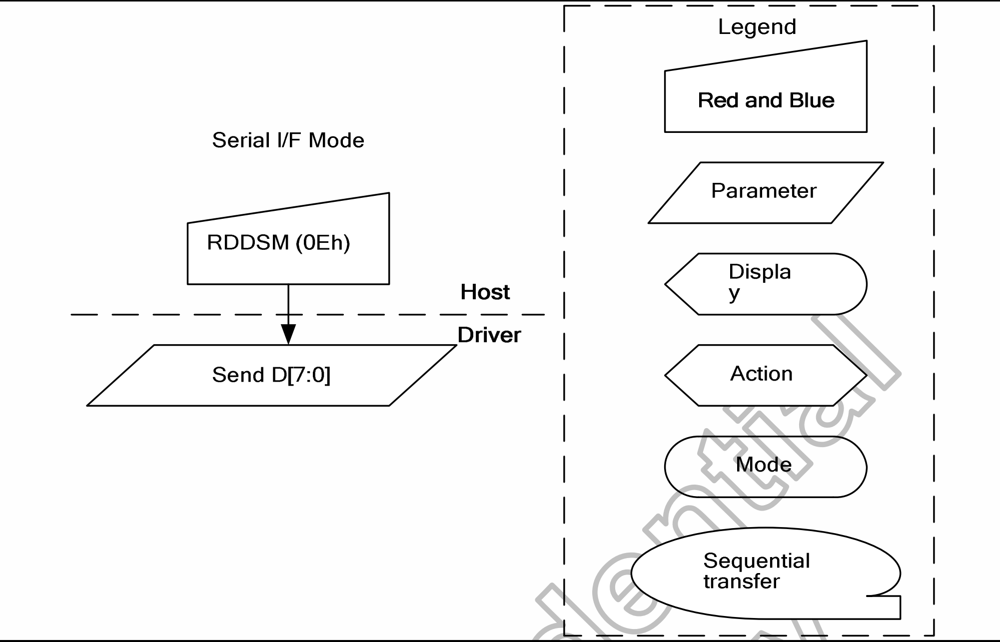
0EH	RDDSM (Read Display Signal Mode)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	0	1	1	1	0	0E
1st parameter	1	D7	D6	D5	D4	D3	D2	0	D0	00
Description	This command indicates the current status of the display described in the table as below: **Bit D7 – Tearing Effect Line On/Off** '0' = Tearing Effect Line Off. '1' = Tearing Effect On. **Bit D6 – Tearing Effect Line Output Mode.** '0' = Mode 1, V-Blanking only. '1' = Mode 2, both H-Blanking and V-Blanking. **Bit D5 – Horizontal Sync. (DPI I/F) On/Off.** '0' = Horizontal Sync. Line is Off ("Low"). '1' = Horizontal Sync. Line is On ("High"). **Bit D4 – Vertical Sync. (DPI I/F) On/Off.** '0' = Vertical Sync. Line is Off ("Low"). '1' = Vertical Sync. Line is On ("High"). **Bit D3 – Pixel Clock (PCLK, DPI I/F) On/Off.** '0' = PCLK line is Off ("Low"). '1' = PCLK line is On ("High"). **Bit D2 – Data Enable (DE, DPI I/F) On/Off.** '0' = DE line is Off ("Low"). '1' = DE line is On ("High"). **Bit D1 – Reserved.** **Bit D0 –Error on DSI** '0' = No Error. '1' = Error.									
Restrictions	-									
Register Availability	Status / Availability: Normal Mode On, Idle Mode Off, Sleep Out — Yes Normal Mode On, Idle Mode On, Sleep Out — Yes Sleep In or Booster Off — Yes									
Default	Status / Default Value: Power On Sequence — 00h S/W Reset — 00h H/W Reset — 00h									

Register Availability:

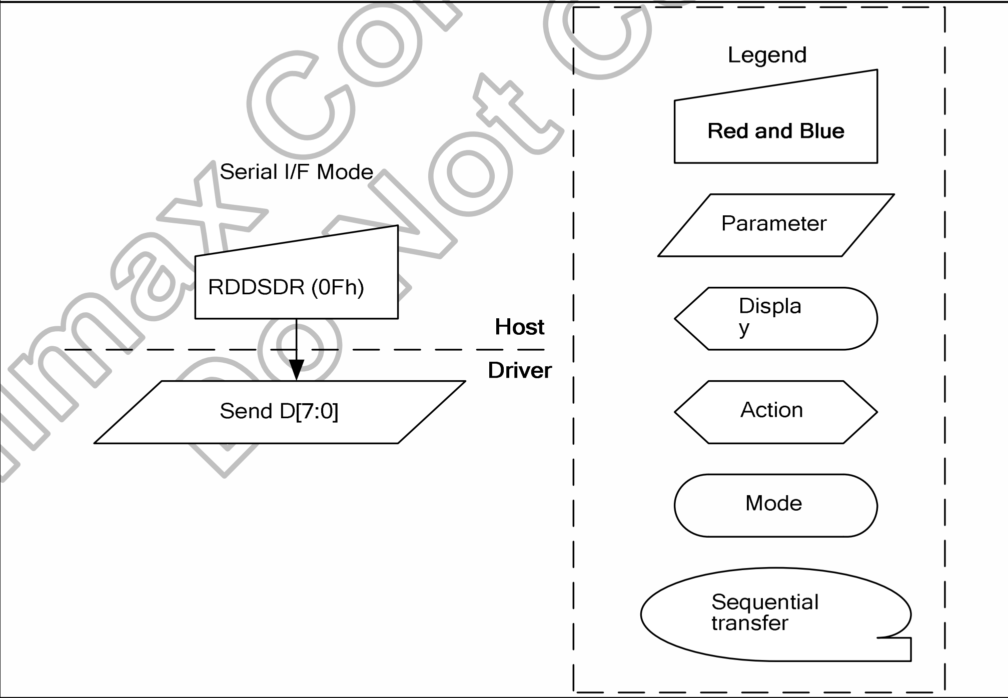
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default:

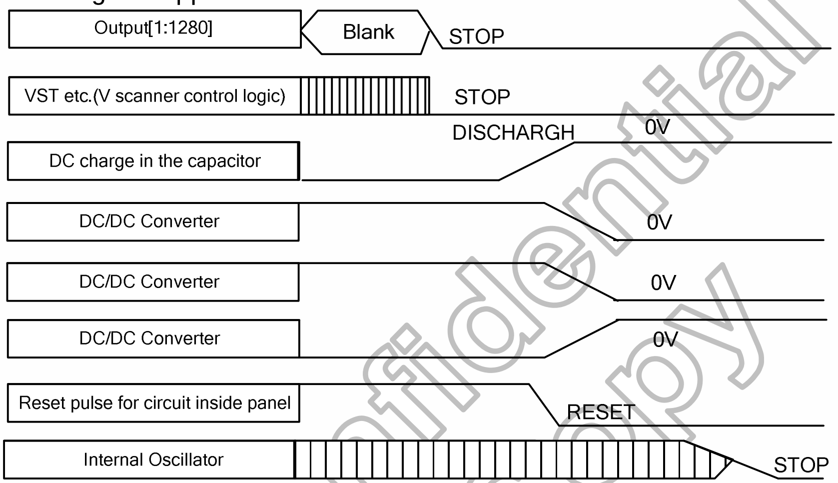
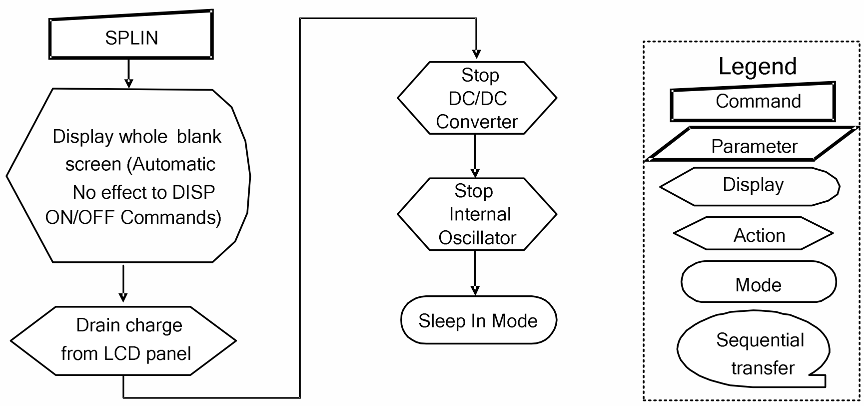
Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart	 Serial I/F Mode RDDSM (0Eh) Host Driver Send D[7:0] Legend Red and Blue Parameter Displa y Action Mode Sequential transfer

6.2.14 Get_diagnostic_result (0Fh)

0FH	RDDSDR (Read Display Self-Diagnostic Result)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	0	1	1	1	1	0F
1st parameter	1	D7	D6	D5	D4	0	0	0	0	00
Description	The display module returns the self-diagnostic results following a Sleep Out command. **Bit D7 – Register Loading Detection** **Bit D6 – Functionality Detection** **Bit D5 – Chip Attachment Detection** Set to '0' if feature unimplemented. **Bit D4 – Display Glass Break Detection** Set to '0' if feature unimplemented. Bits D[3:0] – Reserved.									
Restrictions	-									
Register Availability	Status / Availability: Normal Mode On, Idle Mode Off, Sleep Out: Yes Normal Mode On, Idle Mode On, Sleep Out: Yes Sleep In or Booster Off: Yes									
Default	Status / Default Value: Power On Sequence: 00h S/W Reset: 00h H/W Reset: 00h									
Flow Chart										

6.2.15 Enter_sleep_mode (10h)

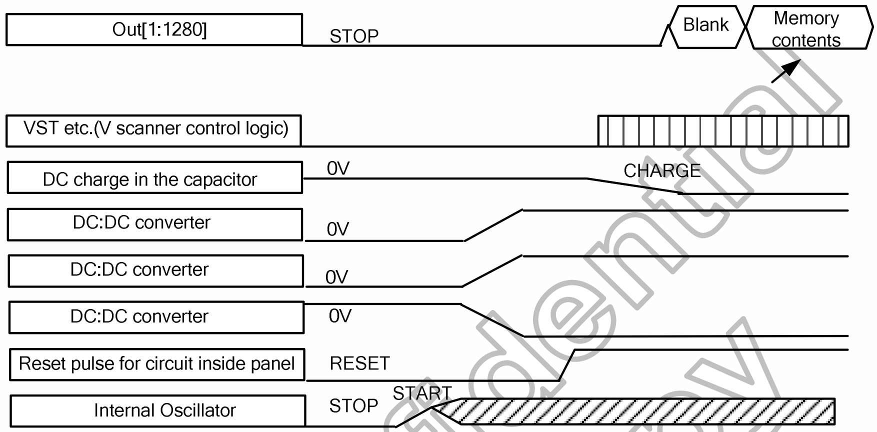
10H	SLPIN (Sleep In)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	1	0	0	0	0	10
Parameter	NO PARAMETER									
Description	This command causes the LCD module to enter the minimum power consumption mode. In this mode the DC/DC converter is stopped, Internal oscillator is stopped, and panel scanning is stopped.  									
Restriction	This command has no effect when module is already in sleep in mode.Sleep In Mode can only be left by the Sleep Out Command (11h). It will be necessary to wait 5msec before sending next command, this is to allow time for the supply voltages and clock circuits to stabilize. It will be necessary to wait 120msec after sending Sleep Out command (when in Sleep In Mode) before Sleep In command can be sent. The host processor continues to send PCLK, HSYNC, and VSYNC and DE signals to HX8399-A for two frames after this command is sent.									
Register Availability	Status: Availability Normal Mode On, Idle Mode Off, Sleep Out: Yes Normal Mode On, Idle Mode On, Sleep Out: Yes Sleep In or Booster Off: Yes									
Default	N/A									
Flow Chart	It takes 120msec to get into Sleep In mode after SLPIN command issued.  									

6.2.16 Exit_sleep_omde (11h)

11H	SLPOUT (Sleep Out)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	1	0	0	0	1	11
Parameter	NO PARAMETER									

Description

This command turns off sleep mode. In this mode the DC/DC converter is enabled, Internal oscillator is started, and panel scanning is started.



Restriction

This command has no effect when module is already in sleep out mode. Sleep Out Mode can only be left by the Sleep In Command (10h). It will be necessary to wait 5msec before sending next command, this is to allow time for the supply voltages and clock circuits to stabilize. The display module loads all display supplier's factory default values to the registers during this 5msec and there cannot be any abnormal visual effect on the display image if factory default and register values are same when this load is done and when the display module is already Sleep Out –mode. The display module is doing self-diagnostic functions during this 5msec. It will be necessary to alit 120msec after sending Sleep In command (when in Sleep Out mode) before Sleep Out command can be sent.
The host processor continues to send PCLK, HSYNC, and VSYNC and DE signals to HX8399-A for two frames after this command is sent.

Register Availability

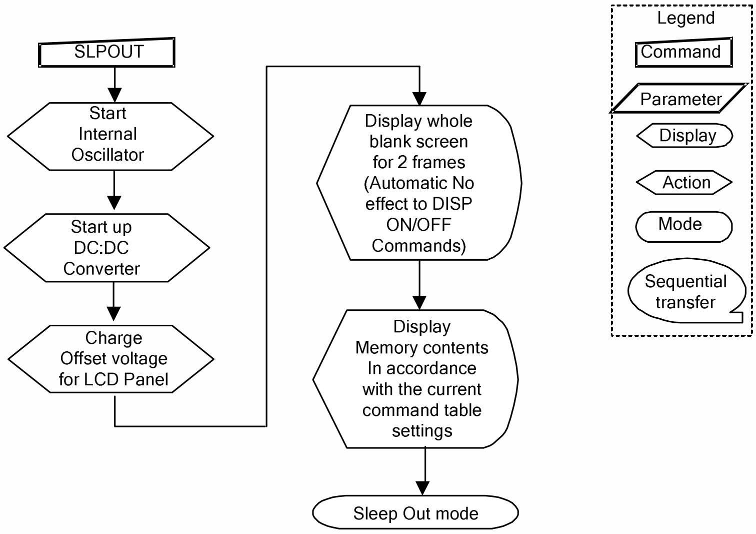
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

N/A

Flow Chart

It takes 120msec to become Sleep Out mode after SLPOUT command issued.



6.2.17 Enter_normal_mode (13h)

13H	NORON (Normal Display Mode On)								
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	1	0	0	1	1	13
Parameter	NO PARAMETER									
Description	This command returns the display to normal mode.									
Restriction	This command has no effect when Normal Display mode is active.									
Register Availability	Status: Normal Mode On, Idle Mode Off, Sleep Out — Availability: Yes Status: Normal Mode On, Idle Mode On, Sleep Out — Availability: Yes Status: Sleep In or Booster Off — Availability: Yes									
Default	Status: Power On Sequence — Default Value: Normal display mode Status: S/W Reset — Default Value: Normal display mode Status: H/W Reset — Default Value: Normal display mode									
Flow Chart	-									

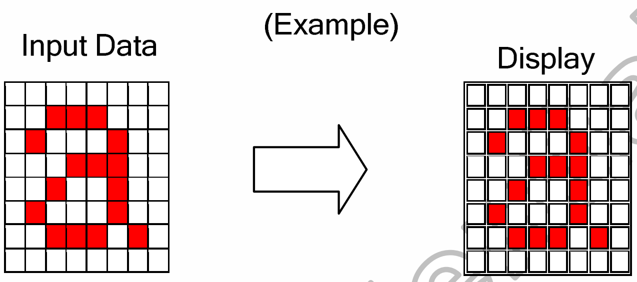
Register Availability:

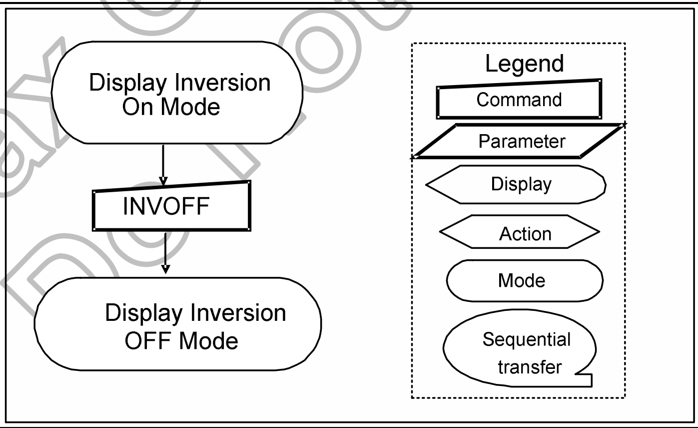
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default:

Status	Default Value
Power On Sequence	Normal display mode
S/W Reset	Normal display mode
H/W Reset	Normal display mode

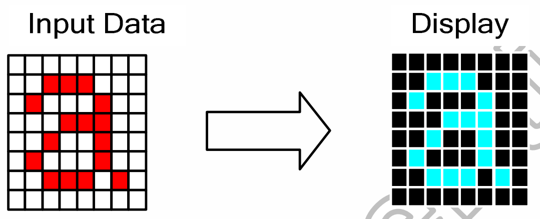
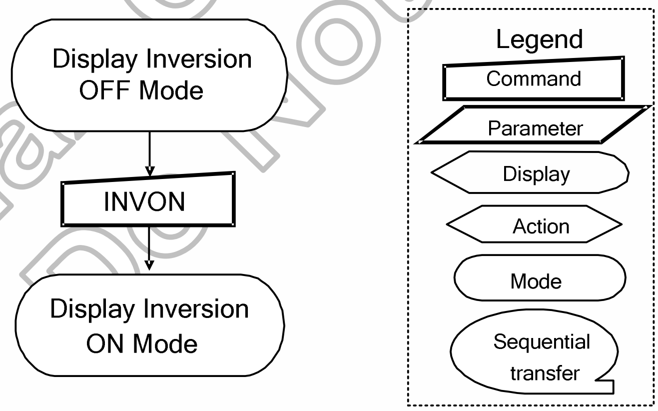
6.2.18 Exit_inversion_mode (20h)

<table>
<tr><th rowspan="2">20H</th><th colspan="10">INVOFF (Display Inversion Off)</th></tr>
<tr><th>DCX</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>20</td></tr>
<tr><td>Parameter</td><td colspan="10">NO PARAMETER</td></tr>
<tr><td>Description</td><td colspan="10">This command is used to recover from display inversion mode.
This command does not change any other status.


</td></tr>
<tr><td>Restriction</td><td colspan="10">This command has no effect when module is already in inversion off mode.</td></tr>
<tr><td>Register Availability</td><td colspan="10">
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In or Booster Off</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td colspan="10">
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>Normal display mode</td></tr>
<tr><td>S/W Reset</td><td>Normal display mode</td></tr>
<tr><td>H/W Reset</td><td>Normal display mode</td></tr>
</table>
</td></tr>
<tr><td>Flow Chart</td><td colspan="10">


</td></tr>
</table>

6.2.19 Enter_inversion_mode (21h)

21H	INVON (Display Inversion On)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	1	0	0	0	0	1	21
Parameter	NO PARAMETER									
Description	This command is used to enter into display inversion mode. This command does not change any other status. (Example) 									
Restriction	This command has no effect when module is already in inversion on mode.									
Register Availability	See table below									
Default	See table below									
Flow Chart										

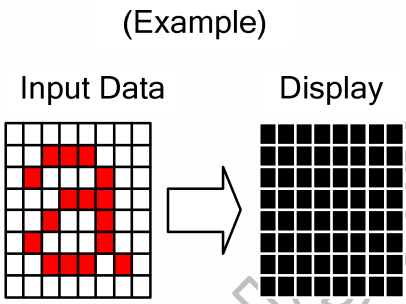
Register Availability:

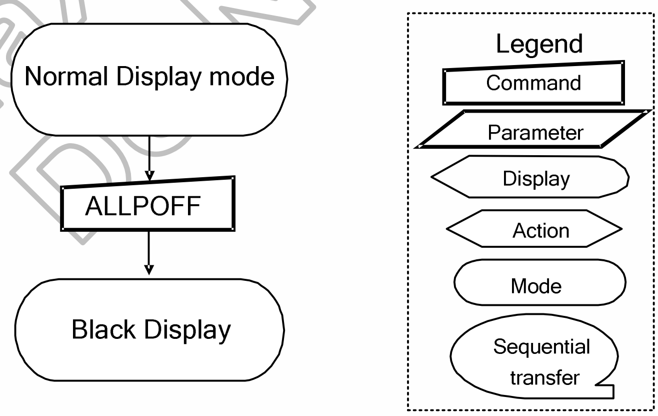
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default:

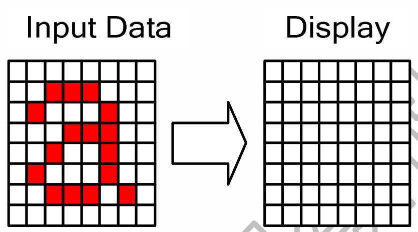
Status	Default Value
Power On Sequence	Normal display mode
S/W Reset	Normal display mode
H/W Reset	Normal display mode

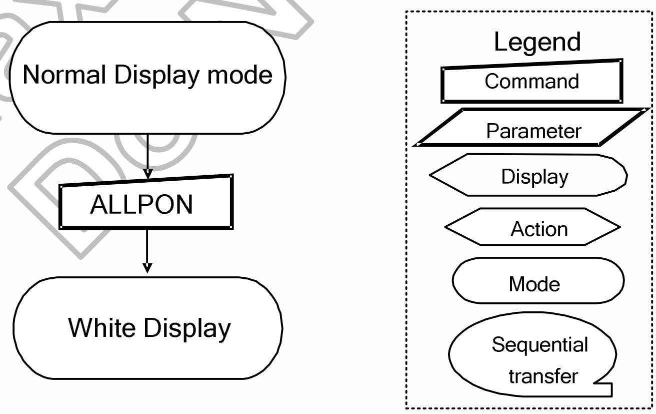
6.2.20 All_Pixel_Off (22h)

<table>
<tr><th rowspan="2">22H</th><th colspan="10">ALLPOFF (All Pixel Off)</th></tr>
<tr><th>DCX</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>22</td></tr>
<tr><td>Parameter</td><td colspan="10">NO PARAMETER</td></tr>
<tr><td>Description</td><td colspan="10">This command turns the display panel black in ‘Sleep Out’ –mode and a status of the ‘Display On/Off’ –register can be ‘on’ or ‘off’.
This command does not change any other status.


‘All Pixels On’ or ’Normal Display Mode On’ – commands are used to leave this mode. The display is showing the input data after ‘Normal Display Mode On’ command.</td></tr>
<tr><td>Restriction</td><td colspan="10">This command has no effect when module is already in inversion on mode.</td></tr>
<tr><td>Register Availability</td><td colspan="10"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="10"><table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Normal display mode</td></tr><tr><td>S/W Reset</td><td>Normal display mode</td></tr><tr><td>H/W Reset</td><td>Normal display mode</td></tr></table></td></tr>
<tr><td>Flow Chart</td><td colspan="10">
</td></tr>
</table>

6.2.21 All_Pixel_On (23h)

<table>
<tr><th rowspan="2">23H</th><th colspan="10">ALLPON(All Pixel On)</th></tr>
<tr><th>DCX</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>23</td></tr>
<tr><td>Parameter</td><td colspan="10">NO PARAMETER</td></tr>
<tr><td>Description</td><td colspan="10">This command turns the display panel white in ‘Sleep out ‘ –mode and a status of the ‘Display On/Off’ –register can be ‘on’ or ‘off’.
This command does not change any other status.
(Example)


‘All Pixels Off’ or ’Normal Display Mode On’ – commands are used to leave this mode. The display is showing the input data after ‘Normal Display Mode On’ command.</td></tr>
<tr><td>Restriction</td><td colspan="10">This command has no effect when module is already in inversion on mode.</td></tr>
<tr><td>Register Availability</td><td colspan="10"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="10"><table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Normal display mode</td></tr><tr><td>S/W Reset</td><td>Normal display mode</td></tr><tr><td>H/W Reset</td><td>Normal display mode</td></tr></table></td></tr>
<tr><td>Flow Chart</td><td colspan="10">
</td></tr>
</table>

6.2.22 Set_gamma_curve (26h)

26H	GAMSET (Gamma Set)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	1	0	0	1	1	0	26
Parameter	1	GC7	GC6	GC5	GC4	GC3	GC2	GC1	GC0	01

Description

This command is used to select the desired Gamma curve for the current display. A maximum of 4 fixed gamma curves can be selected. The curves are defined in Curve Correction Power Supply Circuit. The curve is selected by setting the appropriate bit in the parameter as described in the Table:

GC[7:0]	Parameter	Curve selected
01h	GC0	Gamma Curve 1
02h	GC1	Setting Inhibit
04h	GC2	Setting Inhibit
08h	GC3	Setting Inhibit

Note: All other values are undefined.

Restriction

Values of GC[7:0] not shown in table above are invalid and will not change the current selected Gamma curve until valid value is received.

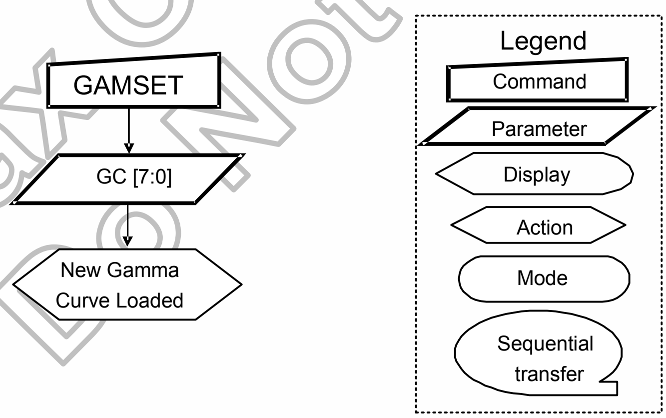
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

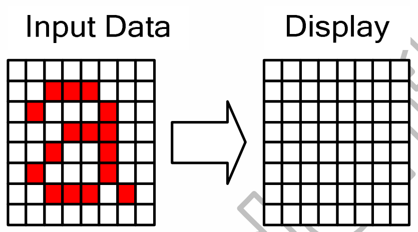
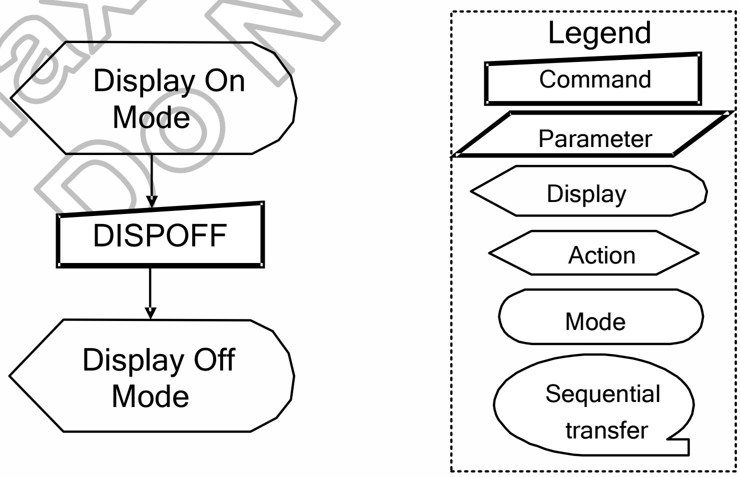
Default

Status	Default Value
Power On Sequence	01h
S/W Reset	01h
H/W Reset	01h

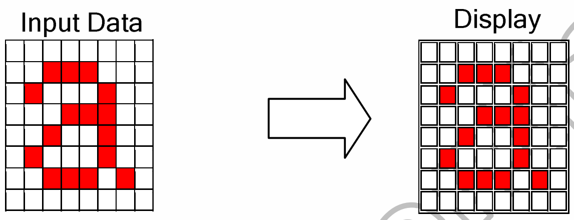
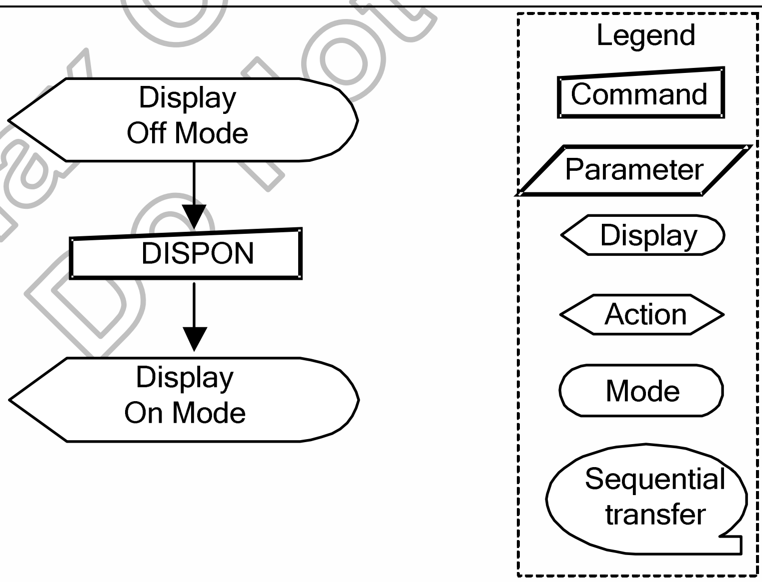
Flow Chart



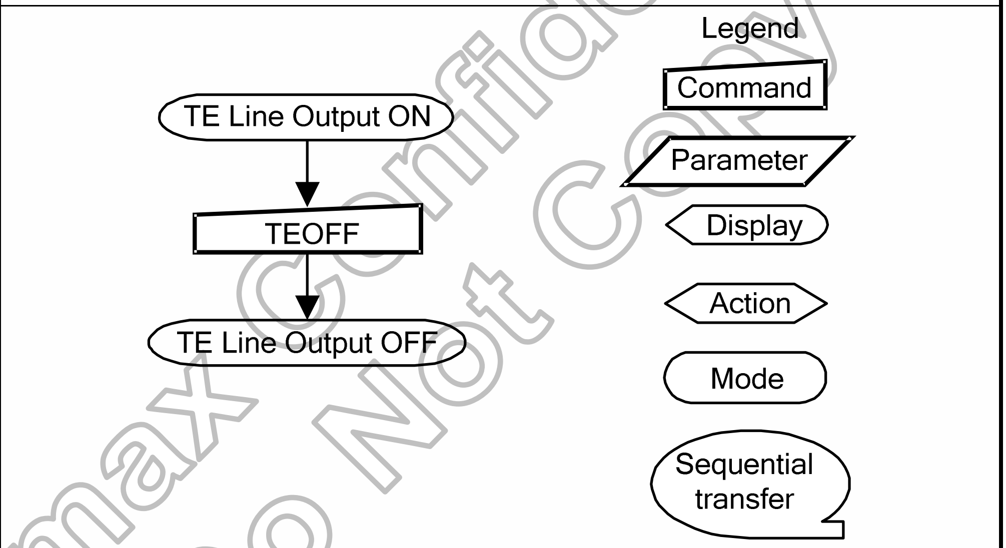
6.2.23 Set_display_off (28h)

<table>
<tr><th rowspan="2">28H</th><th colspan="10">DISPOFF (Display Off)</th></tr>
<tr><th>DCX</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>28</td></tr>
<tr><td>Parameter</td><td colspan="10">NO PARAMETER</td></tr>
<tr><td>Description</td><td colspan="10">This command is used to enter into DISPLAY OFF mode. In this mode, the output from DPI I/F is disabled and blank page inserted.
This command does not change any other status.
There will be no abnormal visible effect on the display.
(Example)

</td></tr>
<tr><td>Restriction</td><td colspan="10">This command has no effect when module is already in display off mode.</td></tr>
<tr><td>Register Availability</td><td colspan="10"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="10"><table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Display off</td></tr><tr><td>S/W Reset</td><td>Display off</td></tr><tr><td>H/W Reset</td><td>Display off</td></tr></table></td></tr>
<tr><td>Flow Chart</td><td colspan="10">
</td></tr>
</table>

6.2.24 Set_display_on (29h)

<table>
<tr><th rowspan="2">29H</th><th colspan="10">DISPON (Display On)</th></tr>
<tr><th>DCX</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>29</td></tr>
<tr><td>Parameter</td><td colspan="10">NO PARAMETER</td></tr>
<tr><td>Description</td><td colspan="10">This command is used to recover from DISPLAY OFF mode. Output from DPI I/F is enabled.
This command does not change any other status.
(Example)

</td></tr>
<tr><td>Restriction</td><td colspan="10">This command has no effect when module is already in display on mode.</td></tr>
<tr><td>Register Availability</td><td colspan="10"><table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In or Booster Off</td><td>Yes</td></tr>
</table></td></tr>
<tr><td>Default</td><td colspan="10"><table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>Display on</td></tr>
<tr><td>S/W Reset</td><td>Display on</td></tr>
<tr><td>H/W Reset</td><td>Display on</td></tr>
</table></td></tr>
<tr><td>Flow Chart</td><td colspan="10">
</td></tr>
</table>

6.2.25 Tearing effect line off (34h)

34H	TEOFF (Tearing Effect Line OFF)									
	DCX	**D7**	**D6**	**D5**	**D4**	**D3**	**D2**	**D1**	**D0**	**HEX**
Command	0	0	0	1	1	0	1	0	0	34
Parameter	NO PARAMETER									
Description	This command is used to turn OFF (Active Low) the Tearing Effect output signal from the TE signal line.									
Restriction	This command has no effect when Tearing Effect output is already OFF.									
Register Availability	Status: Normal Mode On, Idle Mode Off, Sleep Out — Availability: Yes Status: Normal Mode On, Idle Mode On, Sleep Out — Availability: Yes Status: Sleep In or Booster Off — Availability: Yes									
Default	Status: Power On Sequence — Default Value: Off Status: S/W Reset — Default Value: Off Status: H/W Reset — Default Value: Off									
Flow Chart										

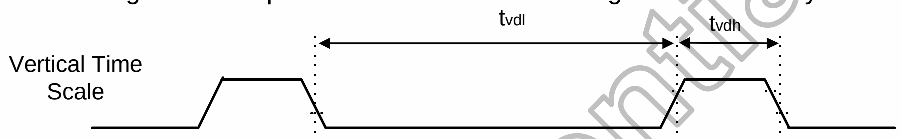
6.2.26 Set_tear_on (35h)

35H	TEON (Tearing Effect Line ON)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	1	1	0	1	0	1	35
Parameter	1	X	X	X	X	X	X	X	M	00

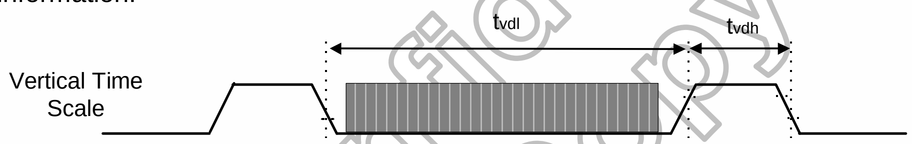
Description

This command is used to turn ON the Tearing Effect output signal from the TE signal line.

The Tearing Effect Line On has one parameter which describes the mode of the Tearing Effect Output Line. (X=Don't Care).
When M=0:
The Tearing Effect Output line consists of V-Blanking information only:



When M=1:
The Tearing Effect Output Line consists of both V-Blanking and H-Blanking information:



Note: M=1 is not applicable for this project, so it is set to "0".

Restriction

This command has no effect when Tearing Effect output is already ON.

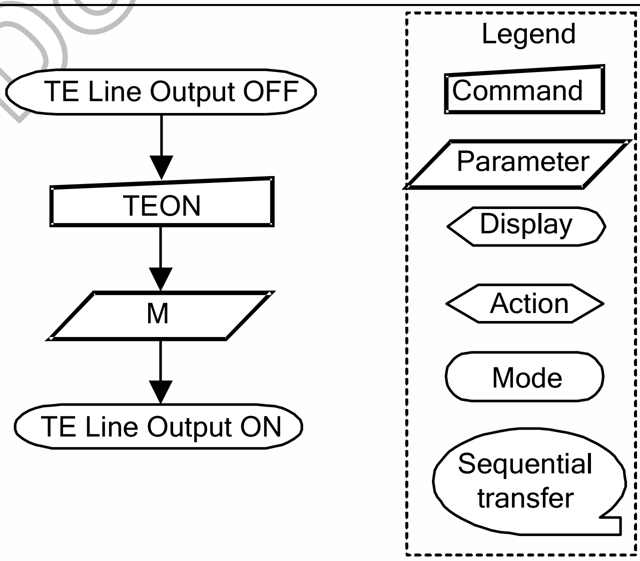
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	Off
S/W Reset	Off
H/W Reset	Off

Flow Chart



6.2.27 Set_address_mode (36h)

36H	MADCTL (Memory Access Control)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	1	1	0	1	1	0	36
1st parameter	1	D7	D6	X	X	D3	D2	D1	D0	00

Description

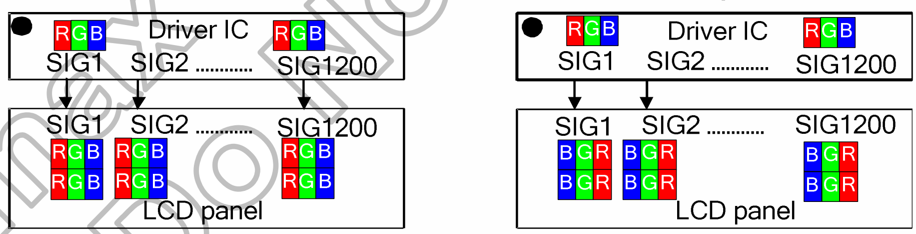
This command defines the display scanning direction of LCD.
This command makes no change on the other driver status.

Bit Assignment

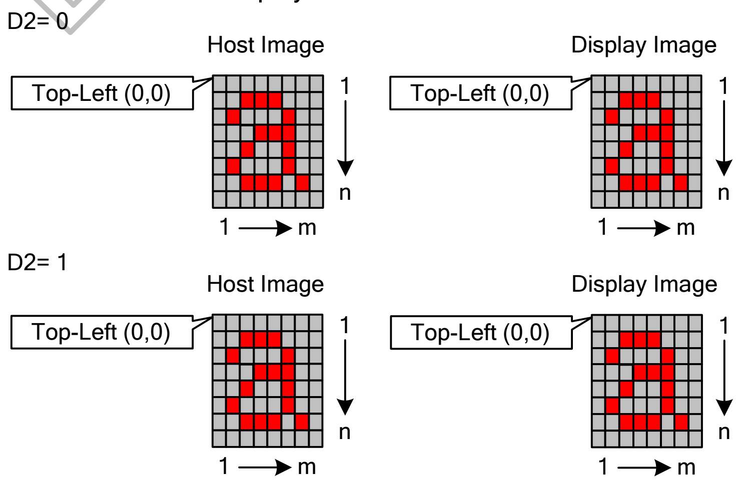
Bit	Name	Description
D7	Page Address Order	LCD vertical updating order direction control
D6	Column Address Order	LCD horizontal updating order direction control
D5	Page/Column Selection	This bit is not applicable for this project, so it is set to "0".
D4	Line Address Order	This bit is not applicable for this project, so it is set to "0".
D3	RGB-BGR Order (BGR)	Colour selector switch control (0=RGB colour filter panel, 1=BGR colour filter panel)
D2	Display Data Latch Order	LCD horizontal refresh direction control
D1	Flip Horizontal (Source scan sequence)	Select the Source driver scan direction on panel module
D0	Flip Vertical (Gate scan sequence)	Select the Gate driver scan direction on panel module

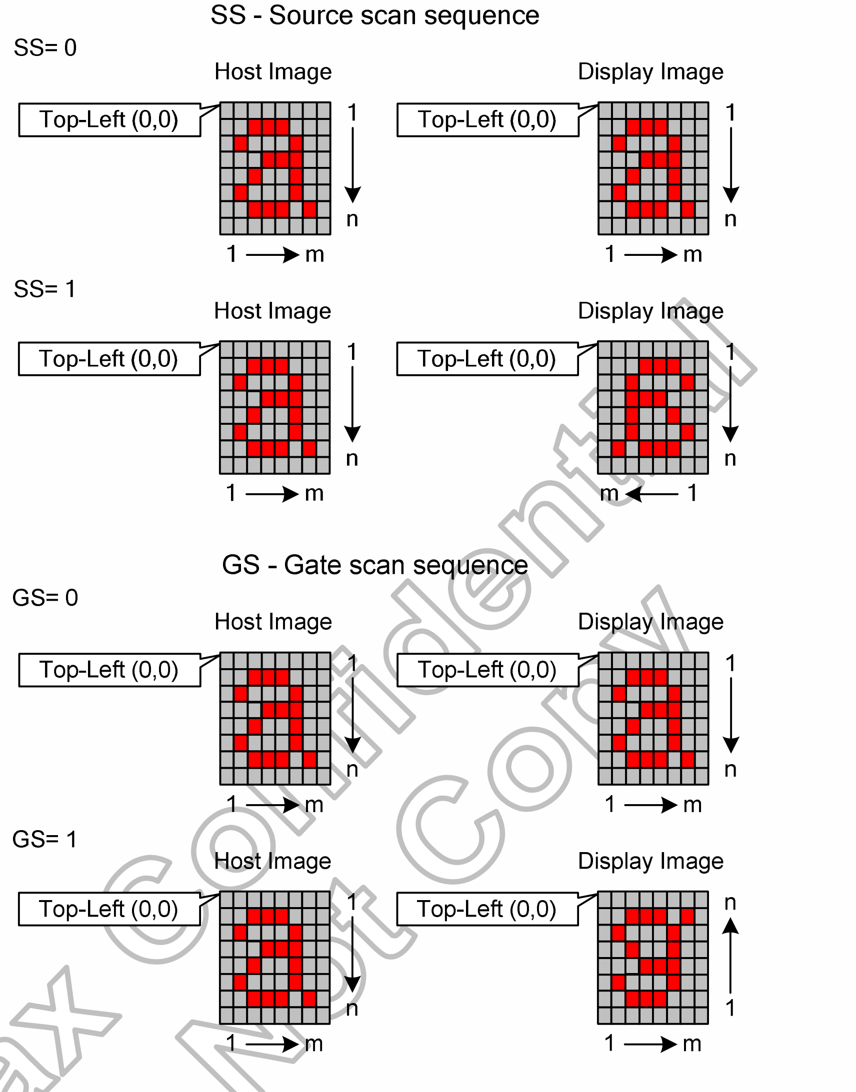
RGB-BGR Order

D3= 0 D3= 1



Display Data Latch Order





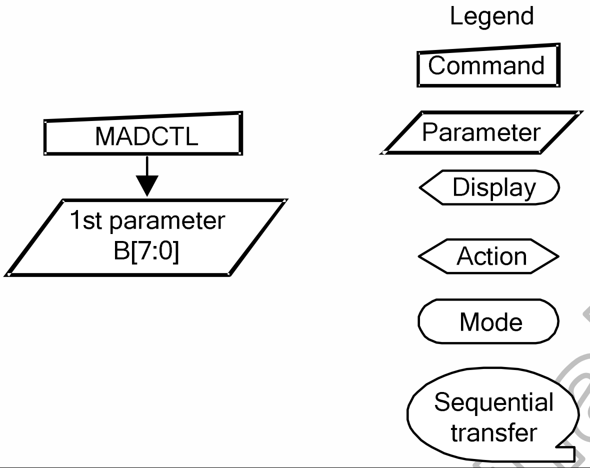
Restriction	D5 and D4 are set to ‘0’ internally.

Register Availability

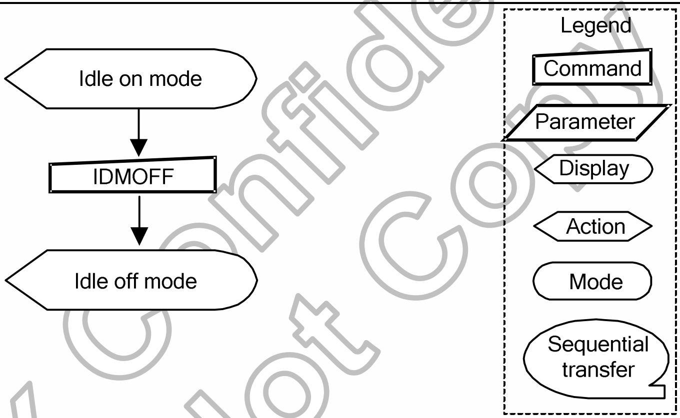
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

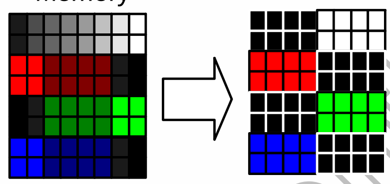
Status	Default Value
Power On Sequence	00h
S/W Reset	No Change
H/W Reset	00h

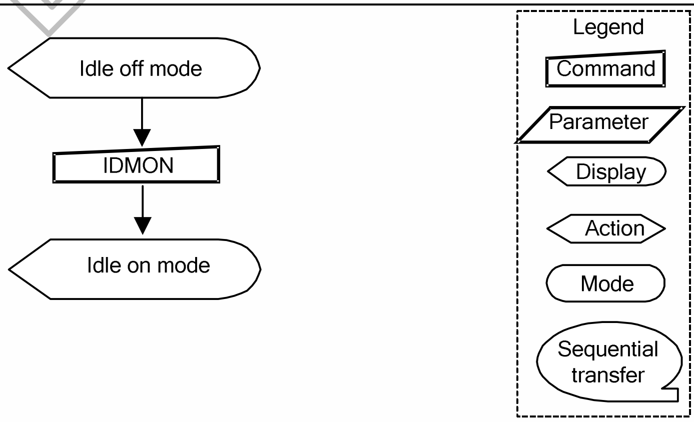
Flow Chart	Legend Command MADCTL Parameter Display 1st parameter B[7:0] Action Mode Sequential transfer

6.2.28 Idle mode off (38h)

<table>
<tr><th rowspan="2">38H</th><th colspan="10">IDMOFF (Idle mode off)</th></tr>
<tr><th>DCX</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>38</td></tr>
<tr><td>Parameter</td><td colspan="10">NO PARAMETER</td></tr>
<tr><td>Description</td><td colspan="10">This command is used to recover from Idle mode on. In the idle off mode, LCD can display maximum 16.7M colours.</td></tr>
<tr><td>Restriction</td><td colspan="10">This command has no effect when module is already in idle off mode.</td></tr>
<tr><td>Register Availability</td><td colspan="10"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="10"><table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Off</td></tr><tr><td>S/W Reset</td><td>Off</td></tr><tr><td>H/W Reset</td><td>Off</td></tr></table></td></tr>
<tr><td>Flow Chart</td><td colspan="10">
</td></tr>
</table>

6.2.29 Enter_Idle_mode (39h)

<table>
<tr><th rowspan="2">39H</th><th colspan="10">IDMON (Idle mode on)</th></tr>
<tr><th>DCX</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>39</td></tr>
<tr><td>Parameter</td><td colspan="10">NO PARAMETER</td></tr>
<tr><td>Description</td><td colspan="10">This command is used to enter into Idle mode on. In the idle on mode, colour expression is reduced. The primary and the secondary colours using MSB of each R, G and B, 8 colour depth data is displayed.
(Example)


Display Colour
<table>
<tr><th></th><th>R7 – R0</th><th>G7 – G0</th><th>B7 – B0</th></tr>
<tr><td>Black</td><td>0XXXXX</td><td>0XXXXX</td><td>0XXXXX</td></tr>
<tr><td>Blue</td><td>0XXXXX</td><td>0XXXXX</td><td>1XXXXX</td></tr>
<tr><td>Red</td><td>1XXXXX</td><td>0XXXXX</td><td>0XXXXX</td></tr>
<tr><td>Magent</td><td>1XXXXX</td><td>0XXXXX</td><td>1XXXXX</td></tr>
<tr><td>Green</td><td>0XXXXX</td><td>1XXXXX</td><td>0XXXXX</td></tr>
<tr><td>Cyan</td><td>0XXXXX</td><td>1XXXXX</td><td>1XXXXX</td></tr>
<tr><td>Yellow</td><td>1XXXXX</td><td>1XXXXX</td><td>0XXXXX</td></tr>
<tr><td>White</td><td>1XXXXX</td><td>1XXXXX</td><td>1XXXXX</td></tr>
</table>
X=don’t care</td></tr>
<tr><td>Restriction</td><td colspan="10">This command has no effect when module is already in idle on mode.</td></tr>
<tr><td>Register Availability</td><td colspan="10">
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In or Booster Off</td><td>Yes</td></tr>
</table></td></tr>
<tr><td>Default</td><td colspan="10">
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>Off</td></tr>
<tr><td>S/W Reset</td><td>Off</td></tr>
<tr><td>H/W Reset</td><td>Off</td></tr>
</table></td></tr>
<tr><td>Flow Chart</td><td colspan="10">
</td></tr>
</table>

6.2.30 Set_pixel_format (3Ah)

3A H	COLMOD (Interface Pixel Format)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	1	1	1	0	1	0	3A
1st parameter	1	X	D6	D5	D4	X	X	X	X	70

Description

This command is used to define the format of RGB picture data.
D6~D4 : DPI Pixel format Definition.
Bits D7, D[3:0] : Reserved.
The formats are shown in the table:

Pixel Format	D6	D5	D4
Not Defined	0	0	0
Not Defined	0	0	1
Not Defined	0	1	0
Not Defined	0	1	1
Not Defined	1	0	0
16 Bit/Pixel	1	0	1
18 Bit/Pixel	1	1	0
24 Bit/Pixel	1	1	1

If the setting is not used then the corresponding bits in the parameter returned

Restriction

There is no visible effect until the image data is written.

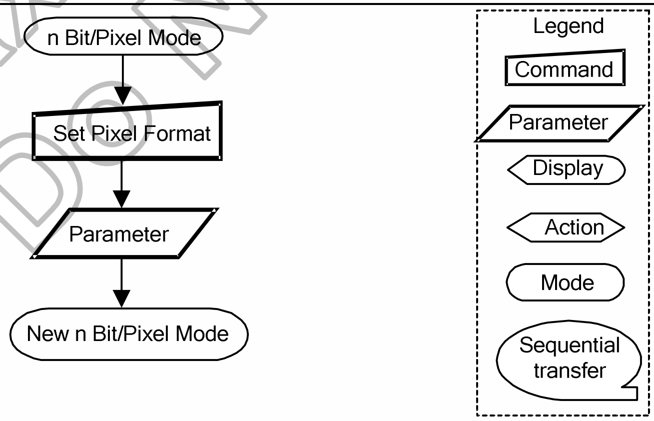
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

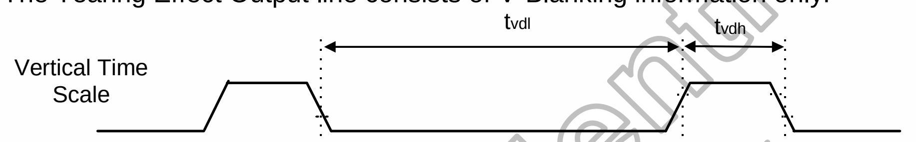
Default

Status	Default Value
Power On Sequence	70h
S/W Reset	70h
H/W Reset	70h

Flow Chart



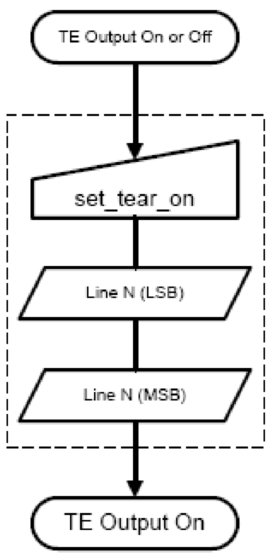
6.2.31 Set tear scan lines (44h)

<table>
<tr><th rowspan="2">44H</th><th colspan="10">TESL (Tear Effect Scan Lines)</th></tr>
<tr><th>DCX</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>44</td></tr>
<tr><td>1st parameter</td><td>1</td><td colspan="8">TELINE[15:8]</td><td>00</td></tr>
<tr><td>2nd parameter</td><td>1</td><td colspan="8">TELINE[7:0]</td><td>00</td></tr>
<tr><td>Description</td><td colspan="10">This command is turns on the display module’s Tearing Effect output signal on the TE signal
Line when the display module reacfes line TELINE. The TE signal is not affected by changing MADCTL bit B4.
The Tearing Effect Line On has one parameter which describes the mode of the Tearing Effect Output Line.
The Tearing Effect Output line consists of V-Blanking information only:

Note: That TELINE=0 is equivalent to TEMODE=0. The Tearing Effect Output Line shall be active low when the display module is in Sleep mode.</td></tr>
<tr><td>Restriction</td><td colspan="10">The command has no effect when Tearing Effect output is already ON.</td></tr>
<tr><td>Register Availability</td><td colspan="10">

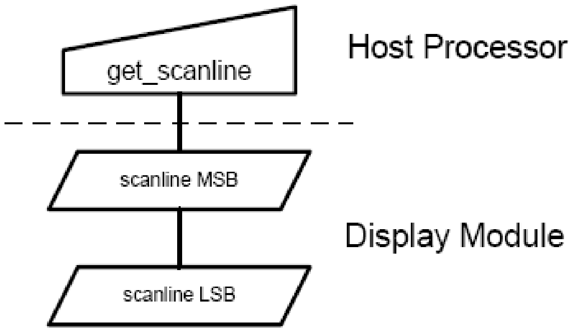
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

</td></tr>
<tr><td>Default</td><td colspan="10">

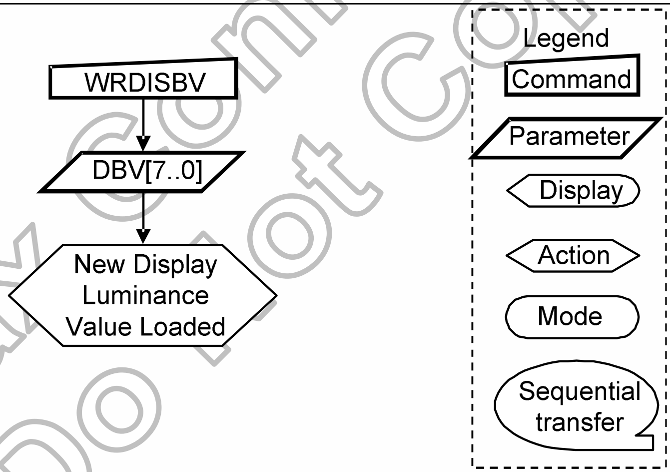
Status	Default Value
Power On Sequence	0000h
S/W Reset	0000h
H/W Reset	0000h

</td></tr>
<tr><td>Flow Chart</td><td colspan="10"></td></tr>
</table>

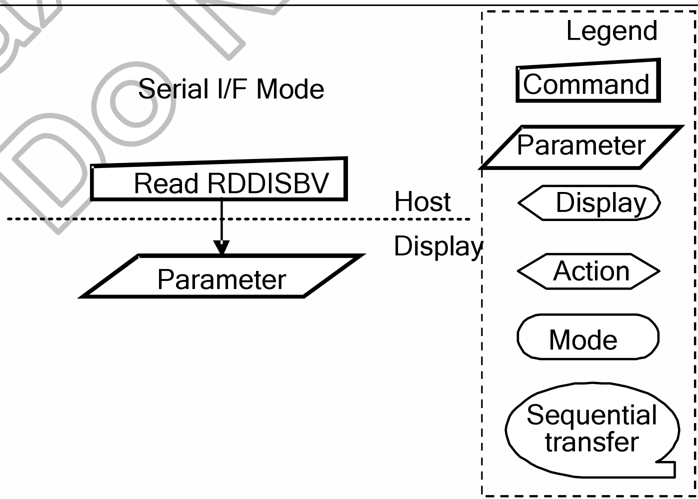
6.2.32 Get the current scanline(45h)

45H	GETSCAN (Get the current scanline)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	1	0	0	0	1	0	1	45
1st parameter	1	SLN[15:8]								00
2nd parameter	1	SLN[7:0]								00F
Description	The display module returns the current scanline, N, used to update the display device. The total number of scanlines on a display device is defined as VSYNC + VBP + VACT + VFP. The first scanline is defined as the first line of V Sync and is denoted as Line 0. When in Sleep Mode, the value returned by get_scanline is undefined.									
Restriction	-									
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table>									
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>0000h</td></tr><tr><td>S/W Reset</td><td>0000h</td></tr><tr><td>H/W Reset</td><td>0000h</td></tr></table>									
Flow Chart										

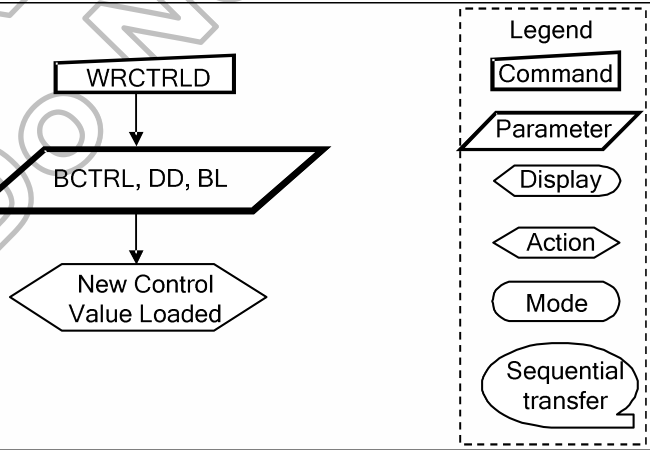
6.2.33 Write display brightness (51h)

<table>
<tr><th rowspan="2">51H</th><th colspan="10">WRDISBV (Write Display Brightness)</th></tr>
<tr><th>DCX</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>51</td></tr>
<tr><td>1st parameter</td><td>1</td><td colspan="8">DBV[7:0]</td><td>00</td></tr>
<tr><td>Description</td><td colspan="10">This command is used to adjust the brightness value of the display.
It should be checked what the relationship between this written value and output brightness of the display is. This relationship is defined on the display module specification.
In principle relationship is that 00h value means the lowest brightness and FFh value means the highest brightness.
See chapter “5.12.3 Brightness Control Block”.</td></tr>
<tr><td>Restriction</td><td colspan="10">-</td></tr>
<tr><td>Register Availability</td><td colspan="10"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="10"><table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>S/W Reset</td><td>00h</td></tr><tr><td>H/W Reset</td><td>00h</td></tr></table></td></tr>
<tr><td>Flow Chart</td><td colspan="10">
</td></tr>
</table>

6.2.34 Read display brightness value (52h)

<table>
<tr><th rowspan="2">52H</th><th colspan="10">RDDISBV (Read Display Brightness Value)</th></tr>
<tr><th>DCX</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>52</td></tr>
<tr><td>1st parameter</td><td>1</td><td colspan="8">DBV[7:0]</td><td>00</td></tr>
<tr><td>Description</td><td colspan="10">This command returns the brightness value of the display.
It should be checked what the relationship between this returned value and output brightness of the display. This relationship is defined on the display modulespecification is.
In principle the relationship is that 00h value means the lowest brightness and FFh value means the highest brightness.
See chapters: “5.12.3 Brightness Control Block”, and “6.2.33 Write Display Brightness (51h)”
DBV[7:0] is reset when display is in sleep-in mode.
DBV[7:0] is ‘00h’ when bit BCTRL of “6.2.35 Write CTRL Display (53h)” command is ‘0’.
DBV[7:0] is manual set brightness specified with “6.2.35 Write CTRL Display (53h)” command when bit BCTRL is ‘1’.
When bit BCTRL of “6.2.35 Write CTRL Display (53h)” command is ‘1’ and bit C1/C0 of “6.2.37 Write Content Adaptive Brightness Control (55h)” are ‘0’, DBV[7:0] output is the brightness value specified with “6.2.33 Write Display Brightness (51h)” command.</td></tr>
<tr><td>Restriction</td><td colspan="10">-</td></tr>
<tr><td>Register Availability</td><td colspan="10"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="10"><table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>S/W Reset</td><td>00h</td></tr><tr><td>H/W Reset</td><td>00h</td></tr></table></td></tr>
<tr><td>Flow Chart</td><td colspan="10"></td></tr>
</table>

6.2.35 Write CTRL display (53h)

<table>
<tr><th rowspan="2">53H</th><th colspan="10">WRCTRLD (Write Control Display)</th></tr>
<tr><th>DCX</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>53</td></tr>
<tr><td>1st parameter</td><td>1</td><td>X</td><td>X</td><td>BCTRL</td><td>X</td><td>DD</td><td>BL</td><td>X</td><td>X</td><td>00</td></tr>
<tr><td>Description</td><td colspan="10">This command is used to control display brightness.
BCTRL: Brightness Control Block On/Off, This bit is always used to switch brightness for display.
0 = Off (Brightness registers are 00h, DBV[7..0])
1 = On (Brightness registers are active, according to the other parameters.)
DD: Display Dimming(Only for manual brightness setting)
DD = 0: Display Dimming is off
DD = 1: Display Dimming is on
BL: Backlight Control On/Off
0 = Off (Completely turn off backlight circuit. Control lines must be low.)
1 = On
Dimming function is adapted to the brightness registers for display when bit BCTRL is changed at DD=1, e.g. BCTRL: 0 -> 1 or 1-> 0.
When BL bit change from “On” to “Off”, backlight is turned off without gradual dimming, even if dimming-on (DD=1) are selected.
X = Don’t care.</td></tr>
<tr><td>Restriction</td><td colspan="10">-</td></tr>
<tr><td>Register Availability</td><td colspan="10"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="10"><table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>S/W Reset</td><td>00h</td></tr><tr><td>H/W Reset</td><td>00h</td></tr></table></td></tr>
<tr><td>Flow Chart</td><td colspan="10">
</td></tr>
</table>

6.2.36 Read CTRL value display (54h)

54H	RDCTRLD (Read Control Value Display)									
	DCX	**D7**	**D6**	**D5**	**D4**	**D3**	**D2**	**D1**	**D0**	**HEX**
Command	0	0	1	0	1	0	1	0	0	54
1st parameter	1	0	0	BCTRL	0	DD	BL	0	0	00
Description	This command returns ambient light and brightness control values, see chapter: “6.2.35 Write CTRL Display (53h)”. **BCTRL:** Brightness Control Block On/Off, This bit is always used to switch brightness for display. 0 = Off 1 = On **DD:** Display Dimming. DD = 0: Display Dimming is off DD = 1: Display Dimming is on **BL:** Backlight Control On/Off 0 = Off (completely turn off backlight circuit) 1 = On									
Restriction	-									

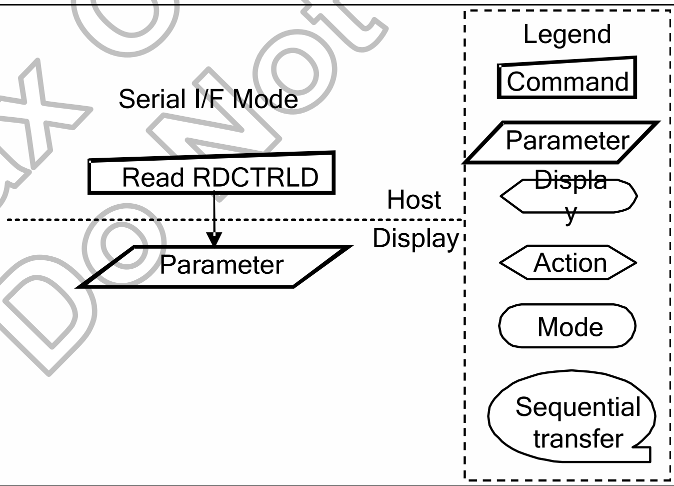
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



6.2.37 Write content adaptive brightness control (55h)

55 H	WRCABC (Write Content Adaptive Brightness Control)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	1	0	1	0	1	0	1	55
1st parameter	1	X	X	X	X	X	X	C[1:0]		00

Description

This command is used to set parameters for image content based adaptive brightness control functionality.
There is possible to use 4 different modes for content adaptive image functionality, which are defined on a table below. See chapter "5.12 Content Adaptive Brightness Control (CABC)".

C1	C0	Function
0	0	Off
0	1	User Interface Image
1	0	Still Picture
1	1	Moving Image

Restriction

-

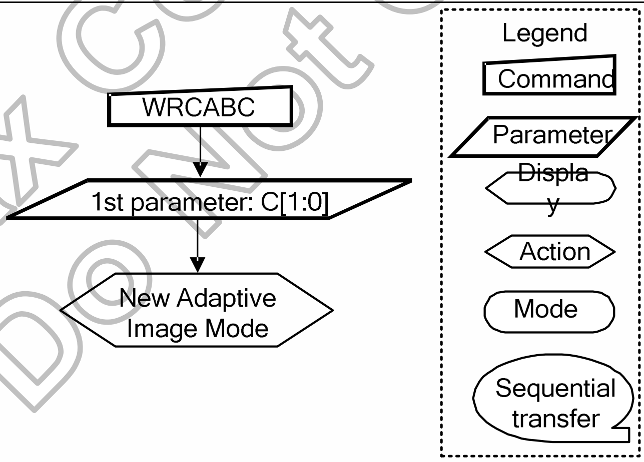
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



6.2.38 Read content adaptive brightness control (56h)

56H	RDCABC (Read Content Adaptive Brightness Control)								
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	1	0	1	0	1	1	0	56
1st parameter	1	0	0	0	0	0	0	C1	C0	00

Description

This command is used to set parameters for image content based adaptive brightness control functionality.
There is possible to use 4 different modes for content adaptive image functionality, which are defined on a table below. See chapter “5.12 Content Adaptive Brightness Control (CABC)”.

C1	C0	Function
0	0	Off
0	1	User Interface Image
1	0	Still Picture
1	1	Moving Image

Restriction

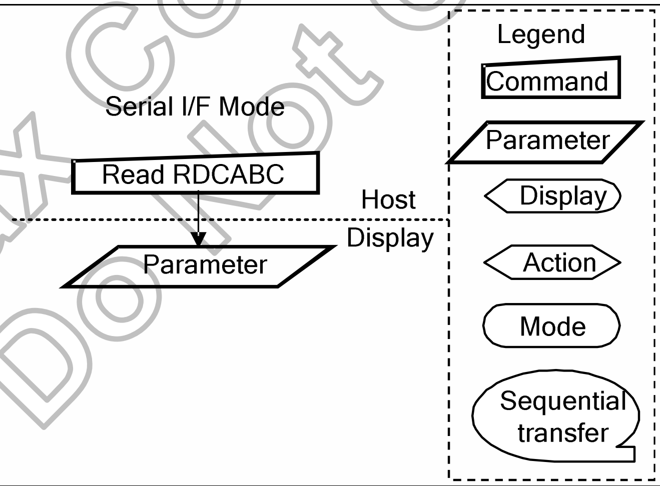
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

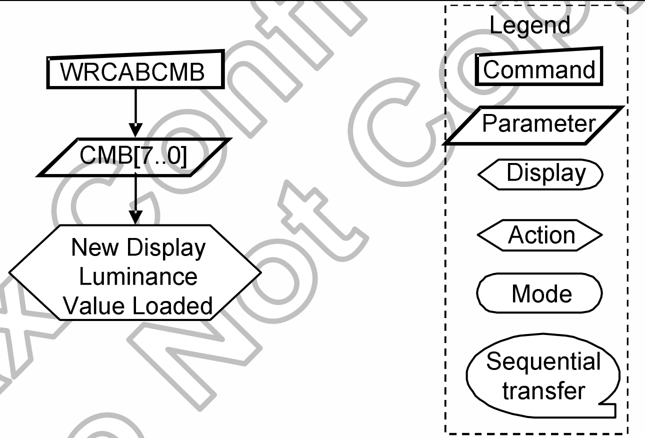
Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

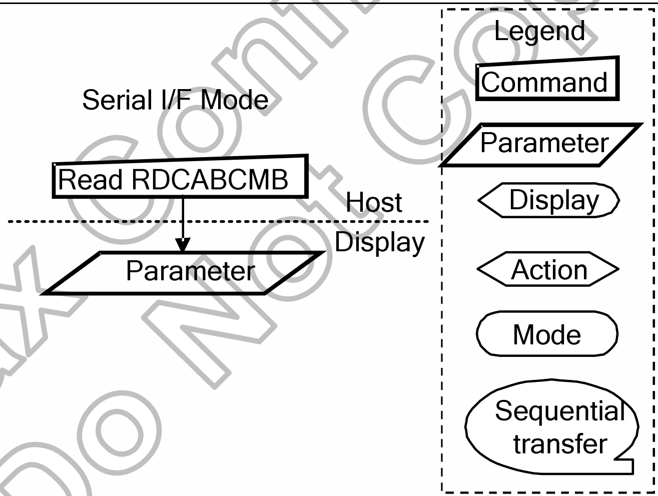
Flow Chart



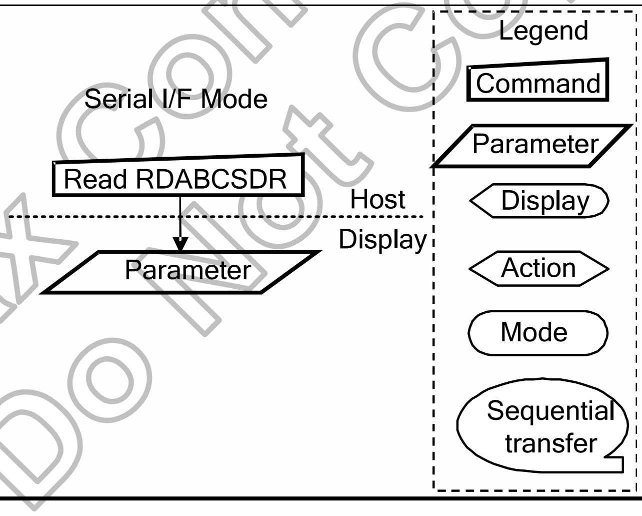
6.2.39 Write CABC minimum brightness (5Eh)

5E H	WRCABCMB (Write CABC minimum brightness)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	1	0	1	1	1	1	0	5E
1st parameter	1	CMB[7:0]								00
Description	This command is used to set the minimum brightness value of the display for CABC function. In principle relationship is that 00h value means the lowest brightness for CABC and FFh value means the highest brightness for CABC. See chapter “5.12.4 Minimum brightness setting of CABC function”.									
Restriction	-									
Register Availability	Status: Availability Normal Mode On, Idle Mode Off, Sleep Out: Yes Normal Mode On, Idle Mode On, Sleep Out: Yes Sleep In or Booster Off: Yes									
Default	Status: Default Value Power On Sequence: 00h S/W Reset: 00h H/W Reset: 00h									
Flow Chart										

6.2.40 Read CABC minimum brightness (5Fh)

<table>
<tr><th rowspan="2">5FH</th><th colspan="10">RDCABCMB (Read CABC minimum brightness)</th></tr>
<tr><th>DCX</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>5F</td></tr>
<tr><td>1st parameter</td><td>1</td><td colspan="8">CMB[7:0]</td><td>00</td></tr>
<tr><td>Description</td><td colspan="10">This command returns the minimum brightness value of CABC function.
In principle the relationship is that 00h value means the lowest brightness and FFh value means the highest brightness.
See chapter “5.12.4 Minimum brightness setting of CABC function”.
CMB[7:0] is CABC minimum brightness specified with “6.2.39 Write CABC minimum brightness (5Eh)” command.</td></tr>
<tr><td>Restriction</td><td colspan="10">-</td></tr>
<tr><td>Register Availability</td><td colspan="10"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="10"><table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>S/W Reset</td><td>00h</td></tr><tr><td>H/W Reset</td><td>00h</td></tr></table></td></tr>
<tr><td>Flow Chart</td><td colspan="10"></td></tr>
</table>

6.2.41 Read automatic brightness control self-diagnostic result (68h)

<table>
<tr><th rowspan="2">68H</th><th colspan="10">RDABCSDR (Read Automatic Brightness Control Self-Diagnostic Result)</th></tr>
<tr><th>DCX</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>68</td></tr>
<tr><td>1st parameter</td><td>1</td><td colspan="2">D[7:6]</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>00</td></tr>
<tr><td>Description</td><td colspan="10">This command indicates the status of the display self-diagnostic results for automatic brightness control after Sleep Out –command as described in the table below:
Bit D7 – Register Loading Detection.
See section “5.9.1 Register loading Detection”.
Bit D6 – Functionality Detection.
See section “5.9.2 Functionality Detection “.
Bits D[5:0] are for future use and are set to ‘0’.</td></tr>
<tr><td>Restriction</td><td colspan="10">-</td></tr>
<tr><td>Register Availability</td><td colspan="10"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="10"><table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>S/W Reset</td><td>00h</td></tr><tr><td>H/W Reset</td><td>00h</td></tr></table></td></tr>
<tr><td>Flow Chart</td><td colspan="10"></td></tr>
</table>

6.2.42 Read_DDB_start (A1h)

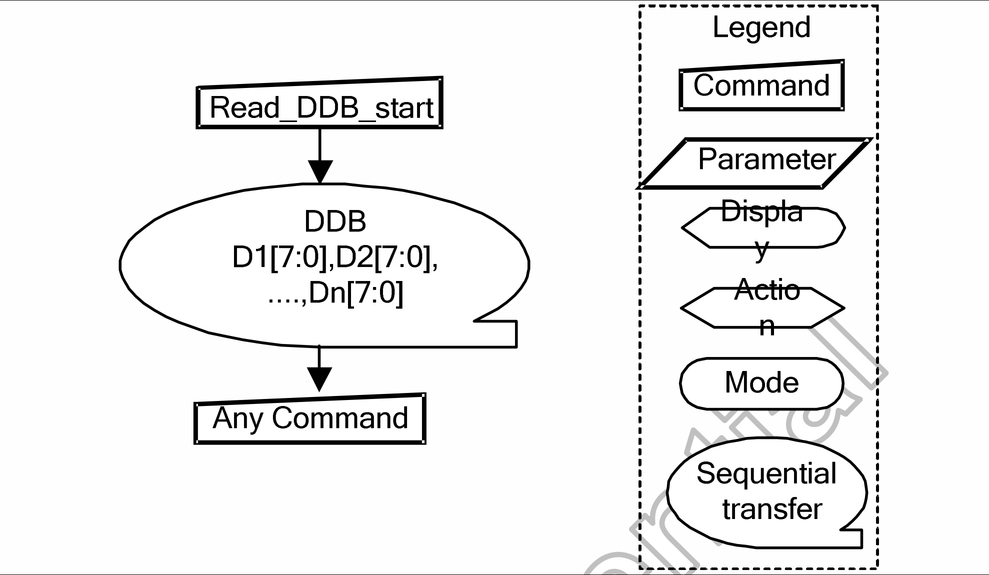
A1H	Read_DDB_start									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	0	1	0	0	0	0	1	A1
1st parameter	1	xx	xx	xx	xx	xx	xx	xx	xx	xx
2nd parameter	1	xx	xx	xx	xx	xx	xx	xx	xx	xx
:	1	xx	xx	xx	xx	xx	xx	xx	xx	xx
Nth parameter	1	xx	xx	xx	xx	xx	xx	xx	xx	xx
Description	This command reads identifying and descriptive information from the peripheral. This information is organized in the Device Descriptor Block (DDB) stored on the peripheral. The response to this command returns a sequence of bytes that may be any length up to 64K bytes. Note that the returned sequence of bytes does not necessarily correspond to the entire DDB; it may be a portion of a larger block of data. The format of returned data is as follows: Parameter 1: LS (least significant) byte of Supplier ID. Supplier ID is a unique value assigned to each peripheral supplier by the MIPI organization. Parameter 2: MS (most significant) byte of Supplier ID. Parameter 3: LS (least significant) byte of Supplier Elective Data. This is a byte of information that is determined by the supplier. It could include model number or revision information, for example. Parameter 4: MS (most significant) byte of Supplier Elective Data Parameter 5: single-byte *Escape or Exit Code* (EEC). The code is interpreted as follows: - FFh – Exit code – there is no more data in the Descriptor Block - 00h – Escape code – there is supplier-proprietary data in the Descriptor Block (does not conform to any MIPI standard) - Any other value – there is DDB data in the Descriptor Block. The format and interpretation of this data is documented in *MIPI Alliance Standard for Device Descriptor Block (DDB)*. DDBs may contain many more data fields providing information about the peripheral. In a DSI system, read activity takes the form of two separate transactions across the bus: first the read command read_DDB_start from host processor to peripheral, which includes the bus turn-around token. The peripheral then takes control of the bus and returns the requested data. The peripheral response to read_DDB_start is a Long Packet type, so its length may be up to 64K bytes unless limited by a previous set_max_return_size command. The response to a read_DDB_start command always starts at the beginning of the Device Descriptor Block. After receiving the first packet and processing the returned DDB data, the host processor may initiate a read_DDB_continue command to access the next portion of the DDB. A read_DDB_continue command begins the next read at the location following the last byte of the previous data read from the DDB. Subsequent read_DDB_continue commands can be used to read a DDB or supplier-proprietary block of arbitrary size. There is, however, no obligation to read the entire block. The host processor may choose to stop reading after completion of any read_DDB_xxx command.									
Restrictions	-									

Register Availability:

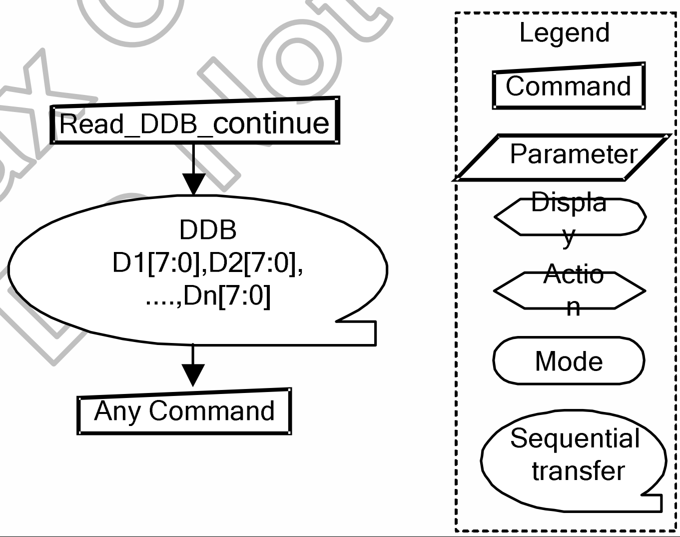
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default:

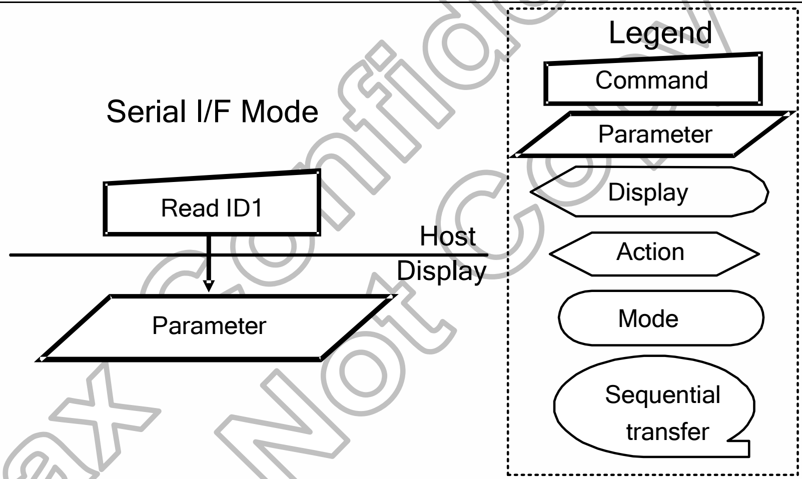
Status	Default Value
Power On Sequence	PA1st~4th is OTP value, PA5th is FFh
S/W Reset	PA1st~4th is OTP value, PA5th is FFh
H/W Reset	PA1st~4th is OTP value, PA5th is FFh

Flow Chart	 Read_DDB_start DDB D1[7:0],D2[7:0], ...,Dn[7:0] Any Command Legend Command Parameter Displa y Actio n Mode Sequential transfer

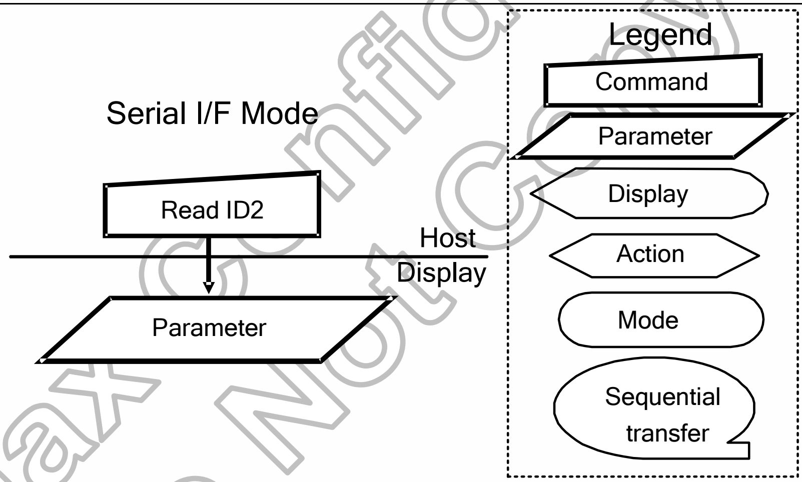
6.2.43 Read_DDB_continue (A8h)

<table>
<tr><th rowspan="2">A8H</th><th colspan="10">Read_DDB_continue</th></tr>
<tr><th>DCX</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>A8</td></tr>
<tr><td>1st parameter</td><td>1</td><td>xx</td><td>xx</td><td>xx</td><td>xx</td><td>xx</td><td>xx</td><td>xx</td><td>xx</td><td>xx</td></tr>
<tr><td>2nd parameter</td><td>1</td><td>xx</td><td>xx</td><td>xx</td><td>xx</td><td>xx</td><td>xx</td><td>xx</td><td>xx</td><td>xx</td></tr>
<tr><td>:</td><td>1</td><td>xx</td><td>xx</td><td>xx</td><td>xx</td><td>xx</td><td>xx</td><td>xx</td><td>xx</td><td>xx</td></tr>
<tr><td>Nth parameter</td><td>1</td><td>xx</td><td>xx</td><td>xx</td><td>xx</td><td>xx</td><td>xx</td><td>xx</td><td>xx</td><td>xx</td></tr>
<tr><td>Description</td><td colspan="10">A read_DDB_start command should be executed at least once before a read_DDB_continue command to define the read location. Otherwise, data read with a read_DDB_continue command is undefined.</td></tr>
<tr><td>Restrictions</td><td colspan="10">-</td></tr>
<tr><td>Register Availability</td><td colspan="10"><table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In or Booster Off</td><td>Yes</td></tr>
</table></td></tr>
<tr><td>Default</td><td colspan="10"><table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>Without A1h read, 1st~4th read is the same as A8h 1st~4th OTP value, after 5th read is FFh.</td></tr>
<tr><td>S/W Reset</td><td>Without A1h read, 1st~4th read is the same as A8h 1st~4th OTP value, after 5th read is FFh.</td></tr>
<tr><td>H/W Reset</td><td>Without A1h read, 1st~4th read is the same as A8h 1st~4th OTP value, after 5th read is FFh.</td></tr>
</table></td></tr>
<tr><td>Flow Chart</td><td colspan="10">
</td></tr>
</table>

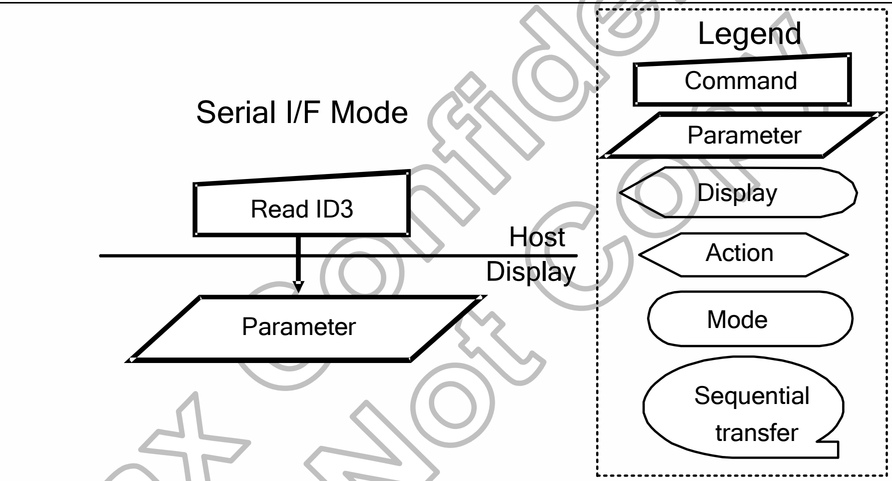
6.2.44 Read ID1 (DAh)

DAH	RDID1 (Read ID1)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	0	1	1	0	1	0	DA
1st parameter	1	ID1[7:0]								83
Description	This read byte identifies the LCD module's manufacturer. It is specified by display supplier and for xx is defined as xxHEX.									
Restriction	-									
Register Availability	Status / Availability: Normal Mode On, Idle Mode Off, Sleep Out: Yes Normal Mode On, Idle Mode On, Sleep Out: Yes Sleep In or Booster Off: Yes									
Default	Status / Default Value: Power On Sequence: OTP value S/W Reset: OTP value H/W Reset: OTP value									
Flow Chart										

6.2.45 Read ID2 (DBh)

<table>
<tr><th rowspan="2">DBH</th><th colspan="10">RDID2 (Read ID2)</th></tr>
<tr><th>DCX</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>DB</td></tr>
<tr><td>1st parameter</td><td>1</td><td colspan="8">ID2[7:0]</td><td>99</td></tr>
<tr><td>Description</td><td colspan="10">This read byte is used to track the LCD module/driver version. It is defined by display supplier and changes each time a revision is made to the display, material or construction specifications.</td></tr>
<tr><td>Restriction</td><td colspan="10">-</td></tr>
<tr><td>Register Availability</td><td colspan="10"><table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In or Booster Off</td><td>Yes</td></tr>
</table></td></tr>
<tr><td>Default</td><td colspan="10"><table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>OTP value</td></tr>
<tr><td>S/W Reset</td><td>OTP value</td></tr>
<tr><td>H/W Reset</td><td>OTP value</td></tr>
</table></td></tr>
<tr><td>Flow Chart</td><td colspan="10"></td></tr>
</table>

6.2.46 Read ID3 (DCh)

<table>
<tr><th rowspan="2">DCH</th><th colspan="10">RDID3 (Read ID3)</th></tr>
<tr><th>DCX</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>DC</td></tr>
<tr><td>1st parameter</td><td>1</td><td colspan="8">ID3[7:0]</td><td>0A</td></tr>
<tr><td>Description</td><td colspan="10">This read byte identifies the LCD module/driver.</td></tr>
<tr><td>Restriction</td><td colspan="10">-</td></tr>
<tr><td>Register Availability</td><td colspan="10"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="10"><table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>OTP value</td></tr><tr><td>S/W Reset</td><td>OTP value</td></tr><tr><td>H/W Reset</td><td>OTP value</td></tr></table></td></tr>
<tr><td>Flow Chart</td><td colspan="10">
</td></tr>
</table>

6.3 User Define Command Description

6.3.1 SETSEQUENCE: Set Sequence (B0h)

B0H	SETSEQUENCE(Set Sequence)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	0	1	1	0	0	0	0	B0
1st parameter	1	-	AUTO_OPT[6:0]							00
2nd parameter	1	-	-	-	-	-	-	-	OSC_EN	00
3rd parameter	1	-	VSN_EN	VSP_EN	VGL_EN	VGH_EN	VCL_EN	-	STB	7D
4th parameter	1	-	-	-	-	GON	DTE	D[1:0]		0C

Description

This command is used for DCS command auto sequence and manual mode debug use, please don't access this command in initial code.

AUTO_OPT[6:0]: Internal option, not open. Please set 00h.

OSC_EN: Enable internal oscillator. “1”: Enable; “0”: Disable.

STB: When STB = “1”, the HX8399-A enters the standby mode, where all display operation stops, suspend all the internal operations. But the internal R-C oscillator stop or not is determined by OSC_EN bit. To minimize the standby power, please set OSC_EN to 0. During the standby mode, only the following process can be executed.

a. Exit the Standby mode (STB = “0”)
b. Enable or disable the oscillator
c. Software reset

VCL_EN : ON/OFF the operation of VCL charge bump circuit in manual power on mode. Using state machine SLPOUT CMD to power on this bit must set “1”.

VCL_EN	Operation of VCL charge bump circuit
0	OFF
1	ON

VGH_EN: ON/OFF the operation of VGH charge bump circuit in manual power on mode. Using state machine SLPOUT CMD to power on this bit must set “1”.

VGH_EN	Operation of VGH charge bump circuit
0	OFF
1	ON

VGL_EN : ON/OFF the operation of VGL charge bump circuit in manual power on mode. Using state machine SLPOUT CMD to power on this bit must set “1”.

VGL_EN	Operation of VGL charge bump circuit
0	OFF
1	ON

VSP_EN: ON/OFF the operation of VSP circuit in manual power on mode. Using state machine SLPOUT CMD to power on this bit must set “1”.

VSP_EN	Operation of VSP DC/DC circuit
0	OFF
1	ON

VSN_EN: ON/OFF the operation of VSN circuit in manual power on mode. Using state machine SLPOUT CMD to power on this bit must set “1”.

VSN_EN	Operation of VSN DC/DC circuit
0	OFF
1	ON

GON: GIP control signal enable selection

GON	GIP Enable
0	GIP Off
1	GIP On

DTE: Source output enable selection

DTE	Source Output
0	Source output off
1	Source output on

D[1:0]: Setting Source driver output

D1	D0	Source Output	HX8399-A Internal Operations
0	0	Halt	Halt
0	1	Blanking or GND	GIP initial sequence
1	0	V255 or V0	Blanking
1	1	Normal	Operate

Restrictions: SETEXTC turn on to enable this command.

Register Availability:

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

6.3.2 SETPOWER: Set power (B1h)

B1H	SETPOWER(Set power related setting)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	0	1	1	0	0	0	1	B1
1st parameter	1	-	-	-	-	-	-	-	DSTB	00
2nd parameter	1	VSP_FBOFF	APF_EN	DD_TU	GASVCI_OPT[1:0]		AP[2:0]			64
3rd parameter	1	-	VCI_LDOS[1:0]		VRHP[4:0]					32
4th parameter	1	-	DT[1:0]		VRHN[4:0]					32
5th parameter	1	VSN_REGS[3:0]				VSP_REGS[3:0]				44
6th parameter	1	-	-	-	DUTY_ADJ_DLY[1:0]		XDK[2:0]			09
7th parameter	1	FS0[3:0]				FS1[3:0]				22
8th parameter		FS2[3:0]				FS3[3:0]				22
9th parameter	1	VCL[2:0]			BTP[4:0]					71
10th parameter	1	CLK_OPT[2:0]			BTN[4:0]					F1
11st parameter	1	VGHS[7:0]								E4
12ndparameter	1	VGLS[7:0]								E4
13rd parameter	1	DCDIVS[3:0]				DCDIV[3:0]				06
14th parameter	1	DCPS[3:0]				DCP[3:0]				FB
15th parameter	1	DCNS[3:0]				DCN[3:0]				FC
16th parameter	1	DTPS[3:0]				DTP[3:0]				05
17th parameter	1	DTNS[3:0]				DTN[3:0]				07
18th parameter	1	MAXDUTY[3:0]				MINDUTY[3:0]				E3
19th parameter	1	RPRIORITY[1:0]		STEP[3:0]				EN_VSN_CLAMP	EN_DUTY_ADJ	06

Description

This command is used to set related setting of power.

DSTB: Set '1' to enter deep standby mode for saving power in SLPIN mode. User must enter SLPIN mode before enter deep standby mode and leave deep stand by mode before SLPOUT.

AP[2:0]: Adjust the amount of fixed current from the fixed current source for the operational amplifier in the power supply circuit. When the amount of fixed current is increased, the LCD driving capacity and the display quality are high, but the current consumption is increased. This is a tradeoff, Adjust the fixed current by considering both the display quality and the current consumption. During no display operation, when AP[2:0] = 000, the current consumption can be reduced by stopping the operations of operational amplifier and step-up circuit.

AP2	AP1	AP0	Constant Current of Operational Amplifier
0	0	0	Stop
0	0	1	0.5µA
0	1	0	1.0µA
0	1	1	1.5µA
1	0	0	2.0µA
1	0	1	2.5µA
1	1	0	3.0µA
1	1	1	3.5µA

GASVCI_OPT[1:0]: Set the threshold voltage of GAS function(APF_EN =1)

GASVCI_OPT1	GASVCI_OPT0	GAS threshold voltage
0	0	1.73V
0	1	2.15V
1	0	2.36V
1	1	2.58V

DD_TU: Set DD_TU="1", VDDD will increase at Sleep In mode.

APF_EN: Abnormal power-off detection enable(GAS function). "1": Enable.

VSP_FBOFF: VSP voltage feedback to control VSP pumping clock operation. "1" no feecdback. For HX5186 mode, no effect for PFM circuit.
When AUTO_XDK=0, set VSP at fixed pumping ratio

VCI_LDOS[1:0]: Set the regulated voltage of VDD3 in external VSP mode.
(PCCS[2:0]= 010 or 011)

VCI_LDOS 1	VCI_LDOS 0	VDD3 threshold voltage
0	0	VDDDX1.6
0	1	VDDDX2
1	0	VDDDX2.2
1	1	VDDDX2.5

DT[1:0]: Delay time of power on and power off sequence.

DT1	DT0	Delay time of power on and power off sequence
0	0	5ms
0	1	10ms
1	0	15ms
1	1	20ms

VRHP[4:0]: VSPR regulator output control setting for source data output driving.

VRHP[4:0]					VSPR Voltage
0	0	0	0	0	Inhibited
0	0	0	0	1	3.1V
0	0	0	1	0	3.2V
0	0	0	1	1	3.3V
0	0	1	0	0	3.4V
0	0	1	0	1	3.5V
0	0	1	1	0	3.6V
0	0	1	1	1	3.7V
0	1	0	0	0	3.8V
0	1	0	0	1	3.9V
0	1	0	1	0	4.0V
0	1	0	1	1	4.1V
0	1	1	0	0	4.2V
0	1	1	0	1	4.3V
0	1	1	1	0	4.4V
0	1	1	1	1	4.5V
1	0	0	0	0	4.6V
1	0	0	0	1	4.7V
1	0	0	1	0	4.8V
1	0	0	1	1	4.9V
1	0	1	0	0	5.0V
1	0	1	0	1	5.1V
1	0	1	1	0	5.2V
1	0	1	1	1	5.3V
1	1	0	0	0	5.4V
1	1	0	0	1	5.5V
1	1	0	1	0	5.6V
1	1	0	1	1	5.7V
1	1	1	0	0	5.8V
1	1	1	0	1	5.9V
1	1	1	1	0	6.0V
1	1	1	1	1	6.1V

VRHN[4:0]: VSNR regulator output control setting for source data output driving

VRHN[4:0]					VSNR Voltage
0	0	0	0	0	Inhibited
0	0	0	0	1	-3.1V
0	0	0	1	0	-3.2V
0	0	0	1	1	-3.3V
0	0	1	0	0	-3.4V
0	0	1	0	1	-3.5V
0	0	1	1	0	-3.6V
0	0	1	1	1	-3.7V
0	1	0	0	0	-3.8V
0	1	0	0	1	-3.9V
0	1	0	1	0	-4.0V
0	1	0	1	1	-4.1V
0	1	1	0	0	-4.2V
0	1	1	0	1	-4.3V
0	1	1	1	0	-4.4V
0	1	1	1	1	-4.5V
1	0	0	0	0	-4.6V
1	0	0	0	1	-4.7V
1	0	0	1	0	-4.8V
1	0	0	1	1	-4.9V
1	0	1	0	0	-5.0V
1	0	1	0	1	-5.1V
1	0	1	1	0	-5.2V
1	0	1	1	1	-5.3V
1	1	0	0	0	-5.4V
1	1	0	0	1	-5.5V
1	1	0	1	0	-5.6V
1	1	0	1	1	-5.7V
1	1	1	0	0	-5.8V
1	1	1	0	1	-5.9V
1	1	1	1	0	-6.0V
1	1	1	1	1	-6.1V

VSP_REGS[3:0]: VSP_REG regulator output control setting.

VSP_REGS[3:0]				VSP_REG Voltage
0	0	0	0	4.6V
0	0	0	1	4.7V
0	0	1	0	4.8V
0	0	1	1	4.9V
0	1	0	0	5.0V
0	1	0	1	5.1V
0	1	1	0	5.2V
0	1	1	1	5.3V
1	0	0	0	5.4V
1	0	0	1	5.5V
1	0	1	0	5.6V
1	0	1	1	5.7V
1	1	0	0	5.8V
1	1	0	1	5.9V
1	1	1	0	6.0V
1	1	1	1	6.1V

VSN_REGS[3:0]: VSN_REG regulator output control setting.

VSN_REGS[3:0]				VSN_REG Voltage
0	0	0	0	-4.6V
0	0	0	1	-4.7V
0	0	1	0	-4.8V
0	0	1	1	-4.9V
0	1	0	0	-5.0V
0	1	0	1	-5.1V
0	1	1	0	-5.2V
0	1	1	1	-5.3V
1	0	0	0	-5.4V
1	0	0	1	-5.5V
1	0	1	0	-5.6V
1	0	1	1	-5.7V
1	1	0	0	-5.8V
1	1	0	1	-5.9V
1	1	1	0	-6.0V
1	1	1	1	-6.1V

DUTY_ADJ_DLY[1:0]: Delay time option for enable auto duty adjustment.
00: 0us
01:100us(default)
10:200us
11:300us

XDK[2:0]: Setting charge pump mode of VSP Voltage

XDK2	XDK1	XDK0	VSP
0	0	0	inhibit
0	0	1	X2
0	1	0	X1.5
0	1	1	inhibit
1	0	0	inhibit
1	0	1	inhibit
1	1	0	X2.5
1	1	1	X3

FS0[3:0]: Set the operating frequency of the step-up circuit for VSP voltage generation. (Fosc_pump=5MHz)

FS03	FS02	FS01	FS00	Operation Frequency of Step-up Circuit
0	0	0	0	Fosc_pump/2
0	0	0	1	Fosc_pump/4
0	0	1	0	Fosc_pump/8
0	0	1	1	Fosc_pump/16
0	1	0	0	Fosc_pump/32
0	1	0	1	Fosc_pump/48
0	1	1	0	Fosc_pump/64
0	1	1	1	Fosc_pump/80
1	0	0	0	Fosc_pump/96
1	0	0	1	Fosc_pump/112
1	0	1	0	Fosc_pump/128
1	0	1	1	Fosc_pump/144
1	1	0	0	Fosc_pump/160
1	1	0	1	Fosc_pump/176
1	1	1	0	Fosc_pump/192
1	1	1	1	Fosc_pump/208

FS1[3:0]: Set the operating frequency of the step-up circuit for VGH voltage generation. (Fosc_pump=5MHz)

FS13	FS12	FS11	FS10	Operation Frequency of Step-up Circuit
0	0	0	0	Fosc_pump/32
0	0	0	1	Fosc_pump/64
0	0	1	0	Fosc_pump/96
0	0	1	1	Fosc_pump/128
0	1	0	0	Fosc_pump/160
0	1	0	1	Fosc_pump/192
0	1	1	0	Fosc_pump/224
0	1	1	1	Fosc_pump/256
1	0	0	0	Hsync*4
1	0	0	1	Hsync*2
1	0	1	0	Hsync
1	0	1	1	Hsync/2
1	1	0	0	Hsync/4
1	1	0	1	Hsync/8
1	1	1	0	Hsync/16
1	1	1	1	Inhibited

FS2[3:0]: Adjust the charge pump frequency of internal VCL. (Fosc_pump=5MHz)

FS23	FS22	FS21	FS20	Operation Frequency of Step-up Circuit
0	0	0	0	Fosc_pump/32
0	0	0	1	Fosc_pump/64
0	0	1	0	Fosc_pump/96
0	0	1	1	Fosc_pump/128
0	1	0	0	Fosc_pump/160
0	1	0	1	Fosc_pump/192
0	1	1	0	Fosc_pump/224
0	1	1	1	Fosc_pump/256
1	0	0	0	Hsync*4
1	0	0	1	Hsync*2
1	0	1	0	Hsync
1	0	1	1	Hsync/2
1	1	0	0	Hsync/4
1	1	0	1	Hsync/8
1	1	1	0	Hsync/16
1	1	1	1	Inhibited

FS3[3:0]: Set the operating frequency of the step-up circuit for VGL voltage generation. (Fosc_pump=5MHz)

FS33	FS32	FS31	FS30	Charge pump frequency of VCL
0	0	0	0	Fosc_pump/32
0	0	0	1	Fosc_pump/64
0	0	1	0	Fosc_pump/96
0	0	1	1	Fosc_pump/128
0	1	0	0	Fosc_pump/160
0	1	0	1	Fosc_pump/192
0	1	1	0	Fosc_pump/224
0	1	1	1	Fosc_pump/256
1	0	0	0	Hsync*4
1	0	0	1	Hsync*2
1	0	1	0	Hsync
1	0	1	1	Hsync/2
1	1	0	0	Hsync/4
1	1	0	1	Hsync/8
1	1	1	0	Hsync/16
1	1	1	1	Inhibited

VCL[2:0]: Set the power level of VCL voltage.

VCL2	VCL1	VCL0	VCL Voltage
0	0	0	-2.5V
0	0	1	-2.6V
0	1	0	-2.7V
0	1	1	-2.8V
1	0	0	-2.9V
1	0	1	-3V
1	1	0	-3.1V
1	1	1	-VDD3

BTP[4:0]: Switch the output factor for DC/DC circuit for VSP voltage generation. The LCD drive voltage level VSP can be selected according to the characteristic of liquid crystal which panel used.

BTP4	BTP3	BTP2	BTP1	BTP0	VSP Voltage
0	0	0	0	0	3.00V
0	0	0	0	1	3.15V
0	0	0	1	0	3.30V
0	0	0	1	1	3.45V
0	0	1	0	0	3.60V
		:			:
1	0	0	0	0	5.40V
1	0	0	0	1	5.55V
1	0	0	1	0	5.70V
1	0	0	1	1	5.85V
1	0	1	0	0	6.00V
1	0	1	0	1	6.15V
1	0	1	1	0	6.30V
1	0	1	1	1	6.45V
1	1	0	0	0	6.60V
1	1	0	0	1	6.75V
		Others			Inhibited

BTN[4:0]: Switch the output factor of DC/DC circuit for VSN voltage generation. The LCD drive voltage level VSN can be selected according to the characteristic of liquid crystal which panel used.

BTN4	BTN3	BTN2	BTN1	BTN0	VSN Voltage
0	0	0	0	0	-3.00V
0	0	0	0	1	-3.15V
0	0	0	1	0	-3.30V
0	0	0	1	1	-3.45V
0	0	1	0	0	-3.60V
		:			:
1	0	0	0	0	-5.40V
1	0	0	0	1	-5.55V
1	0	0	1	0	-5.70V
1	0	0	1	1	-5.85V
1	0	1	0	0	-6.00V
1	0	1	0	1	-6.15V
1	0	1	1	0	-6.30V
1	0	1	1	1	-6.45V
1	1	0	0	0	-6.60V
1	1	0	0	1	-6.75V
		Others			Inhibited

CLK_OPT0: The pumping clock of VGH will reset with Hsync when CLK_OPT0 = 1.

CLK_OPT1: The pumping clock of VCL will reset with Hsync when CLK_OPT1 = 1.

CLK_OPT2: The pumping clock of VGL will reset with Hsync when CLK_OPT2 = 1.

VGHS[7:6]: Specify the VGH voltage source.

VGHS7	VGHS6	VGH Voltage
0	0	VDD3-VSN
0	1	VSP-VSN
1	0	VSP-VSN+VDD3
1	1	2*VSP -VSN

VGHS[5:0]: VGHO regulator output voltage setting. The LCD drive voltage level VGHO can be selected according to the characteristic of liquid crystal which panel used.

VGHS5	VGHS4	VGHS3	VGHS2	VGHS1	VGHS0	VGHO Voltage
0	0	0	0	0	0	6.0V
0	0	0	0	0	1	6.1V
0	0	0	0	1	0	6.2V
0	0	0	0	1	1	6.3V
			:			:
1	0	0	1	0	0	9.6V
			:			:
1	1	1	1	0	1	12.1V
1	1	1	1	1	0	12.2v
1	1	1	1	1	1	12.3V

VGLS[7:6]: Specify the VGL voltage source.

VGLS7	VGLS6	VGL Voltage
0	0	VSN-VDD3
0	1	VSN-VSP
1	0	2*VSN-VDD3
1	1	2*VSN-VSP

VGLS[5:0]: VGLO regulator output voltage setting. The LCD drive voltage level VGLO can be selected according to the characteristic of liquid crystal which panel used.

VGLS5	VGLS4	VGLS3	VGLS2	VGLS1	VGLS0	VGLO Voltage
0	0	0	0	0	0	-5.0V
0	0	0	0	0	1	-5.1V
0	0	0	0	1	0	-5.2V
0	0	0	0	1	1	-5.3V
			:			:
1	0	0	1	0	0	-8.6V
			:			:
1	1	1	1	0	1	-11.1V
1	1	1	1	1	0	-11.2V
1	1	1	1	1	1	-11.3V

DCDIV[3:0]: Set the normal operate frequency FoscD of DC/DC converter circuit during normal mode. (Fosc=73MHz)

DCDIV[3:0]				Normal operate frequency of DC/DC converter(foscD)
0	0	0	0	Fosc/1
0	0	0	1	Fosc/2
0	0	1	0	Fosc/3
0	0	1	1	Fosc/4
0	1	0	0	Fosc/5
0	1	0	1	Fosc/6
0	1	1	0	Fosc/7
0	1	1	1	Fosc/8
1	0	0	0	Fosc/9

1	0	0	1	Fosc/10
1	0	1	0	Fosc/11
1	0	1	1	Fosc/12
1	1	0	0	Fosc/13
1	1	0	1	Fosc/14
1	1	1	0	Fosc/15
1	1	1	1	Fosc/16

DCDIVS[3:0]: Set the soft start operate frequency FoscD of DC/DC converter circuit during normal mode. (Fosc=73MHz)

DCDIVS[3:0]				Normal operate frequency of DC/DC converter(foscD)
0	0	0	0	Fosc/1
0	0	0	1	Fosc/2
0	0	1	0	Fosc/3
0	0	1	1	Fosc/4
0	1	0	0	Fosc/5
0	1	0	1	Fosc/6
0	1	1	0	Fosc/7
0	1	1	1	Fosc/8
1	0	0	0	Fosc/9
1	0	0	1	Fosc/10
1	0	1	0	Fosc/11
1	0	1	1	Fosc/12
1	1	0	0	Fosc/13
1	1	0	1	Fosc/14
1	1	1	0	Fosc/15
1	1	1	1	Fosc/16

DCP[3:0]: Operation VSP frequency of DC/DC clock.

DCP[3:0]				Operation Frequency of DC/DC Clock
0	0	0	0	1/foscD
0	0	0	1	2/foscD
0	0	1	0	3/foscD
0	0	1	1	4/foscD
0	1	0	0	5/foscD
0	1	0	1	6/foscD
0	1	1	0	7/foscD
0	1	1	1	8/foscD
1	0	0	0	9/foscD
1	0	0	1	10/foscD
1	0	1	0	11/foscD
1	0	1	1	12/foscD
1	1	0	0	13/foscD
1	1	0	1	14/foscD
1	1	1	0	15/foscD
1	1	1	1	foscD/16

DCPS[3:0]: Soft start VSP frequency of DC/DC clock.

DCPS[3:0]				Operation Frequency of DC/DC Clock
0	0	0	0	1/foscD
0	0	0	1	2/foscD
0	0	1	0	3/foscD
0	0	1	1	4/foscD
0	1	0	0	5/foscD
0	1	0	1	6/foscD
0	1	1	0	7/foscD
0	1	1	1	8/foscD

1	0	0	0	9/foscD
1	0	0	1	10/foscD
1	0	1	0	11/foscD
1	0	1	1	12/foscD
1	1	0	0	13/foscD
1	1	0	1	14/foscD
1	1	1	0	15/foscD
1	1	1	1	16/foscD

DCN[3:0]: Operation VSN frequency of DC/DC clock.

DCN[3:0]				Operation Frequency of DC/DC Clock
0	0	0	0	1/foscD
0	0	0	1	2/foscD
0	0	1	0	3/foscD
0	0	1	1	4/foscD
0	1	0	0	5/foscD
0	1	0	1	6/foscD
0	1	1	0	7/foscD
0	1	1	1	8/foscD
1	0	0	0	9/foscD
1	0	0	1	10/foscD
1	0	1	0	11/foscD
1	0	1	1	12/foscD
1	1	0	0	13/foscD
1	1	0	1	14/foscD
1	1	1	0	15/foscD
1	1	1	1	16/foscD

DCNS[3:0]: Soft start VSN frequency of DC/DC clock.

DCNS[3:0]				Operation Frequency of DC/DC Clock
0	0	0	0	1/foscD
0	0	0	1	2/foscD
0	0	1	0	3/foscD
0	0	1	1	4/foscD
0	1	0	0	5/foscD
0	1	0	1	6/foscD
0	1	1	0	7/foscD
0	1	1	1	8/foscD
1	0	0	0	9/foscD
1	0	0	1	10/foscD
1	0	1	0	11/foscD
1	0	1	1	12/foscD
1	1	0	0	13/foscD
1	1	0	1	14/foscD
1	1	1	0	15/foscD
1	1	1	1	16/foscD

DTP[3:0]: For PFM circuit:Set the operating duty cycle of DC/DC clock for VSP.

DTP[3:0]				Operation duty of VSP
0	0	0	0	1/foscD
0	0	0	1	2/foscD
0	0	1	0	3/foscD
0	0	1	1	4/foscD
0	1	0	0	5/foscD
0	1	0	1	6/foscD
0	1	1	0	7/foscD
0	1	1	1	8/foscD
1	0	0	0	9/foscD

1	0	0	1	10/foscD
1	0	1	0	11/foscD
1	0	1	1	12/foscD
1	1	0	0	13/foscD
1	1	0	1	14/foscD
1	1	1	0	15/foscD
1	1	1	1	16/foscD

DTPS[3:0]: For PFM circuit. Set the soft start operating duty cycle of DC/DC circuit.

DTPS[3:0]				Soft start operation duty of VSP
0	0	0	0	1/foscD
0	0	0	1	2/foscD
0	0	1	0	3/foscD
0	0	1	1	4/foscD
0	1	0	0	5/foscD
0	1	0	1	6/foscD
0	1	1	0	7/foscD
0	1	1	1	8/foscD
1	0	0	0	9/foscD
1	0	0	1	10/foscD
1	0	1	0	11/foscD
1	0	1	1	12/foscD
1	1	0	0	13/foscD
1	1	0	1	14/foscD
1	1	1	0	15/foscD
1	1	1	1	16/foscD

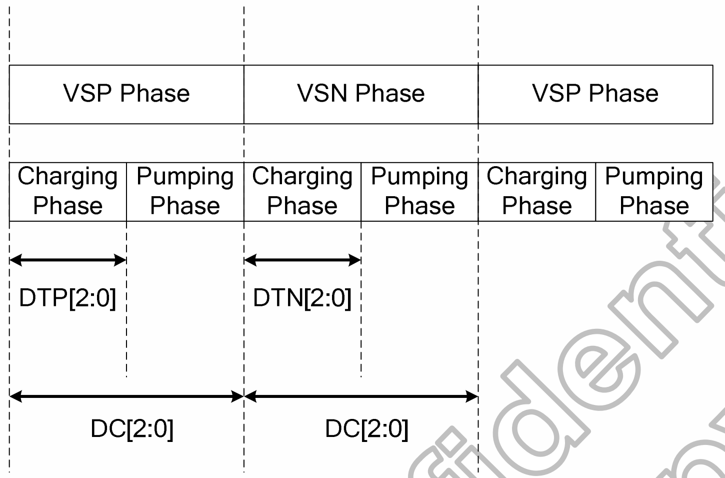
DTN[3:0]: For PFM circuit. Set the operating duty cycle of DC/DC clock for VSN.

DTN[3:0]				Operation duty of VSN
0	0	0	0	1/foscD
0	0	0	1	2/foscD
0	0	1	0	3/foscD
0	0	1	1	4/foscD
0	1	0	0	5/foscD
0	1	0	1	6/foscD
0	1	1	0	7/foscD
0	1	1	1	8/foscD
1	0	0	0	9/foscD
1	0	0	1	10/foscD
1	0	1	0	11/foscD
1	0	1	1	12/foscD
1	1	0	0	13/foscD
1	1	0	1	14/foscD
1	1	1	0	15/foscD
1	1	1	1	16/foscD

DTNS[3:0]: For PFM circuit. Set the soft start operating duty cycle of DC/DC circuit.

DTNS[3:0]				Soft start operation duty of VSN
0	0	0	0	1/foscD
0	0	0	1	2/foscD
0	0	1	0	3/foscD
0	0	1	1	4/foscD
0	1	0	0	5/foscD
0	1	0	1	6/foscD
0	1	1	0	7/foscD
0	1	1	1	8/foscD
1	0	0	0	9/foscD
1	0	0	1	10/foscD

1	0	1	0	11/foscD
1	0	1	1	12/foscD
1	1	0	0	13/foscD
1	1	0	1	14/foscD
1	1	1	0	15/foscD
1	1	1	1	16/foscD



MAXDUTY[3:0]: For PFM circuit, set the max duty of DTP[3:0], DTN[3:0] when auto duty adjustment application.

MAXDUTY[3:0]				Max. duty of DTP[3:0], DTN[3:0]
0	0	0	0	1/foscD
0	0	0	1	2/foscD
0	0	1	0	3/foscD
0	0	1	1	4/foscD
0	1	0	0	5/foscD
0	1	0	1	6/foscD
0	1	1	0	7/foscD
0	1	1	1	8/foscD
1	0	0	0	9/foscD
1	0	0	1	10/foscD
1	0	1	0	11/foscD
1	0	1	1	12/foscD
1	1	0	0	13/foscD
1	1	0	1	14/foscD
1	1	1	0	15/foscD
1	1	1	1	16/foscD

MINDUTY[3:0]: For PFM circuit, set the min duty of DTP[3:0], DTN[3:0] when auto duty adjustment application.

MINDUTY[3:0]				Min. duty of DTP[3:0], DTN[3:0]
0	0	0	0	1/foscD
0	0	0	1	2/foscD
0	0	1	0	3/foscD
0	0	1	1	4/foscD
0	1	0	0	5/foscD
0	1	0	1	6/foscD
0	1	1	0	7/foscD
0	1	1	1	8/foscD
1	0	0	0	9/foscD
1	0	0	1	10/foscD
1	0	1	0	11/foscD
1	0	1	1	12/foscD

1	1	0	0	13/foscD
1	1	0	1	14/foscD
1	1	1	0	15/foscD
1	1	1	1	16/foscD

RPRIORITY[1:0]: Priority setting when pump VSP and pump VSN request come together.
00:alternately(default)
01:VSN 1st priority
10:VSP 1st priority
11:alternately

STEP[3:0]: Set step of auto duty adjustment.

STEP[3:0]				Adjust width of duty
0	0	0	0	1/foscD
0	0	0	1	2/foscD
0	0	1	0	3/foscD
0	0	1	1	4/foscD
0	1	0	0	5/foscD
0	1	0	1	6/foscD
0	1	1	0	7/foscD
0	1	1	1	8/foscD
1	0	0	0	9/foscD
1	0	0	1	10/foscD
1	0	1	0	11/foscD
1	0	1	1	12/foscD
1	1	0	0	13/foscD
1	1	0	1	14/foscD
1	1	1	0	15/foscD
1	1	1	1	16/foscD

EN_VSN_CLAMP: Enable VSN clamp function. ‘1’: Enable; ‘0’: Disable.

EN_DUTY_ADJ: Enable auto duty adjustment. ‘1’: Enable; ‘0’: Disable.

Restrictions: SETEXTC turn on to enable this command.

Register Availability:

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

6.3.3 SETDISP: Set display related register (B2h)

B2H	SETDISP(Set display related register)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	0	1	1	0	0	1	0	B2
1st parameter	1	ZZ_LR	ZZ_EO	-	-	-	NW[2:0]			80
2nd parameter	1	MUX_SEL	-	-	-	TGS[3:0]				00
3rd parameter	1	-	-	H_RES[2:0]			V_RES[1:0]			00
4th parameter	1	NL[7:0]								7F
5th parameter	1	BP [7:0]								0B
6th parameter	1	FP [7:0]								11
7th parameter	1	RTN[7:0]								23
8th parameter	1	RTN_GAS[7:0]								4D
9th parameter	1	-	INIT_SD_SEL	INIT_VCOM_SEL	-	INIT_SET[3:0]				01
10th parameter	1	-	END_SD_SEL	END_VCOM_SEL	-	END_SET[3:0]				01

Description

This command is used to set display related register

NW[2:0]: Inversion type setting.

NW2	NW1	NW0	Inversion type
0	0	0	Column inversion
0	0	1	1-dot inversion
0	1	0	2-dot inversion
0	1	1	4-dot inversion
1	0	0	8-dot inversion
1	0	1	Zig-zag inversion
1	1	0	Pixel-column inversion
1	1	1	Inhibited

ZZ_LR: Zig-zag Left / Right mode selection.

ZZ_LR	Zig-zag Left / Right mode selection
0	S1~SR1
1	SL1~S1200

ZZ_EO: Zig-zag Odd / Even mode selection.

ZZ_EO	Zig-zag Odd / Even mode selection
0	Odd-line shift
1	Even-line shift

MUX_SEL: Source driver switch select.
0: Setting 1:3 MUX
1: Setting 2:6 MUX

TGS[3:0]: Source switch sequence select.

TGS3	TGS2	TGS1	TGS0	TG sequence
0	0	0	0	SW1→ SW2→ SW3
0	0	0	1	SW2→ SW3→ SW1
0	0	1	0	SW2→ SW1→ SW3
0	0	1	1	SW3→ SW1→ SW2
Others				inhibited

H_RES[2:0], V_RES[1:0]: Resolution selection.

H_RES[2:0]	V_RES[1:0]	Resolution	Source channels
000	00	1080RGBX1920 dot	S1 ~ S540 , S661 ~ S1200
000	10	1080RGBX1440 dot	S1 ~ S540 , S661 ~ S1200
001	11	1024RGBX1280 dot	S1 ~ S512 , S689 ~ S1200
001	01	1024RGBX1600 dot	S1 ~ S512 , S689 ~ S1200
010	01	960RGBX1600 dot	S1 ~ S480 , S721 ~ S1200
010	10	960RGBX1440 dot	S1 ~ S480 , S721 ~ S1200
010	11	960RGBX1280 dot	S1 ~ S480 , S721 ~ S1200
011	01	900RGBX1600 dot	S1 ~ S450 , S751 ~ S1200
011	10	900RGBX1440 dot	S1 ~ S450 , S751 ~ S1200
100	00	1200RGBX1920 dot	S1 ~ S1200
101	11	720RGBx1280 dot	S1 ~ S360 , S841 ~ S1200
110	11	800RGBx1280 dot	S1 ~ S400 , S801 ~ S1200
111	-	Inhibited	-

NL[7:0]: Setting the number of lines to drive the LCD at an interval of 8 lines. The number of lines must be the same or more than the number of lines necessary for the size of the liquid crystal panel. If user want to use this function, please set V_RES[1:0]=00.

NL[7:0]								Lines
0	0	0	0	0	0	0	0	904
0	0	0	0	0	0	0	1	912
:								:
0	1	1	1	1	1	1	1	1920
Others								Inibited

BP[7:0] : Specify the amount of scan line for back porch(BP) in blanking.

FP[7:0]: Specify the amount of scan line for front porch (FP) in blanking.

FP[7:0] / BP[7:0]	Number of front porch/ back porch Lines
8h'00	2 lines
8h'01	3 lines
8h'02	4 lines
8h'03	5 lines
8h'04	6 lines
8h'05	7 lines
:	:
8h'FB	253 lines
8h'FC	254 lines
8h'FD	255 lines
8h'FE	256 lines
8h'FF	257 lines

Note: Set BP[7:0] = VS + VBP – 2, and FP[7:0] = VFP – 2.

RTN[7:0]: A cycle time of line width in blanking.

(1 clock period= 1/8MHz)

RTN[7:0]	Clock per Line
8h'00	150 clocks
8h'01	151 clocks
8h'02	152 clocks
:	:
8'hFD	403 clocks
8'hFE	404 clocks
8'hFF	405 clocks

	RTN_GAS[7:0]: no function. **INIT_SD_SEL/END_SD_SEL:** Source driver voltage control at D[1:0]=01

INIT_SD_SEL /END_SD_SEL	Voltage level
0	GND
1	Blanking

INIT_VCOM_SEL/END_VCOM_SEL: VCOM voltage control at D[1:0]=01

INIT/END_VCOM_SEL	Voltage level
0	GND
1	VCMC_F/VCMC_B

INIT_SET/END_SET[3:0]: GIP control at D[1:0]=01, GIP state set by RD8h

INIT_SET[3:0]/ END_SET[3:0]	Time or Frame	By time/frame
0000	0 frame	By frame
0001	1 frame	By frame
0010	2 frames	By frame
0011	3 frames	By frame
0100~0111	inhibited	inhibited
1000	4ms	By time
1001	8ms	By time
1010	12ms	By time
1011	16ms	By time
1100	20ms	By time
1101	24ms	By time
1110	28ms	By time
1111	32ms	By time

Note: INIT means SLPin to SLPOUT
END means SLPout to SLPIN

Restrictions	SETEXTC turn on to enable this command.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

6.3.4 SETCYC: Set display waveform cycles (B4h)

B4H	SETCYC(Set panel driving timing)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	0	1	1	0	1	0	0	B4
1st parameter	1	GEN_ON[7:0]								00
2nd parameter	1	GEN_OFF[7:0]								FF
3rd parameter	1	SPON[7:0]								03
4th parameter	1	SPOFF[7:0]								38
5th parameter	1	CON[7:0]								05
6th parameter	1	COFF[7:0]								36
7th parameter	1	CON1[7:0]								05
8th parameter	1	COFF1[7:0]								36
9th parameter	1	N_t1[7:0]								00
10th parameter	1	N_t2[7:0]								09
11st parameter	1	N_t3[7:0]								01
12nd parameter	1	N_t4[7:0]								01
13rd parameter	1	N_t5[7:0]								00
14th parameter	1	N_t6[7:0]								10
15th parameter	1	N_t7[7:0]								01
16th parameter	1	N_t8[7:0]								02
17th parameter	1	N_t9[7:0]								02
18th parameter	1	SAP[3:0]				EQT[3:0]				71
19th parameter	1	N_t10[7:0]								00
20th parameter	1	SON[7:0]								00
21st parameter	1	SOFF[7:0]								FF
22nd parameter	1	SPON_MPU[7:0]								03
23rd parameter	1	SPOFF_MPU[7:0]								38
24th parameter	1	CON_MPU[7:0]								05
25th parameter	1	COFF_MPU[7:0]								36
26th parameter	1	CON1_MPU[7:0]								05
27th parameter	1	COFF1_MPU[7:0]								36
28th parameter	1	N_t1_MPU[7:0]								00
29th parameter	1	N_t2_MPU[7:0]								09
30th parameter	1	N_t3_MPU[7:0]								01
31st parameter	1	N_t4_MPU[7:0]								01
32nd parameter	1	N_t5_MPU[7:0]								00
33rd parameter	1	N_t6_MPU[7:0]								10
34th parameter	1	N_t7_MPU[7:0]								01
35th parameter	1	N_t8_MPU[7:0]								02
36th parameter	1	N_t9_MPU[7:0]								02
37th parameter	1	-	-	-	-	EQT_MPU[3:0]				01
38th parameter	1	N_t10_MPU[7:0]								00
39th parameter	1	SON_MPU[7:0]								00
40th parameter	1	SOFF_MPU[7:0]								FF

Description

This command is used to set display waveform cycles.

SAP[3:0]: Set Current of Operational Amplifier for source driver

SAP3	SAP2	SAP1	SAP0	Fixed Current of Operational Amplifier
0	0	0	0	1 * Iref
0	0	0	1	2 * Iref
0	0	1	0	3 * Iref
0	0	1	1	4 * Iref
0	1	0	0	5 * Iref
:				
1	1	1	1	16 * Iref

GEN_ON[7:0]: Gamma OP turned on timing (in-house function not open).

GEN_OFF[7:0]: Gamma OP turned off timing (in-house function not open).

SPON[7:0]: Fine tune the Start and End signal delay from original starting point.
(1 OSC CLK = 1/8MHz)

SPON[7:0]	Start / END signal output start delay
0x00h	0 x OSC CLK
0x01h	1 x OSC CLK
0x02h	2 x OSC CLK
0x03h	3 x OSC CLK
:	
0xFEh	254 OSC CLK
0xFFh	255 OSC CLK

SPOFF[7:0]: Fine tune the Start and End signal ending point.

SPOFF[7:0]	Start / END signal output end delay
0x00h	0 x OSC CLK
0x01h	1 x OSC CLK
0x02h	2 x OSC CLK
0x03h	3 x OSC CLK
:	
0xFEh	254 x OSC CLK
0xFFh	255 x OSC CLK

Note: When output Start / End signal width is 1- Hsync only, set SPON[7:0] < SPOFF[7:0]

CON[7:0]/CON1[7:0]: Fine tune the Clock signal delay from original starting point.

CON[7:0]/CON1[7:0]	Clock signal output start delay
0x00h	0 x OSC CLK
0x01h	1 x OSC CLK
0x02h	2 x OSC CLK
0x03h	3 x OSC CLK
:	
0xFEh	254 OSC CLK
0xFFh	255 OSC CLK

COFF[7:0]/COFF1[7:0]: Fine tune the Clock signal ending point.

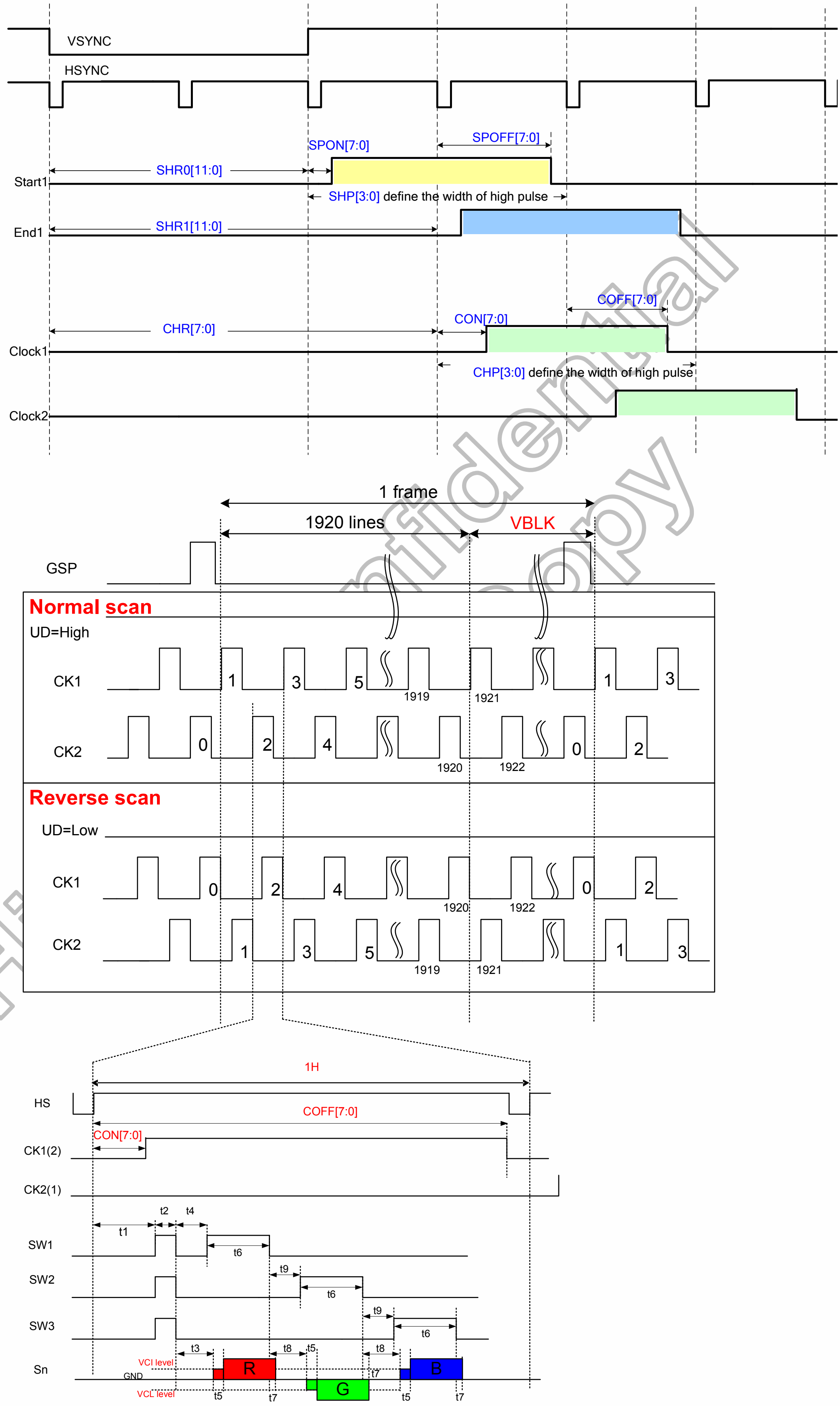
COFF[7:0]/COFF1[7:0]	Clock signal output end delay
0x00h	0 x OSC CLK
0x01h	1 x OSC CLK
0x02h	2 x OSC CLK
0x03h	3 x OSC CLK
:	
0xFEh	254 OSC CLK
0xFFh	255 OSC CLK

Note: When output Clock signal width is 1- Hsync only, set COFF[7:0] ≧ CON[7:0] + 2

SON[7:0]: Source OP turn on time.

SOFF[7:0]: Source OP turn off time.

SON[7:0]/SOFF[7:0]	Source OP turn on/off time
0x00h	0 x OSC CLK
0x01h	1 x OSC CLK
0x02h	2 x OSC CLK
:	
0xFEh	254 OSC CLK
0xFFh	255 OSC CLK


VSYNC
HSYNC
SPON[7:0]
SPOFF[7:0]
SHR0[11:0]
Start1
SHP[3:0] define the width of high pulse
End1
SHR1[11:0]
COFF[7:0]
CON[7:0]
CHR[7:0]
Clock1
CHP[3:0] define the width of high pulse
Clock2
1 frame
1920 lines
VBLK
GSP
Normal scan
UD=High
CK1
CK2
Reverse scan
UD=Low
1H
HS
COFF[7:0]
CON[7:0]
CK1(2)
CK2(1)
SW1
SW2
SW3
Sn
VCI level
GND
VCL level
R
G
B

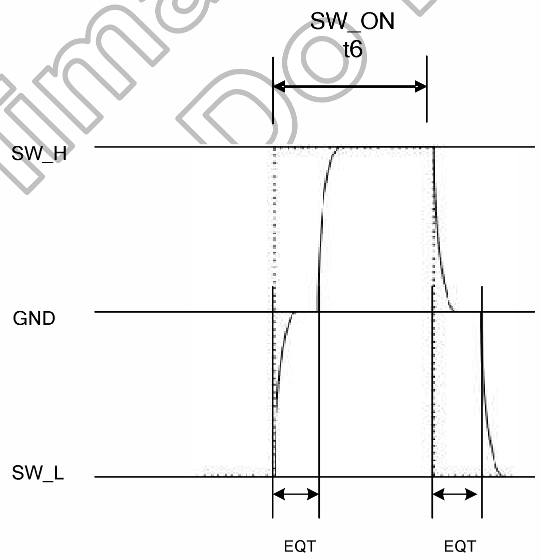
N_t1~t9[7:0]: The timing definition of t1~t9.

N_t1~9[7:0]								Clock cycles
0	0	0	0	0	0	0	0	0 x OSC CLK
0	0	0	0	0	0	0	1	1 x OSC CLK
0	0	0	0	0	0	1	0	2 x OSC CLK
0	0	0	0	0	0	1	1	3 x OSC CLK
0	0	0	0	0	1	0	0	4 x OSC CLK
:	:	:	:	:	:	:	:	:
1	1	1	1	1	1	0	0	252 x OSC CLK
1	1	1	1	1	1	0	1	253 x OSC CLK
1	1	1	1	1	1	1	0	254 x OSC CLK
1	1	1	1	1	1	1	1	255 x OSC CLK

N_t10[7:0]: Reserved.

EQT[3:0]: Equalizing period of TG output.

EQT3	EQT2	EQT1	EQT0	OSC Clock cycles
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3
0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7
1	0	0	0	8
1	0	0	1	9
1	0	1	0	10
1	0	1	1	11
1	1	0	0	12
1	1	0	1	13
1	1	1	0	14
1	1	1	1	15



SPON_MPU[7:0]: Fine tune the Start and End signal delay from original starting point for blanking frame.

SPOFF_MPU[7:0]: Fine tune the Start and End signal ending point for blanking frame.

CON_MPU[7:0]/CON1_MPU[7:0]: Fine tune the Clock signal delay from original starting

	point for blanking frame. **COFF_MPU[7:0]/COFF1_MPU[7:0]:** Fine tune the Clock signal ending point for blanking frame. **SON_MPU[7:0]:** Source OP turn on time for blanking frame. **SOFF_MPU[7:0]:** Source OP turn off time for blanking frame. **N_t1~t9_MPU[7:0]:** The timing definition of t1~t9 for blanking frame. **N_t10_MPU[7:0]:** Reserved. **EQT_MPU[3:0]:** Equalizing period of TG output for blanking frame.
Restrictions	-
Register Availability	see table below

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

6.3.5 SETVCOM: Set VCOM voltage (B6h)

B6 H	SETVCOM (Set VCOM Voltage)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	0	1	1	0	1	1	0	B6
1st parameter	1	VCMC_F[7:0]								46
2nd parameter	1	VCMC_B[7:0]								46
3rd parameter	1	VCOM_TIMES[2:0]			-	-	-	VCMC_B8	VCMC_F8	E0

Description

This command is used to set VCOM Voltage.

VCMC_F[8:0]: Forward scan VCOM voltage control.

VCMC_B[8:0]: Backward scan VCOM voltage control.

VCMC_F[8:0]/VCMC_B[8:0]									VCOM
0	0	0	0	0	0	0	0	0	-0.30V
0	0	0	0	0	0	0	0	1	-0.31V
0	0	0	0	0	0	0	1	0	-0.32V
0	0	0	0	0	0	0	1	1	-0.33V
:									:
1	0	1	1	1	0	0	0	0	-3.98V
1	0	1	1	1	0	0	0	1	-3.99V
1	0	1	1	1	0	0	1	0	-4.00V
Others									Inhibited
1	1	1	1	1	1	1	1	0	VSSA
1	1	1	1	1	1	1	1	1	HZ

VCOM_TIMES[2:0]: Read the VCOM OTP programmed times.

VCOM_TIMES[2:0]	VCOM OTP Programmed Times
111	No programmed
011	VCOM has been programmed 1 time
001	VCOM has been programmed 2 times
000	VCOM has been programmed 3 times

Restrictions

SETEXTC turn on to enable this command.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

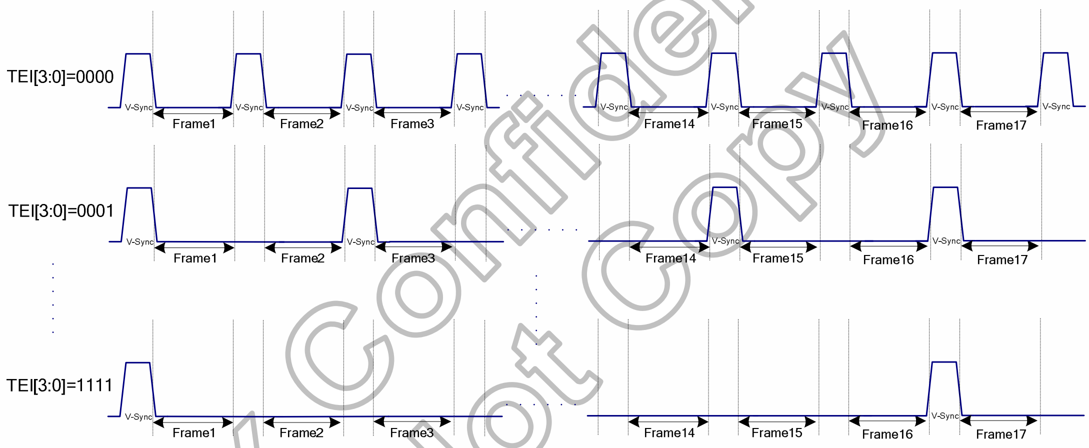
6.3.6 SETTE: Set internal TE function (B7h)

B7H	SETTE (Set internal TE function)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	0	1	1	0	1	1	1	B7
1st parameter	1	TEI[3:0]				-	TEP[10:8]			00
2nd parameter	1	TEP[7:0]								00

Description

TEI[3:0]: Set the output interval of TE signal according to the display data rewrite cycle and data transfer rate.

TEI3	TEI2	TEI1	TEI0	Output Interval
0	0	0	0	1 frame
0	0	0	1	2 frames
0	0	1	0	3 frames
:				:
1	1	1	0	15 frames
1	1	1	1	16 frames



TEP[10:0]: Set the output position of frame cycle signal. TE can be used as the trigger signal for frame synchronous write operation.
Make sure the setting restriction 11'h000 ≤ TEP[10:0] ≤ Numbers of Line-1.

TEP[10:0]	Output position
000h	0th line
001h	1st line
002h	2nd line
003h	3rd line
:	:
780h	1919th line
781h	1920th line
Others	Inhibited

Restrictions: SETEXTC turn on to enable this command.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

6.3.7 SETEXTC: Set extension command (B9h)

<table>
<tr><th rowspan="2">B9H</th><th colspan="10">SETEXTC (Set extended command set)</th></tr>
<tr><th>DCX</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>B9</td></tr>
<tr><td>1st parameter</td><td>1</td><td colspan="8">EXTC1[7:0]</td><td>00</td></tr>
<tr><td>2nd parameter</td><td>1</td><td colspan="8">EXTC2[7:0]</td><td>00</td></tr>
<tr><td>3rd parameter</td><td>1</td><td colspan="8">EXTC3[7:0]</td><td>00</td></tr>
<tr><td>Description</td><td colspan="10">This command is used to set extended command set access enable.
<table><tr><th>Extend cmd</th><th>Command description</th></tr><tr><td>Enable</td><td>After command (B9h), must write 3 parameters (FFh, 83h, 99h) by order</td></tr><tr><td>Disable(default)</td><td>After command(B9h), write 3 parameters (xxh,xxh,xxh) any value except (FFh, 83h, 99h)</td></tr></table></td></tr>
<tr><td>Restrictions</td><td colspan="10">-</td></tr>
<tr><td>Register Availability</td><td colspan="10"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table></td></tr>
</table>

6.3.8 SETMIPI: Set MIPI control (BAh)

BAH	SETMIPI(Set MIPI control)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	0	1	1	1	0	1	0	BA
1st parameter	1	-	DSISETUP0[6:0]							03
2nd parameter	1	DSISETUP1[7:0]								82

Description

This command is used to set MIPI DSI Related Setting.

DSISETUP0[6]	Tx Type: Define the LP-TX BTA behavior when there are error. 0: only BTA Error(default) 1: BTA Read + Error
DSISETUP0[5]	CD_disable: Define the contention detection (LP-CD) function. 0 : LP-CD function enable 1: LP-CD function disable
DSISETUP0[4]	Tx_OscDiv: LP-TX clock (T_{LPX}) selection. 0 : 50ns (10MHz)(osc_clk) 1 :100ns (5MHz)(osc_clk/2)
DSISETUP0[3:2]	vc_main: Define the main function Virtual Channel ID.
DSISETUP0[1:0]	LAN_NUM: Define the DSI lane number 00 : 1-lane 01 : 2-lane 01 : 3-lane 11 : 4-lane

DSISETUP1[7]	CRC_enable: Enable RX CRC Check.
DSISETUP1[6]	ECC_ignore: Define the RX behavior when error occurring. 0 : the transmission will be broken when there are ECC or CRC error 1 : the transmission will keep when there are ECC or CRC error
DSISETUP1[5]	RstTrig: Define the reset trigger function(46h). 0 : Disable reset trigger 1 : same as HW_RESET function
DSISETUP1[4]	Reserved
DSISETUP1[3:2]	Txe_Wait: BTA from Tx into Rx overlap waiting time counter(T_{TA-GO}) 00: TA-go = 2 Tlpx 01: TA-go = 4 Tlpx 10: TA-go = 6 Tlpx 11: TA-go = 8 Tlpx
DSISETUP1[1:0]	Txs_Wait: BTA from Rx into Tx overlap waiting time counter(T_{TA-GET}) 00: Disable, no wait time 01: TA-get = 2 Tlpx 10: TA-get = 4 Tlpx 11: TA-get = 6 Tlpx

Restrictions

SETEXTC turn on to enable this command.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

6.3.9 SETOTP: Set OTP (BBh)

BBH	SETOTP(Set OTP Related Setting)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	0	1	1	1	0	1	1	BB
1st parameter	1	OTP_KEY0[7:0]								00
2nd parameter	1	OTP_KEY1[7:0]								00
3rd parameter	1	OTP_MASK[7:0]								00
4th parameter		INTVPP_EN	-	-	-	-	-	-	-	00
5th parameter	1	OTP_INDEX[7:0]								00
6th parameter	1	OTP_LOAD_DISABLE	OTP_TEST	OTP_POR	OTP_PWE	OTP_PTM[1:0]		OTP_PWR_SEL	OTP_PROG	00
7th parameter	1	OTP_DATA[7:0]								-

Description

This command is used to set OTP related setting.

OTP_KEY0[7:0], OTP_KEY1[7:0]	Description	Note
OTP_KEY0[7:0] = 0xAAh OTP_KEY1[7:0] = 0x55h	Enter OTP program mode	
OTP_KEY0[7:0] = 0x00h OTP_KEY1[7:0] = 0x00h	Leave OTP program mode	
Other value	Invalid	If HX8399-A operate on OTP program mode, Then keep on OTP program mode. If HX8399-A operate on non-OTP program mode, Then keep on non-OTP program mode.

OTP_MASK[7:0]: Bit programming mask, “1” means this bit can’t be programmed.

OTP_INDEX[7:0]: Set index of OTP table for programming.

NTVPP_EN: OTP_PWR power selected.
“0” : External OTP_PWR is selected when programmed.
“1” : Internal OTP_PWR (VGH short to OTP_PWR) is selected when programmed.

OTP_PROG: When set to “1’, the register content of OTP index is programmed.

OTP_PWR_SEL: When written to “1”, OTP power voltage is fed to OTP circuit.

OTP_PTM[1:0]: For test mode using. Not open..

OTP_PWE: OTP program write enable, “1” means OTP is able to be programmed.

OTP_POR: Pulse for OTP data read operation.

OTP_TEST: “0”, setting OTP_PROG high will trigger internal state machine.
“1”, setting OTP_PROG high will not trigger internal state machine.

OTP_LOAD_DISABLE: Normally the internal registers are auto-loaded from OTP when the SLPOUT command is received. Nevertheless, if this bit is set to 1, it will disable the auto loading function when the SLPOUT command was received. In general, this bit is used when OTP is not yet programmed.

OTP_DATA[7:0]: Read back the OTP index data.

Restrictions	SETEXTC turn on to enable this command.
Register Availability	Status / Availability: Normal Mode On, Idle Mode Off, Sleep Out — Yes Normal Mode On, Idle Mode On, Sleep Out — Yes Sleep In or Booster Off — Yes

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

6.3.10 SET_BANK: Set register bank (BDh)

BDH	SET_BANK(Set register bank)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	0	1	1	1	1	0	1	BD
1st parameter	1	-	-	-	-	-	-	BANK_INDEX [1:0]		00

Description

Set the register bank for some Commands that beyond 64 parameters

This command is active only for RC1h/RCDh/RD1h.
For example:
Write RC1h, PA44~PA85
Step1: write RBDh = 01h
Step2: write RC1h, PA1~42

Read RC1h, PA86~PA127
Step1: write RBDh = 02h
Step2: read RC1h, PA1~42

BANK INDEX[1:0]	RC1h (SETDGCLUT)	RCDh (SETANYCALLCE)	D1h (SET_SLR)
00(Bank 0)	PA1~43	PA1~56	PA1~53
01(Bank 1)	PA44~85	PA57~112	PA54~67
10(Bank 2)	PA86~127	NA	NA
11(Bank 3)	NA	NA	NA

Restrictions

SETEXTC turn on to enable this command.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

6.3.11 SETDGCLUT: Set DGC LUT (C1h)

C1H	SETDGCLUT (Set DGC LUT)								
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	0	0	0	0	0	1	C1
1st parameter	1	-	-	-	-	-	-	-	DGC_EN	00
2nd parameter	1	D1[7:0]								-
:	1	Dn[7:0]								-
127th parameter	1	D126[7:0]								-

Description

This command is used to set Digital Gamma Curve Look-Up Table.

DGC_EN: Enable the DGC function. “1”: Enable; “0”: Disable.

D1[7:0] ~ D126[7:0]:

LUT	D7	D6	D5	D4	D3	D2	D1	D0	Default
1st	R00_9	R00_8	R00_7	R00_6	R00_5	R00_4	R00_3	R00_2	00h
2nd	R01_9	R01_8	R01_7	R01_6	R01_5	R01_4	R01_3	R01_2	08h
3rd	R02_9	R02_8	R02_7	R02_6	R02_5	R02_4	R02_3	R02_2	10h
:	:	:	:	:	:	:	:	:	:
32th	R31_9	R31_8	R31_7	R31_6	R31_5	R31_4	R31_3	R31_2	F8h
33th	R32_9	R32_8	R32_7	R32_6	R32_5	R32_4	R32_3	R32_2	FFh
34th	R00_1	R00_0	R01_1	R01_0	R02_1	R02_0	R03_1	R03_0	00h
35th	R04_1	R04_0	R05_1	R05_0	R06_1	R06_0	R07_1	R07_0	00h
:	:	:	:	:	:	:	:	:	:
41th	R28_1	R28_0	R29_1	R29_0	R30_1	R30_0	R31_1	R31_0	00h
42th	R32_1	R32_0	0	0	0	0	0	0	00h
43th	G00_9	G00_8	G00_7	G00_6	G00_5	G00_4	G00_3	G00_2	00h
44th	G01_9	G01_8	G01_7	G01_6	G01_5	G01_4	G01_3	G01_2	08h
45th	G02_9	G02_8	G02_7	G02_6	G02_5	G02_4	G02_3	G02_2	10h
:	:	:	:	:	:	:	:	:	:
74th	G31_9	G31_8	G31_7	G31_6	G31_5	G31_4	G31_3	G31_2	F8h
75th	G32_9	G32_8	G32_7	G32_6	G32_5	G32_4	G32_3	G32_2	FFh
76th	G00_1	G00_0	G01_1	G01_0	G02_1	G02_0	G03_1	G03_0	00h
77th	G04_1	G04_0	G05_1	G05_0	G06_1	G06_0	G07_1	G07_0	00h
:	:	:	:	:	:	:	:	:	:
83th	G28_1	G28_0	G29_1	G29_0	G30_1	G30_0	G31_1	G31_0	00h
84th	G32_1	G32_0	0	0	0	0	0	0	00h
85th	B00_9	B00_8	B00_7	B00_6	B00_5	B00_4	B00_3	B00_2	00h
86th	B01_9	B01_8	B01_7	B01_6	B01_5	B01_4	B01_3	B01_2	08h
87th	B02_9	B02_8	B02_7	B02_6	B02_5	B02_4	B02_3	B02_2	10h
:	:	:	:	:	:	:	:	:	:
116th	B31_9	B31_8	B31_7	B31_6	B31_5	B31_4	B31_3	B31_2	F8h
117th	B32_9	B32_8	B32_7	B32_6	B32_5	B32_4	B32_3	B32_2	FFh
118th	B00_1	B00_0	B01_1	B01_0	B02_1	B02_0	B03_1	B03_0	00h
119th	B04_1	B04_0	B05_1	B05_0	B06_1	B06_0	B07_1	B07_0	00h
:	:	:	:	:	:	:	:	:	:
125th	B28_1	B28_0	B29_1	B29_0	B30_1	B30_0	B31_1	B31_0	00h
126th	B32_1	B32_0	0	0	0	0	0	0	00h

Restrictions

SETEXTC turn on to enable this command.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

6.3.12 SETID: Set ID (C3h)

C3H	SETID (Set ID)									
	DCX	**D7**	**D6**	**D5**	**D4**	**D3**	**D2**	**D1**	**D0**	**HEX**
Command	0	1	1	0	0	0	0	1	1	C3
1st parameter	1	ID1[7:0]								83
2nd parameter	1	ID2[7:0]								99
3rd parameter	1	ID3[7:0]								0A
4th parameter	1	ID4[7:0]								00
5th parameter	1	ID_TIMES[2:0]			-	-	-	-	-	E0

Description

ID1[7:0] is used to set ID RDAh value.

ID2[7:0] is used to set ID RDBh value.

ID3[7:0] is used to set ID RDCh value.

ID4[7:0] is used to set the fourth ID.

ID_TIMES[2:0]: Read the ID OTP programmed times.

ID_TIMES[2:0]	ID OTP Programmed Times
111	No programmed
011	ID has been programmed 1 time
001	ID has been programmed 2 times
000	ID has been programmed 3 times

Restrictions

SETEXTC turn on to enable this command.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

6.3.13 SETDDB: Set DDB (C4h)

C4H	SETDDB (Set DDB)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	0	0	0	1	0	0	C4
1st parameter	1	DDB1[7:0]								00
2nd parameter	1	DDB2[7:0]								00
3rd parameter	1	DDB3[7:0]								00
4th parameter	1	DDB4[7:0]								00
Description	This command is used to set CMD RA1h DDB1~4 value.									
Restrictions	SETEXTC turn on to enable this command.									
Register Availability	(see table below)									

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

6.3.14 SETCABC: Set CABC control (C9h)

C9H	SETCABC (Set CABC Control)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	0	0	1	0	0	1	C9
1st Parameter	1	PWM_PERIOD[16]	SEL_PWMCLK[2:0]			SEL_GAIN[1:0]		INVPULS	SEL_BLDUTY	1F
2nd Parameter	1	PWM_PERIOD[15:8]								00
3rd Parameter	1	PWM_PERIOD[7:0]								2E
4th Parameter	1	CABC_FSYNC	DIM_FRAME[6:0]							1E
5th Parameter	1	CABC_DD	-	-	-	CABC_FLM[3:0]				01
6th Parameter	1	CABC_STEP[7:0]								1E
7th Parameter	1	-	SAVEPOWER[6:0]							00

Description

This command is used to set CABC parameter.

SEL_BLDUTY: Backligh pwm output duty on/off control when CABC operation.
'0', The Backligh pwm output duty is 100%.
'1', The Backligh pwm output duty is calculate from CABC operation..

INVPULS: The backlight PWM output polarity select.
'0', The backlight PWM output is low level active.
'1', The backlight PWM output is high level active.

SEL_GAIN[1:0]: CABC gain select.

SEL_GAIN[1:0]	Gain
00	Use 1.00 as CABC calculate gain
01	Use 0.5x CABC calculate gain
10	Use 0.75x as CABC calculate gain
11	Use CABC calculate gain

SEL_PWMCLK[2:0] : Internal PWM_CLK divider for CABC clock.

SEL_PWMCLK[2:0]			Brightness Control Clock
0	0	0	PWM_CLK / 1
0	0	1	PWM_CLK / 2
0	1	0	PWM_CLK / 4
0	1	1	PWM_CLK / 6
1	0	0	PWM_CLK / 8
1	0	1	PWM_CLK / 10
1	1	0	PWM_CLK / 12
1	1	1	PWM_CLK / 14

PWM_PERIOD[16:0]: The backlight PWM output period setting.

When PWM_PERIOD[16]=0, PWM_PERIOD[15:6] setting inhibited.
(Fosc=73MHz, for 1200RGBx1920 Fosc=80MHz)

PWM_PERIOD [5:0]						CABC_PWM_OUT signal frequency
0	0	0	0	0	0	(Fosc/SEL_PWMCLK[2:0])/500
0	0	0	0	0	1	(Fosc/SEL_PWMCLK[2:0])/513
0	0	0	0	1	0	(Fosc/SEL_PWMCLK[2:0])/527
0	0	0	0	1	1	(Fosc/SEL_PWMCLK[2:0])/541
0	0	0	1	0	0	(Fosc/SEL_PWMCLK[2:0])/556
0	0	0	1	0	1	(Fosc/SEL_PWMCLK[2:0])/572
0	0	0	1	1	0	(Fosc/SEL_PWMCLK[2:0])/589
0	0	0	1	1	1	(Fosc/SEL_PWMCLK[2:0])/607
0	0	1	0	0	0	(Fosc/SEL_PWMCLK[2:0])/625

0	0	1	0	0	1	(Fosc/SEL_PWMCLK[2:0])/646
0	0	1	0	1	0	(Fosc/SEL_PWMCLK[2:0])/667
0	0	1	0	1	1	(Fosc/SEL_PWMCLK[2:0])/690
0	0	1	1	0	0	(Fosc/SEL_PWMCLK[2:0])/715
0	0	1	1	0	1	(Fosc/SEL_PWMCLK[2:0])/741
0	0	1	1	1	0	(Fosc/SEL_PWMCLK[2:0])/770
0	0	1	1	1	1	(Fosc/SEL_PWMCLK[2:0])/800
0	1	0	0	0	0	(Fosc/SEL_PWMCLK[2:0])/834
0	1	0	0	0	1	(Fosc/SEL_PWMCLK[2:0])/870
0	1	0	0	1	0	(Fosc/SEL_PWMCLK[2:0])/910
0	1	0	0	1	1	(Fosc/SEL_PWMCLK[2:0])/953
0	1	0	1	0	0	(Fosc/SEL_PWMCLK[2:0])/1000
0	1	0	1	0	1	(Fosc/SEL_PWMCLK[2:0])/1053
0	1	0	1	1	0	(Fosc/SEL_PWMCLK[2:0])/1112
0	1	0	1	1	1	(Fosc/SEL_PWMCLK[2:0])/1177
0	1	1	0	0	0	(Fosc/SEL_PWMCLK[2:0])/1250
0	1	1	0	0	1	(Fosc/SEL_PWMCLK[2:0])/1334
0	1	1	0	1	0	(Fosc/SEL_PWMCLK[2:0])/1429
0	1	1	0	1	1	(Fosc/SEL_PWMCLK[2:0])/1539
0	1	1	1	0	0	(Fosc/SEL_PWMCLK[2:0])/1667
0	1	1	1	0	1	(Fosc/SEL_PWMCLK[2:0])/1819
0	1	1	1	1	0	(Fosc/SEL_PWMCLK[2:0])/2000
0	1	1	1	1	1	(Fosc/SEL_PWMCLK[2:0])/2223
1	0	0	0	0	0	(Fosc/SEL_PWMCLK[2:0])/2500
1	0	0	0	0	1	(Fosc/SEL_PWMCLK[2:0])/2858
1	0	0	0	1	0	(Fosc/SEL_PWMCLK[2:0])/3334
1	0	0	0	1	1	(Fosc/SEL_PWMCLK[2:0])/4000
1	0	0	1	0	0	(Fosc/SEL_PWMCLK[2:0])/5000
1	0	0	1	0	1	(Fosc/SEL_PWMCLK[2:0])/6667
1	0	0	1	1	0	(Fosc/SEL_PWMCLK[2:0])/10000
1	0	0	1	1	1	(Fosc/SEL_PWMCLK[2:0])/20000
1	0	1	0	0	0	(Fosc/SEL_PWMCLK[2:0])/22223
1	0	1	0	0	1	(Fosc/SEL_PWMCLK[2:0])/25000
1	0	1	0	1	0	(Fosc/SEL_PWMCLK[2:0])/28572
1	0	1	0	1	1	(Fosc/SEL_PWMCLK[2:0])/33334
1	0	1	1	0	0	(Fosc/SEL_PWMCLK[2:0])/40000
1	0	1	1	0	1	(Fosc/SEL_PWMCLK[2:0])/50000
1	0	1	1	1	0	(Fosc/SEL_PWMCLK[2:0])/66667
1	0	1	1	1	1	(Fosc/SEL_PWMCLK[2:0])/100000
1	1	0	0	0	0	(Fosc/SEL_PWMCLK[2:0])/200000
Others						Inhibited

When PWM_PERIOD[16]=1:
CABC_PWM_OUT frequency=(Fosc/SEL_PWMCLK[2:0])/(PWM_PERIOD[15:0]+1)
Note: PWM_PERIOD[15:0]=0000h is inhibited.

DIM_FRAME[6:0] : Manual brightness setting dimming period.

CABC_FSYNC: Reset CABC_PWM_OUT signal by each VSYNC.

CABC_DD: CABC dimming function enable bit.
'0', Disable CABC dimming.
'1', Enable CABC dimming.

CABC_STEP[7:0]: CABC step numbers during dimming period.

CABC_FLM[3:0]: CABC dimming frame number for each step.

SAVEPOWER[6:0] : Minimum CABC gain / maximum CABC duty output select.

SAVEPOWER [6:0]							Min. Gain	Max. Duty
0	0	x	x	x	x	x	Reserve	
0	1	0	0	0	0	0	1+0/32	100%
0	1	0	0	0	0	1	1+1/32	96.97%
0	1	0	0	0	1	0	1+2/32	94.12%
0	1	0	0	0	1	1	1+3/32	91.43%
0	1	0	0	1	0	0	1+4/32	88.89%
0	1	0	0	1	0	1	1+5/32	86.49%
0	1	0	0	1	1	0	1+6/32	84.21%
0	1	0	0	1	1	1	1+7/32	82.05%
0	1	0	1	0	0	0	1+8/32	80%
0	1	0	1	0	0	1	1+9/32	78.05%
0	1	0	1	0	1	0	1+10/32	76.19%
0	1	0	1	0	1	1	1+11/32	74.42%
0	1	0	1	1	0	0	1+12/32	72.73%
0	1	0	1	1	0	1	1+13/32	71.11%
0	1	0	1	1	1	0	1+14/32	69.57%
0	1	0	1	1	1	1	1+15/32	68.09%
0	1	1	0	0	0	0	1+16/32	66.67%
0	1	1	0	0	0	1	1+17/32	65.31%
0	1	1	0	0	1	0	1+18/32	64%
0	1	1	0	0	1	1	1+19/32	62.75%
0	1	1	0	1	0	0	1+20/32	61.54%
0	1	1	0	1	0	1	1+21/32	60.38%
0	1	1	0	1	1	0	1+22/32	59.26%
0	1	1	0	1	1	1	1+23/32	58.18%
0	1	1	1	0	0	0	1+24/32	57.14%
0	1	1	1	0	0	1	1+25/32	56.14%
0	1	1	1	0	1	0	1+26/32	55.17%
0	1	1	1	0	1	1	1+27/32	54.24%
0	1	1	1	1	0	0	1+28/32	53.33%
0	1	1	1	1	0	1	1+29/32	52.46%
0	1	1	1	1	1	0	1+30/32	51.61%
0	1	1	1	1	1	1	1+31/32	50.79%
1	0	0	0	0	0	0	1+32/32	50%

For details, please refer to chapter “5.13.2 CABC Block”.

Restriction: SETEXTC turn on to enable this command.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

6.3.15 SETCLOCK (CBh)

CBH	SETCLOCK									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	0	0	1	0	1	1	CB
1st Parameter	1	-	-	-	-	UADJ[3:0]				07

Description

UADJ[3:0]: For user to adjust OSC frequency, default is 73MHz(80MHz for 1200RGBx1920).

UADJ[3:0]				Internal oscillator frequency
0	0	0	0	39.2%
0	0	0	1	49.7%
0	0	1	0	59.2%
0	0	1	1	68.2%
0	1	0	0	75.5%
0	1	0	1	83.1%
0	1	1	0	89.8%
0	1	1	1	96.2%
1	0	0	0	100.0%
1	0	0	1	105.7%
1	0	1	0	110.4%
1	0	1	1	115.0%
1	1	0	0	118.4%
1	1	0	1	122.4%
1	1	1	0	125.8%
1	1	1	1	129.0%

Restrictions: SETEXTC turn on to enable this command

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

6.3.16 SETPANEL (CCh)

<table>
<tr><th rowspan="2">CCH</th><th colspan="10">SETPANEL(Set panel related register)</th></tr>
<tr><th>DCX</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>CC</td></tr>
<tr><td>1st parameter</td><td>1</td><td>-</td><td>-</td><td>-</td><td>-</td><td>SS_PANEL</td><td>GS_PANEL</td><td>REV_PANEL</td><td>BGR_PANEL</td><td>02</td></tr>
<tr><td>Description0</td><td colspan="10">This command is used to set setting of panel related register and make panel module meets below spec from viewpoint of user

BGR_PANEL: The order of <R><G><B> dot color for module supplier, default value is stored in OTP. If color filter of panel is <B><G><R> type, setting BGR_PANEL = 1, if color filter of panel is <R><G><B> type, setting BGR_PANEL = 0. This bit is to make panel module look like a <R><G><B> type panel form the user viewpoint.

REV_PANEL: The REV_PANEL setting is used to select the inversion of the display of all characters and graphics. This setting allows the display of the same data on both normally white and normally black panels.

GS_PANEL: Specify the shift direction of gate driver output.
When GS_PANEL = 0, the panel control signal is normal scan.
When GS_PANEL = 1, the panel control signal is reverse scan.

SS_PANEL: Specify the shift direction of source driver output. When SS_PANEL = 0, the shift direction from S1 to S1200 When SS_PANEL = 1, the shift direction from S1200 to S1.</td></tr>
<tr><td>Restrictions</td><td colspan="10">SETEXTC turn on to enable this command</td></tr>
<tr><td>Register Availability</td><td colspan="10"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table></td></tr>
</table>

6.3.17 SET_SLR_MODE (D0h)

D0H	SET_SLR_MODE									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	0	1	0	0	0	0	D0
1st parameter	1	reg_amb_val[7:0]								80
2nd parameter	1	reg_CTRL_EHLV[7:0]								40
3rd parameter	1	-	-	-	-	slr_sel_opt	slr_demo_on	slr_bps_en	reg_slr_en	00
Description	This command is used for sun light readability. **reg_amb_val[7:0]:** Set larger value for reg_amb_val if brighter ambient light is detected via a sensor. Value ranges from 0 ~ 80. **reg_CTRL_EHLV[7:0]:** Set the enhanced level of SLR.User-defined enhance strength. 40h: normal enhancement; 20h: weak enhancement; 00h: zero enhancement; **slr_sel_opt:** SLR resolution related register choose option. 0: auto set resolution related registers. 1: manual set resolution related registers. **slr_demo_on:** SLR demo mode signal, for demo only. 0: apply SLR normal display mode. 1: enable SLR demo mode, only the right side of the current frame will be enhanced. **slr_bps_en:** SLR bypass signal, for debug only. 0: apply SLR enhancement onto output. 1: ignore SLR enhancement for debugging (output is copied from input). **reg_slr_en:** SLR function enable. 0: Disable. 1: Enable.									
Restrictions	SETEXTC turn on to enable this command.									
Register Availability	(see table below)									

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

6.3.18 SETGIP0: Set GIP Option0 (D3h)

D3H	SETGIP0(Set GIP Option0)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	0	1	0	0	1	1	D3
1st parameter	1	-	-	GIP_MODE[1:0]		EQ_DELAY_HSYNC[3:0]				00
2nd parameter	1	EQ_DELAY[7:0]								00
3rd parameter	1	-	-	-	-	-	-	OVERLAP_OPT	CKV_VGH_OPT	00
4th parameter	1	-	-	GTO[5:0]						00
5th parameter	1	GNO[7:0]								00
6th parameter	1	USER_GIP_GATE[7:0]								08
7th parameter	1	USER_GIP_GATE1[7:0]								08
8th parameter	1	SHR0_3[3:0]				SHR0_2[3:0]				32
9th parameter	1	SHR0_1[3:0]				SHR0[11:8]				10
10th parameter	1	SHR0[7:0]								00
11st parameter	1	-	-	-	-	SHR0_GS[11:8]				00
12nd parameter	1	SHR0_GS[7:0]								00
13rd parameter	1	SHR1_3[3:0]				SHR1_2[3:0]				32
14th parameter	1	SHR1_1[3:0]				SHR1[11:8]				13
15th parameter	1	SHR1[7:0]								C0
16th parameter	1	-	-	-	-	SHR1_GS[11:8]				00
17th parameter	1	SHR1_GS[7:0]								00
18th parameter	1	SHR2_3[3:0]				SHR2_2[3:0]				32
19th parameter	1	SHR2_1[3:0]				SHR2[11:8]				10
20th parameter	1	SHR2[7:0]								08
21st parameter	1	-	-	-	-	SHR2_GS[11:8]				00
22nd parameter	1	SHR2_GS[7:0]								00
23rd parameter	1	SHP[3:0]				SCP[3:0]				23
24th parameter	1	CHR0[7:0]								04
25th parameter	1	CHR0_GS[7:0]								00
26th parameter	1	CHP0[3:0]				CCP0[3:0]				27
27th parameter	1	CHR1[7:0]								04
28th parameter	1	CHR1_GS[7:0]								00
29th parameter	1	CHP1[3:0]				CCP1[3:0]				27
30th parameter	1	vbp_setting[7:0]								00
31st parameter	1	-	-	-	-	vbp_self_learning	-	-	-	00

Description

This command is used for GIP signal setting.

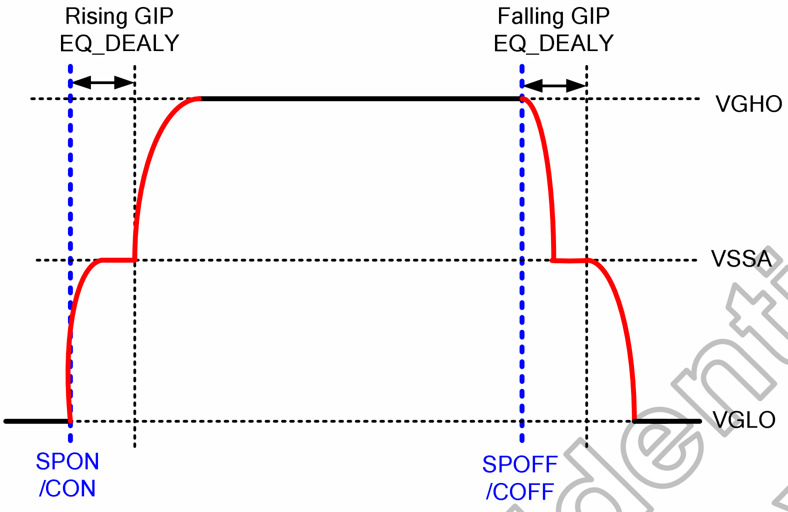
GIP_MODE[1:0]: GIP EQ (pre-charge) type selection

GIP_MODE1	GIP_MODE0	GIP EQ Type
0	0	Both rising / falling EQ
0	1	Only rising edge EQ
1	0	Only falling edge EQ
1	1	EQ Off

EQ_DELAY[7:0]: Set GIP control signal EQ period

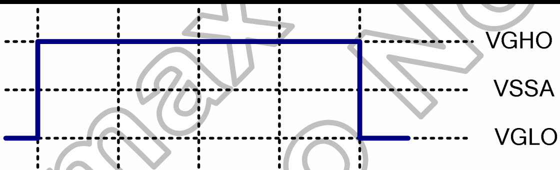
EQ_DELAY [7:0]								GIP EQ Period
0	0	0	0	0	0	0	0	0 OSC clock cycle
0	0	0	0	0	0	0	1	4 OSC clock cycle
0	0	0	0	0	0	1	0	8 OSC clock cycle
0	0	0	0	0	0	1	1	12 OSC clock cycle
0	0	0	0	0	1	0	0	16 OSC clock cycle
:								:
1	1	1	1	1	1	0	0	1008 OSC clock cycle

1	1	1	1	1	1	0	1	1012 OSC clock cycle
1	1	1	1	1	1	1	0	1016 OSC clock cycle
1	1	1	1	1	1	1	1	1020 OSC clock cycle

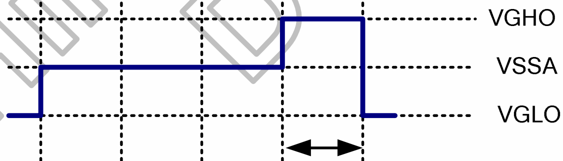


EQ_DELAY_HSYNC[3:0]: Set the EQ period in HSYNC width.

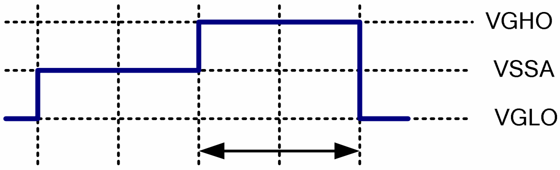
EQ_DELAY_HSYNC[3:0]				EQ Period in HSYNC width
0	0	0	0	Normal Driving
0	0	0	1	1 x Hsync
0	0	1	0	2 x Hsync
0	0	1	1	3 x Hsync
:				:
1	1	1	0	14 x Hsync
1	1	1	1	15 x Hsync



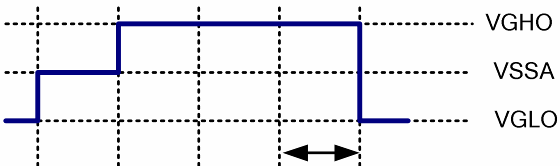
Full Driving



Driving 1-HSYNC, others EQ VSSA



Driving 2-HSYNC, others EQ VSSA



Driving 3-HSYNC, others EQ VSSA

OVERLAP_OPT: Choose GPWR over lap type.

When set '1', GPWR low period longer then high period.
When set '0', GPWR high period longer then low period.

CKV_VGH_OPT: set 1, then all CKV will output VGH during Vsync, the width of CKV will follow the width of Vsync.

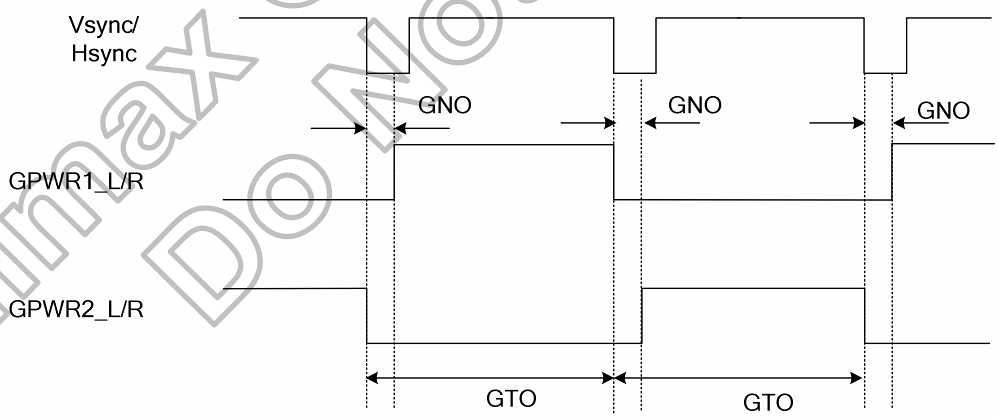
GTO[5:0]: The period of the CLK.

GTO[5:0]	GPWR toggle frequency
6'h00	64 x Frame/Line
6'h01	1 x Frame/ Line
6'h02	2 x Frame/ Line
6'h03	3 x Frame/ Line
:	:
6'h3D	61 x Frame/ Line
6'h3E	62 x Frame/ Line
6'h3F	63 x Frame/ Line

GNO[7:0]: To define the latency of the CLK.

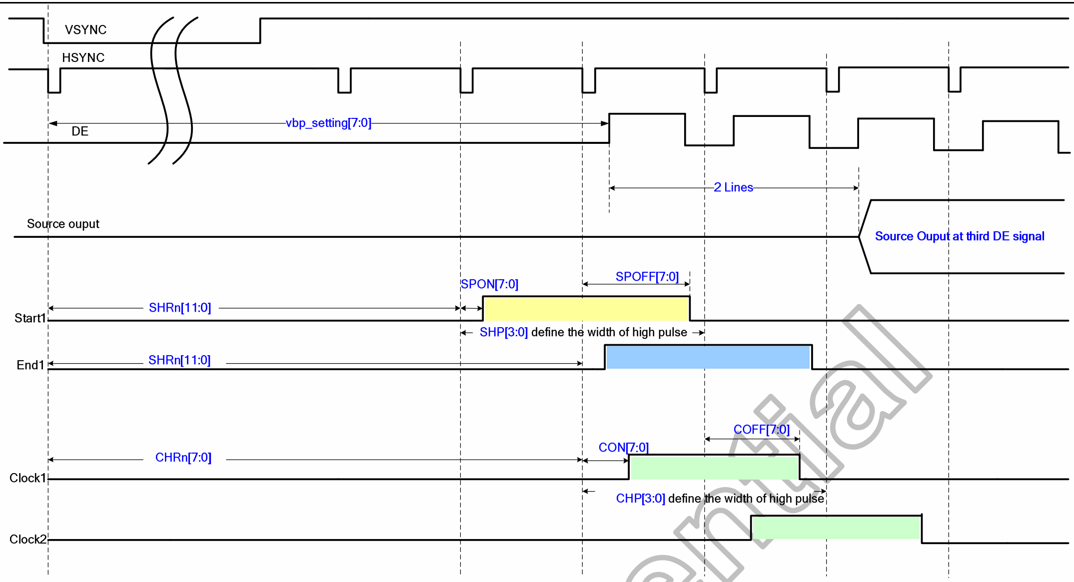
GNO[7:0]	GPWR non-overlap timing
8'h00	0
8'h01	1 x OSC CLK
8'h02	2 x OSC CLK
8'h03	3 x OSC CLK
:	:
8'hFD	253 x OSC CLK
8'hFE	254 x OSC CLK
8'hFF	255 x OSC CLK

GPWR1/2 toggle period by Frame or Line depend on GIP_OPT[4]. Non-overlap time depends on GNO.



USER_GIP_Gate[7:0]: Set the GIP dummy clock numbers for first CKV.

USER_GIP_Gate1[7:0]: Set the GIP dummy clock numbers for second CKV.



Group0:
SHR0[11:0]: Set the 1st Start/End signal delay from VSYNC falling edge when GS=0.
SHR0_GS[11:0]: Set the 1st Start/End signal delay from VSYNC falling edge when GS=1.
SHR0_1 / SHR0_2 / SHR0_3[3:0]: Set the 2nd/3rd/4th Start / End signal delay from the 1st Start/End signal.

Group1:
SHR1[11:0]: Set the 1st Start/End siganl delay from VSYNC falling edge when GS=0.
SHR1_GS[11:0]: Set the 1st Start/End siganl delay from VSYNC falling edge when GS=1.
SHR1_1 / SHR1_2 / SHR1_3[3:0]: Set the 2nd/3rd/4th Start / End signal delay from the 1st Start/End signal.

Group2:
SHR2[11:0]: Set the 1st Start/End signal delay from VSYNC falling edge when GS=0.
SHR2_GS[11:0]: Set the 1st Start/End signal delay from VSYNC falling edge when GS=1.
SHR2_1 / SHR2_2 / SHR2_3[3:0]: Set the 2nd/3rd/4th Start / End signal delay from the 1st Start/End signal.

SHR0/SHR1/SHR2[11:0] SHR0_GS/SHR1_GS/SHR2_GS[11:0]	Start signal output delay
0x000h	2 x Hsync
0x001h	3 x Hsync
0x002h	4 x Hsync
:	:
0xFFEh	4096 x Hsync
0xFFFh	4097 x Hsync

SHR0_1 / SHR1_1 / SHR2_1[3:0] SHR0_2 / SHR1_2 / SHR2_2[3:0] SHR0_3 / SHR1_3 / SHR2_3[3:0]	2nd Start / End signal output delay 3th Start / End signal output delay 4th Start / End signal output delay
0000	0 x Hsync
0001	1 x Hsync
0010	2 x Hsync
:	:
1110	14 x Hsync
1111	15 x Hsync

SCP[3:0]: Numbers of output Start and End signal.

SCP3	SCP2	SCP1	SCP0	Start and End numbers
0	0	0	0	1
0	0	0	1	2
0	0	1	0	3
:				:
1	1	0	1	14
1	1	1	0	15
1	1	1	1	16

SHP[3:0]: Width of Start and End signal high pulse.

SHP3	SHP2	SHP1	SHP0	Start Pulse Width
0	0	0	0	1 x Hsync
0	0	0	1	2 x Hsync
0	0	1	0	3 x Hsync
:				:
1	1	1	0	15 x Hsync
1	1	1	1	16 x Hsync

CHR0[7:0]/CHR1[7:0]: Set the Clock signal delay from VSYNC falling edge when GS=0.
CHR0_GS[7:0]/CHR1_GS[7:0]: Set the Clock signal delay from VSYNC falling edge when GS=1.

CHR0[7:0]/CHR0_GS[7:0] CHR1[7:0]/CHR1_GS[7:0]	Clock signal output delay
0x00h	2 x HSYNC
0x01h	3 x HSYNC
0x02h	4 x HSYNC
:	:
0xFEh	256 x HSYNC
0xFFh	257 x HSYNC

CCP0[3:0]/CCP1[3:0]: Numbers of Output Clock signal.

CCP0[3:0]/CCP1[3:0]				Clock nunbers
0	0	0	0	1
0	0	0	1	2
0	0	1	0	3
:				:
1	1	1	0	15
1	1	1	1	16

CHP0[3:0]/CHP1[3:0]: Width of Clock signal high pulse.

CHP0[3:0]/CHP1[3:0]				Clock signal width
0	0	0	0	1 x Hsync
0	0	0	1	2 x Hsync
0	0	1	0	3 x Hsync
:				:
1	1	1	0	15 x Hsync
1	1	1	1	16 x Hsync

vbp_setting[7:0]: Set Vertical porch lines(from VSYNC falling to first DE signal).

vbp_setting [7:0]	Vertical porch lines
0x00h	1 x HSYNC
0x01h	2 x HSYNC
0x02h	3 x HSYNC
:	:
0xFEh	255 x HSYNC
0xFFh	256 x HSYNC

	vbp_self_learning: Set ‘1’ to enable self-learning.
Restrictions	-
Register Availability	<table><tr><td>Status</td><td>Availability</td></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table>

6.3.19 SETGIP1: Set GIP Option1 (D5h)

D5H	SETGIP1(Set GIP Option1)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	0	1	0	1	0	1	D5
1st parameter	1	-	CGTS_L_INV[1]	COS1_L[5:0]						00
2nd parameter	1	-	CGTS_R_INV[1]	COS1_R[5:0]						00
3rd parameter	1	-	CGTS_L_INV[2]	COS2_L[5:0]						00
4th parameter	1	-	CGTS_R_INV[2]	COS2_R[5:0]						00
5th parameter	1	-	CGTS_L_INV[3]	COS3_L[5:0]						00
6th parameter	1	-	CGTS_R_INV[3]	COS3_R[5:0]						00
7th parameter	1	-	CGTS_L_INV[4]	COS4_L[5:0]						00
8th parameter	1	-	CGTS_R_INV[4]	COS4_R[5:0]						00
9th parameter	1	-	CGTS_L_INV[5]	COS5_L[5:0]						00
10th parameter	1	-	CGTS_R_INV[5]	COS5_R[5:0]						00
11st parameter	1	-	CGTS_L_INV[6]	COS6_L[5:0]						00
12nd parameter	1	-	CGTS_R_INV[6]	COS6_R[5:0]						00
13rd parameter	1	-	CGTS_L_INV[7]	COS7_L[5:0]						00
14th parameter	1	-	CGTS_R_INV[7]	COS7_R[5:0						00
15th parameter	1	-	CGTS_L_INV[8]	COS8_L[5:0]						00
16th parameter	1	-	CGTS_R_INV[8]	COS8_R[5:0]						00
17th parameter	1	-	CGTS_L_INV[9]	COS9_L[5:0]						00
18th parameter	1	-	CGTS_R_INV[9]	COS9_R[5:0]						00
19th parameter	1	-	CGTS_L_INV[10]	COS10_L[5:0]						00
20th parameter	1	-	CGTS_R_INV[10]	COS10_R[5:0]						00
21st parameter	1	-	CGTS_L_INV[11]	COS11_L[5:0]						00
22nd parameter	1	-	CGTS_R_INV[11]	COS11_R[5:0]						00
23rd parameter	1	-	CGTS_L_INV[12]	COS12_L[5:0]						00
24th parameter	1	-	CGTS_R_INV[12]	COS12_R[5:0]						00
25th parameter	1	-	CGTS_L_INV[13]	COS13_L[5:0]						00
26th parameter	1	-	CGTS_R_INV[13]	COS13_R[5:0]						00
27th parameter	1	-	CGTS_L_INV[14]	COS14_L[5:0]						00
28th parameter	1	-	CGTS_R_INV[14]	COS14_R[5:0]						00
29th parameter	1	-	CGTS_L_INV[15]	COS15_L[5:0]						00
30th parameter	1	-	CGTS_R_INV[15]	COS15_R[5:0]						00
31st parameter	1	-	CGTS_L_INV[16]	COS16_L[5:0]						00
32nd parameter	1	-	CGTS_R_INV[16]	COS16_R[5:0]						00
Description	**CGTS_L_INV[16:1]:** Set the corresponding signal CGOUTL_n output polarity,n=1~16 **CGTS_R_INV[16:1]:** Set the corresponding signal CGOUTR_n output polarity,n=1~16 0: Normal output 1: Invert output signal **COSn_L_GS[3:0]:** When GS_Panel=1, select CGOUTL_n output, n=1~16 **COSn_R_GS[3:0]:** When GS_Panel=1, select CGOUTR_n output, n=1~16									

COSn_L[5:0]~ COSn_R[5:0]~ n=1~16	Output Signal	Description
00_0000	CK[0]	**GROUP0:**Gate CLK
00_0001	CK[1]	Gate CLK
00_0010	CK[2]	Gate CLK
00_0011	CK[3]	Gate CLK
00_0100	CK[4]	Gate CLK
00_0101	CK[5]	Gate CLK
00_0110	CK[6]	Gate CLK
00_0111	CK[7]	Gate CLK
00_1000	CK[8]	Gate CLK
00_1001	CK[9]	Gate CLK
00_1010	CK[10]	Gate CLK
00_1011	CK[11]	Gate CLK
00_1100	CK[12]	Gate CLK
00_1101	CK[13]	Gate CLK
00_1110	CK[14]	Gate CLK
00_1111	CK[15]	Gate CLK
01_0000	CK_1[0]	**GROUP1:**Gate CLK
01_0001	CK_1 [1]	Gate CLK
01_0010	CK_1 [2]	Gate CLK
01_0011	CK_1 [3]	Gate CLK
01_0100	CK_1 [4]	Gate CLK
01_0101	CK_1 [5]	Gate CLK
01_0110	CK_1 [6]	Gate CLK
01_0111	CK_1 [7]	Gate CLK
01_1000	1'b0	-
01_1001	1'b1	-
01_1010	PA69_CGOUT3_ PRE	Frame or Line toggle signal GPWR1 FOR LGD
01_1011	PA69_CGOUT4_ PRE	Frame or Line toggle signal GPWR2 FOR LGD
01_1100	AUO_VST_0_D1H	-
01_1101	AUO_VST_1_D1H	-
01_1110	DIR	FOR SMD (SUMSUNG)
01_1111	DIRB	FOR SMD (SUMSUNG)
10_0000	STV[0]	**GROUP 0** SHR0[11:0]
10_0001	STV[1]	SHR0[11:0]+SHR0_1[3:0]
10_0010	STV[2]	SHR0[11:0]+SHR0_2[3:0]
10_0011	STV[3]	SHR0[11:0]+SHR0_3[3:0]
10_0100	STV[4]	**GROUP 1** SHR1[11:0]
10_0101	STV[5]	SHR1[11:0]+SHR1_1[3:0]
10_0110	STV[6]	SHR1[11:0]+SHR1_2[3:0]
10_0111	STV[7]	SHR1[11:0]+SHR1_3[3:0]
10_1000	1'b0	-
10_1001	STV[8]	**GROUP-2** SHR2[11:0]
10_1010	STV[9]	SHR2[11:0]+SHR2_1[3:0]
10_1011	STV[10]	SHR2[11:0]+SHR2_2[3:0]
10_1100	STV[11]	SHR2[11:0]+SHR2_3[3:0]
10_1101	SMD_RST	RST FOR SMD (SUMSUNG)
10_1110	DISP_AREA_LINE_ EXT	DISPLAY AREA PER FRAME FOR HANNSTAR PANEL
10_1111	dgc_vs_post	Internal VSYNC
11_0000	SW1_L/R	Source driver swtich

	11_0001	SW2_L/R	Source driver swtich
	11_0010	SW3_L/R	Source driver swtich
	11_0011	SW1B_L/R	Source driver switch inveter
	11_0100	SW2B_L/R	Source driver switch inveter
	11_0101	SW3B_L/R	Source driver switch inveter
	11_0110	SW_INI	Source drive precharge signal
	11_0111	SW_INI_INV	Source drive precharge signal inverter
	Others	inhibited	-
Restrictions	-		

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

6.3.20 SETGIP2: Set GIP Option2 (D6h)

D6H	SETGIP2(Set GIP Option2)								
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	0	1	0	1	1	0	D6
1st parameter	1	-	CGOUT_L_HIZ_IN[1]	COS1_L_GS[5:0]						00
2nd parameter	1	-	CGOUT_R_HIZ_IN[1]	COS1_R_GS[5:0]						00
3rd parameter	1	-	CGOUT_L_HIZ_IN[2]	COS2_L_GS[5:0]						00
4th parameter	1	-	CGOUT_R_HIZ_IN[2]	COS2_R_GS[5:0]						00
5th parameter	1	-	CGOUT_L_HIZ_IN[3]	COS3_L_GS[5:0]						00
6th parameter	1	-	CGOUT_R_HIZ_IN[3]	COS3_R_GS[5:0]						00
7th parameter	1	-	CGOUT_L_HIZ_IN[4]	COS4_L_GS[5:0]						00
8th parameter	1	-	CGOUT_R_HIZ_IN[4]	COS4_R_GS[5:0]						00
9th parameter	1	-	CGOUT_L_HIZ_IN[5]	COS5_L_GS[5:0]						00
10th parameter	1	-	CGOUT_R_HIZ_IN[5]	COS5_R_GS[5:0]						00
11st parameter	1	-	CGOUT_L_HIZ_IN[6]	COS6_L_GS[5:0]						00
12nd parameter	1	-	CGOUT_R_HIZ_IN[6]	COS6_R_GS[5:0]						00
13rd parameter	1	-	CGOUT_L_HIZ_IN[7]	COS7_L_GS[5:0]						00
14th parameter	1	-	CGOUT_R_HIZ_IN[7]	COS7_R_GS[5:0]						00
15th parameter	1	-	CGOUT_L_HIZ_IN[8]	COS8_L_GS[5:0]						00
16th parameter	1	-	CGOUT_R_HIZ_IN[8]	COS8_R_GS[5:0]						00
17th parameter	1	-	CGOUT_L_HIZ_IN[9]	COS9_L_GS[5:0]						00
18th parameter	1	-	CGOUT_R_HIZ_IN[9]	COS9_R_GS[5:0]						00
19th parameter	1	-	CGOUT_L_HIZ_IN[10]	COS10_L_GS[5:0]						00
20th parameter	1	-	CGOUT_R_HIZ_IN[10]	COS10_R_GS[5:0]						00
21st parameter	1	-	CGOUT_L_HIZ_IN[11]	COS11_L_GS[5:0]						00
22nd parameter	1	-	CGOUT_R_HIZ_IN[11]	COS11_R_GS[5:0]						00
23rd parameter	1	-	CGOUT_L_HIZ_IN[12]	COS12_L_GS[5:0]						00
24th parameter	1	-	CGOUT_R_HIZ_IN[12]	COS12_R_GS[5:0]						00
25th parameter	1	-	CGOUT_L_HIZ_IN[13]	COS13_L_GS[5:0]						00
26th parameter	1	-	CGOUT_R_HIZ_IN[13]	COS13_R_GS[5:0]						00

27th parameter	1	-	CGOUT_L_HIZ_IN[14]	COS14_L_GS[5:0]	00
28th parameter	1	-	CGOUT_R_HIZ_IN[14]	COS14_R_GS[5:0]	00
29th parameter	1	-	CGOUT_L_HIZ_IN[15]	COS15_L_GS[5:0]	00
30th parameter	1	-	CGOUT_R_HIZ_IN[15]	COS15_R_GS[5:0]	00
31st parameter	1	-	CGOUT_L_HIZ_IN[16]	COS16_L_GS[5:0]	00
32nd parameter	1	-	CGOUT_R_HIZ_IN[16]	COS16_R_GS[5:0]	00

Description

CGOUT_L_HIZ_IN[n]: Set the CGOUTL_n output = Hi-Z, n=1~16
CGOUT_R_HIZ_IN[n]: Set the CGOUTR_n output = Hi-Z, n=1~16
0: Normal output
1: Hi-Z

COSn_L_GS[3:0]: When GS_Panel=1, select CGOUTL_n output, n=1~16
COSn_R_GS[3:0]: When GS_Panel=1, select CGOUTR_n output, n=1~16

COSn_L_GS[5:0]~ COSn_R_GS[5:0]~ n=1~16	Output Signal	Description
00_0000	CK[0]	**GROUP0:**Gate CLK
00_0001	CK[1]	Gate CLK
00_0010	CK[2]	Gate CLK
00_0011	CK[3]	Gate CLK
00_0100	CK[4]	Gate CLK
00_0101	CK[5]	Gate CLK
00_0110	CK[6]	Gate CLK
00_0111	CK[7]	Gate CLK
00_1000	CK[8]	Gate CLK
00_1001	CK[9]	Gate CLK
00_1010	CK[10]	Gate CLK
00_1011	CK[11]	Gate CLK
00_1100	CK[12]	Gate CLK
00_1101	CK[13]	Gate CLK
00_1110	CK[14]	Gate CLK
00_1111	CK[15]	Gate CLK
01_0000	CK_1[0]	**GROUP1:**Gate CLK
01_0001	CK_1 [1]	Gate CLK
01_0010	CK_1 [2]	Gate CLK
01_0011	CK_1 [3]	Gate CLK
01_0100	CK_1 [4]	Gate CLK
01_0101	CK_1 [5]	Gate CLK
01_0110	CK_1 [6]	Gate CLK
01_0111	CK_1 [7]	Gate CLK
01_1000	1'b0	-
01_1001	1'b1	-
01_1010	PA69_CGOUT3_PRE	Frame or Line toggle signal GPWR1 FOR LGD
01_1011	PA69_CGOUT4_PRE	Frame or Line toggle signal GPWR2 FOR LGD
01_1100	AUO_VST_0_D1H	-
01_1101	AUO_VST_1_D1H	-
01_1110	DIR	FOR SMD (SUMSUNG)
01_1111	DIRB	FOR SMD (SUMSUNG)

	10_0000	STV[0]	**GROUP 0** SHR0[11:0]
	10_0001	STV[1]	SHR0[11:0]+SHR0_1[3:0]
	10_0010	STV[2]	SHR0[11:0]+SHR0_2[3:0]
	10_0011	STV[3]	SHR0[11:0]+SHR0_3[3:0]
	10_0100	STV[4]	**GROUP 1** SHR1[11:0]
	10_0101	STV[5]	SHR1[11:0]+SHR1_1[3:0]
	10_0110	STV[6]	SHR1[11:0]+SHR1_2[3:0]
	10_0111	STV[7]	SHR1[11:0]+SHR1_3[3:0]
	10_1000	1'b0	-
	10_1001	STV[8]	**GROUP-2** SHR2[11:0]
	10_1010	STV[9]	SHR2[11:0]+SHR2_1[3:0]
	10_1011	STV[10]	SHR2[11:0]+SHR2_2[3:0]
	10_1100	STV[11]	SHR2[11:0]+SHR2_3[3:0]
	10_1101	SMD_RST	RST FOR SMD (SUMSUNG)
	10_1110	DISP_AREA_LINE_ EXT	DISPLAY AREA PER FRAME FOR HANNSTAR PANEL
	10_1111	dgc_vs_post	Internal VSYNC
	11_0000	SW1_L/R	Source driver swtich
	11_0001	SW2_L/R	Source driver swtich
	11_0010	SW3_L/R	Source driver swtich
	11_0011	SW1B_L/R	Source driver switch inveter
	11_0100	SW2B_L/R	Source driver switch inveter
	11_0101	SW3B_L/R	Source driver switch inveter
	11_0110	SW_INI	Source drive precharge signal
	11_0111	SW_INI_INV	Source drive precharge signal inverter
	Others	inhibited	-
Restrictions	-		

Register Availability	

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

6.3.21 SETGIP3: Set GIP Option3 (D8h)

D8H	SETGIP3(Set GIP Option3)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	0	1	1	0	0	0	D8
1st parameter	1	INIT_0_SEL_C GOUT1_L[1:0]		INIT_0_SEL_C GOUT2_L[1:0]		INIT_0_SEL_C GOUT3_L[1:0]		INIT_0_SEL_C GOUT4_L[1:0]		00
2nd parameter	1	INIT_0_SEL_C GOUT5_L[1:0]		INIT_0_SEL_C GOUT6_L[1:0]		INIT_0_SEL_C GOUT7_L[1:0]		INIT_0_SEL_C GOUT8_L[1:0]		00
3rd parameter	1	INIT_0_SEL_C GOUT9_L[1:0]		INIT_0_SEL_C GOUT10_L[1:0]		INIT_0_SEL_C GOUT11_L[1:0]		INIT_0_SEL_C GOUT12_L[1:0]		00
4th parameter	1	INIT_0_SEL_C GOUT13_L[1:0]		INIT_0_SEL_C GOUT14_L[1:0]		INIT_0_SEL_C GOUT15_L[1:0]		INIT_0_SEL_C GOUT16_L[1:0]		00
5th parameter	1	INIT_0_SEL_C GOUT1_R[1:0]		INIT_0_SEL_C GOUT2_R[1:0]		INIT_0_SEL_C GOUT3_R[1:0]		INIT_0_SEL_C GOUT4_R[1:0]		00
6th parameter	1	INIT_0_SEL_C GOUT5_R[1:0]		INIT_0_SEL_C GOUT6_R[1:0]		INIT_0_SEL_C GOUT7_R[1:0]		INIT_0_SEL_C GOUT8_R[1:0]		00
7th parameter	1	INIT_0_SEL_C GOUT9_R[1:0]		INIT_0_SEL_C GOUT10_R[1:0]		INIT_0_SEL_C GOUT11_R[1:0]		INIT_0_SEL_C GOUT12_R[1:0]		00
8th parameter	1	INIT_0_SEL_C GOUT13_R[1:0]		INIT_0_SEL_C GOUT14_R[1:0]		INIT_0_SEL_C GOUT15_R[1:0]		INIT_0_SEL_C GOUT16_R[1:0]		00
9th parameter	1	INIT_1_SEL_C GOUT1_L[1:0]		INIT_1_SEL_C GOUT2_L[1:0]		INIT_1_SEL_C GOUT3_L[1:0]		INIT_1_SEL_C GOUT4_L[1:0]		00
10th parameter	1	INIT_1_SEL_C GOUT5_L[1:0]		INIT_1_SEL_C GOUT6_L[1:0]		INIT_1_SEL_C GOUT7_L[1:0]		INIT_1_SEL_C GOUT8_L[1:0]		00
11st parameter	1	INIT_1_SEL_C GOUT9_L[1:0]		INIT_1_SEL_C GOUT10_L[1:0]		INIT_1_SEL_C GOUT11_L[1:0]		INIT_1_SEL_C GOUT12_L[1:0]		00
12nd parameter	1	INIT_1_SEL_C GOUT13_L[1:0]		INIT_1_SEL_C GOUT14_L[1:0]		INIT_1_SEL_C GOUT15_L[1:0]		INIT_1_SEL_C GOUT16_L[1:0]		00
13rd parameter	1	INIT_1_SEL_C GOUT1_R[1:0]		INIT_1_SEL_C GOUT2_R[1:0]		INIT_1_SEL_C GOUT3_R[1:0]		INIT_1_SEL_C GOUT4_R[1:0]		00
14th parameter	1	INIT_1_SEL_C GOUT5_R[1:0]		INIT_1_SEL_C GOUT6_R[1:0]		INIT_1_SEL_C GOUT7_R[1:0]		INIT_1_SEL_C GOUT8_R[1:0]		00
15th parameter	1	INIT_1_SEL_C GOUT9_R[1:0]		INIT_1_SEL_C GOUT10_R[1:0]		INIT_1_SEL_C GOUT11_R[1:0]		INIT_1_SEL_C GOUT12_R[1:0]		00
16th parameter	1	INIT_1_SEL_C GOUT13_R[1:0]		INIT_1_SEL_C GOUT14_R[1:0]		INIT_1_SEL_C GOUT15_R[1:0]		INIT_1_SEL_C GOUT16_R[1:0]		00
17th parameter	1	END_0_SEL_C GOUT1_L[1:0]		END_0_SEL_C GOUT2_L[1:0]		END_0_SEL_C GOUT3_L[1:0]		END_0_SEL_C GOUT4_L[1:0]		00
18th parameter	1	END_0_SEL_C GOUT5_L[1:0]		END_0_SEL_C GOUT6_L[1:0]		END_0_SEL_C GOUT7_L[1:0]		END_0_SEL_C GOUT8_L[1:0]		00
19th parameter	1	END_0_SEL_C GOUT9_L[1:0]		END_0_SEL_C GOUT10_L[1:0]		END_0_SEL_C GOUT11_L[1:0]		END_0_SEL_C GOUT12_L[1:0]		00
20th parameter	1	END_0_SEL_C GOUT13_L[1:0]		END_0_SEL_C GOUT14_L[1:0]		END_0_SEL_C GOUT15_L[1:0]		END_0_SEL_C GOUT16_L[1:0]		00
21st parameter	1	END_0_SEL_C GOUT1_R[1:0]		END_0_SEL_C GOUT2_R[1:0]		END_0_SEL_C GOUT3_R[1:0]		END_0_SEL_C GOUT4_R[1:0]		00
22nd parameter	1	END_0_SEL_C GOUT5_R[1:0]		END_0_SEL_C GOUT6_R[1:0]		END_0_SEL_C GOUT7_R[1:0]		END_0_SEL_C GOUT8_R[1:0]		00
23rd parameter	1	END_0_SEL_C GOUT9_R[1:0]		END_0_SEL_C GOUT10_R[1:0]		END_0_SEL_C GOUT11_R[1:0]		END_0_SEL_C GOUT12_R[1:0]		00
24th parameter	1	END_0_SEL_C GOUT13_R[1:0]		END_0_SEL_C GOUT14_R[1:0]		END_0_SEL_C GOUT15_R[1:0]		END_0_SEL_C GOUT16_R[1:0]		00
25th parameter	1	END_1_SEL_C GOUT1_L[1:0]		END_1_SEL_C GOUT2_L[1:0]		END_1_SEL_C GOUT3_L[1:0]		END_1_SEL_C GOUT4_L[1:0]		00

26th parameter	1	END_1_SEL_CGOUT5_L[1:0]	END_1_SEL_CGOUT6_L[1:0]	END_1_SEL_CGOUT7_L[1:0]	END_1_SEL_CGOUT8_L[1:0]	00
27th parameter	1	END_1_SEL_CGOUT9_L[1:0]	END_1_SEL_CGOUT10_L[1:0]	END_1_SEL_CGOUT11_L[1:0]	END_1_SEL_CGOUT12_L[1:0]	00
28th parameter	1	END_1_SEL_CGOUT13_L[1:0]	END_1_SEL_CGOUT14_L[1:0]	END_1_SEL_CGOUT15_L[1:0]	END_1_SEL_CGOUT16_L[1:0]	00
29th parameter	1	END_1_SEL_CGOUT1_R[1:0]	END_1_SEL_CGOUT2_R[1:0]	END_1_SEL_CGOUT3_R[1:0]	END_1_SEL_CGOUT4_R[1:0]	00
30th parameter	1	END_1_SEL_CGOUT5_R[1:0]	END_1_SEL_CGOUT6_R[1:0]	END_1_SEL_CGOUT7_R[1:0]	END_1_SEL_CGOUT8_R[1:0]	00
31st parameter	1	END_1_SEL_CGOUT9_R[1:0]	END_1_SEL_CGOUT10_R[1:0]	END_1_SEL_CGOUT11_R[1:0]	END_1_SEL_CGOUT12_R[1:0]	00
32nd parameter	1	END_1_SEL_CGOUT13_R[1:0]	END_1_SEL_CGOUT14_R[1:0]	END_1_SEL_CGOUT15_R[1:0]	END_1_SEL_CGOUT16_R[1:0]	00
33rd parameter	1	END_2_SEL_CGOUT1_L[1:0]	END_2_SEL_CGOUT2_L[1:0]	END_2_SEL_CGOUT3_L[1:0]	END_2_SEL_CGOUT4_L[1:0]	00
34th parameter	1	END_2_SEL_CGOUT5_L[1:0]	END_2_SEL_CGOUT6_L[1:0]	END_2_SEL_CGOUT7_L[1:0]	END_2_SEL_CGOUT8_L[1:0]	00
35th parameter	1	END_2_SEL_CGOUT9_L[1:0]	END_2_SEL_CGOUT10_L[1:0]	END_2_SEL_CGOUT11_L[1:0]	END_2_SEL_CGOUT12_L[1:0]	00
36th parameter	1	END_2_SEL_CGOUT13_L[1:0]	END_2_SEL_CGOUT14_L[1:0]	END_2_SEL_CGOUT15_L[1:0]	END_2_SEL_CGOUT16_L[1:0]	00
37th parameter	1	END_2_SEL_CGOUT1_R[1:0]	END_2_SEL_CGOUT2_R[1:0]	END_2_SEL_CGOUT3_R[1:0]	END_2_SEL_CGOUT4_R[1:0]	00
38th parameter	1	END_2_SEL_CGOUT5_R[1:0]	END_2_SEL_CGOUT6_R[1:0]	END_2_SEL_CGOUT7_R[1:0]	END_2_SEL_CGOUT8_R[1:0]	00
39th parameter	1	END_2_SEL_CGOUT9_R[1:0]	END_2_SEL_CGOUT10_R[1:0]	END_2_SEL_CGOUT11_R[1:0]	END_2_SEL_CGOUT12_R[1:0]	00
40th parameter	1	END_2_SEL_CGOUT13_R[1:0]	END_2_SEL_CGOUT14_R[1:0]	END_2_SEL_CGOUT15_R[1:0]	END_2_SEL_CGOUT16_R[1:0]	00
41st parameter	1	GAS_0_SEL_CGOUT1_L[1:0]	GAS_0_SEL_CGOUT2_L[1:0]	GAS_0_SEL_CGOUT3_L[1:0]	GAS_0_SEL_CGOUT4_L[1:0]	00
42nd parameter	1	GAS_0_SEL_CGOUT5_L[1:0]	GAS_0_SEL_CGOUT6_L[1:0]	GAS_0_SEL_CGOUT7_L[1:0]	GAS_0_SEL_CGOUT8_L[1:0]	00
43rd parameter	1	GAS_0_SEL_CGOUT9_L[1:0]	GAS_0_SEL_CGOUT10_L[1:0]	GAS_0_SEL_CGOUT11_L[1:0]	GAS_0_SEL_CGOUT12_L[1:0]	00
44th parameter	1	GAS_0_SEL_CGOUT13_L[1:0]	GAS_0_SEL_CGOUT14_L[1:0]	GAS_0_SEL_CGOUT15_L[1:0]	GAS_0_SEL_CGOUT16_L[1:0]	00
45th parameter	1	GAS_0_SEL_CGOUT1_R[1:0]	GAS_0_SEL_CGOUT2_R[1:0]	GAS_0_SEL_CGOUT3_R[1:0]	GAS_0_SEL_CGOUT4_R[1:0]	00
46th parameter	1	GAS_0_SEL_CGOUT5_R[1:0]	GAS_0_SEL_CGOUT6_R[1:0]	GAS_0_SEL_CGOUT7_R[1:0]	GAS_0_SEL_CGOUT8_R[1:0]	00
47th parameter	1	GAS_0_SEL_CGOUT9_R[1:0]	GAS_0_SEL_CGOUT10_R[1:0]	GAS_0_SEL_CGOUT11_R[1:0]	GAS_0_SEL_CGOUT12_R[1:0]	00
48th parameter	1	GAS_0_SEL_CGOUT13_R[1:0]	GAS_0_SEL_CGOUT14_R[1:0]	GAS_0_SEL_CGOUT15_R[1:0]	GAS_0_SEL_CGOUT16_R[1:0]	00
Description	**INIT means SLPIN to SLPOUT:** **INIT_0_SEL_CGOUTn_L:** Set CGOUTL_n output state of first frame or first time interval when D[1:0]=01. **INIT_0_SEL_CGOUTn_R:** Set CGOUTR_n output state of first frame or first time interval when D[1:0]=01. n=1~16					

	INIT_1_SEL_CGOUTn_L: Set CGOUTL_n output state of second frame or second time interval when D[1:0]=01. **INIT_1_SEL_CGOUTn_R:** Set CGOUTR_n output state of second frame or second time interval when D[1:0]=01. n=1~16 **END means SLPOUT to SLPIN** **END_0_SEL_CGOUTn_L:** Set CGOUTL_n output state of first frame or first time interval when D[1:0]=01. **END_0_SEL_CGOUTn_R:** Set CGOUTR_n output state of first frame or first time interval when D[1:0]=01. n=1~16 **END_1_SEL_CGOUTn_L:** Set CGOUTL_n output state of second frame or second time interval when D[1:0]=01. **END_1_SEL_CGOUTn_R:** Set CGOUTR_n output state of second frame or second time interval when D[1:0]=01. n=1~16 **END_2_SEL_CGOUTn_L:** Set CGOUTL_n output state of third frame or third time interval when D[1:0]=01. **END_2_SEL_CGOUTn_R:** Set CGOUTR_n output state of third frame or third time interval when D[1:0]=01. n=1~16 **GAS means abnormal power off** **GAS_0_SEL_CGOUTn_L:** When abnormal power off happens,set CGOUTL_n output state. **GAS_0_SEL_CGOUTn_R:** When abnormal power off happens,set CGOUTR_n output state. n=1~16 GIP output state: “00”: Keep normal GIP output. “01”: Keep normal GIP output. “10”: Fixed VGL. “11”: Fixed VGH.
Restrictions	-
Register Availability	(see table below)

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

6.3.22 SETGPO (D9h)

D9H	SETGPO									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	0	1	1	0	0	1	D9
1st parameter	1	-	-	-	-	TE_GPO[3:0]				00
2nd parameter	1	-	-	-	-	TE1_GPO[3:0]				01
3rd parameter	1	-	-	-	-	CABC_GPO[3:0]				02
4th parameter	1	-	-	-	-	SDO_GPO[3:0]				07

Description

TE_GPO[3:0]: Set the output pin TE.
TE1_GPO[3:0]: Set the output pin TE1.
CABC_GPO[3:0]: : Set the output pin CABC_PWM_OUT.
SDO_GPO[3:0]: Set the output pin SDO.

TE_GPO[3:0]/ TE1_GPO[3:0]/ CABC_GPO[3:0]/ SDO_GPO[3:0]				TE/ TE1/ CABC_PWM_OUT/ SDO Output signal
0	0	0	0	TE
0	0	0	1	TE1
0	0	1	0	CABC
0	0	1	1	HSYNC
0	1	0	0	VSYNC
0	1	0	1	DE
0	1	1	0	DSI_ERR_RPT
0	1	1	1	SDO
1	0	0	0	STB
Others				Inhibited

Restrictions: SETEXTC turn on to enable this command

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

6.3.23 SETGAMMA: Set gamma curve related setting (E0h)

E0H	SETGAMMA (Set Gamma Curve Related Setting)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	1	0	0	0	0	0	E0
1st Parameter	1	-	-	VRP0[5:0]						04
2nd parameter	1	-	-	VRP1[5:0]						0C
3rd parameter	1	-	-	VRP2[5:0]						0D
4th parameter	1	-	-	VRP3[5:0]						0A
5th parameter	1	-	-	VRP4[5:0]						15
6th parameter	1	-	-	VRP5[5:0]						21
7th parameter	1	-	PRP0[6:0]							0D
8th parameter	1	-	PRP1[6:0]							19
9th parameter	1	-	-	-	PKP0[4:0]					0A
10th parameter	1	-	-	-	PKP1[4:0]					0C
11st parameter	1	-	-	-	PKP2[4:0]					0F
12nd parameter	1	-	-	-	PKP3[4:0]					13
13rd parameter	1	-	-	-	PKP4[4:0]					1A
14th parameter	1	-	-	-	PKP5[4:0]					14
15th parameter	1	-	-	-	PKP6[4:0]					15
16th parameter	1	-	-	-	PKP7[4:0]					0D
17th parameter	1	-	-	-	PKP8[4:0]					13
18th parameter	1	-	-	-	PKP_EX0[4:0]					0A
19th parameter	1	-	-	-	PKP_EX1[4:0]					16
20th parameter	1	-	-	-	PKP_EX2[4:0]					06
21st parameter	1	-	-	-	PKP_EX3[4:0]					0F
22nd parameter	1	-	-	VRN0[5:0]						04
23rd parameter	1	-	-	VRN1[5:0]						0C
24th parameter	1	-	-	VRN2[5:0]						0D
25th parameter	1	-	-	VRN3[5:0]						0A
26th parameter	1	-	-	VRN4[5:0]						15
27th parameter	1	-	-	VRN5[5:0]						21
28th parameter	1	-	PRN0[6:0]							0D
29th parameter	1	-	PRN1[6:0]							19
30th parameter	1	-	-	-	PKN0[4:0]					06
31st parameter	1	-	-	-	PKN1[4:0]					0C
32nd parameter	1	-	-	-	PKN2[4:0]					0F
33rd parameter	1	-	-	-	PKN3[4:0]					13
34th parameter	1	-	-	-	PKN4[4:0]					16
35th parameter	1	-	-	-	PKN5[4:0]					14
36th parameter	1	-	-	-	PKN6[4:0]					15
37th parameter	1	-	-	-	PKN7[4:0]					0D
38th parameter	1	-	-	-	PKN8[4:0]					13
39th parameter	1	-	-	-	PKN_EX0[4:0]					07
40th parameter	1	-	-	-	PKN_EX1[4:0]					16
41st parameter	1	-	-	-	PKN_EX2[4:0]					06
42nd parameter	1	-	-	-	PKN_EX3[4:0]					11
Description	This command is to set gamma register.									

Register Groups	Positive Polarity	Negative Polarity	Description
Center Adjustment	PRP0 6-0	PRN0 6-0	88-to-1 selector (voltage level of grayscale 52)
	PRP1 6-0	PRN1 6-0	88-to-1 selector (voltage level of grayscale 203)
Macro Adjustment	PKP0 4-0	PKN0 4-0	32-to-1 selector (voltage level of grayscale 12)
	PKP1 4-0	PKN1 4-0	32-to-1 selector (voltage level of grayscale 28)
	PKP2 4-0	PKN2 4-0	32-to-1 selector (voltage level of grayscale 76)
	PKP3 4-0	PKN3 4-0	32-to-1 selector (voltage level of grayscale 100)

		PKP4 4-0	PKN4 4-0	32-to-1 selector (voltage level of grayscale 131)
		PKP5 4-0	PKN5 4-0	32-to-1 selector (voltage level of grayscale 155)
		PKP6 4-0	PKN6 4-0	32-to-1 selector (voltage level of grayscale 179)
		PKP7 4-0	PKN7 4-0	32-to-1 selector (voltage level of grayscale 227)
		PKP8 4-0	PKN8 4-0	32-to-1 selector (voltage level of grayscale 243)
		PKP_EX0 4-0	PKN_EX0 4-0	32-to-1 selector (voltage level of grayscale 20)
		PKP_EX1 4-0	PKN_EX1 4-0	32-to-1 selector (voltage level of grayscale 40)
		PKP_EX2 4-0	PKN_EX2 4-0	32-to-1 selector (voltage level of grayscale 215)
		PKP_EX2 4-0	PKN_EX2 4-0	32-to-1 selector (voltage level of grayscale 235)
	Offset Adjustment	VRP0 5-0	VRN0 5-0	64-to-1 selector (voltage level of grayscale 0)
		VRP1 5-0	VRN1 5-0	64-to-1 selector (voltage level of grayscale 4)
		VRP2 5-0	VRN2 5-0	64-to-1 selector (voltage level of grayscale 8)
		VRP3 5-0	VRN3 5-0	64-to-1 selector (voltage level of grayscale 247)
		VRP4 5-0	VRN4 5-0	64-to-1 selector (voltage level of grayscale 251)
		VRP5 5-0	VRN5 5-0	64-to-1 selector (voltage level of grayscale 255)
Restriction	SETEXTC turn on to enable this command.			

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

6.3.24 SETCHEMODE_DYN (E4h)

E4H	SETCHEMODE_DYN (Set color enhancement mode, dynamic)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	1	0	0	1	0	0	E4
1st parameter	1	-	-	-	-	-	-	-	DYN_CHEH_EN	01
2nd parameter	1	HUE_MODE[1:0]		SE_MODE[1:0]		BE_MODE[1:0]		CE_MODE[1:0]		00

Description

This command is to set color enhanment and dynamic mode.

DYN_CHE_EN: Select the color enhancement reload mode. 0: static mode, 1: dynamic mode.

CE_MODE[1:0]: Set color (saturation) enhancement.

BE_MODE[1:0]: Set brightness enhancement.

SE_MODE[1:0]: Set sharpness enhancement.

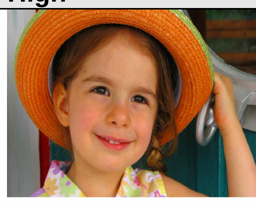
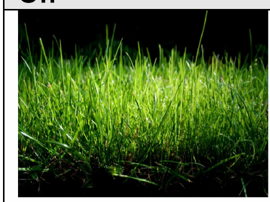
HUE_MODE[1:0]: Set hue enhancement.

In Static mode:
Enhancement level selection: when SE/BE/CE/HUE is turn on, the enhancement effect depends on the gobal gain curve set by user in command RE5h and RE6h.

CE_MODE[1:0]/ BE_MODE[1:0]/ SE_MODE[1:0]/ HUE_MODE[1:0]		Enhance
0	0	Off
0	1	On
1	0	
1	1	

In Dynamic mode:
Enhance level selection. wWhen SE/BE/CE/HUE turn on, the enhancement gain setting will be read from the ROM table. Three sets of enhancement gain are provide for selection.

CE_MODE[1:0]/ BE_MODE[1:0]/ SE_MODE[1:0]/ HUE_MODE[1:0]		Enhance
0	0	Off
0	1	Low
1	0	Medium
1	1	High

Color Enhancemet			
Off	Low	Medium	High
			
Brightness Enhancement			
Off	Low	Medium	High
			
Sharpness Enhancement			
Off	Low	Medium	High

<table>
<tr><td></td><td>   </td></tr>
<tr><td>Restrictions</td><td>SETEXTC turn on to enable this command.</td></tr>
<tr><td>Register Availability</td><td><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table></td></tr>
</table>

6.3.25 SETI2C_SA: Set I2C Slave Address (E8h)

E8H	SETI2C_SA(Set I2C Slave Address)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	1	0	1	0	0	0	E8
1st parameter	1	-	I2C_SA[6:0]							00

Description: This register is for setting I2C slave address.

I2C_SA[6 :0]	Slave address (A6-A0)
000_0000	000_0000
000_0001~000_0111	Reserved
000_1000	000_1000
:	:
111_0110	111_0110
111_0111	111_0111
111_1xxx	Reserved

Restrictions: SETEXTC turn on to enable this command

Register Availability:

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

6.3.26 SETCNCD/GETCNCD (FDh)

FDH	SETCNCD/GETCNCD (Set/Get Continue Command)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	1	1	1	1	0	1	FD
1st parameter	1	WR_CMD_CN[7:0]								-
Description	This function is use to instead of Register-Content interface mode. The parameter for SETCNCD will continue to write or read from the last command address automatically.									
Restrictions	SETEXTC turn on to enable this command									
Register Availability	(see table below)									

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

6.3.27 SETREADINDEX: Set SPI Read Index (FEh)

FEH	SET SPI READ INDEX (Set SPI READ Command Address)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	1	1	1	1	1	0	FE
1st parameter	1	CMD_ADD[7:0]								-
Description	SET SPI Read Command Address for User Define Command.									
Restrictions	SETEXTC turn on to enable this command									
Register Availability										

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

6.3.28 GETSPIREAD: SPI Read Command Data (FFh)

FFH	GETSPIREAD (Read Command Data)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	1	1	1	1	1	1	FF
1st parameter	1	CMD_DATA1[7:0]								-
:	1	:								-
nth parameter	1	CMD_DATAn[7:0]								-
Description	Read SPI Command Data for User Define Command.									
Restrictions	SETEXTC turn on to enable this command.									
Register Availability	Status: Normal Mode On, Idle Mode Off, Sleep Out — Yes; Normal Mode On, Idle Mode On, Sleep Out — Yes; Sleep In or Booster Off — Yes									

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

7. Layout Recommendation

7.1 Maximum layout resistance

Name	Type	Maximum series resistance	Unit
VDD1	Power supply	5	Ω
VDD3	Power supply	5	Ω
VSSD	Power supply	5	Ω
VSSA	Power supply	5	Ω
HS_VCC	Power supply	5	Ω
HS_VSS	Power supply	5	Ω
VSSAC	Power supply	5	Ω
VPP	Power supply	20	Ω
IM[2:0]	Input	100	Ω
DSWAP[1:0], PNSWAP	Input	100	Ω
SCL, DCX, CSX, RESX	Input	100	Ω
HSYNC, VSYNC, DE, PCLK	Input	100	Ω
SDI_SDA	Input	100	Ω
SDO	Output	100	Ω
DB[7:0]	Input + Output	100	Ω
CABC_PWM_OUT, TE, TE1, VCSW1, VCSW2	Output	100	Ω
VCOM	Output	10	Ω
HS_CLKP, HS_CLKN	Input	6	Ω
HS_D0P, HS_D0N	Input + Output	6	Ω
HS_D1P, HS_D1N	Input	6	Ω
HS_D2P, HS_D2N	Input	6	Ω
HS_D3P, HS_D3N	Input	6	Ω
PCCS[2:0]	Input	100	
VDDD	Capacitor Connection	5	Ω
VSP	Capacitor Connection	10	Ω
VSN	Capacitor Connection	10	Ω
VSP_REG, VSN_REG	Capacitor Connection	10	Ω
VSPR, VSNR	Capacitor Connection	20	Ω
VGH, VGL, VGHO_L, VGHO_R, VGLO_L, VGLO_R, VCL	Capacitor Connection	10	Ω
HS_LDO	Capacitor Connection	10	Ω
OSC	Input	100	Ω
C31P, C31N, C41P, C41N	Capacitor Connection	5	Ω
C21P, C21N	Capacitor Connection	5	Ω
TEST[2:0]	Input	100	Ω
VTESTOUTP, VTESTOUTN	Output	100	Ω

Table 7.1: Maximum Layout Resistance

7.2 External Components Connection

HX5186-A/B/C mode:

<table>
<tr><th>Pad Name</th><th>Symbol</th><th>Connection</th><th>Typical Component Value</th></tr>
<tr><td>VCOM</td><td>C1</td><td>Connect to Capacitor (Max 6V): VCOM ---(-)----| |--- (+)----- VSSA</td><td>1.0 μF</td></tr>
<tr><td>VGH</td><td>C2</td><td>Connect to Capacitor (Max 25V): VGH ---(+)----| |--- (-)----- VSSA</td><td>1.0 μF</td></tr>
<tr><td rowspan="2">VGL</td><td>C3</td><td>Connect to Capacitor (Max 25V): VGL ---(-)----| |--- (+)----- VSSA</td><td>1.0 μF</td></tr>
<tr><td>D1</td><td>Connect to Schottky Diode(VR≥30V): VSSA ---(-)----∫◄--- (+)---- VGL</td><td>VF < 0.4V / 20mA @ 25°C, VR ≥30V (Recommended diode: RB521S-30)</td></tr>
<tr><td>VCL</td><td>C4</td><td>Connect to Capacitor (Max 6V): VCL ---(-)----| |--- (+)----- VSSA</td><td>2.2 μF</td></tr>
<tr><td>C21P – C21N</td><td>C5</td><td>Connect to Capacitor (Max 16V): C21P ---(+)----| |--- (-)-----C21N</td><td>1.0 μF</td></tr>
<tr><td>C31P – C31N</td><td>C6</td><td>Connect to Capacitor (Max 16V): C31P ---(+)----| |--- (-)-----C31N</td><td>1.0 μF</td></tr>
<tr><td>C41P – C41N</td><td>C7</td><td>Connect to Capacitor (Max 6V): C41P ---(+)----| |--- (-)-----C41N</td><td>2.2 μF</td></tr>
<tr><td>VDDD</td><td>C8</td><td>Connect to Capacitor (Max 6V): VDDD ---(+)----| |--- (-)-----VSSA</td><td>2.2 μF</td></tr>
<tr><td>VSP</td><td>C9</td><td>Connect to Capacitor (Max 10V):VSP ---(+)----| |--- (-)-----VSSA</td><td>2.2 μF</td></tr>
<tr><td>VSN</td><td>C10</td><td>Connect to Capacitor (Max 10V):VSN ---(-)----| |--- (+)-----VSSA</td><td>2.2 μF</td></tr>
<tr><td>VSP_REG</td><td>C11</td><td>Connect to Capacitor (Max 10V): VSP_REG ---(+)----| |--- (-)-----VSSA</td><td>2.2 μF</td></tr>
<tr><td>VSN_REG</td><td>C12</td><td>Connect to Capacitor (Max 10V): VSN_REG ---(-)----| |--- (+)-----VSSA</td><td>2.2 μF</td></tr>
<tr><td>VGHO_L/R</td><td>C13</td><td>Connect to Capacitor (Max 16V): VGHO_L/R ---(+)----| |--- (-)-----VSSA</td><td>1.0 μF</td></tr>
<tr><td>VGLO_L/R</td><td>C14</td><td>Connect to Capacitor (Max 16V): VGLO_L/R ---(-)----| |--- (+)-----VSSA</td><td>1.0 μF</td></tr>
<tr><td>VDD3</td><td>C15</td><td>Connect to Capacitor (Max 10V): VDD3 ---(+)----| |--- (-)-----VSSA</td><td>2.2 μF</td></tr>
<tr><td>VDD1</td><td>C16</td><td>Connect to Capacitor (Max 6V): VDD1 ---(+)----| |--- (-)-----VSSA</td><td>2.2 μF</td></tr>
<tr><td>HS_VCC</td><td>C17</td><td>Connect to Capacitor (Max 6V): HS_VCC ---(+)----| |--- (-)-----VSSA</td><td>2.2 μF</td></tr>
<tr><td>HS_LDO</td><td>C18</td><td>Connect to Capacitor (Max 6V): HS_LDO ---(+)----| |--- (-)----HS_VSS</td><td>2.2 μF</td></tr>
<tr><td>HX5186-A/B/C</td><td>U1</td><td>Please refer HX5186-A/B/C datasheet</td><td>-</td></tr>
<tr><td>HX5186-A</td><td>C19</td><td>Please refer HX5186-A datasheet</td><td>1.0uF</td></tr>
<tr><td>HX5186-A</td><td>C20</td><td>Please refer HX5186-A datasheet</td><td>1.0uF</td></tr>
<tr><td>HX5186-A</td><td>C21</td><td>Please refer HX5186-A datasheet</td><td>1.0uF</td></tr>
</table>

Table 7.2: HX5186-A/B/C mode external components

PFM Type A mode:

Pad Name	Symbol	Connection	Typical Component Value
VCOM	C1	Connect to Capacitor (Max 6V): VCOM ---(-)----| |--- (+)----- VSSA	1.0 μF
VGH	C2	Connect to Capacitor (Max 25V): VGH ---(+)----| |--- (-)----- VSSA	1.0 μF
VGL	C3	Connect to Capacitor (Max 25V): VGL ---(-)----| |--- (+)----- VSSA	1.0 μF
	D1	Connect to Schottky Diode(VR≥30V): VSSA ---(-)----∫◄--- (+)---- VGL	VF < 0.4V / 20mA @ 25°C, VR ≥30V (Recommended diode: RB521S-30)
VCL	C4	Connect to Capacitor (Max 6V): VCL ---(-)----| |--- (+)----- VSSA	2.2 μF
C21P – C21N	C5	Connect to Capacitor (Max 16V): C21P ---(+)----| |--- (-)-----C21N	1.0 μF
C31P – C31N	C6	Connect to Capacitor (Max 16V): C31P ---(+)----| |--- (-)-----C31N	1.0 μF
C41P – C41N	C7	Connect to Capacitor (Max 6V): C41P ---(+)----| |--- (-)-----C41N	2.2 μF
VDDD	C8	Connect to Capacitor (Max 6V): VDDD ---(+)----| |--- (-)-----VSSA	2.2 μF
VSP	C9	Connect to Capacitor (Max 10V):VSP ---(+)----| |--- (-)-----VSSA	2.2 μF
VSN	C10	Connect to Capacitor (Max 10V):VSN ---(-)----| |--- (+)-----VSSA	2.2 μF
VSP_REG	C11	Connect to Capacitor (Max 10V): VSP_REG ---(+)----| |--- (-)-----VSSA	2.2 μF
VSN_REG	C12	Connect to Capacitor (Max 10V): VSN_REG ---(-)----| |--- (+)-----VSSA	2.2 μF
VGHO_L/R	C13	Connect to Capacitor (Max 16V): VGHO_L/R ---(+)----| |--- (-)-----VSSA	1.0 μF
VGLO_L/R	C14	Connect to Capacitor (Max 16V): VGLO_L/R ---(-)----| |--- (+)-----VSSA	1.0 μF
VDD3	C15	Connect to Capacitor (Max 10V): VDD3 ---(+)----| |--- (-)-----VSSA	2.2 μF
VDD1	C16	Connect to Capacitor (Max 6V): VDD1 ---(+)----| |--- (-)-----VSSA	2.2 μF
HS_VCC	C17	Connect to Capacitor (Max 6V): HS_VCC ---(+)----| |--- (-)-----VSSA	2.2 μF
HS_LDO	C18	Connect to Capacitor (Max 6V): HS_LDO ---(+)----| |--- (-)----HS_VSS	2.2 μF
PFM	L1	Inductiance, reference PFM Type A connection	-
PFM	SW1	MOS switch, reference PFM Type A connection	-
PFM	SW2	MOS switch, reference PFM Type A connection	-
PFM	D2	Schottkey diode, reference PFM Type A connection	VF < 0.4V / 20mA @ 25°C, VR ≥30V (Recommended diode: RB521S-30)
PFM	D3	Schottkey diode, reference PFM Type A connection	VF < 0.4V / 20mA @ 25°C, VR ≥30V (Recommended diode: RB521S-30)

Table 7.3: PFM Type A mode external components

PFM Type B mode:

Pad Name	Symbol	Connection	Typical Component Value
VCOM	C1	Connect to Capacitor (Max 6V): VCOM ---(-)----| |--- (+)----- VSSA	1.0 μF
VGH	C2	Connect to Capacitor (Max 25V): VGH ---(+)----| |--- (-)----- VSSA	1.0 μF
VGL	C3	Connect to Capacitor (Max 25V): VGL ---(-)----| |--- (+)----- VSSA	1.0 μF
	D1	Connect to Schottky Diode(VR≥30V): VSSA ---(-)----⌠◄--- (+)---- VGL	VF < 0.4V / 20mA @ 25°C, VR ≥30V (Recommended diode: RB521S-30)
VCL	C4	Connect to Capacitor (Max 6V): VCL ---(-)----| |--- (+)----- VSSA	2.2 μF
C21P – C21N	C5	Connect to Capacitor (Max 16V): C21P ---(+)----| |--- (-)-----C21N	1.0 μF
C31P – C31N	C6	Connect to Capacitor (Max 16V): C31P ---(+)----| |--- (-)-----C31N	1.0 μF
C41P – C41N	C7	Connect to Capacitor (Max 6V): C41P ---(+)----| |--- (-)-----C41N	2.2 μF
VDDD	C8	Connect to Capacitor (Max 6V): VDDD ---(+)----| |--- (-)-----VSSA	2.2 μF
VSP	C9	Connect to Capacitor (Max 10V):VSP ---(+)----| |--- (-)-----VSSA	2.2 μF
VSN	C10	Connect to Capacitor (Max 10V):VSN ---(-)----| |--- (+)-----VSSA	2.2 μF
VSP_REG	C11	Connect to Capacitor (Max 10V): VSP_REG ---(+)----| |--- (-)-----VSSA	2.2 μF
VSN_REG	C12	Connect to Capacitor (Max 10V): VSN_REG ---(-)----| |--- (+)-----VSSA	2.2 μF
VGHO_L/R	C13	Connect to Capacitor (Max 16V): VGHO_L/R ---(+)----| |--- (-)-----VSSA	1.0 μF
VGLO_L/R	C14	Connect to Capacitor (Max 16V): VGLO_L/R ---(-)----| |--- (+)-----VSSA	1.0 μF
VDD3	C15	Connect to Capacitor (Max 10V): VDD3 ---(+)----| |--- (-)-----VSSA	2.2 μF
VDD1	C16	Connect to Capacitor (Max 6V): VDD1 ---(+)----| |--- (-)-----VSSA	2.2 μF
HS_VCC	C17	Connect to Capacitor (Max 6V): HS_VCC ---(+)----| |--- (-)-----VSSA	2.2 μF
HS_LDO	C18	Connect to Capacitor (Max 6V): HS_LDO ---(+)----| |--- (-)----HS_VSS	2.2 μF
PFM	L1	Inductiance, reference PFM Type B connection	-
PFM	SW2	MOS switch, reference PFM Type B connection	-
PFM	D3	Schottkey diode, reference PFM Type B connection	VF < 0.4V / 20mA @ 25°C, VR ≥30V (Recommended diode: RB521S-30)

Table 7.4: PFM Type B mode external components

PFM Type C mode:

Pad Name	Symbol	Connection	Typical Component Value
VCOM	C1	Connect to Capacitor (Max 6V): VCOM ---(-)----| |--- (+)----- VSSA	1.0µF
VGH	C2	Connect to Capacitor (Max 25V): VGH ---(+)----| |--- (-)----- VSSA	1.0 µF
VGL	C3	Connect to Capacitor (Max 25V): VGL ---(-)----| |--- (+)----- VSSA	1.0 µF
	D1	Connect to Schottky Diode(VR≥30V): VSSA ---(-)----|◄--- (+)---- VGL	VF < 0.4V / 20mA @ 25°C, VR ≥30V (Recommended diode: RB521S-30)
VCL	C4	Connect to Capacitor (Max 6V): VCL ---(-)----| |--- (+)----- VSSA	2.2 µF
C21P – C21N	C5	Connect to Capacitor (Max 16V): C21P ---(+)----| |--- (-)-----C21N	1.0 µF
C31P – C31N	C6	Connect to Capacitor (Max 16V): C31P ---(+)----| |--- (-)-----C31N	1.0 µF
C41P – C41N	C7	Connect to Capacitor (Max 6V): C41P ---(+)----| |--- (-)-----C41N	2.2 µF
VDDD	C8	Connect to Capacitor (Max 6V): VDDD ---(+)----| |--- (-)-----VSSA	2.2 µF
VSP	C9	Connect to Capacitor (Max 10V):VSP ---(+)----| |--- (-)-----VSSA	2.2 µF
VSN	C10	Connect to Capacitor (Max 10V):VSN ---(-)----| |--- (+)-----VSSA	2.2 µF
VSP_REG	C11	Connect to Capacitor (Max 10V): VSP_REG ---(+)----| |--- (-)-----VSSA	2.2 µF
VSN_REG	C12	Connect to Capacitor (Max 10V): VSN_REG ---(-)----| |--- (+)-----VSSA	2.2 µF
VGHO_L/R	C13	Connect to Capacitor (Max 16V): VGHO_L/R ---(+)----| |--- (-)-----VSSA	1.0 µF
VGLO_L/R	C14	Connect to Capacitor (Max 16V): VGLO_L/R ---(-)----| |--- (+)-----VSSA	1.0 µF
VDD3	C15	Connect to Capacitor (Max 10V): VDD3 ---(+)----| |--- (-)-----VSSA	2.2 µF
VDD1	C16	Connect to Capacitor (Max 6V): VDD1 ---(+)----| |--- (-)-----VSSA	2.2 µF
HS_VCC	C17	Connect to Capacitor (Max 6V): HS_VCC ---(+)----| |--- (-)-----VSSA	2.2 µF
HS_LDO	C18	Connect to Capacitor (Max 6V): HS_LDO ---(+)----| |--- (-)----HS_VSS	2.2 µF
PFM	L1	Inductiance, reference PFM Type C connection	-
PFM	SW1	MOS switch, reference PFM Type C connection	-
PFM	D2	Schottkey diode, reference PFM Type C connection	VF < 0.4V / 20mA @ 25°C, VR ≥30V (Recommended diode: RB521S-30)
PFM	D3	Schottkey diode, reference PFM Type C connection	VF < 0.4V / 20mA @ 25°C, VR ≥30V (Recommended diode: RB521S-30)
PFM	D4	Schottkey diode, reference PFM Type C connection	VF < 0.4V / 20mA @ 25°C, VR ≥30V (Recommended diode: RB521S-30)
PFM	C22	Capacitor, reference PFM Type C connection	1.0uF

Table 7.5: PFM Type C mode external components

External VSP/VSN mode:

<table>
<tr><th>Pad Name</th><th>Symbol</th><th>Connection</th><th>Typical Component Value</th></tr>
<tr><td>VCOM</td><td>C1</td><td>Connect to Capacitor (Max 6V): VCOM ---(-)----| |--- (+)----- VSSA</td><td>1.0 µF</td></tr>
<tr><td>VGH</td><td>C2</td><td>Connect to Capacitor (Max 25V): VGH ---(+)----| |--- (-)----- VSSA</td><td>1.0 µF</td></tr>
<tr><td rowspan="2">VGL</td><td>C3</td><td>Connect to Capacitor (Max 25V): VGL ---(-)----| |--- (+)----- VSSA</td><td>1.0 µF</td></tr>
<tr><td>D1</td><td>Connect to Schottky Diode(VR≥30V): VSN ---(-)----∫◄--- (+)---- VGL</td><td>VF < 0.4V / 20mA @ 25°C, VR ≥30V (Recommended diode: RB521S-30)</td></tr>
<tr><td>VCL</td><td>C4</td><td>Connect to Capacitor (Max 6V): VCL ---(-)----| |--- (+)----- VSSA</td><td>2.2 µF</td></tr>
<tr><td>C21P – C21N</td><td>C5</td><td>Connect to Capacitor (Max 16V): C21P ---(+)----| |--- (-)-----C21N</td><td>1.0 µF</td></tr>
<tr><td>C31P – C31N</td><td>C6</td><td>Connect to Capacitor (Max 16V): C31P ---(+)----| |--- (-)-----C31N</td><td>1.0 µF</td></tr>
<tr><td>C41P – C41N</td><td>C7</td><td>Connect to Capacitor (Max 6V): C41P ---(+)----| |--- (-)-----C41N</td><td>2.2 µF</td></tr>
<tr><td>VDDD</td><td>C8</td><td>Connect to Capacitor (Max 6V): VDDD ---(+)----| |--- (-)-----VSSA</td><td>2.2 µF</td></tr>
<tr><td>VSP</td><td>C9</td><td>Connect to Capacitor (Max 10V):VSP ---(+)----| |--- (-)-----VSSA</td><td>2.2 µF</td></tr>
<tr><td>VSN</td><td>C10</td><td>Connect to Capacitor (Max 10V):VSN ---(-)----| |--- (+)-----VSSA</td><td>2.2 µF</td></tr>
<tr><td>VSP_REG</td><td>C11</td><td>Connect to Capacitor (Max 10V): VSP_REG ---(+)----| |--- (-)-----VSSA</td><td>2.2 µF</td></tr>
<tr><td>VSN_REG</td><td>C12</td><td>Connect to Capacitor (Max 10V): VSN_REG ---(-)----| |--- (+)-----VSSA</td><td>2.2 µF</td></tr>
<tr><td>VGHO_L/R</td><td>C13</td><td>Connect to Capacitor (Max 16V): VGHO_L/R ---(+)----| |--- (-)-----VSSA</td><td>1.0 µF</td></tr>
<tr><td>VGLO_L/R</td><td>C14</td><td>Connect to Capacitor (Max 16V): VGLO_L/R ---(-)----| |--- (+)-----VSSA</td><td>1.0 µF</td></tr>
<tr><td>VDD3</td><td>C15</td><td>Connect to Capacitor (Max 10V): VDD3 ---(+)----| |--- (-)-----VSSA</td><td>2.2 µF</td></tr>
<tr><td>VDD1</td><td>C16</td><td>Connect to Capacitor (Max 6V): VDD1 ---(+)----| |--- (-)-----VSSA</td><td>2.2 µF</td></tr>
<tr><td>HS_VCC</td><td>C17</td><td>Connect to Capacitor (Max 6V): HS_VCC ---(+)----| |--- (-)-----VSSA</td><td>2.2 µF</td></tr>
<tr><td>HS_LDO</td><td>C18</td><td>Connect to Capacitor (Max 6V): HS_LDO ---(+)----| |--- (-)----HS_VSS</td><td>2.2 µF</td></tr>
</table>

Table 7.6: External VSP/VSN mode external components

External VDD3/VSP/VSN mode:

<table>
<tr><th>Pad Name</th><th>Symbol</th><th>Connection</th><th>Typical Component Value</th></tr>
<tr><td>VCOM</td><td>C1</td><td>Connect to Capacitor (Max 6V): VCOM ---(-)----| |--- (+)----- VSSA</td><td>2.2 µF</td></tr>
<tr><td>VGH</td><td>C2</td><td>Connect to Capacitor (Max 25V): VGH ---(+)----| |--- (-)----- VSSA</td><td>1.0 µF</td></tr>
<tr><td rowspan="2">VGL</td><td>C3</td><td>Connect to Capacitor (Max 25V): VGL ---(-)----| |--- (+)----- VSSA</td><td>1.0 µF</td></tr>
<tr><td>D1</td><td>Connect to Schottky Diode(VR≥30V): VSN---(-)----∫◄--- (+)---- VGL</td><td>VF < 0.4V / 20mA @ 25°C, VR ≥30V (Recommended diode: RB521S-30)</td></tr>
<tr><td>VCL</td><td>C4</td><td>Connect to Capacitor (Max 6V): VCL ---(-)----| |--- (+)----- VSSA</td><td>2.2 µF</td></tr>
<tr><td>C21P – C21N</td><td>C5</td><td>Connect to Capacitor (Max 16V): C21P ---(+)----| |--- (-)-----C21N</td><td>1.0 µF</td></tr>
<tr><td>C31P – C31N</td><td>C6</td><td>Connect to Capacitor (Max 16V): C31P ---(+)----| |--- (-)-----C31N</td><td>1.0 µF</td></tr>
<tr><td>C41P – C41N</td><td>C7</td><td>Connect to Capacitor (Max 6V): C41P ---(+)----| |--- (-)-----C41N</td><td>2.2 µF</td></tr>
<tr><td>VDDD</td><td>C8</td><td>Connect to Capacitor (Max 6V): VDDD ---(+)----| |--- (-)-----VSSA</td><td>2.2 µF</td></tr>
<tr><td>VSP</td><td>C9</td><td>Connect to Capacitor (Max 10V):VSP ---(+)----| |--- (-)-----VSSA</td><td>2.2 µF</td></tr>
<tr><td>VSN</td><td>C10</td><td>Connect to Capacitor (Max 10V):VSN ---(-)----| |--- (+)-----VSSA</td><td>2.2 µF</td></tr>
<tr><td>VSP_REG</td><td>C11</td><td>Connect to Capacitor (Max 10V): VSP_REG ---(+)----| |--- (-)-----VSSA</td><td>2.2 µF</td></tr>
<tr><td>VSN_REG</td><td>C12</td><td>Connect to Capacitor (Max 10V): VSN_REG ---(-)----| |--- (+)-----VSSA</td><td>2.2 µF</td></tr>
<tr><td>VGHO_L/R</td><td>C13</td><td>Connect to Capacitor (Max 16V): VGHO_L/R ---(+)----| |--- (-)-----VSSA</td><td>1.0 µF</td></tr>
<tr><td>VGLO_L/R</td><td>C14</td><td>Connect to Capacitor (Max 16V): VGLO_L/R ---(-)----| |--- (+)-----VSSA</td><td>1.0 µF</td></tr>
<tr><td>VDD3</td><td>C15</td><td>Connect to Capacitor (Max 10V): VDD3 ---(+)----| |--- (-)-----VSSA</td><td>2.2 µF</td></tr>
<tr><td>VDD1</td><td>C16</td><td>Connect to Capacitor (Max 6V): VDD1 ---(+)----| |--- (-)-----VSSA</td><td>2.2 µF</td></tr>
<tr><td>HS_VCC</td><td>C17</td><td>Connect to Capacitor (Max 6V): HS_VCC ---(+)----| |--- (-)-----VSSA</td><td>2.2 µF</td></tr>
<tr><td>HS_LDO</td><td>C18</td><td>Connect to Capacitor (Max 6V): HS_LDO ---(+)----| |--- (-)----HS_VSS</td><td>2.2 µF</td></tr>
</table>

Table 7.7: External VDD3/VSP/VSN mode external components

8. Electrical Characteristics

8.1 Absolute maximum ratings

The absolute maximum ratings are list on Table 8.1. When used out of the absolute maximum ratings, the LSI may be permanently damaged. Using the LSI within the following electrical characteristics limit is strongly recommended for normal operation. If these electrical characteristic conditions are exceeded during normal operation, the LSI will malfunction and cause poor reliability.

Item	Symbol	Unit	Value	Note
Power Supply Voltage 1	VDD1~ VSSD	V	-0.3 to +3.6	Note(1),(2)
Power Supply Voltage 2	VDD3 ~ VSSA	V	-0.3 to +5.5	Note(1) (3)
Power Supply Voltage 3	HS_VCC ~ HS_VSS	V	-0.3 to +3.6	Note(1) (4)
Power Supply Voltage 4	VSP ~ VSSA	V	-0.3 to +6.6	Note(5)
Power Supply Voltage 5	VSSA ~ VSN	V	0 to -6.6	Note(6)
Power Supply Voltage 6	VGH ~ VSSA	V	-0.3 to +25	Note(7)
Power Supply Voltage 7	VSSA ~ VGL	V	0 to -16	Note(8)
Operating Temperature	Topr	°C	-40 to +85	Note(9)
Storage Temperature	Tstg	°C	-55 to +110	Note(10)

Note: (1) VDD1, VSSD must be maintained.
(2) To make sure VDD1 ≥ VSSD.
(3) To make sure VDD3≥ VSSA.
(4) To make sure HS_VCC ≥ HS_VSS.
(5) To make sure VSP ≥ VSSA.
(6) To make sure VSSA ≥ VSN
(7) To make sure VGH ≥ VSSA.
(8) To make sure VSSA ≥ VGL
VGH +|VGL| < 30V
(9) For die and wafer products, specified up to +85℃.
(10) This temperature specifications apply to the TCP package.

Table 8.1: Absolute maximum rating

8.2 DC characteristics

(VDD3=2.5 ~ 3.6V, VDD1=1.65~3.3V, T_A=-40 ~ 85 °C)

Item	Symbol	Test Condition	Min.	Typ.	Max.	Unit
Input high voltage	V_{IH}	VDD1= 1.65 ~ 3.3V VDD3= 2.5 ~ 3.3V	0.7 VDD1	-	VDD1	V
Input low voltage	V_{IL}		0	-	0.3 VDD1	V
VPP	V_{IH} V_{IL}	VPP	7.25V	7.5V	7.75V	V
Output high voltage (SDO, CABC_PWM_OUT)	V_{OH1}	I_{OH} = -1.0 mA	0.8 VDD1	-	VDD1	V
Output low voltage (SDO, CABC_PWM_OUT)	V_{OL1}	VDD1= 1.65 ~ 2.4V I_{OL} = 1.0 mA	0	-	0.2 VDD1	V
Logic High level input current	I_{IH}	VSYNC, HSYNC	-	-	1	uA
		RESX, DCX, CSX, SCL	-	-	1	uA
	I_{IHD}	DB[7…0], SDI_SDA, DCX	-	-	1	uA
		DB[7…0]	-	-	1	uA
Logic Low level input current	I_{IL}	VSYNC, HSYNC	-1	-		uA
		RESX, DCX, CSX, SCL	-1	-		uA
	I_{ILD}	DB[7…0], SDI_SDA, DCX	-1	-		uA
		DB[7…0]	-1	-		uA
Current consumption standby mode (VDD3-VSSA)	$I_{ST(VDD)}$	VDD3=2.8V, VDD1=1.8V, HS_VCC, T_A =25°C	-	TBD	TBD	uA
Current consumption standby mode (VDD1– VSSD)	$I_{ST(VDD1)}$		-	TBD	TBD	uA
Current consumption standby mode (HS_VCC– HS_VSS)	$I_{ST(HS\_VCC)}$		-	TBD	TBD	uA

Note: 1. The VPP pin is open on normal mode and in used while OTP programming condition.

Table 8.2: DC characteristic

8.3 AC characteristics

8.3.1 I2C AC characteristics

Characteristics of SDA and SCL bus lines for I2C-bus devices

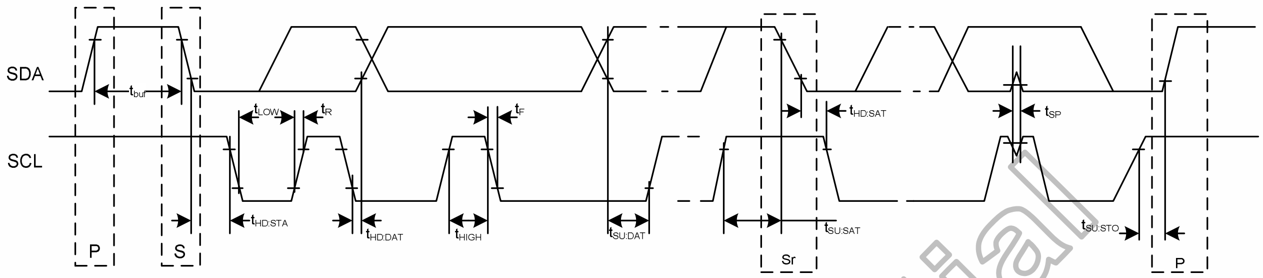


Figure 8.1 I2C timing

Parameter	Symbol	Standard-Mode I2C-BUS		Fast-Mode I2C-BUS		Unit
		Min.	Max.	Min.	Max.	
SCL clock frequency	f_{SCL}	0	100	0	400	KHz
Bus free time between STOP and START condition	t_{BUF}	4.7	-	1.3	-	μs
Hold time (repeated) START condition. After this period, the first clock pulse is generated	$t_{HD:STA}$	4.0	-	0.6	-	μs
LOW period of the SCL clock	t_{LOW}	4.7	-	1.3	-	μs
HIGH period of the SCL clock	t_{HIGH}	4.0	-	0.6	-	μs
Set-up time for a repeated START condition	$t_{SU:STA}$	4.7	-	0.6	-	μs
Data hold time	$t_{HD:DAT}$	0	-	0	0.9	μs
Data set-up time	$t_{SU:DAT}$	250	-	100	-	ns
Rise time of both SDA and SCL signals	t_R	-	1000	$20+0.1\,C_b$	300	ns
Fall time of both SDA and SCL signals	t_F	-	300	$20+0.1\,C_b$	300	ns
Set-up time for STOP condition	$t_{SU:STO}$	4.0	-	0.6	-	μs
Capacitive load for each bus line.	C_b	-	400	-	400	pF

Note: (1) All values are referred to VIH (0.7xVCCIO) and VIL (0.3xVCCIO) level.

(2) A device must internally provide a hold time of at least 300ns for the SDA signal (referred to the VIH of the SCL signal) in order to bridge the undefined region of the falling edge of SCL.

(3) The maximum $t_{HD:DAT}$ has only to be met if the device does not stretch the LOW period (t_{LOW}) of the SCL signal.

(4) A fast-mode I2C-bus device can be used in a standard-mode I2C-bus system, but the requirement $t_{SU:DAT} \geqq$ 250ns must then be met. This will automatically be the case if the device does not stretch the LOW period of the SCL signal. If such a device does stretch the LOW period of the SCL signal, it must output the next data bit to the SDA line $t_{R\,max} + t_{SU:DAT}$ = 1000+250=1250ns (according to the standard-mode I2C-bus specification) before the SCL line is released.

(5) C_b = total capacitance of one bus line in pF.

Table 8.3 I2C timing spec.

8.3.2 DBI Type C interface characteristics

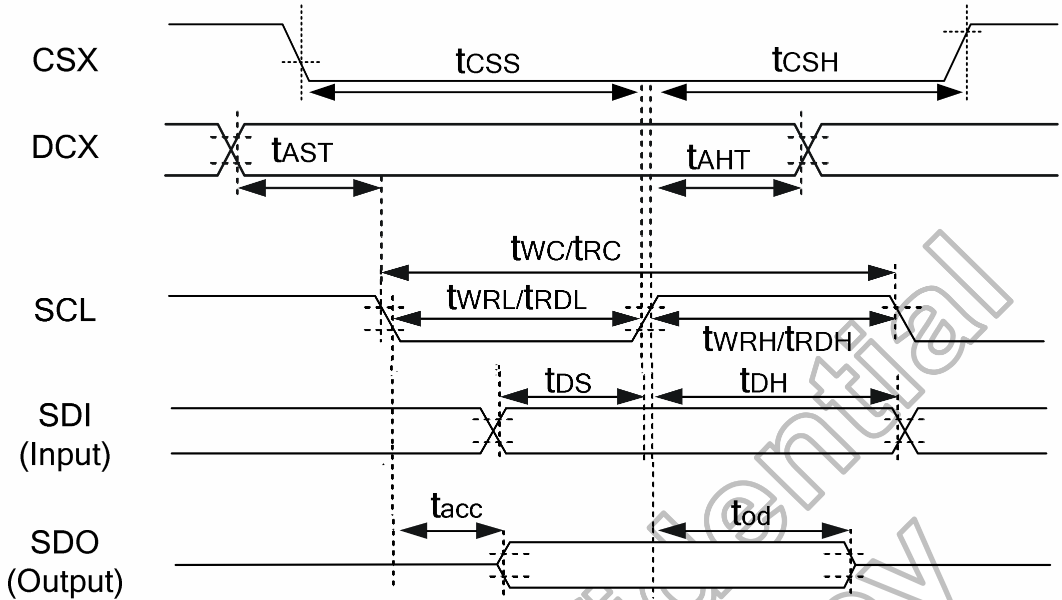


Figure 8.2: DBI Type C interface characteristics

(VSSA=0V, VDD1=1.8V, VDD3=2.8V, T_A = 25℃)

Signal	Symbol	Parameter	Min.	Max.	Unit	Description
CSX	tCSS tCSH	Chip select setup time (Write) Chip select setup time (Read)	40 40	- -	ns	-
DCX	tAST tAHT	Address setup time Address hold time (Write/Read)	10 10	- -	ns	-
SCL (Write)	tWC tWRH tWRL	Write cycle Control pulse "H" duration Control pulse "L" duration	100 40 40	- - -	ns	-
SCL (Read)	tRC tRDH tRDL	Read cycle Control pulse "H" duration Control pulse "L" duration	150 60 60	- - -	ns	-
SDI/SDO (Input)	tDS tDT	Data setup time Data hold time	30 30	- -	ns	For maximum CL=30pF For minimum CL=8pF
SDI/SDO (Output)	tRACC tOD	Read access time Output disable time	10 10	- 50	ns	

Note: The input signal rise time and fall time (tr, tf) is specified at 15 ns or less.
Logic high and low levels are specified as 30% and 70% of VDD1 for Input signals.

Table 8.4: DBI Type C interface characteristics

8.3.3 DSI D-PHY electrical characteristics

8.3.3.1 The Electrical Characteristics of D-PHY Layer

In general, the DSI - PHY may contain the following electrical functions: High-Speed Receiver (HS-RX), Low Power Transmitter (LP-TX), a Low-Power Receiver (LP-RX), and the Low-Power Contention Detector (LP-CD). Figure 8.3 shows the complete set of electrical functions required for a fully featured PHY transceiver.

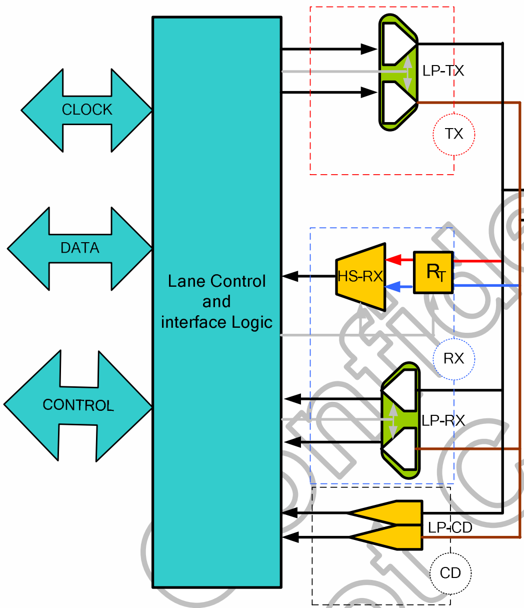


Figure 8.3: Electrical functions of a fully D-PHY transceiver

Where, the HS receiver utilize low-voltage swing differential signaling for signal transmission. The LP transmitter and LP receiver serve as a low power signaling mechanism. The Figure 8.4 shows both the HS and LP signal levels on the left and right sides, respectively.

Because the HS signaling levels are below the LP low-level input threshold, Lane switches between Low-Power and High-Speed mode during normal operation.

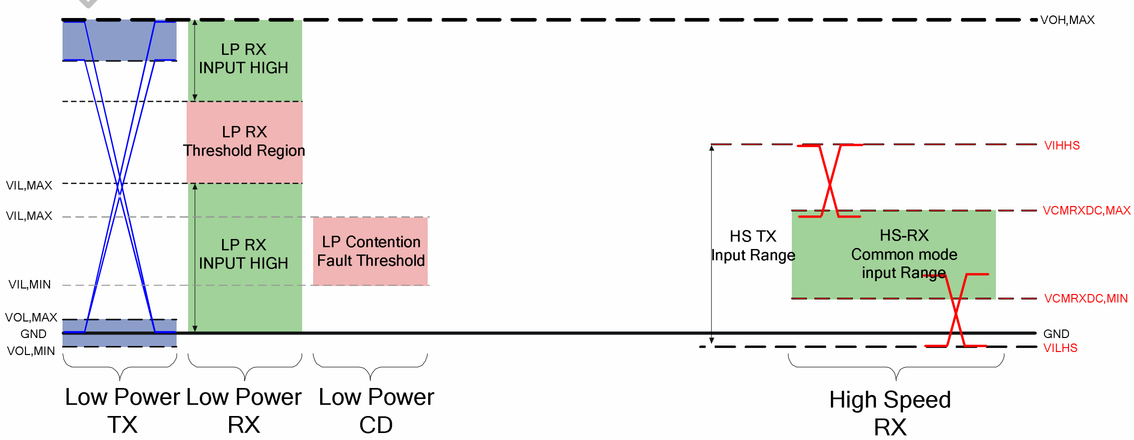


Figure 8.4: Shows both the HS and LP signal levels

8.3.3.2 The Electrical Characteristics of Low-Power Transmitter

The Low-Power transmitter shall be a slew-rate controlled push-pull driver. It is used for driving the Lines in all Low-Power operating modes It is therefore important that the static power consumption of a LP transmitter be as low as possible. Under tables list DC and AC characteristic for LP-TX

Parameter	Description	Min.	Typ.	Max.	Unit	Note
V_{OL}	Thevenin output low level	-50	-	50	mV	-
V_{OH}	Thevenin output high level	1.1	1.2	1.3	V	-
Z_{OLP}	Output impedance of LP-TX	110	-	-	Ω	(1)

Note: (1)Though no maximum value for Z_{OLP} is specified, the LP transmitter output impedance shall ensure the t_{RLP}/t_{FLP} specification is met.

Table 8.5: LP Transmitter DC Specifications

Parameter	Description	Min.	Typ.	Max.	Unit	Note
t_{RLP}/t_{FLP}	15%-85% rise time and fall time	-	-	25	ns	(1)
$\delta V/\delta t_{SR}$	Slew rate @ CLOAD = 0pF	30	-	500	mV/ns	(1),(3),(5),(6)
	Slew rate @ CLOAD = 5pF	-	-	300	mV/ns	(1),(3),(5),(6)
	Slew rate @ CLOAD = 20pF	-	-	250	mV/ns	(1),(3),(5),(6)
	Slew rate @ CLOAD = 70pF	-	-	150	mV/ns	(1),(3),(5),(6)
	Slew rate @ CLOAD = 0 to 70pF (Falling Edge Only)	30	-	-	mV/ns	(1),(2),(3)
	Slew rate @ CLOAD = 0 to 70pF (Rising Edge Only)	30	-	-	mV/ns	(1),(3),(7)
	Slew rate @ CLOAD = 0 to 70pF (Rising Edge Only)	30 – 0.075 * (VO,INST- 700)	-	-	mV/ns	(1),(8),(9)
C_{LOAD}	Load capacitance	-	-	70	pF	-

Note: (1) CLOAD includes the low-frequency equivalent transmission line capacitance. The capacitance of TX and RX are assumed to always be <10pF. The distributed line capacitance can be up to 50pF for a transmission line with 2ns delay.
(2) When the output voltage is between 400 mV and 930 mV.
(3) Measured as average across any 50 mV segment of the output signal transition.
(4) This parameter value can be lower than TLPX due to differences in rise vs. fall signal slopes and trip levels and mismatches between Dp and Dn LP transmitters.
(5) This value represents a corner point in a piecewise linear curve.
(6) When the output voltage is in the range specified by VPIN(absmax).
(7) When the output voltage is between 400 mV and 700 mV.
(8) Where VO,INST is the instantaneous output voltage, VDP or VDN, in millivolts.
(9) When the output voltage is between 700 mV and 930 mV.

Table 8.6: LP Transmitter AC Specifications

8.3.3.3 The Electrical Characteristics of Receiver

This part will contain two parts which High-Speed Receiver and Low-Power Receiver. Because their have differential DC and AC characteristic, describe HS-RX first then describe LP-RX.

8.3.3.4 High-Speed Receiver

The HS receiver is a differential line receiver. It contains a switch-able parallel input termination, ZID, between the positive input pin Dp and the negative input pin Dn. Under Tables list DC and AC characteristic for HS-RX.

Parameter	Description	Min.	Typ.	Max.	Unit	Note
V_{IDTH}	Differential input high threshold	-	-	70	mV	-
V_{IDTL}	Differential input low threshold	-70	-	-	mV	-
V_{ILHS}	Single-ended input low voltage	-40	-	-	mV	(1)
V_{IHHS}	Single-ended input high voltage	-	-	460	mV	(1)
V_{CMRXDC}	Common-mode voltage HS receive mode	70	-	330	mV	(1),(2)
Z_{ID}	Differential input impedance	80	100	125	Ω	-

Note: (1) Excluding possible additional RF interference of 100mV peak sine wave beyond 450MHz.
(2) This table value includes a ground difference of 50mV between the transmitter and the receiver, the static common-mode level tolerance and variations below 450MHz

Table 8.7: HS Receiver DC Specifications

Parameter	Description	Min.	Typ.	Max.	Unit	Note
$\Delta V_{CMRX(HF)}$	Common mode interference beyond 450 MHz	-	-	100	mV_{PP}	(1)
C_{CM}	Common mode termination	-	-	60	pF	(2)

Note: (1) ΔVCMRX(HF) is the peak amplitude of a sine wave superimposed on the receiver inputs.
(2) For higher bit rates a 14pF capacitor will be needed to meet the common-mode return loss specification.

Table 8.8: HS Receiver AC Specifications

8.3.3.5 Low-Power Receiver

The low power receiver is an un-terminated, single-ended receiver circuit. The LP receiver is used to detect the Low-Power state on each pin. For high robustness, the LP receiver shall filter out noise pulses and RF interference. It is recommended the implementer optimize the LP receiver design for low power. The LP receiver shall reject any input glitch when the glitch is smaller than eSPIKE. The filter shall allow pulses wider than TMIN to propagate through the LP receiver. The related diagram shows as Figure 8.5 Input Glitch Rejection of Low-Power Receivers. Besides, under tables list DC and AC characteristic for LP-RX.

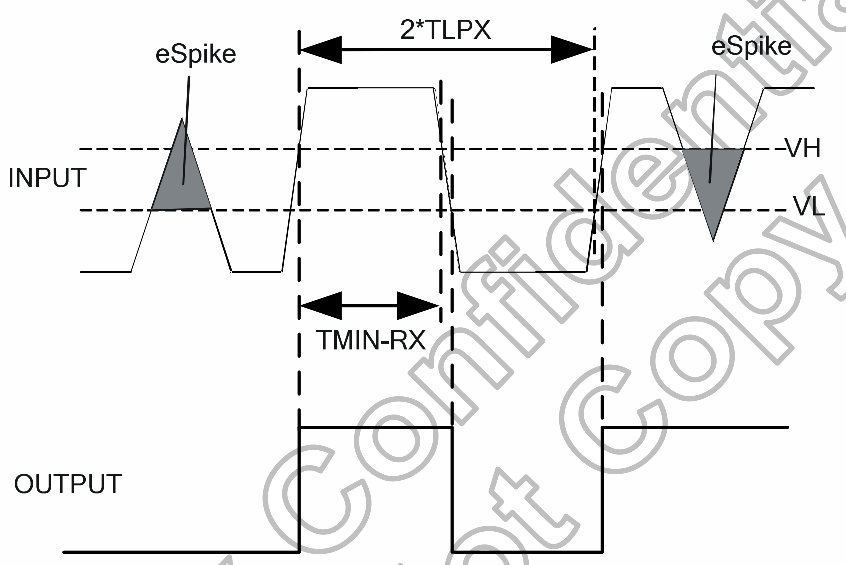


Figure 8.5: Input Glitch Rejections of Low-Power Receivers

Parameter	Description	Min.	Typ.	Max.	Unit	Note
V_{IL}	Logic 0 input threshold	-	-	550	mV	-
V_{IH}	Logic 1 input threshold	880	-	-	mV	-

Table 8.9: LP Receiver DC Specifications

Parameter	Description	Min.	Typ.	Max.	Unit	Note
e_{SPIKE}	Input pulse rejection	-	-	300	V.ps	1, 2, 3
T_{MIN}	Minimum pulse width response	20	-	-	ns	4
V_{INT}	Peak-to-peak interference voltage	-	-	200	mV	-
f_{INT}	Interference frequency	450	-	-	MHz	-

Note: (1) Time-voltage integration of a spike above VIL when being in LP-0 state or below VIH when being in LP-1 state
(2) An impulse less than this will not change the receiver state.
(3) In addition to the required glitch rejection, implementers shall ensure rejection of known RF-interferers.
(4) An input pulse greater than this shall toggle the output.

Table 8.10: LP Receiver AC Specifications

8.3.3.6 Line Contention Detection

Contention can be inferred from any of the following conditions:

A. An LP high fault shall be detected when the LP transmitter is driving high and the pin voltage is less than VIL.
B. An LP low fault shall be detected when the LP transmitter is driving low and the pad pin voltage is greater than VILF.

Parameter	Description	Min.	Typ.	Max.	Unit	Note
V_{IHCD}	Logic 1 contention threshold	450	-	-	mV	-
V_{ILCD}	Logic 0 contention threshold	-	-	200	mV	-

Table 8.11: Contention Detector DC Specifications

8.3.3.7 High-Speed Data-Clock Timing

This section specifies the required timings on the high-speed signaling interface independent of the electrical characteristics of the signal. The PHY is a source synchronous interface in the Forward direction. In either the Forward or Reverse signaling modes there shall be only one clock source. In the Reverse direction, Clock is sent in the Forward direction and one of four possible edges is used to launch the data.

The Master side of the Link shall send a differential clock signal to the Slave side to be used for data sampling. This signal shall be a DDR (half-rate) clock and shall have one transition per data bit time. All timing relationships required for correct data sampling are defined relative to the clock transitions. Therefore, implementations may use frequency spreading modulation on the clock to reduce EMI.

The DDR clock signal shall maintain a quadrature phase relationship to the data signal. Data shall be sampled on both the rising and falling edges of the Clock signal. The term “rising edge” means “rising edge of the differential signal, i.e. CP – CN, and similarly for “falling edge”. Therefore, the period of the Clock signal shall be the sum of two successive instantaneous data bit times. This relationship is shown in Figure 8.6.

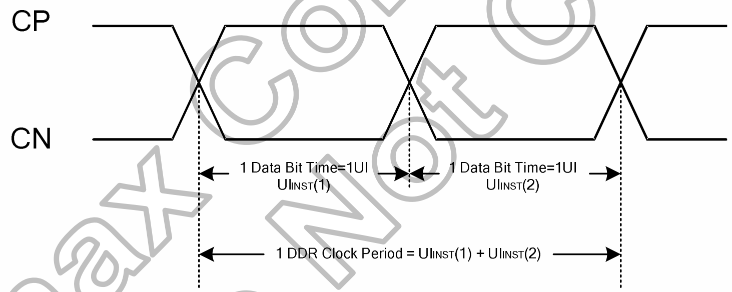


Figure 8.6: DDR Clock Definition

The same clock source is used to generate the DDR Clock and launch the serial data. Since the Clock and Data signals propagate together over a channel of specified skew, the Clock may be used directly to sample the Data lines in the receiver. Such a system can accommodate large instantaneous variations in UI.

The allowed instantaneous UI variation can cause large, instantaneous data rate variations. Therefore, devices shall either accommodate these instantaneous variations with appropriate FIFO logic outside of the PHY or provide an accurate clock source to the Lane Module to eliminate these instantaneous variations.

The UIINST specifications for the Clock signal are summarized in Table 8.12.

Parameter	Symbol	Min.	Typ.	Max.	Unit	Note
UI instantaneous	UI_{INST}	-	-	12.5	ns	(1), (2), (3)

Note: (1) This value corresponds to a minimum 80 Mbps data rate.
(2) The minimum UI shall not be violated for any single bit period, i.e., any DDR half cycle within a data burst.
(3) Maximum total bit rate is 4Gbps of 4 data lanes 24-bit data format/ 3Gbps of 4 data lane 18-bit data format/ 2.67Gbps of 4 data lane 16-bit data format.

Table 8.12: Re verse HS Data Transmission Timing Parameters

The timing relationship of the DDR Clock differential signal to the Data differential signal is shown in Figure 8.7. Data is launched in a quadrature relationship to the clock such that the Clock signal edge may be used directly by the receiver to sample the received data.

The transmitter shall ensure that a rising edge of the DDR clock is sent during the first payload bit of a transmission burst such that the first payload bit can be sampled by the receiver on the rising clock edge, the second bit can be sampled on the falling edge, and all following bits can be sampled on alternating rising and falling edges.

All timing values are measured with respect to the actual observed crossing of the Clock differential signal. The effects due to variations in this level are included in the clock to data timing budget.

Receiver input offset and threshold effects shall be accounted as part of the receiver setup and hold parameters.

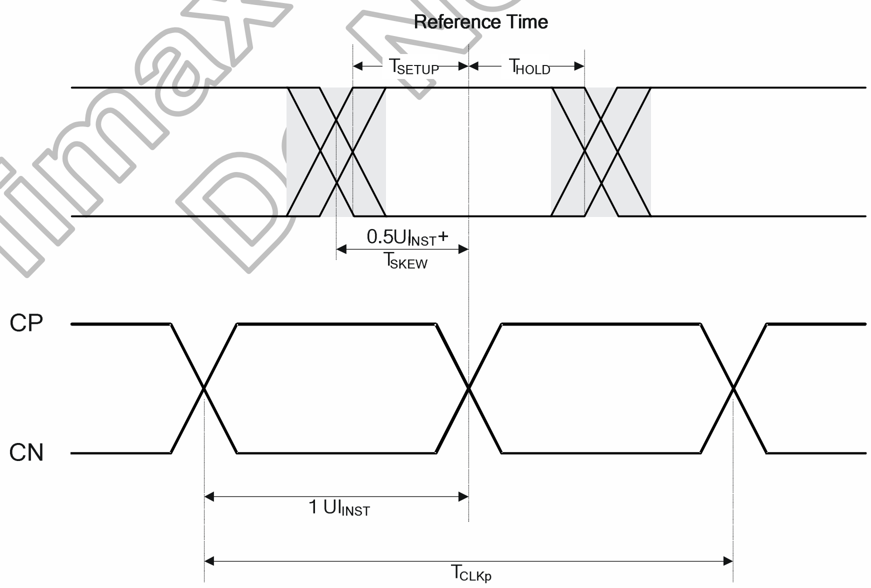


Figure 8.7: Data to Clock Timing Definitions

8.3.3.8 Data-Clock Timing Specifications

The Data-Clock timing specifications are shown in Table 8.13. Implementers shall specify a value UIINST,MIN that represents the minimum instantaneous UI possible within a High-Speed data transfer for a given implementation. Parameters in Table 8.13 are specified as a part of this value. The skew specification, TSKEW[TX], is the allowed deviation of the data launch time to the ideal ½UIINST displaced quadrature clock edge. The setup and hold times, TSETUP[RX] and THOLD[RX], respectively, describe the timing relationships between the data and clock signals. TSETUP[RX] is the minimum time that data shall be present before a rising or falling clock edge and THOLD[RX] is the minimum time that data shall remain in its current state after a rising or falling clock edge. The timing budget specifications for a receiver shall represent the minimum variations observable at the receiver for which the receiver will operate at the maximum specified acceptable bit error rate.

The intent in the timing budget is to leave 0.2*UIINST, i.e. ±0.1*UIINST for degradation contributed by the interconnect.

Parameter	Symbol	Min.	Typ.	Max.	Unit	Note
Data to Clock Setup Time [Receiver]	$T_{SETUP[RX]}$	0.2	-	-	UIINST	1
Clock to Data Hold Time [Receiver]	$T_{HOLD[RX]}$	0.2	-	-	UIINST	1

Note: (1) Total setup and hold window for receiver of 0.4*UIINST.

Table 8.13: Data to Clock Timing Specifications

8.3.4 Timings for DSI Video mode

8.3.4.1 Vertical Timings

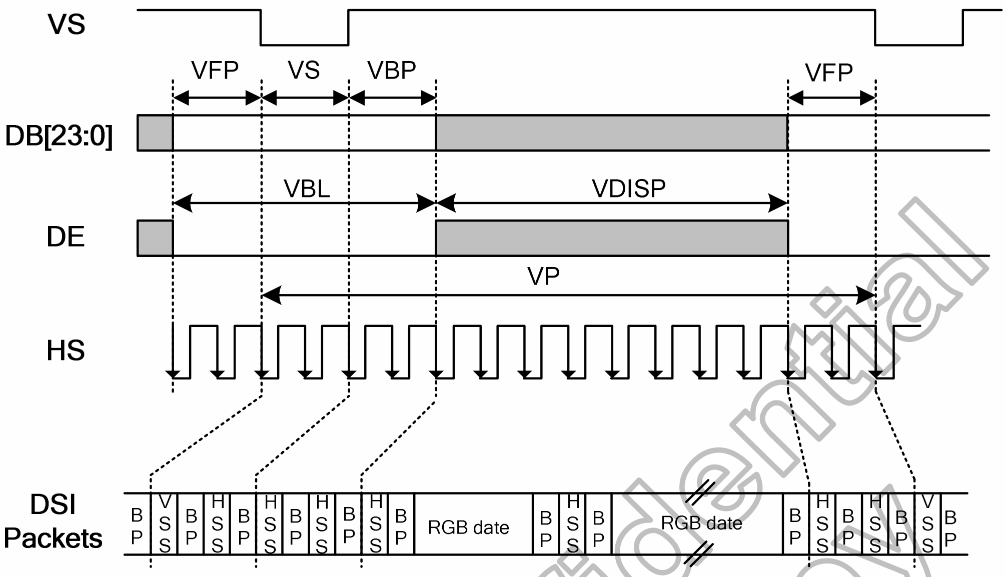


Figure 8.8: Vertical Timings for DSI I/F

Resolution=1200x1920/1080x1920(VSSA=0V, VDD1=1.8V, VDD3=2.8V, T_A=25℃)

Item	Symbol	Condition	Min.	Typ.	Max.	Unit
Vertical cycle	**VP**	**-**	**1926**	**-**	**-**	**Line**
Vertical low pulse width	**VS**	**-**	**2**	**-**	**Note(1)**	**Line**
Vertical front porch	**VFP**	**-**	**2**	**-**	**-**	**Line**
Vertical back porch	**VBP**	**-**	**2**	**-**	**Note(1)**	**Line**
Vertical data start point	**-**	**VS+VBP**	**4**	**-**	**Note(1)**	**Line**
Vertical blanking period	**VBL**	**VS+VBP+VFP**	**6**	**-**	**-**	**Line**
Vertical active area	**-**	**VDISP**	**-**	**1920**	**-**	**Line**
Vertical Refresh rate	**VRR**	**-**	**-**	**60**	**-**	**Hz**

Resolution=1080x1440/960x1440/900x1440 (VSSA=0V, VDD1=1.8V, VDD3=2.8V, T_A=25℃)

Item	Symbol	Condition	Min.	Typ.	Max.	Unit
Vertical cycle	**VP**	**-**	**1446**	**-**	**-**	**Line**
Vertical low pulse width	**VS**	**-**	**2**	**-**	**Note(1)**	**Line**
Vertical front porch	**VFP**	**-**	**2**	**-**	**-**	**Line**
Vertical back porch	**VBP**	**-**	**2**	**-**	**Note(1)**	**Line**
Vertical data start point	**-**	**VS+VBP**	**4**	**-**	**Note(1)**	**Line**
Vertical blanking period	**VBL**	**VS+VBP+VFP**	**6**	**-**	**-**	**Line**
Vertical active area	**-**	**VDISP**	**-**	**1440**	**-**	**Line**
Vertical Refresh rate	**VRR**	**-**	**-**	**60**	**-**	**Hz**

Resolution=1024x1600/960x1600/900x1600 (VSSA=0V, VDD1=1.8V, VDD3=2.8V, T_A=25℃)

Item	Symbol	Condition	Min.	Typ.	Max.	Unit
Vertical cycle	**VP**	**-**	**1606**	**-**	**-**	**Line**
Vertical low pulse width	**VS**	**-**	**2**	**-**	**Note(1)**	**Line**
Vertical front porch	**VFP**	**-**	**2**	**-**	**-**	**Line**
Vertical back porch	**VBP**	**-**	**2**	**-**	**Note(1)**	**Line**
Vertical data start point	**-**	**VS+VBP**	**4**	**-**	**Note(1)**	**Line**
Vertical blanking period	**VBL**	**VS+VBP+VFP**	**6**	**-**	**-**	**Line**
Vertical active area	**-**	**VDISP**	**-**	**1600**	**-**	**Line**
Vertical Refresh rate	**VRR**	**-**	**-**	**60**	**-**	**Hz**

Resolution=1024x1280/960x1280/800x1280/720x1280 (VSSA=0V, VDD1=1.8V, VDD3=2.8V, T_A=25℃)

Item	Symbol	Condition	Min.	Typ.	Max.	Unit
Vertical cycle	**VP**	**-**	**1286**	**-**	**-**	**Line**
Vertical low pulse width	**VS**	**-**	**2**	**-**	**Note(1)**	**Line**
Vertical front porch	**VFP**	**-**	**2**	**-**	**-**	**Line**
Vertical back porch	**VBP**	**-**	**2**	**-**	**Note(1)**	**Line**
Vertical data start point	**-**	**VS+VBP**	**4**	**-**	**Note(1)**	**Line**
Vertical blanking period	**VBL**	**VS+VBP+VFP**	**6**	**-**	**-**	**Line**
Vertical active area	**-**	**VDISP**	**-**	**1280**	**-**	**Line**
Vertical Refresh rate	**VRR**	**-**	**-**	**60**	**-**	**Hz**

Note: (1) The VS and VBP pulse width are related to GSP and GCK timing. The GSP and GCK must be set at corresponding position for LCD normal display. Also refer to setion 6.3.5 SETCYC.

Table 8.14: Vertical Timings for DSI I/F

8.3.4.2 Horizontal Timings

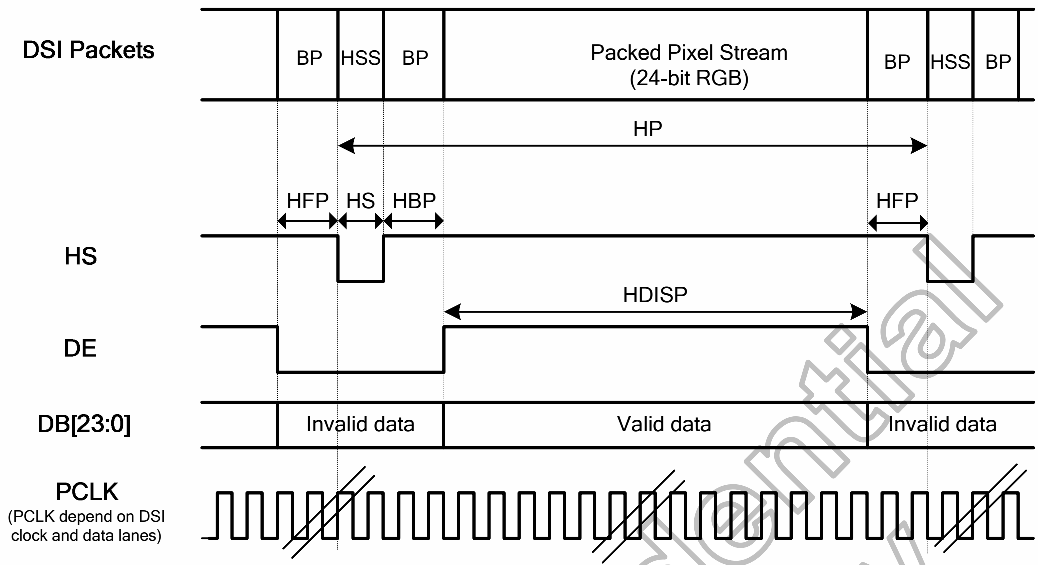


Figure 8.9: Horizontal Timing for DSI Video mode I/F

Resolution=1200x1920 (VSSA=0V, VDD1=1.8V, VDD3=HS_VCC=2.8V, T_A=25℃)

Item	Symbol	Condition	Min.	Typ.	Max.	Unit
HS cycle	**HP**	**-**	**1215**	**-**	**-**	**DCK**
HS low pulse width	**HS**	**-**	**5**	**-**	**-**	**DCK**
Horizontal back porch	**HBP**	**-**	**5**	**-**	**-**	**DCK**
Horizontal front porch	**HFP**	**-**	**5**	**-**	**-**	**DCK**
Horizontal data start point	**-**	**HS+HBP**	**10**	**-**	**-**	**DCK**
Horizontal blanking period	**HBLK**	**HS+HBP+HFP**	**15**	**-**	**-**	**DCK**
Horizontal active area	**HDISP**	**-**	**-**	**1200**	**-**	**DCK**

Resolution=1080x1920/1080x1440 (VSSA=0V, VDD1=1.8V, VDD3=HS_VCC=2.8V, T_A=25℃)

Item	Symbol	Condition	Min.	Typ.	Max.	Unit
HS cycle	**HP**	**-**	**1095**	**-**	**-**	**DCK**
HS low pulse width	**HS**	**-**	**5**	**-**	**-**	**DCK**
Horizontal back porch	**HBP**	**-**	**5**	**-**	**-**	**DCK**
Horizontal front porch	**HFP**	**-**	**5**	**-**	**-**	**DCK**
Horizontal data start point	**-**	**HS+HBP**	**10**	**-**	**-**	**DCK**
Horizontal blanking period	**HBLK**	**HS+HBP+HFP**	**15**	**-**	**-**	**DCK**
Horizontal active area	**HDISP**	**-**	**-**	**1080**	**-**	**DCK**

Resolution=1024x1600/1024x1280 (VSSA=0V, VDD1=1.8V, VDD3=2.8V, T_A=25℃)

Item	Symbol	Condition	Min.	Typ.	Max.	Unit
HS cycle	**HP**	**-**	**1039**	**-**	**-**	**DCK**
HS low pulse width	**HS**	**-**	**5**	**-**	**-**	**DCK**
Horizontal back porch	**HBP**	**-**	**5**	**-**	**-**	**DCK**
Horizontal front porch	**HFP**	**-**	**5**	**-**	**-**	**DCK**
Horizontal data start point	**-**	**HS+HBP**	**10**	**-**	**-**	**DCK**
Horizontal blanking period	**HBLK**	**HS+HBP+HFP**	**15**	**-**	**-**	**DCK**
Horizontal active area	**HDISP**	**-**	**-**	**1024**	**-**	**DCK**

Resolution=960x1600/960x1440/960x1280 (VSSA=0V, VDD1=1.8V, VDD3=2.8V, T_A=25℃)

Item	Symbol	Condition	Min.	Typ.	Max.	Unit
HS cycle	HP	-	975	-	-	DCK
HS low pulse width	HS	-	5	-	-	DCK
Horizontal back porch	HBP	-	5	-	-	DCK
Horizontal front porch	HFP	-	5	-	-	DCK
Horizontal data start point	-	HS+HBP	10	-	-	DCK
Horizontal blanking period	HBLK	HS+HBP+HFP	15	-	-	DCK
Horizontal active area	HDISP	-	-	960	-	DCK

Resolution=900x1600/900x1440 (VSSA=0V, VDD1=1.8V, VDD3=2.8V, T_A=25℃)

Item	Symbol	Condition	Min.	Typ.	Max.	Unit
HS cycle	HP	-	915	-	-	DCK
HS low pulse width	HS	-	5	-	-	DCK
Horizontal back porch	HBP	-	5	-	-	DCK
Horizontal front porch	HFP	-	5	-	-	DCK
Horizontal data start point	-	HS+HBP	10	-	-	DCK
Horizontal blanking period	HBLK	HS+HBP+HFP	15	-	-	DCK
Horizontal active area	HDISP	-	-	900	-	DCK

Resolution=800x1280 (VSSA=0V, VDD1=1.8V, VDD3=2.8V, T_A=25℃)

Item	Symbol	Condition	Min.	Typ.	Max.	Unit
HS cycle	HP	-	815	-	-	DCK
HS low pulse width	HS	-	5	-	-	DCK
Horizontal back porch	HBP	-	5	-	-	DCK
Horizontal front porch	HFP	-	5	-	-	DCK
Horizontal data start point	-	HS+HBP	10	-	-	DCK
Horizontal blanking period	HBLK	HS+HBP+HFP	15	-	-	DCK
Horizontal active area	HDISP	-	-	800	-	DCK

Resolution=720x1280 (VSSA=0V, VDD1=1.8V, VDD3=2.8V, T_A=25℃)

Item	Symbol	Condition	Min.	Typ.	Max.	Unit
HS cycle	HP	-	735	-	-	DCK
HS low pulse width	HS	-	5	-	-	DCK
Horizontal back porch	HBP	-	5	-	-	DCK
Horizontal front porch	HFP	-	5	-	-	DCK
Horizontal data start point	-	HS+HBP	10	-	-	DCK
Horizontal blanking period	HBLK	HS+HBP+HFP	15	-	-	DCK
Horizontal active area	HDISP	-	-	720	-	DCK

Table 8.15: Horizontal Timings for DSI Video mode I/F

8.3.5 Reset input timing

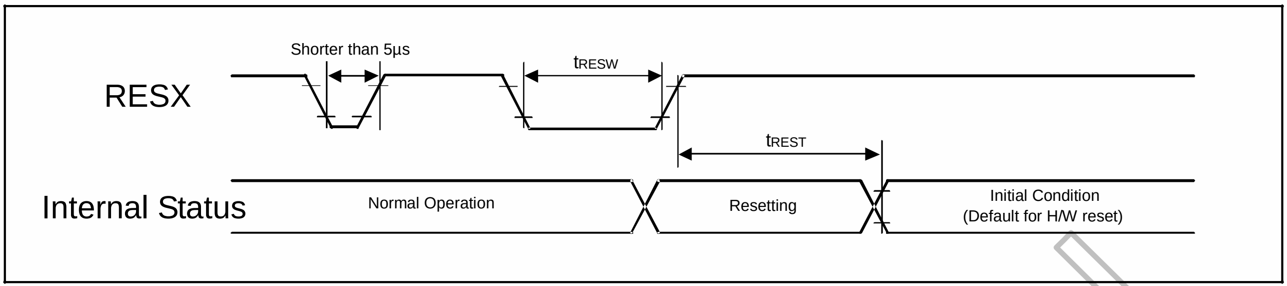


Figure 8.10: Reset input timing

Symbol	Parameter	Related pins	Min.	Typ.	Max.	Note	Unit
t_{RESW}	Reset low pulse width(1)	RESX	10	-	-	-	µs
t_{REST}	Reset complete time(2)	-	-	-	5	HX8399-A reset execute period when reset is applied during Sleep In mode	ms
		-	-	-	120	HX8399-A reset execute period when reset is applied during Sleep Out mode	ms

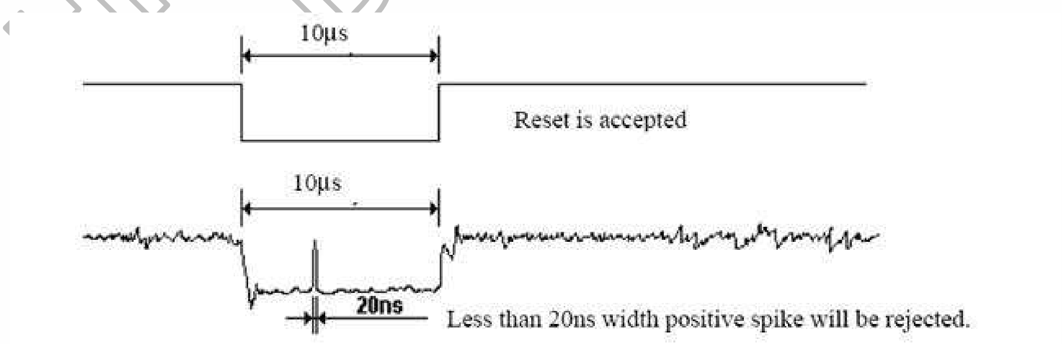
Note: (1) Spike due to an electrostatic discharge on RESX line does not cause irregular system reset according to the table below.

RESX Pulse	Action
Shorter than 5 µs	Reset Rejected
Longer than 10 µs	Reset
Between 5 µs and 10 µs	Reset Start

(2) During the resetting period, the display will be blanked (The display is entering blanking sequence, which maximum time is 120 ms, when Reset Starts in Sleep Out –mode. The display remains the blank state in Sleep In –mode) and then returns to Default condition for H/W reset.

(3) During Reset Complete Time, OTP will be latched to internal register during this period. This loading is done every time when there is H/W reset complete time (tREST) within 5ms after a rising edge of RESX.

(4) Spike Rejection also applies during a valid reset pulse as shown below:



(5) When Reset is applied during Sleep In Mode.

(6) When Reset is applied during Sleep Out Mode.

(7) It is necessary to wait 5msec after releasing RESX before sending commands. Also Sleep Out command cannot be sent for 120msec.

Table 8.16: Reset timing

9. Ordering Information

Part No.	Package
HX8399-A000 PDxxx	PD: mean COG xxx: mean chip thickness (μm), (default: 250 μm)